

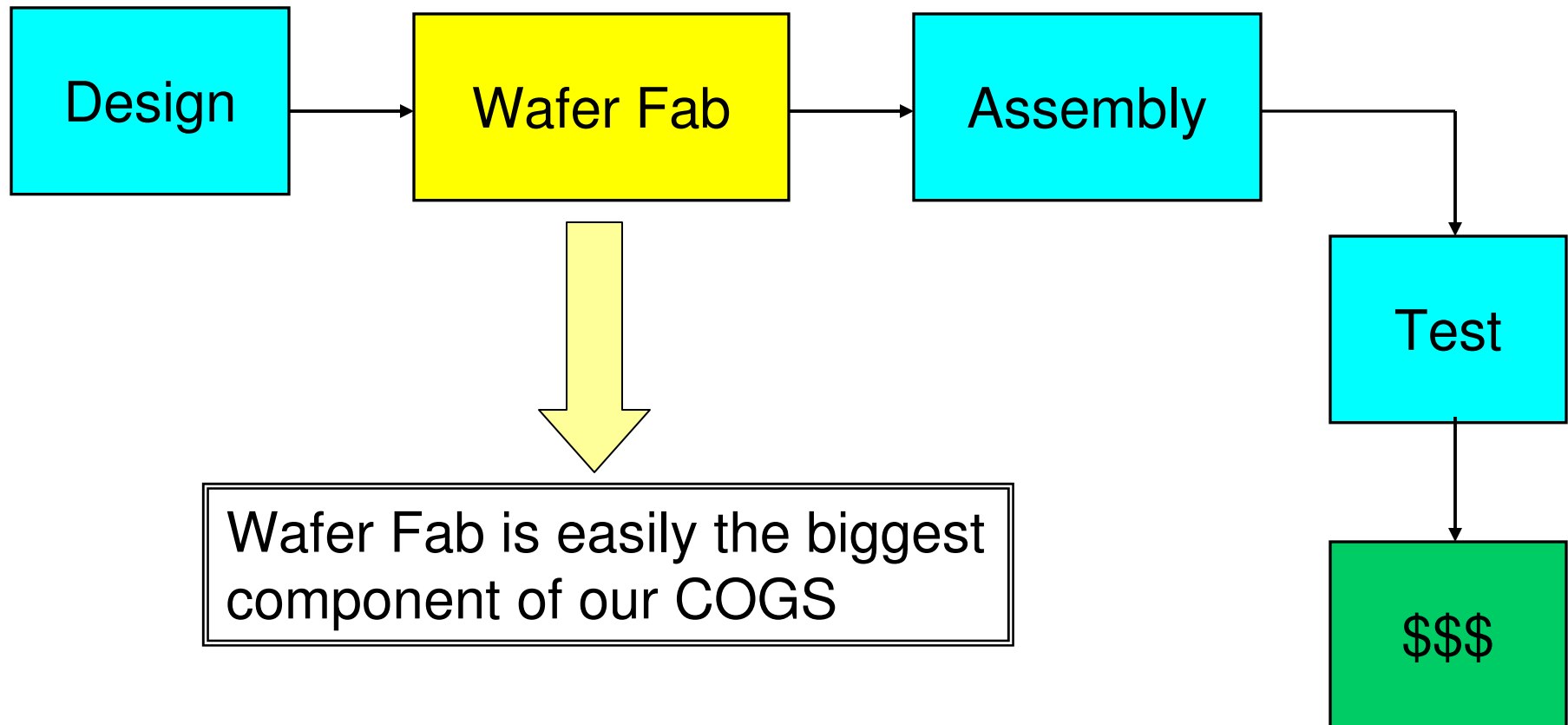


CMOS processing

With help from David Szmyd, Silicon Labs



Where Does Wafer Fab Fit into Product Flow?



SILICON LABORATORIES

What is a Wafer Fab?

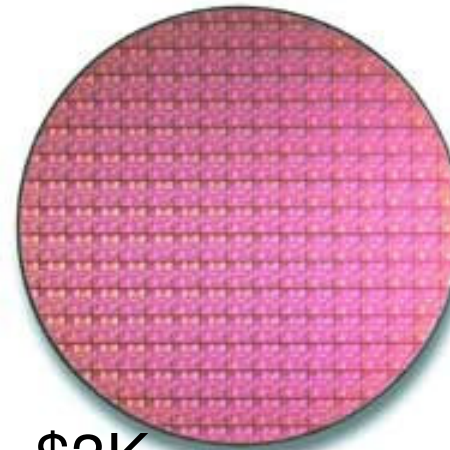


\$3B

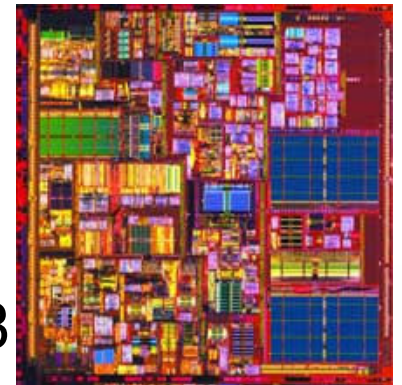


\$3M

3 decades/step



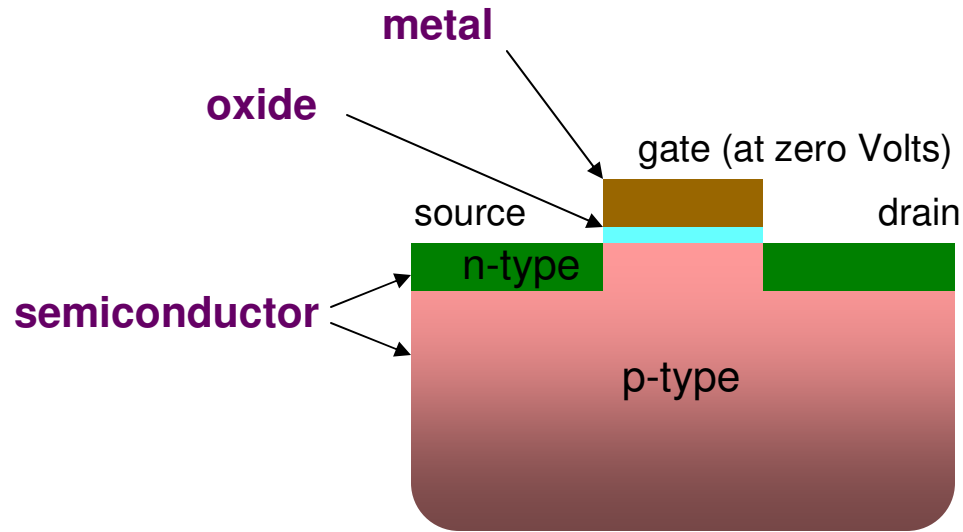
\$3K



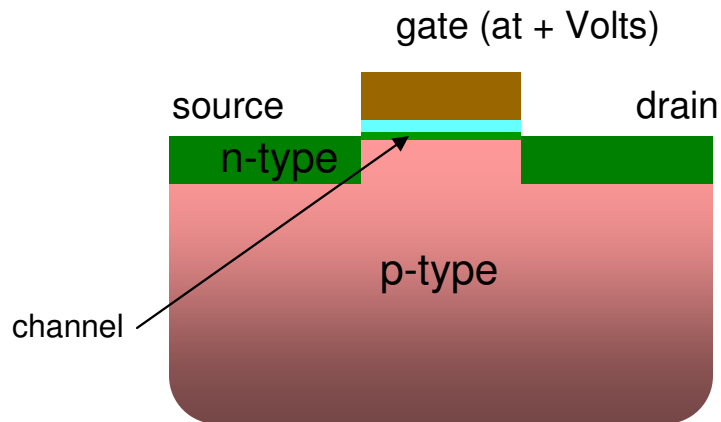
\$3

SILICON LABORATORIES

MOS Transistor Cross Section



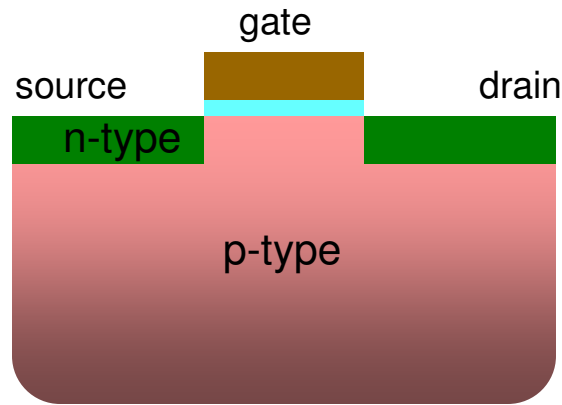
Electrons in source cannot flow to the drain because p-type region is a barrier.
Transistor is OFF.



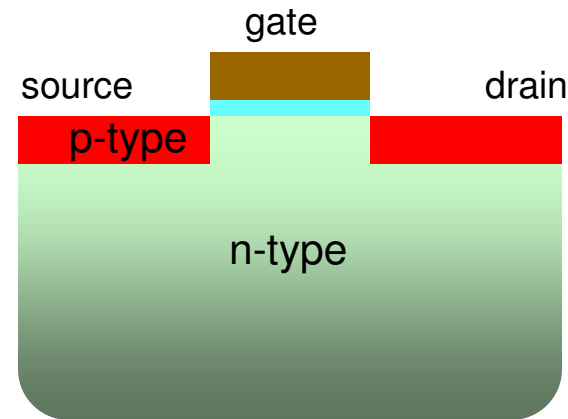
Apply positive voltage to gate. Attracts electrons to oxide, forming n-type channel. Now, electrons have a continuous path from source to drain.
Transistor is ON.

CMOS: Complementary MOS

NMOS: S, D and channel are n-type



PMOS: S, D and channel are p-type

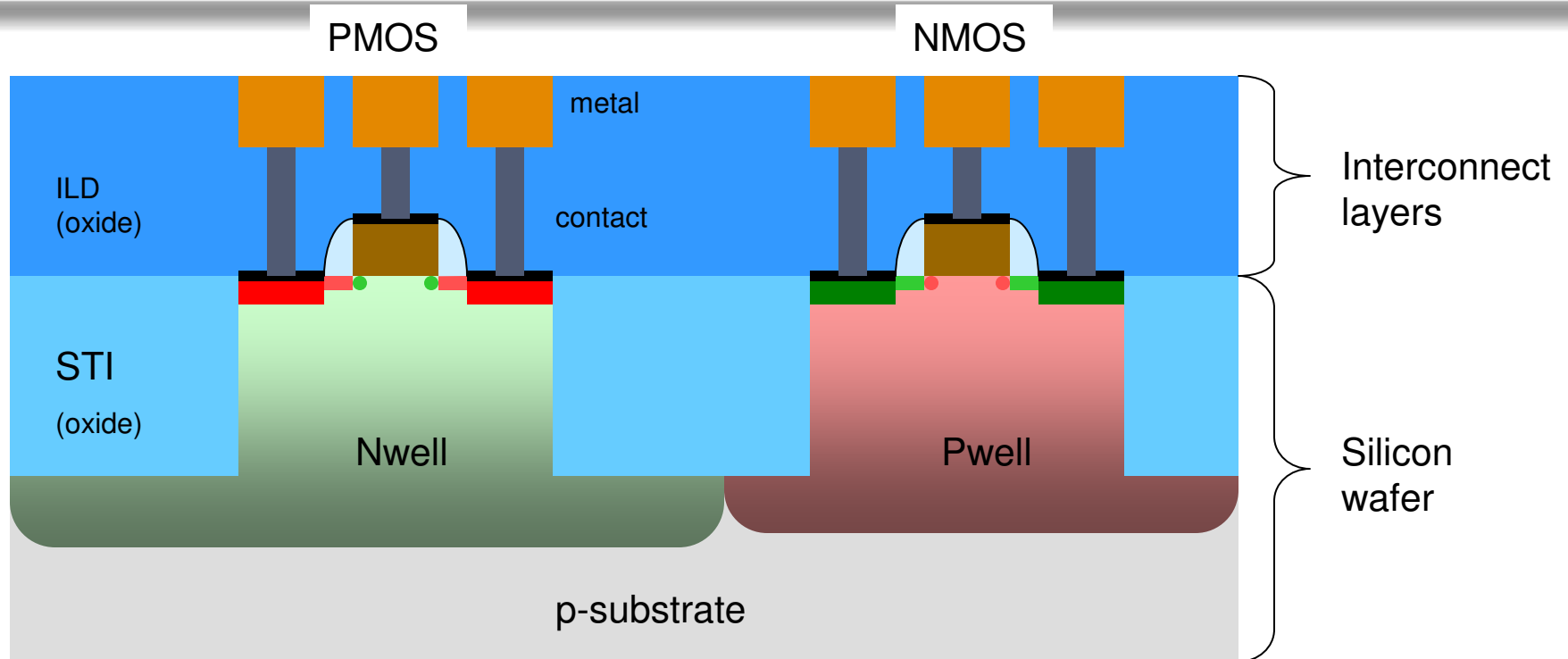


- Can combine NMOS and PMOS so that when one is on, the other is off.
 - No current flows because one device is always off. Saves power!!
 - Exception: Current flows only when devices are switching.
 - Devices are complementary → CMOS.



SILICON LABORATORIES

CMOS Integration



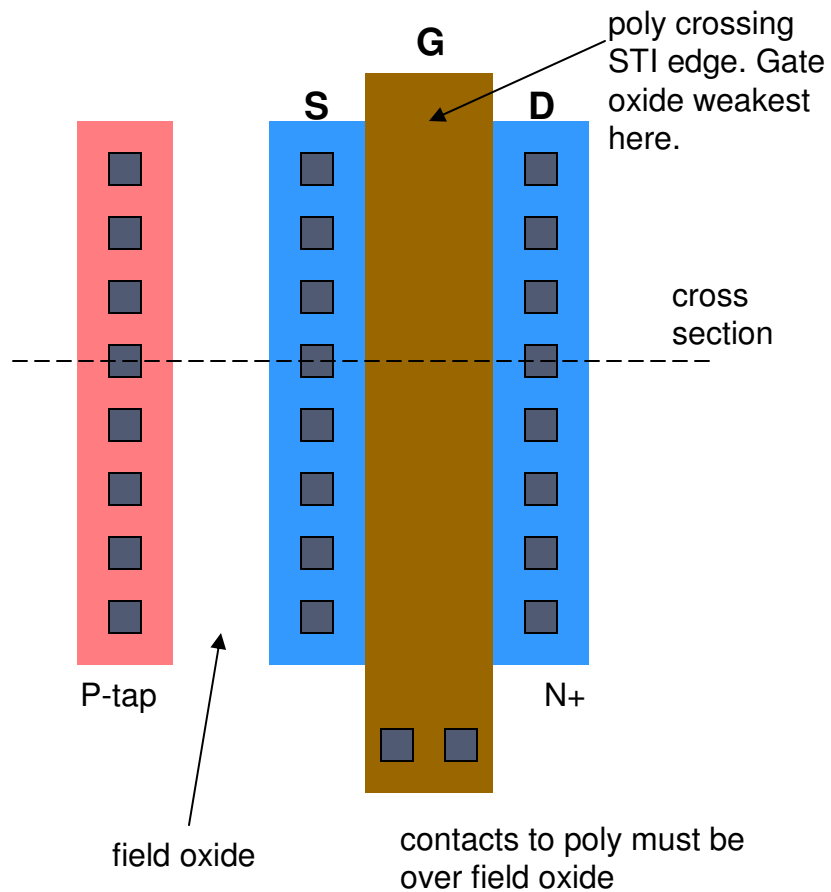
- Devices are built into a common p-type substrate (wafer).
- Shallow Trench Isolation (STI) provides electrical isolation between devices.
- Metal and contacts provide access to the device terminals S, D, G.
- Multiple levels of metal lines are routed to interconnect the devices → form a circuit on a chip.
- Capacitors, resistors and inductors can also be integrated.



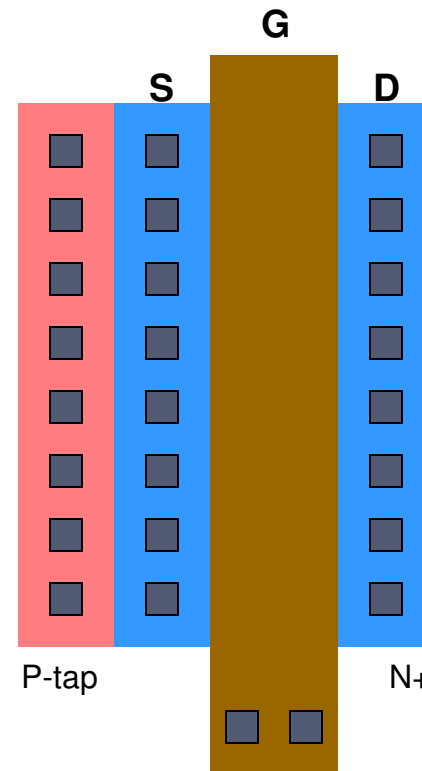
SILICON LABORATORIES

Transistor Layout

NMOS with separate well tap



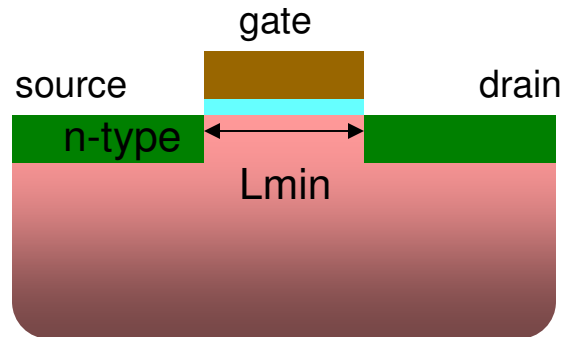
NMOS with well tap shorted to source by the silicide



SILICON LABORATORIES

Technology Nodes

Technology defined by minimum allowed gate length.



Shorter gates → faster transistors (100 GHz) and denser circuits. Each node increases density by ~2x.

0.5μm → 0.35μm → 0.25μm → 0.18μm → 0.13μm

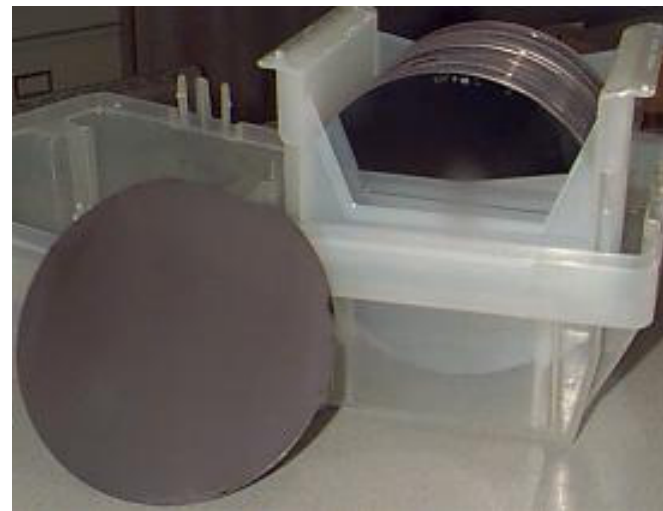
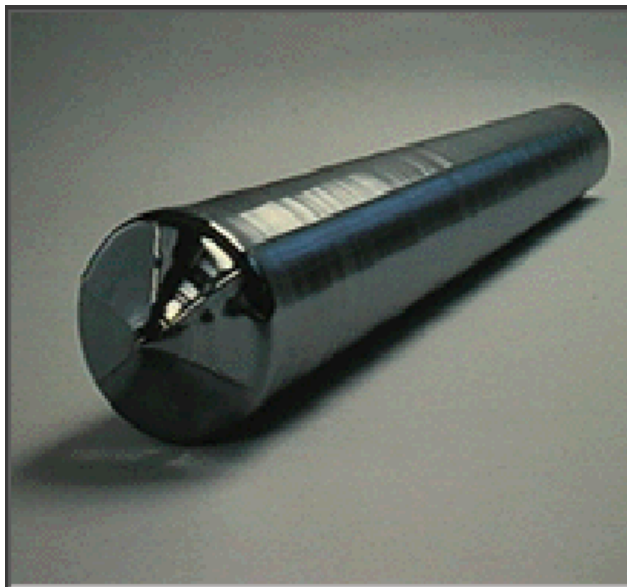
- At finer nodes, all features shrink: contact size, metal width, oxide thickness, etc.
- In 0.13μm, gate oxide thickness is only 20Å (about a dozen SiO₂ molecules).

Part	Function	Tech	Contact	M1 L/S	# metals	# masks	Size
chip1	example1	0.5μm	0.5μm	0.6/0.6μm	3	16	4.62 mm ²
chip2	Example2	0.35μm	0.4μm	0.5/0.45μm	4	21	6.07 mm ²
chip3	Example3	0.35μm	0.4μm	0.5/0.45μm	4	30	8.95 mm ²
chip4	Example4	0.25μm	0.3μm	0.32/0.32μm	4	23	5.5 mm ²
chip5	Example5	0.18μm	0.22μm	0.23/0.23μm	5	23	6.51 mm ²
chip6	Example6	0.13μm	0.16μm	0.16/0.18μm	8	37	2.93 mm ²

SILICON LABORATORIES

Wafer Fabrication

- ◆ Armed with photomasks and starting substrates, the foundry can now fabricate the wafers.
 - Starting substrates are cut and polished from huge single crystals of silicon. Not done by the foundries.



SILICON LABORATORIES

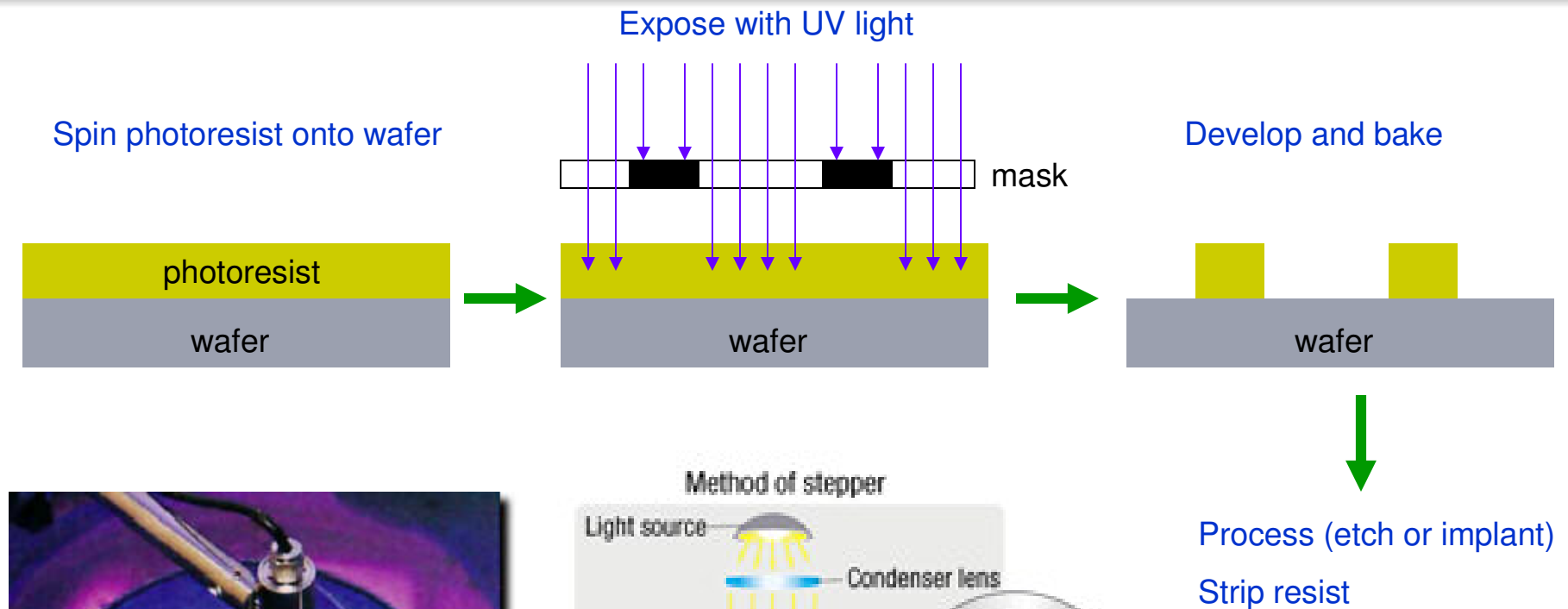
Basic Process Steps

- ◆ Photolithography – transfer mask pattern to wafer
- ◆ Implant – shoot impurities into the silicon
- ◆ Diffusion – anneal implant damage, grow oxide
- ◆ Deposition – deposit layers (oxides, metals, etc.)
- ◆ Etch – remove unwanted material
- ◆ CMP – chemo-mechanical polishing. Removes unwanted material by polishing, leaving the wafer flat.

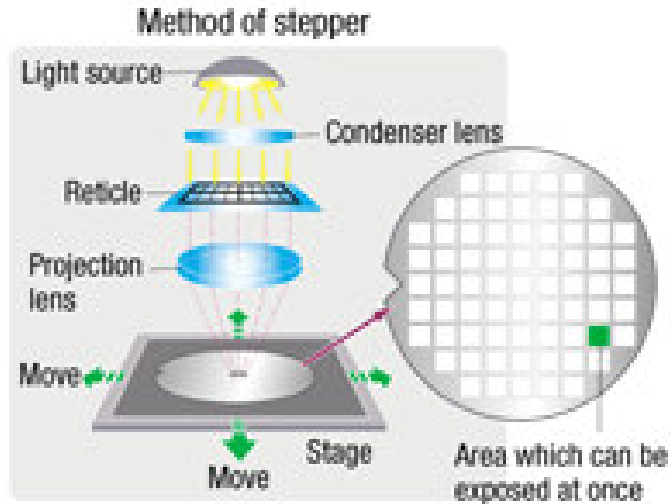


SILICON LABORATORIES

Photolithography



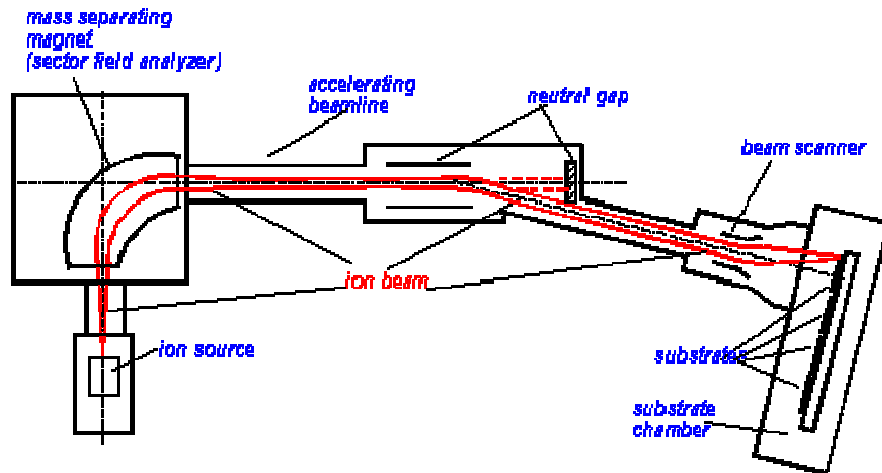
Photoresist Application
(Ontrak)



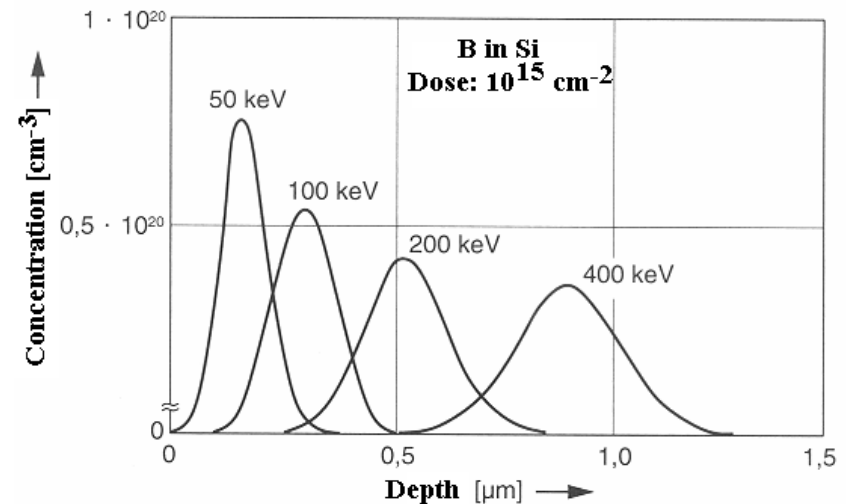
SILICON LABORATORIES

Ion Implantation

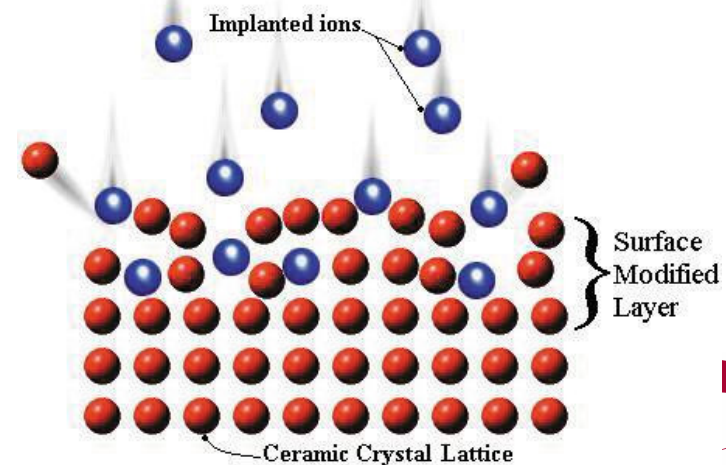
Ionize and accelerate impurity atoms into the silicon. 100eV – 1 MeV.



Profile is Gaussian. Peak is not at surface



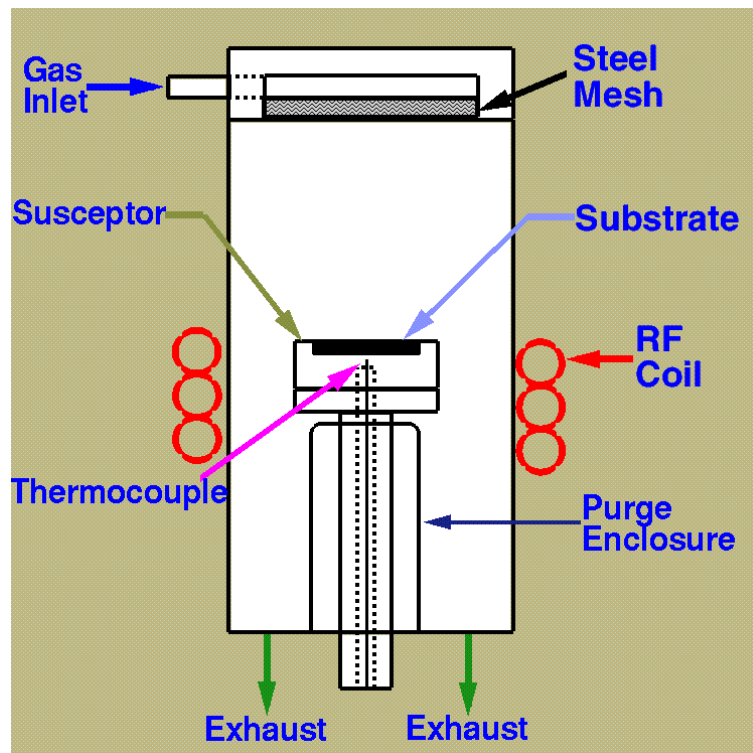
Must anneal crystal damage



SILICON LABORATORIES

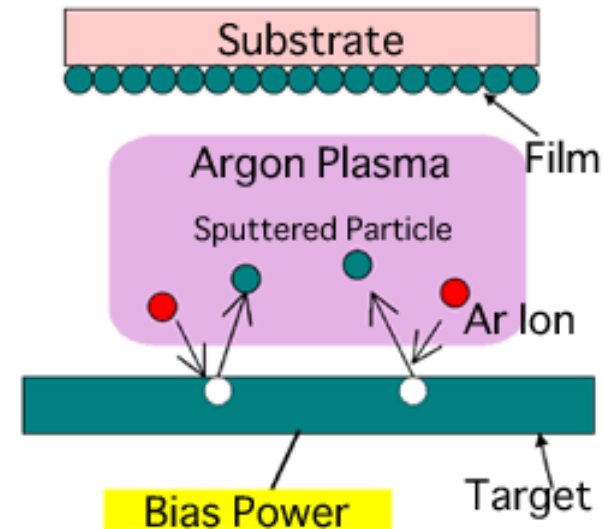
Deposition

Chemical Vapor Deposition



- React source gases inside chamber.
- RF plasma (usually)
- Reduced pressure
- Used for oxides and polysilicon, epitaxy.

Sputter material from a target onto the wafer



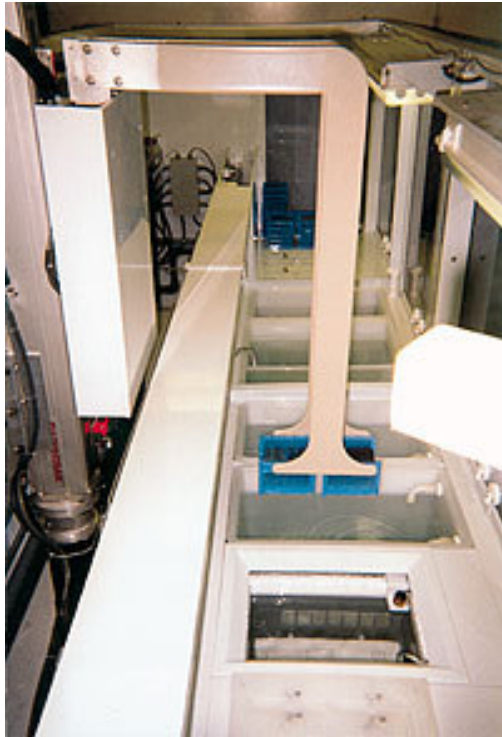
- RF plasma + magnetron.
- Argon ions physically dislodge target atoms.
- Good for depositing metals.
- Can add O_2 for reactive sputtering.



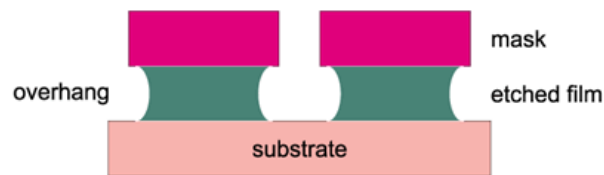
SILICON LABORATORIES

Etch

Wet Etch

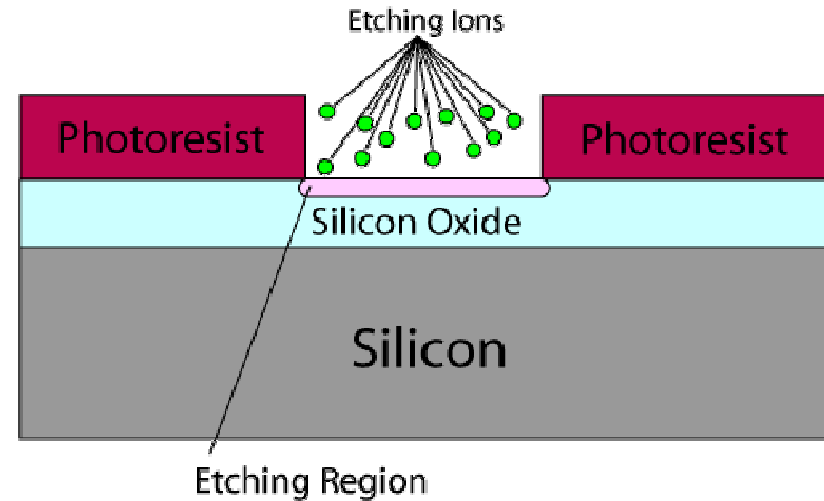


- Dip wafers boat in HF acid to remove oxide
- Isotropic

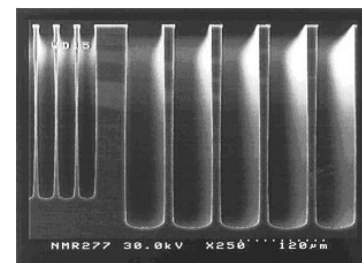


ISOTROPIC ETCHING

RIE – Reactive Ion Etch



- Use reactive ions like Cl^- or F^- to etch material.
- Single wafer tool.
- Want high selectivity.
- Usually want high anisotropy.



SILICON LABORATORIES

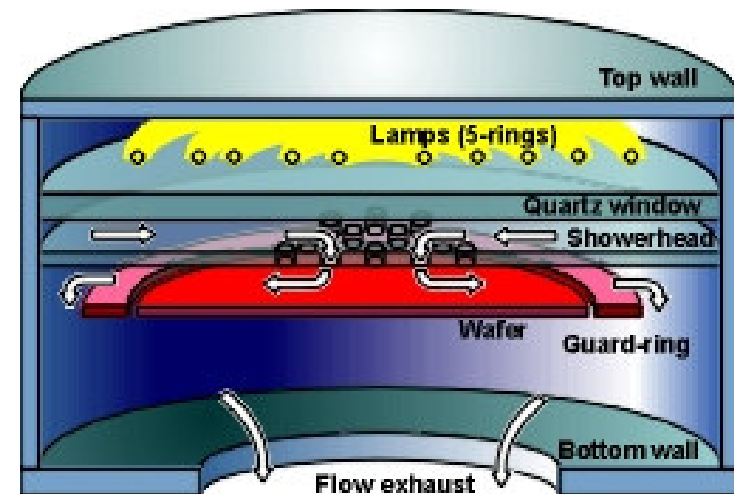
Diffusion

Furnace



- Used for long anneal or alloy
- Grow gate oxide
- Batch mode

RTP- Rapid Thermal Process



- Heat wafer extremely fast with IR lamps. Soak time of a few seconds.
- Used for impurity activation. Impurities have no time to move.
- Uniform
- Can grow films
- Single wafer tool

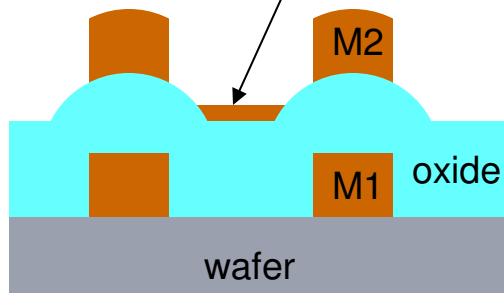


SILICON LABORATORIES

CMP: Chemo-Mechanical Polishing

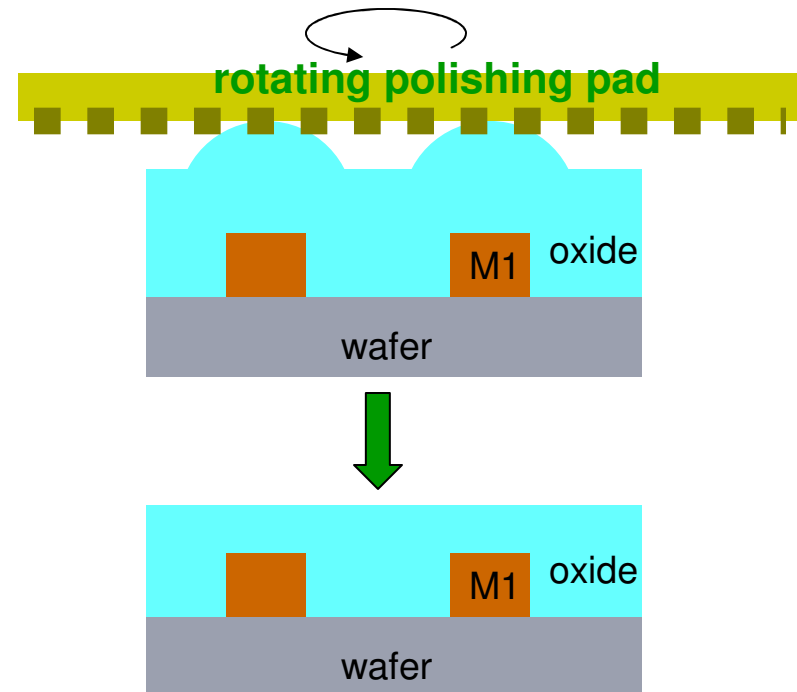
Nonplanar wafer surface

hard to etch away metal in valleys

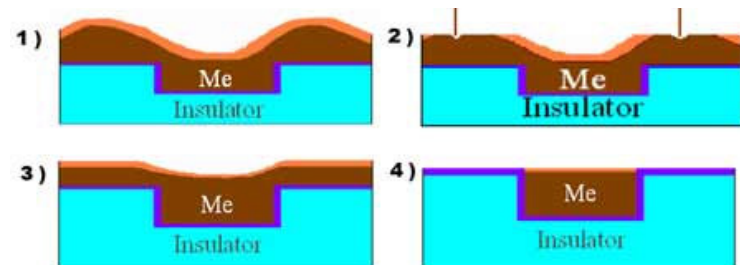


- Lowers yields.
- Limits number of interconnect layers to 2 or 3.
- Limits spacing between lines.

Polish oxide flat first



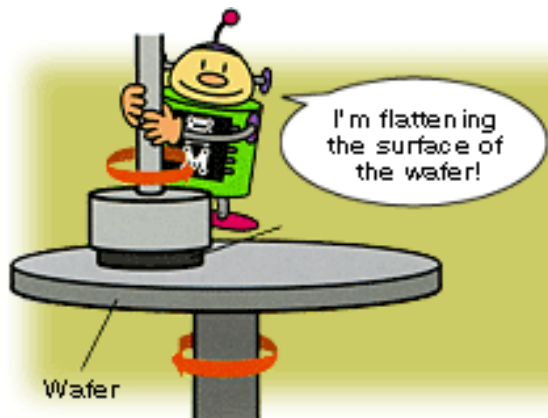
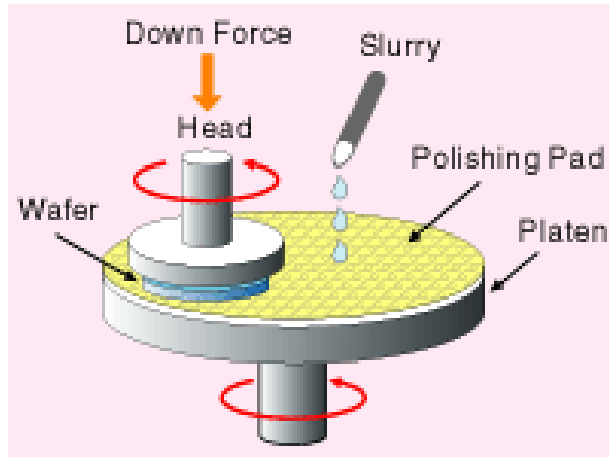
• Can also polish metal



SILICON LABORATORIES

CMP Tools

CMP Tool



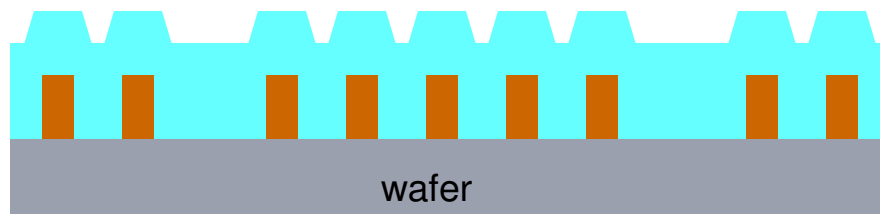
- Dirty - kept apart from other tools

CMP Density Limitation → DFM

Large open areas get overpolished

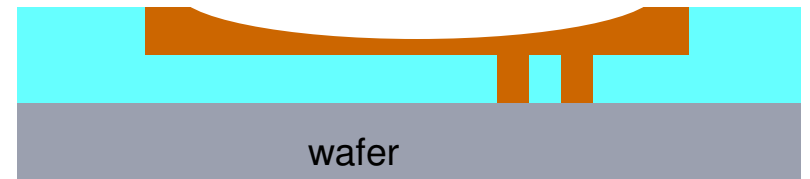
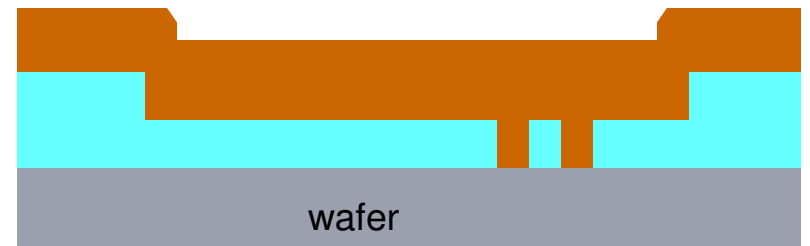


Solution. Add dummy lines to meet minimum density requirements.

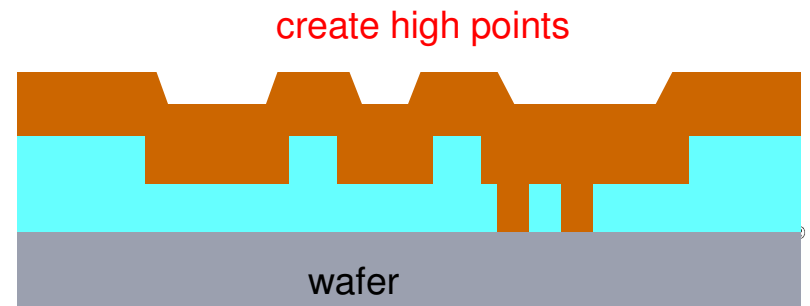


- can be done automatically
- increases parasitic capacitance

Wide metal lines get overpolished



Solution: Restrict width. Slot line.



create high points

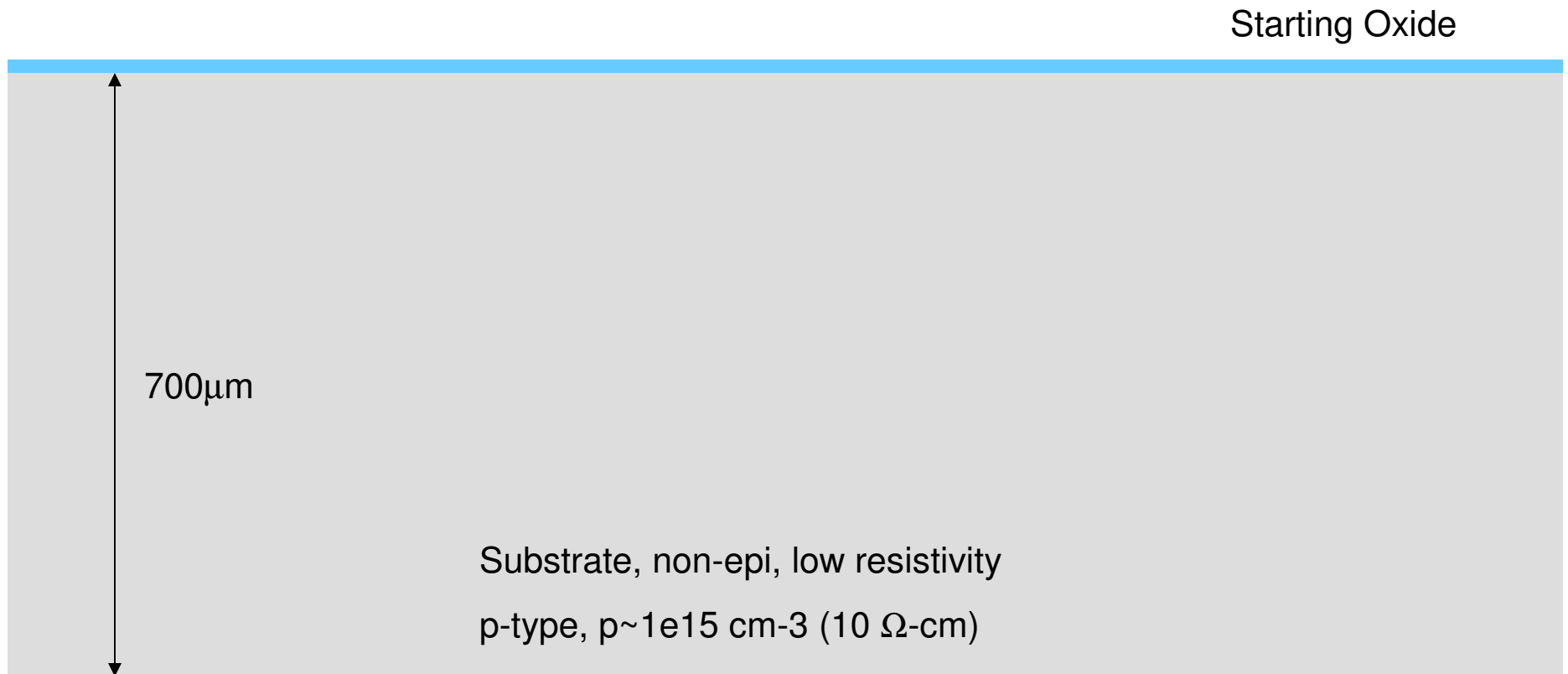
Process Flow

- ◆ Integrate all the previous process steps top make realize a fully processed wafer with an array of functional die.
- ◆ 0.13um process with Cu interconnect.
- ◆ Two halves
 - Frontend: from bare wafer to transistors with S/D/G electrodes.
 - Backend: interconnect layers (metals and vias) for wiring the devices.



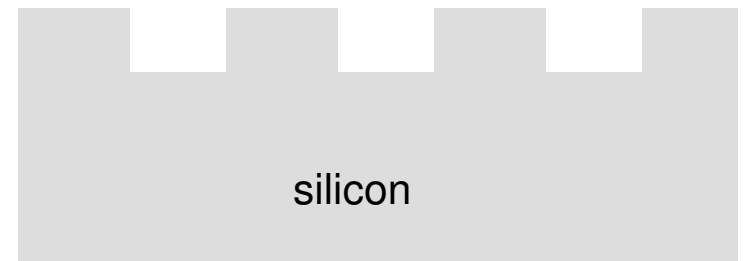
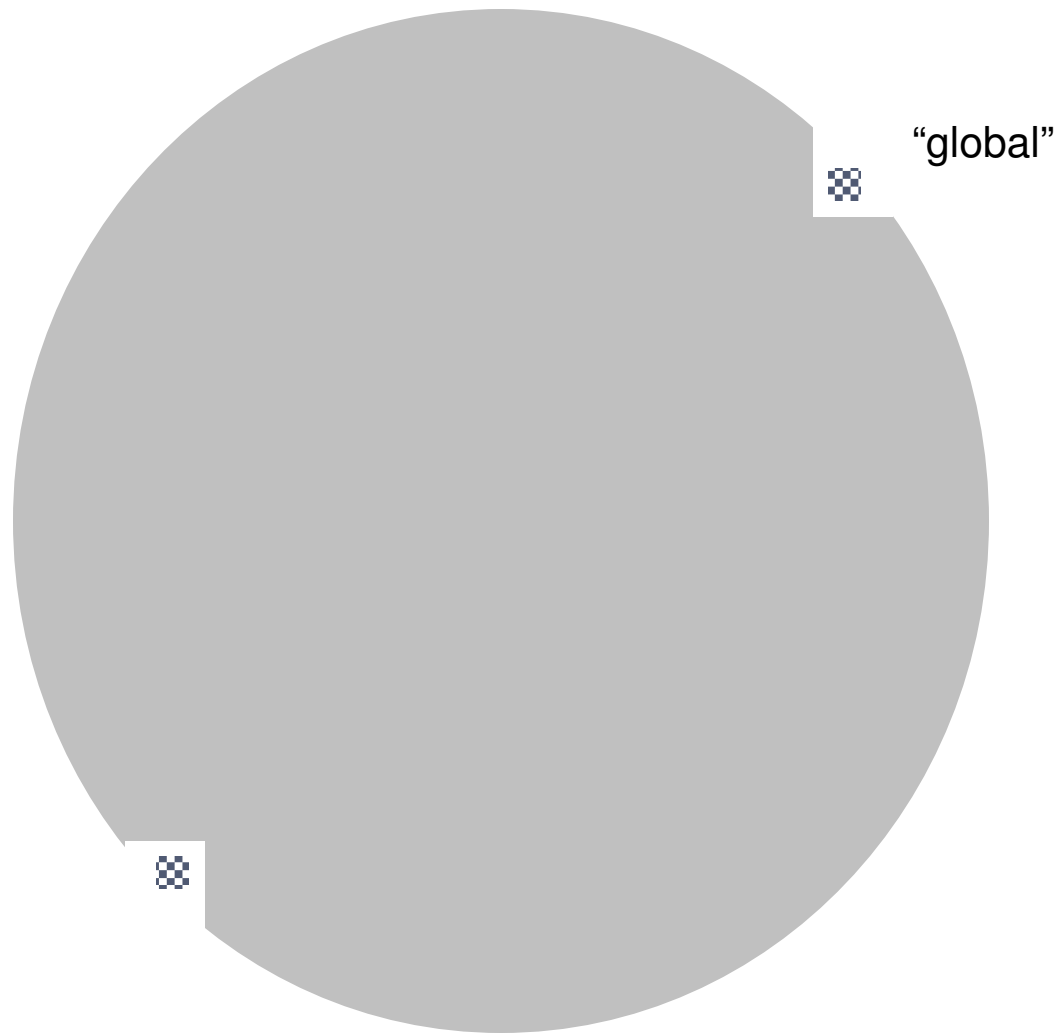
SILICON LABORATORIES

Starting Wafer



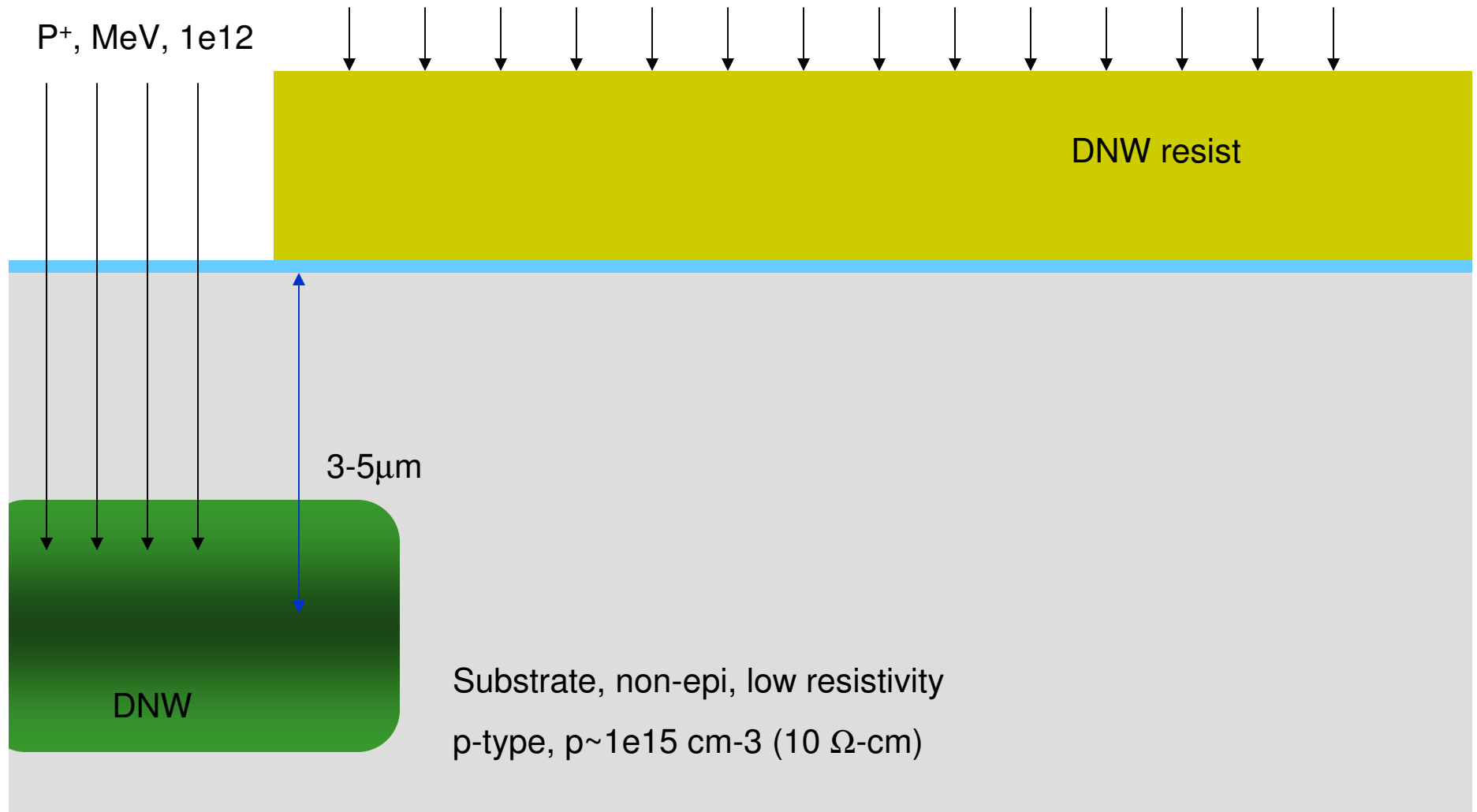
SILICON LABORATORIES

Etch Global Alignment Marks



SILICON LABORATORIES

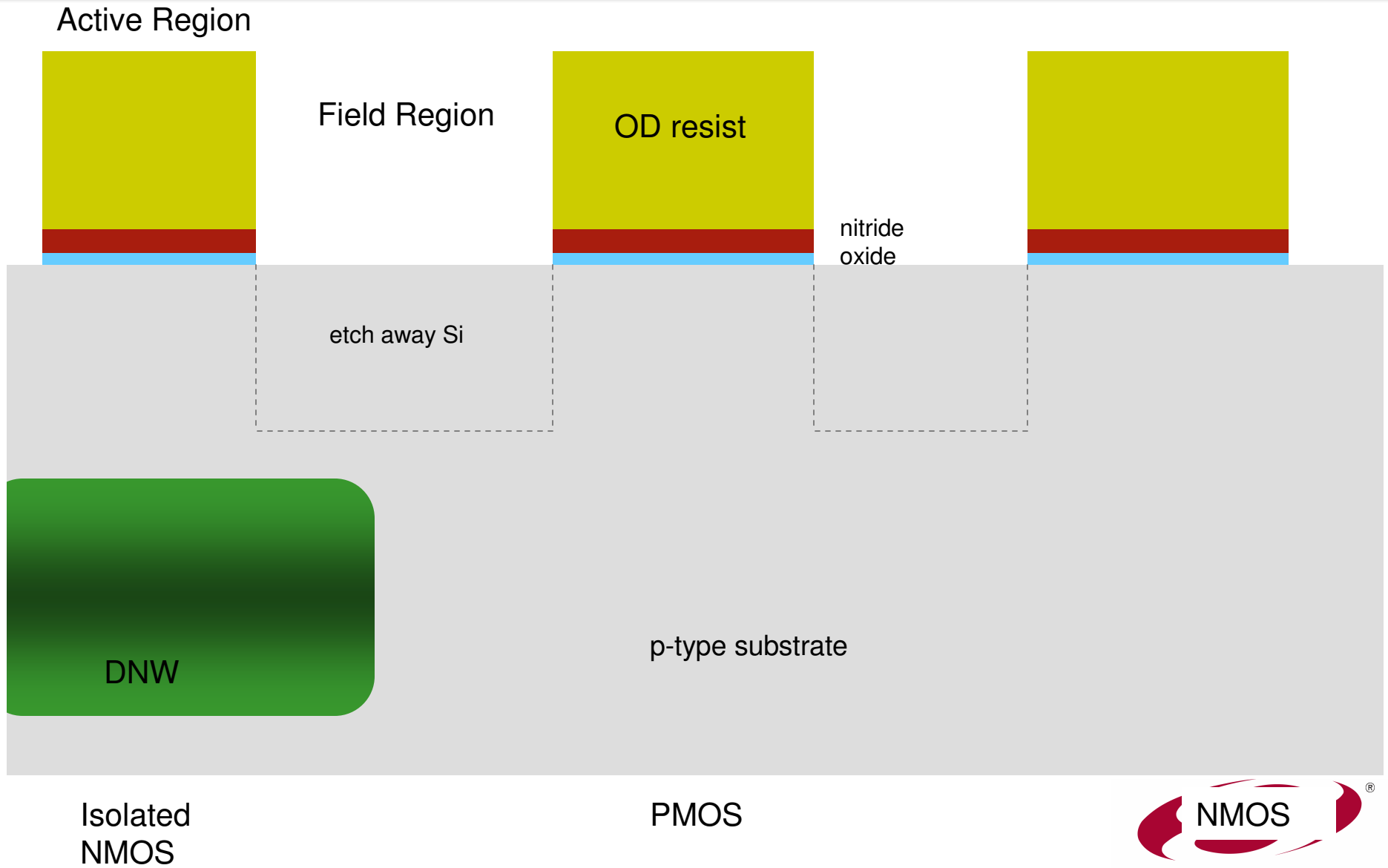
DNW Mask: Deep Nwell → Isolation



Isolated
NMOS

PMOS

OD Mask/Etch: Oxide Definition → Field Oxide

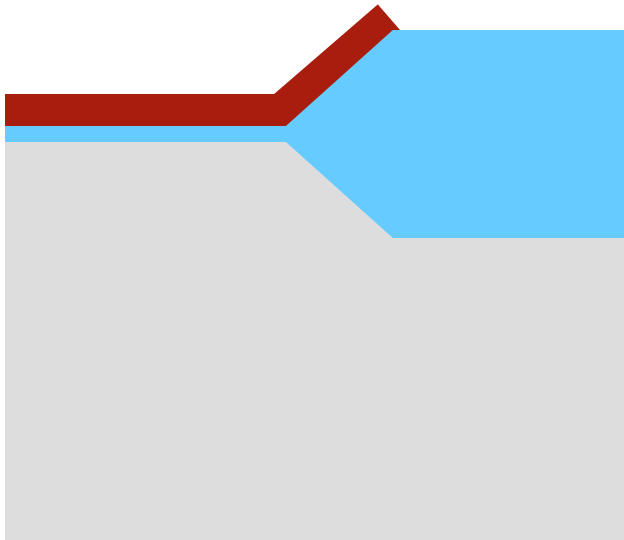


Field Oxide: STI or LOCOS

LOCOS

(Local Oxidation of Silicon)

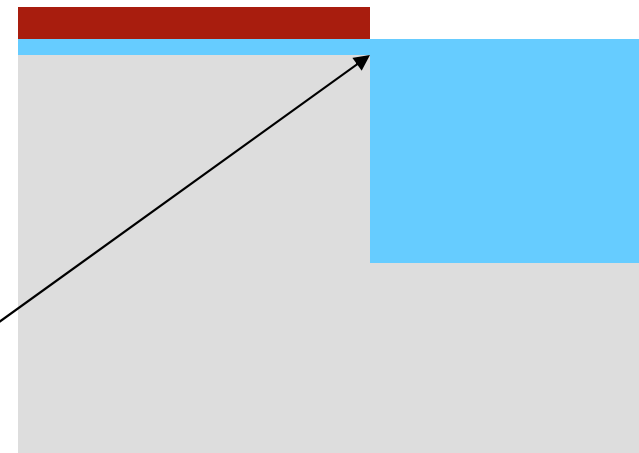
- High temperature oxidation
- Bird's beak → edge ill-defined
- Coarse pitch. No narrow field oxide
- Partially Recessed. Non-planar
- Old, 0.35um and higher



NEW: STI

(Shallow Trench Isolation)

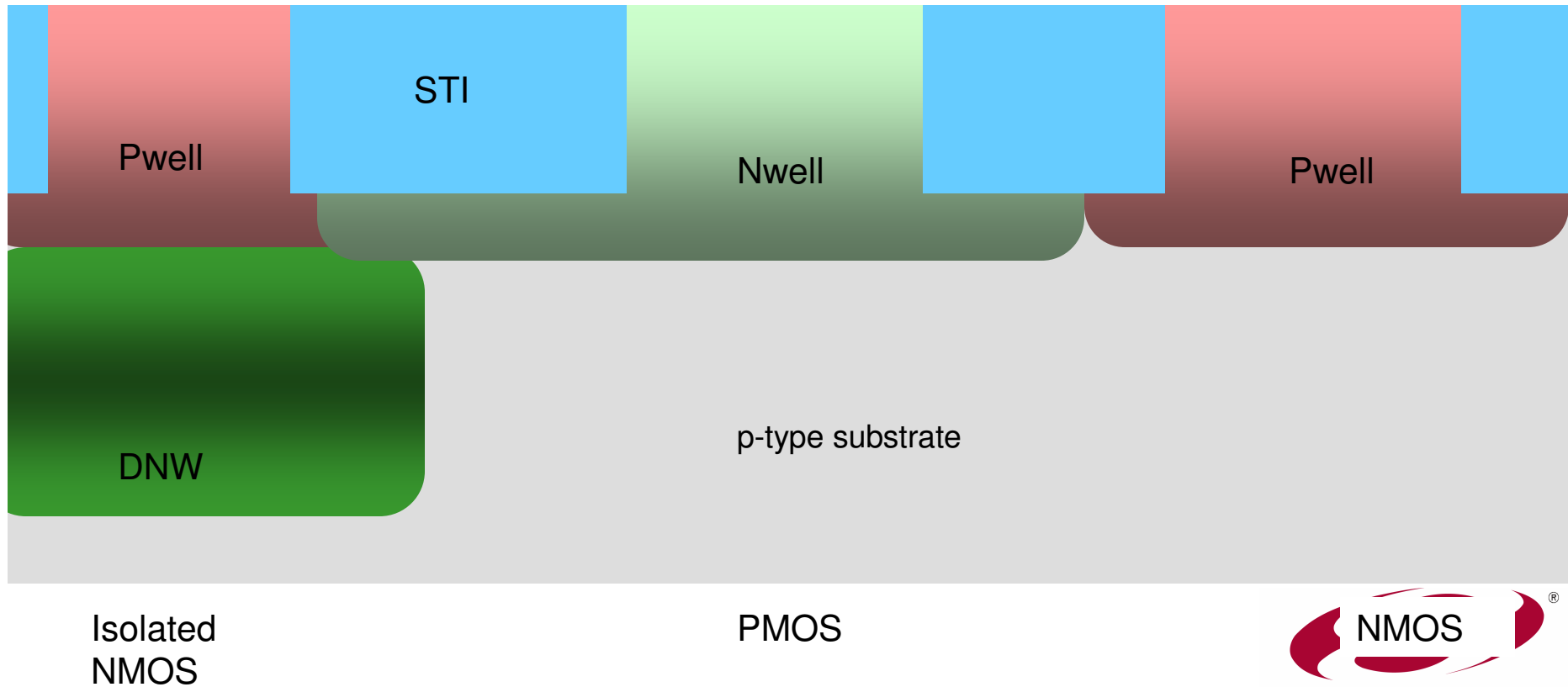
- Si etch, Oxide overfill, CMP back
- Flat, planar
- Low temperature
- Fine pitch
- New, 0.25um and lower



Need corner rounding, but not too much

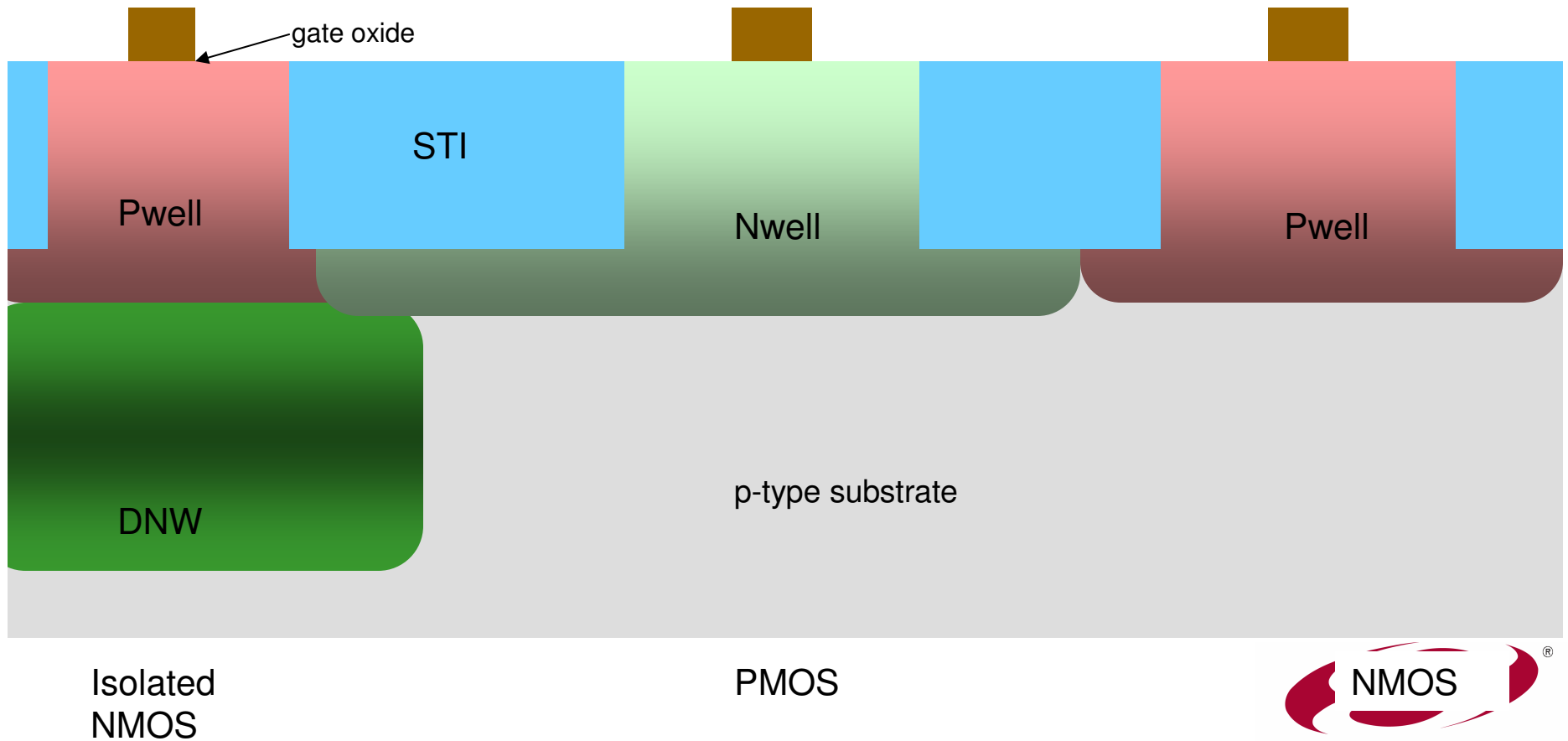
PW and NW Masks: Nwell and Pwell Formation

- 2-3 implants/mask, B/In for Pwell, As/P for Nwell
- If process has dual Vdd, then core and I/O have separate wells
- Also, Vt adjust implants for LV and HV flavors of core transistors

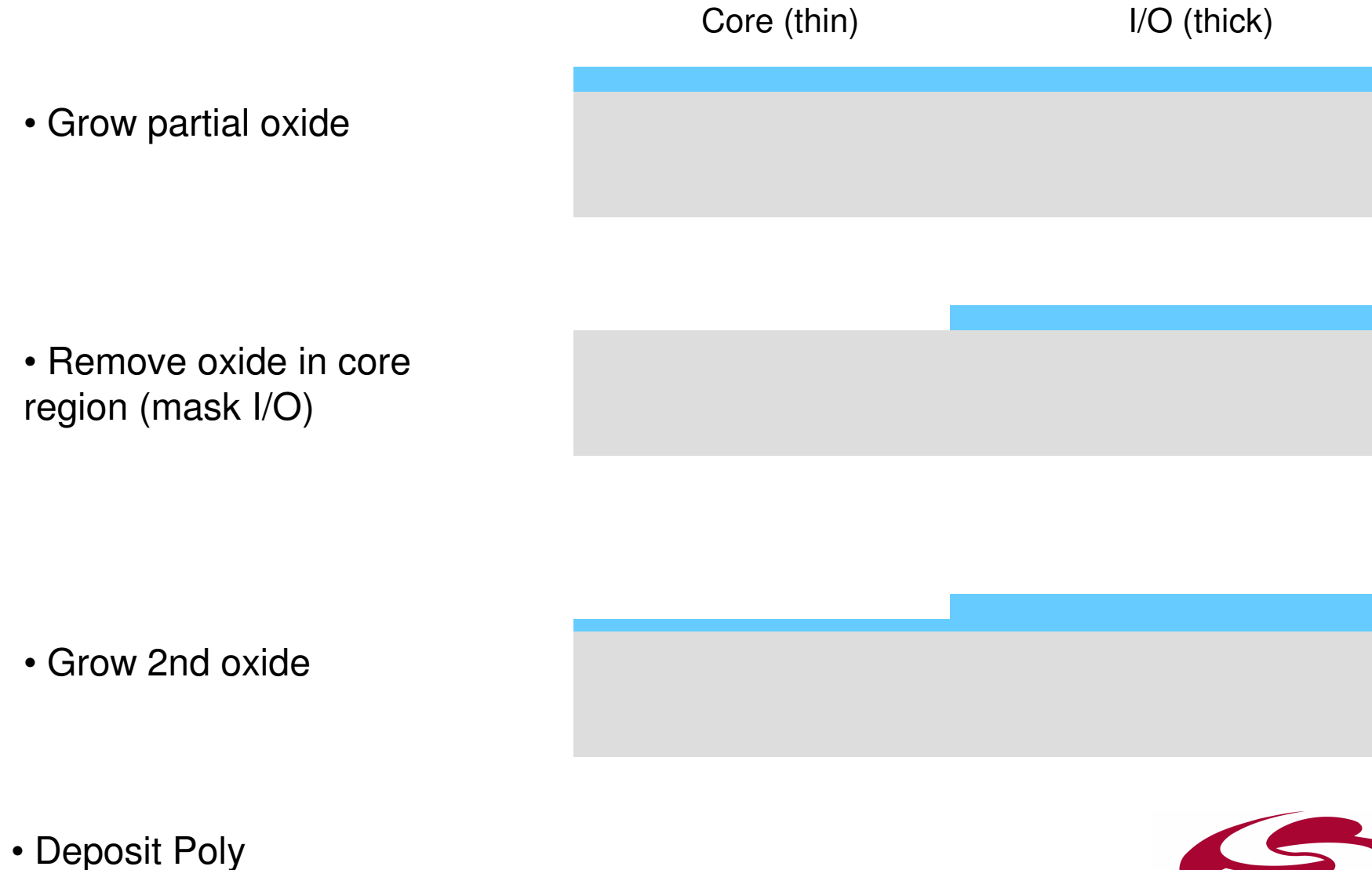


Gate Oxide and Poly

- Grow gate oxide (could be dual oxide)
- Deposit Polysilicon (gate material)
- Etch Poly. Width is the transistor L. Control is critical (CD = Critical Dimension)



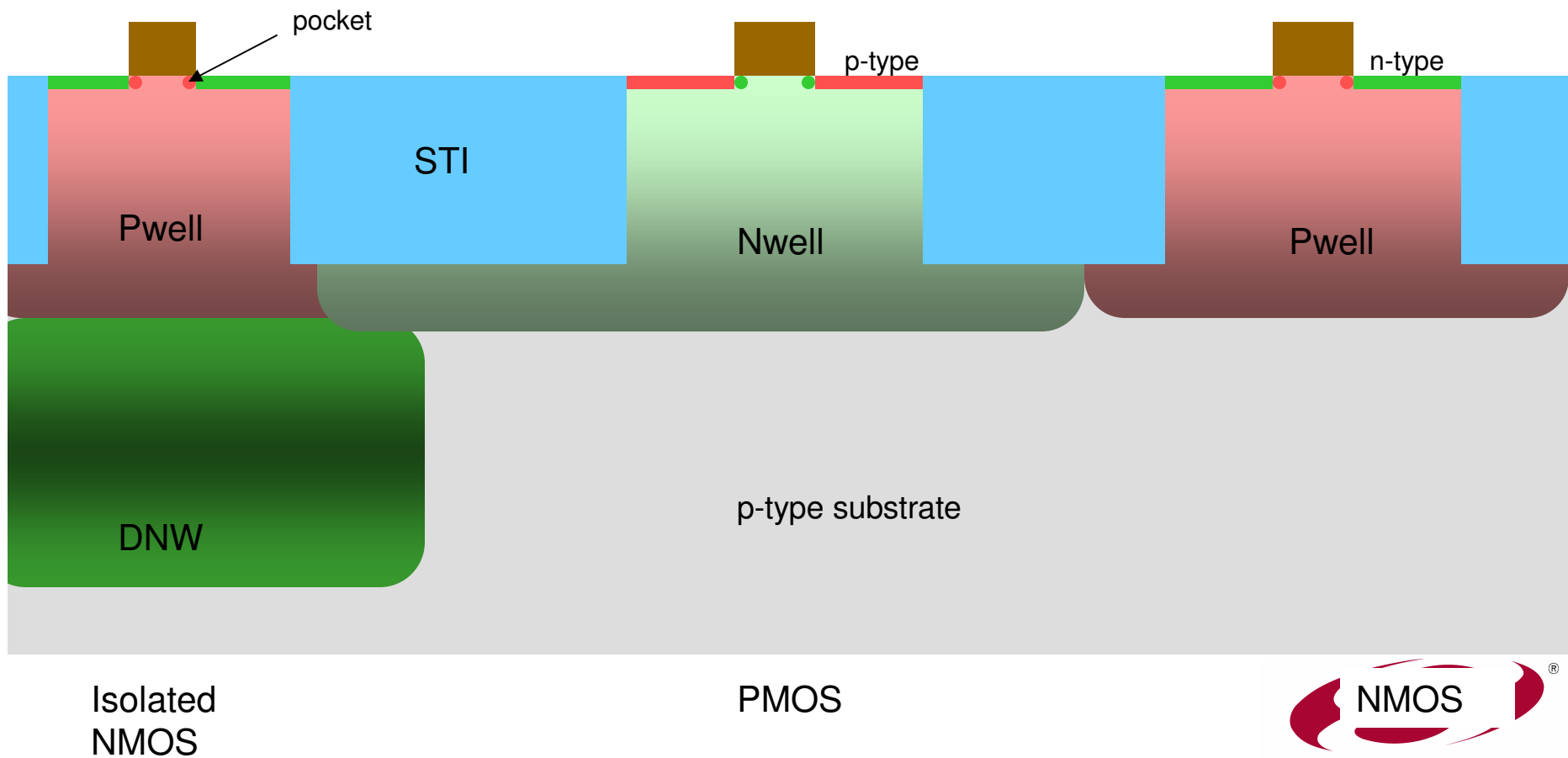
Dual Gate Oxide: Core and I/O Transistors



SILICON LABORATORIES

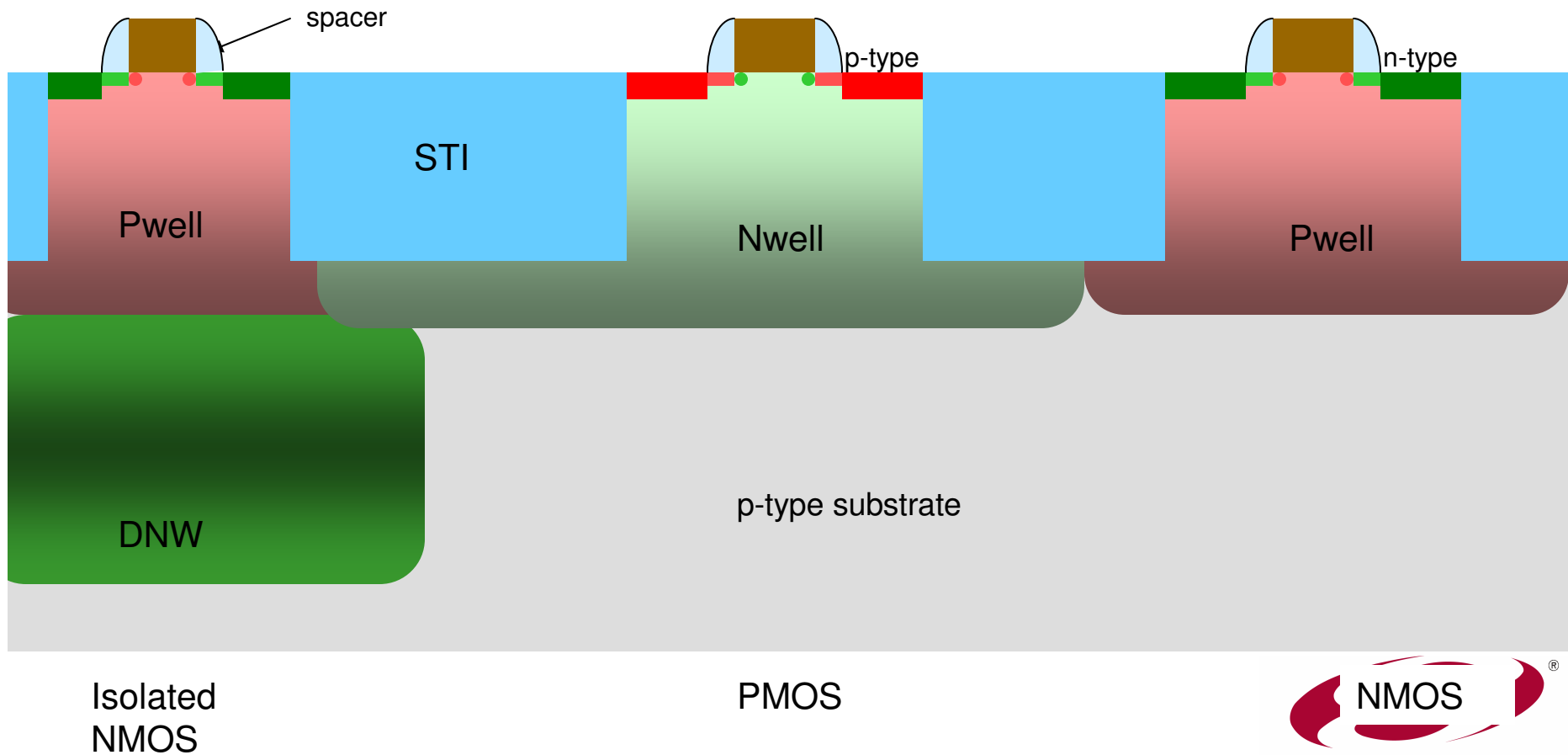
LDD and Pocket Implants

- Self-aligned to gate poly.
- Pocket: extra well doping to prevent V_t rolloff. Angled to get under gat poly.
- LDD: Lightly-Doped Drain. Reduces electric field and hot-carriers at drain.



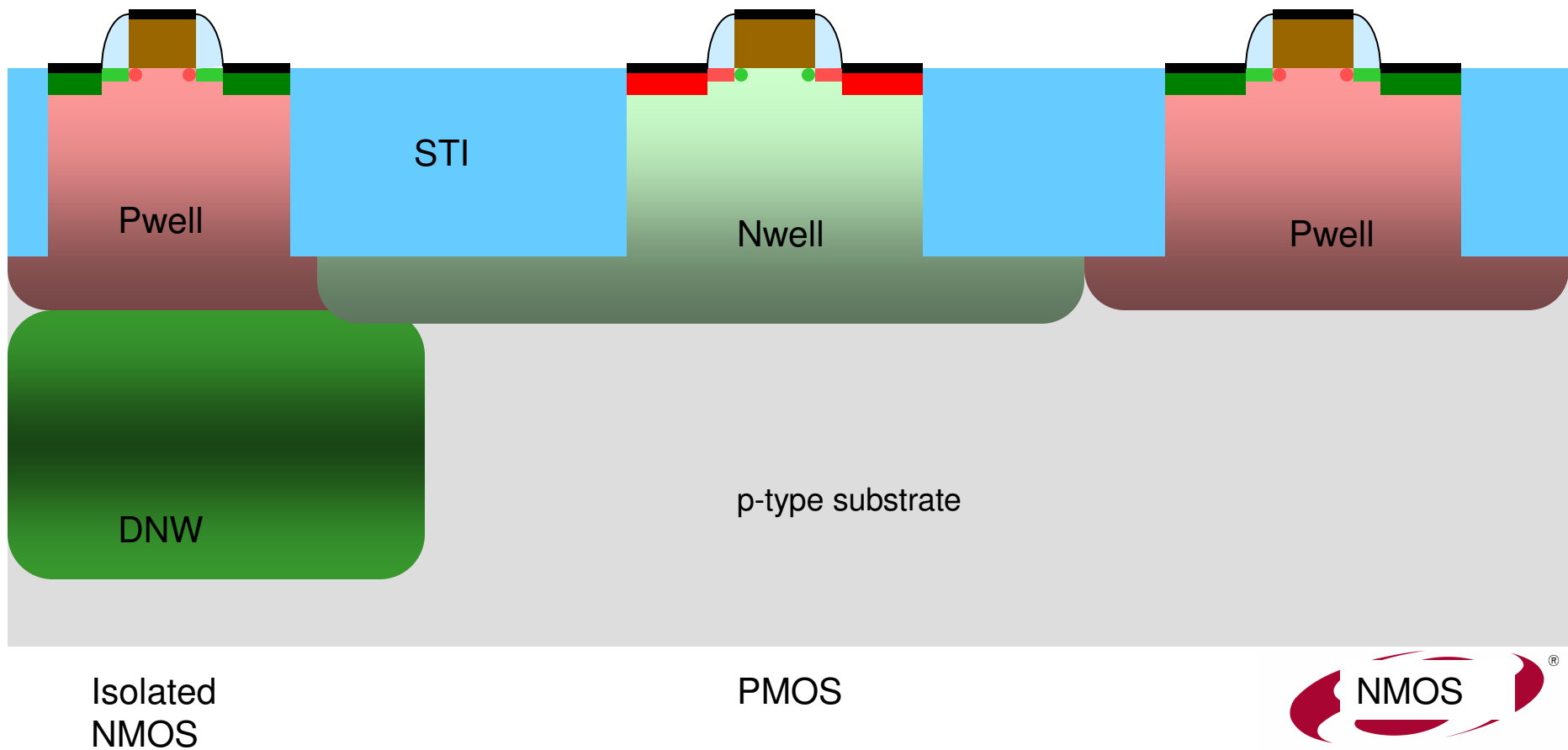
Spacer and Extrinsic Drain

- Spacer protects LDD. About $0.1\mu\text{m} - 0.2\mu\text{m}$ wide.
- Spacer: TEOS only or nitride/oxide bilayer stack.
- Heavy extrinsic implants for low Rseries. Also dopes the poly.



Silicide Formation

- Self-aligned to diffusion regions and poly.
- TiSi (0.25um and above), CoSi (0.18um to 90nm), NiSi (65nm)
- Deposit → React → Strip → Convert to low resistance phase



Prevent Silicide with RPO Layer

- ◆ Before sputtering silicide metal, deposit oxide.
- ◆ Remove oxide where silicide is needed. Block oxide etch with RPO mask for resistors.

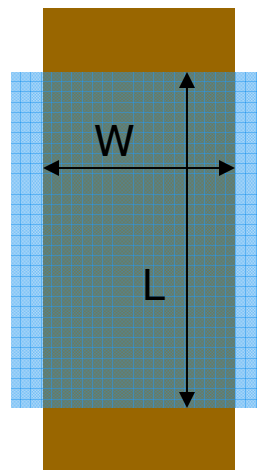
before strip



after conversion



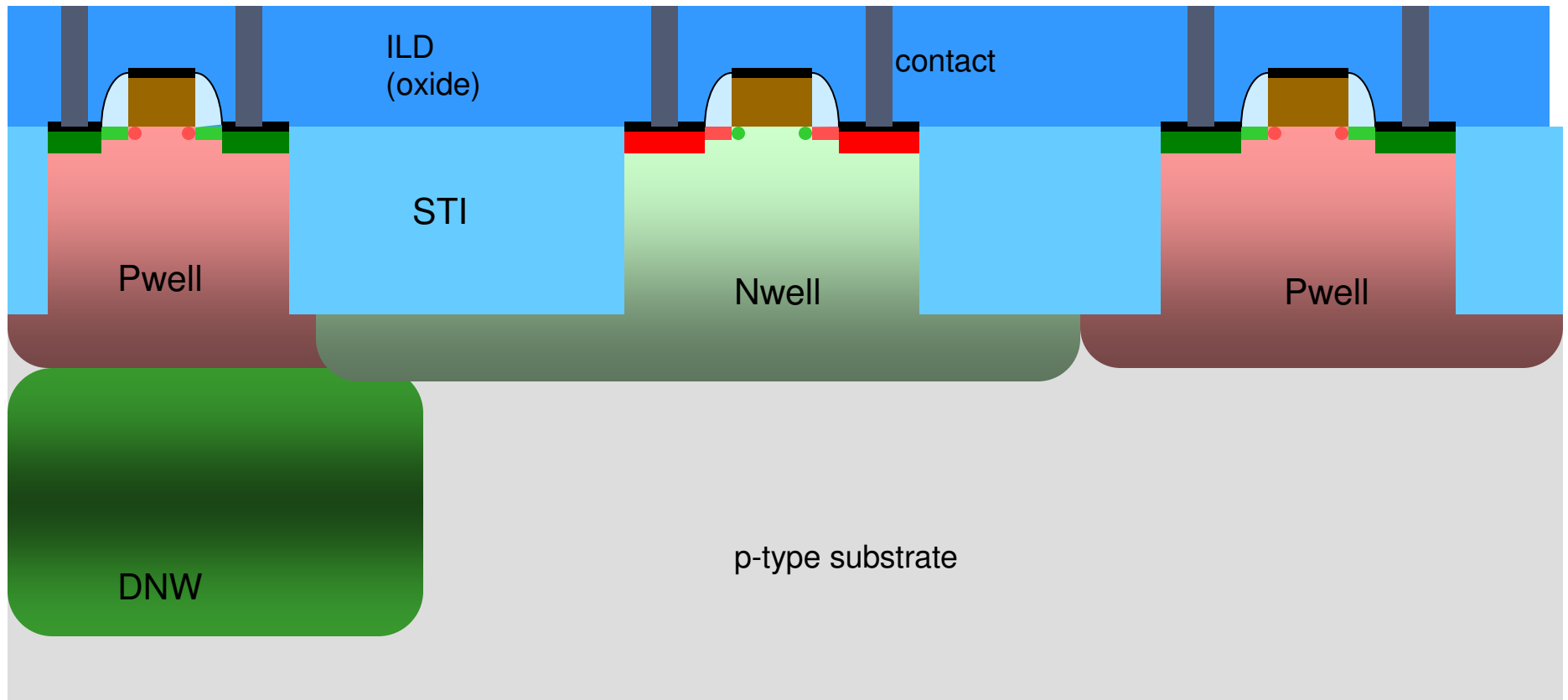
resistor top view



SILICON LABORATORIES

Contact Formation

- Deposit ILD oxide and CMP flat
- Usually W-plug
- Etch hole → Deposit W → CMP flat

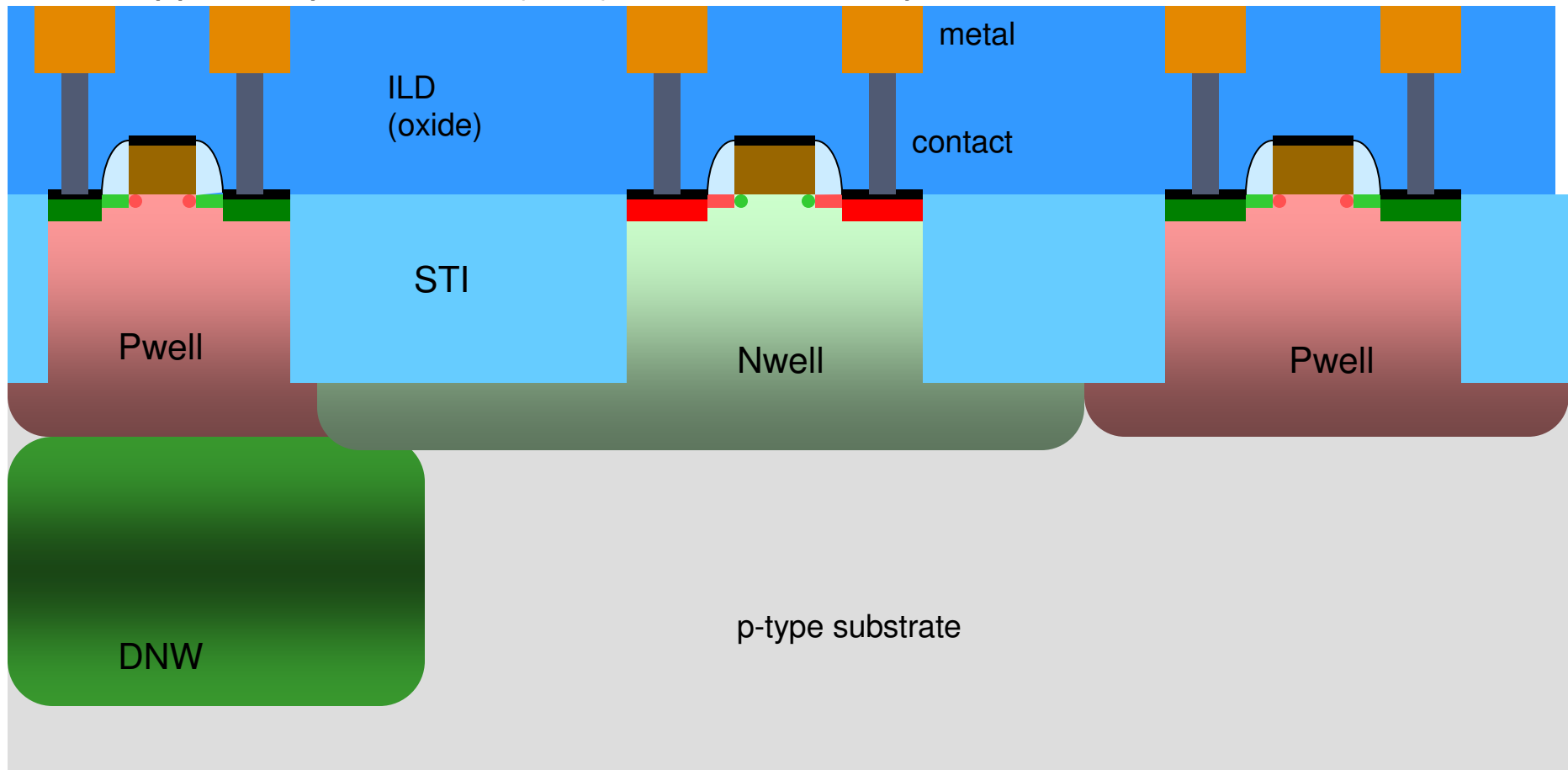


Isolated
NMOS

PMOS

Metal1

- Aluminum: Deposit metal, mask and etch.
- Copper: Deposit oxide (IMD), etch trench, deposit metal, CMP flat



Isolated
NMOS

PMOS



Processing of Al and Cu Metal Lines

Aluminum

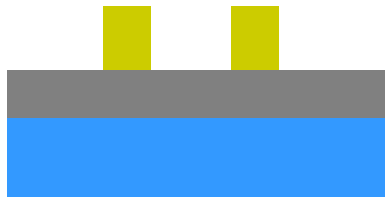
start with flat oxide



sputter Al film

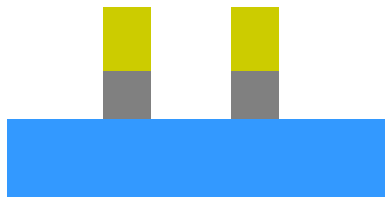


pattern photoresist



etch Al

hard to etch fine lines



strip photoresist

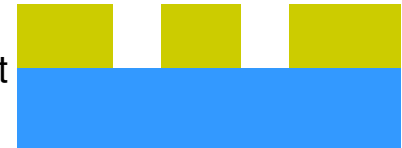


Copper

start with flat oxide



pattern photoresist



etch trenches in oxide, strip resist

easy to etch oxide



electroplate Cu



CMP Cu flat



Processing of W and Cu Vias

Tungsten

start with flat oxide
on top Al



pattern photoresist



etch holes in
oxide, strip resist



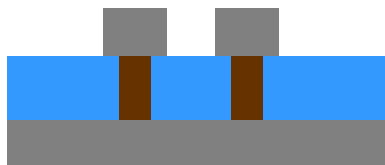
deposit W



CMP W flat



deposit and
pattern Al lines



Copper

start with flat oxide
on top Cu



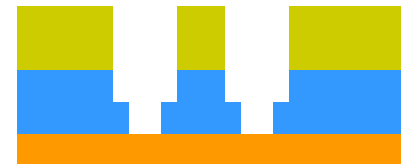
pattern photoresist



etch holes to etch
stop layer, strip
resist



pattern and etch
metal line trenches



electroplate Cu



CMP Cu flat



Passivation and bondpads

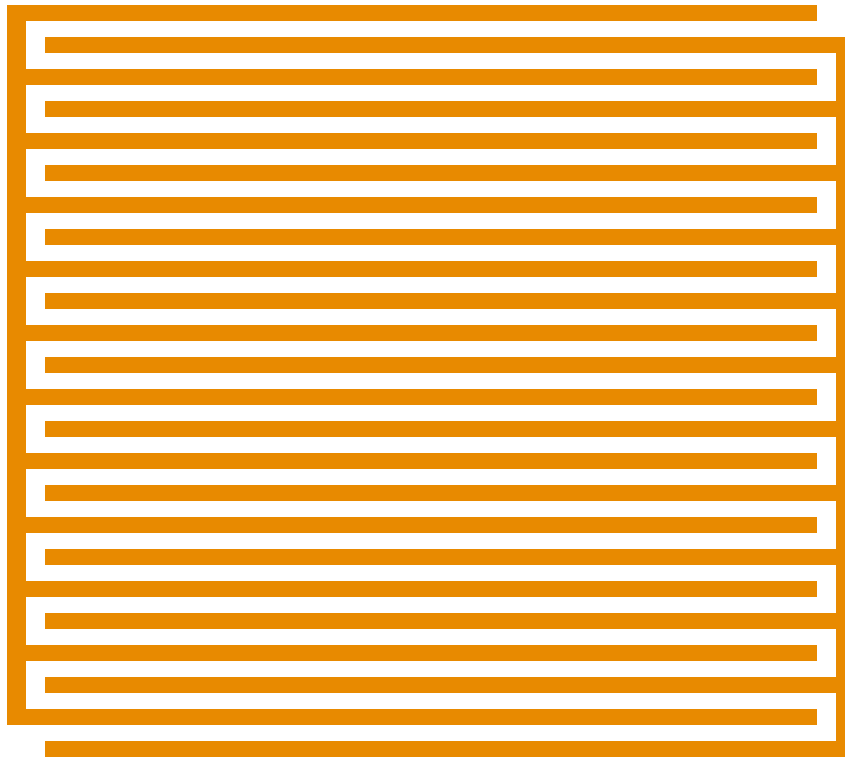
- Repeat metal and via processing steps to complete interconnect.
- Deposit passivation (nitride) and etch.
- For Cu, deposit Al wirebond layer and etch.
- For P49, route Al layer to bump (redistribution of bondpads)



SILICON LABORATORIES

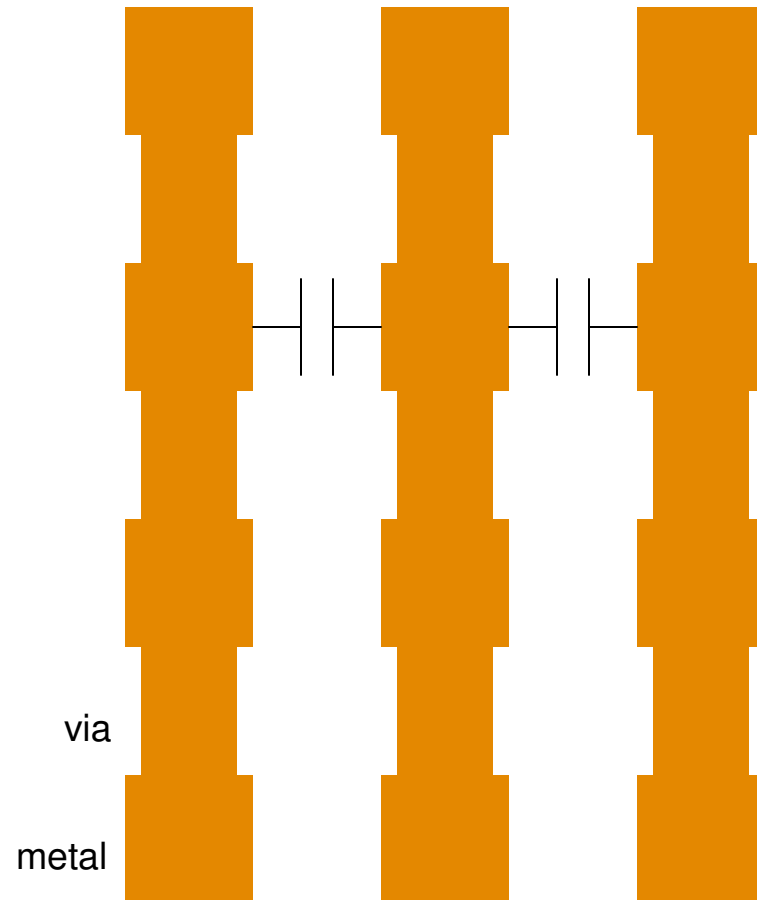
Metal Finger Capacitors

Top View



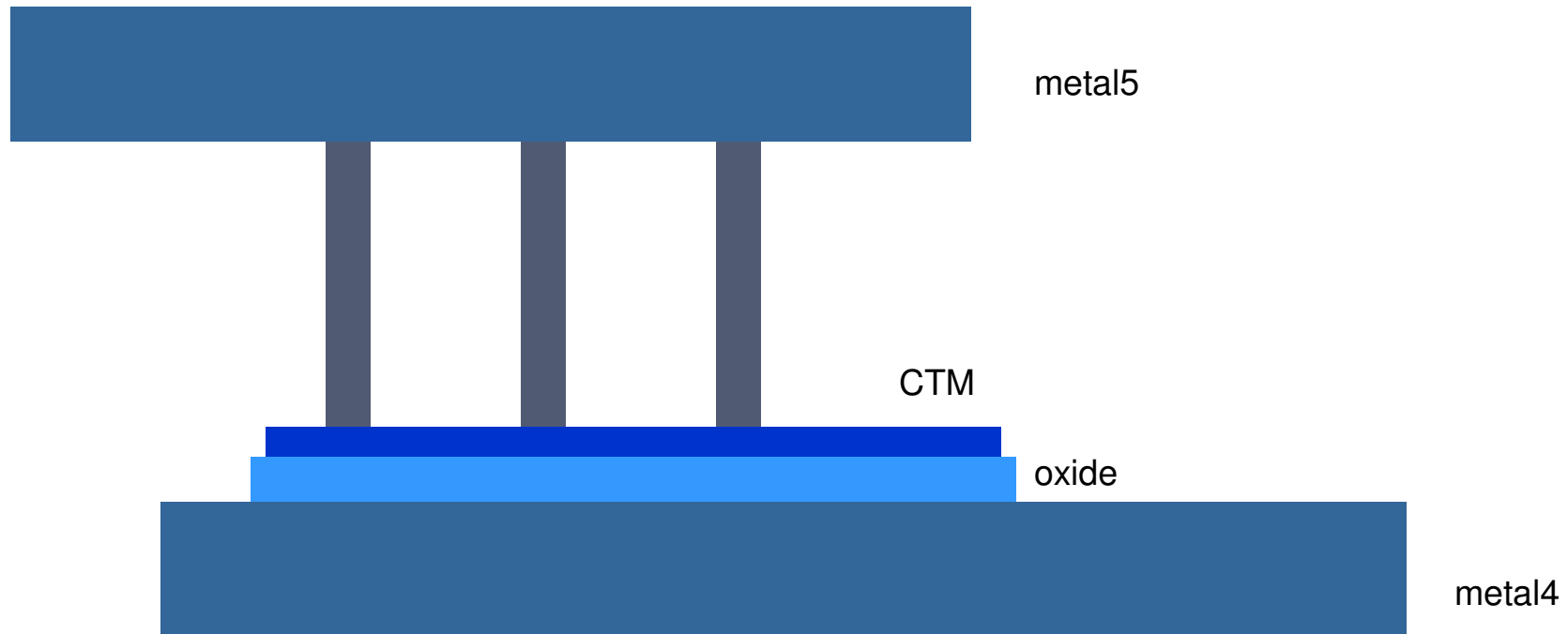
M1-M5 stacked and tied together with vias

Cross Section (4 metal stack)



SILICON LABORATORIES

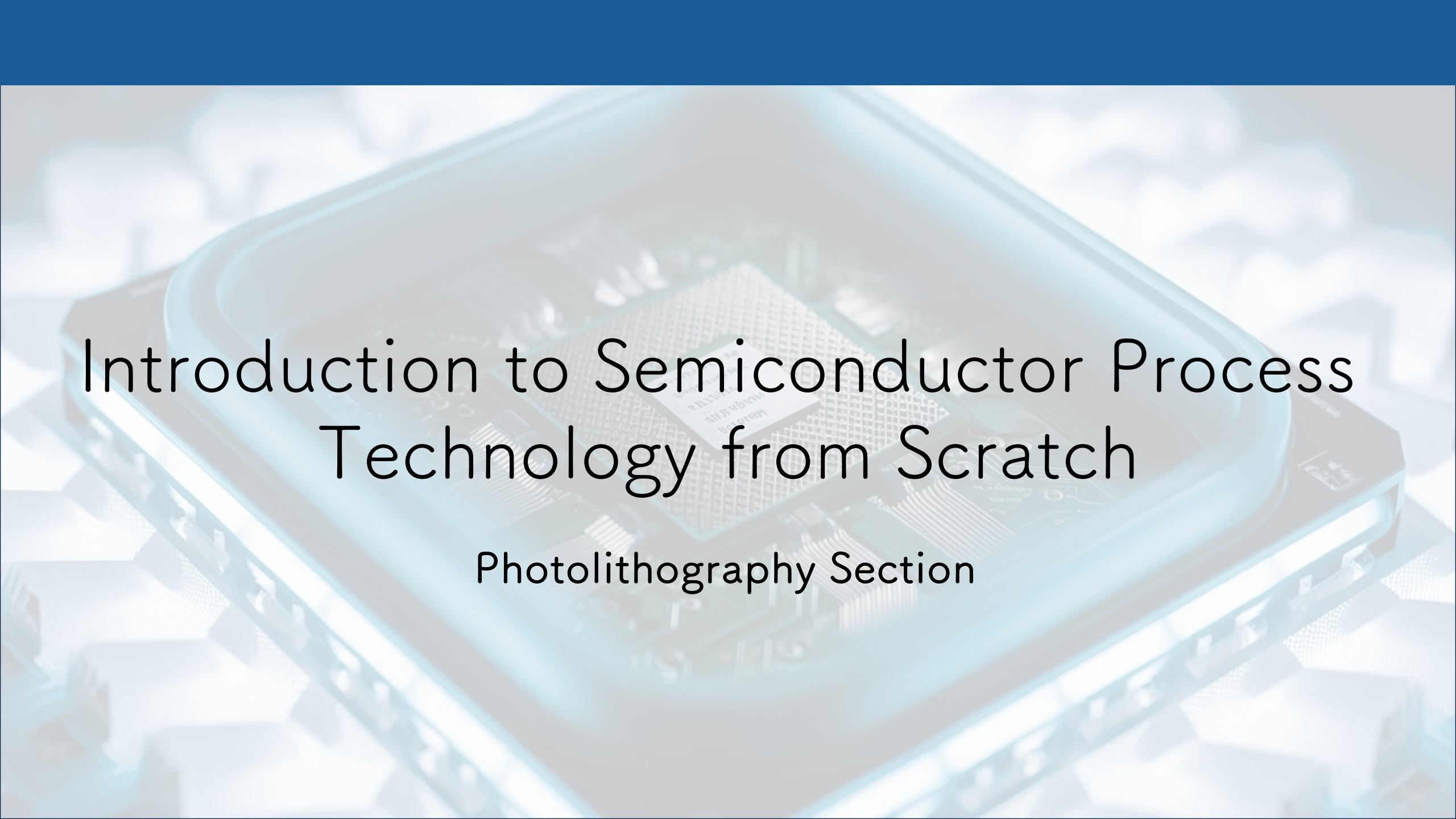
MIM Capacitor



- ◆ Above MIMcap uses existing metal for bottom plate. This style found in 0.18um and higher. Requires one mask for CTM
- ◆ For 0.13um process, cannot use copper as a bottom plate. Instead, separate CBM layer is added. Two masks required.



SILICON LABORATORIES

The background image is a close-up, high-angle photograph of a semiconductor chip mounted on a circuit board. The chip is square with a grid of pins on its underside. It is surrounded by various electronic components and traces on the board. A semi-transparent blue overlay covers the entire image, creating a modern, technological feel.

Introduction to Semiconductor Process Technology from Scratch

Photolithography Section

Overview of the Photolithography Process

- Summary of the photolithography process
- Position of the photolithography step within the overall semiconductor process
- Process flow of photolithography and the role of each step
- Recap

About Photoresist

- Components and types of photoresist
- Reaction mechanism of photoresist
- Recap

Coating Process

- Position of the coating process within the overall flow
- Coating Method Part 1: Spin Coating
- Coating Method Part 2: Coating Sequence
- Types of Coating Methods:
 - Static Coating, Dynamic Coating
- Impact of coating distribution variation on products
- Reasons for product impact:
- Cause 1: Non-uniform exposure dose
- Cause 2: Focus misalignment during exposure
- Cause 3: Variation in development process
- Key physical parameters determining coating distribution
- Recap

Pre-Bake

- Position of the pre-bake step within the overall flow
- Purpose of the pre-bake process
- Types of pre-bake methods
- Impact of pre-bake temperature
- Recap

Exposure Process 1

- Position of the exposure process within the overall flow
- Purpose of the exposure process
- About photomasks
- Light used for exposure
- Parameters related to wavelength
- Definition of pattern resolution
- OK/NG criteria for pattern resolution
- What does “depends on the next process” mean?
- Relationship between NA, pattern resolution,
and depth of focus
- What is NA (Numerical Aperture)?
- Relationship between NA and pattern resolution – Formula
- Relationship between NA and resolution
 - When NA is large
- Relationship between NA and resolution
 - When NA is small
- Relationship between NA and depth of focus – Formula

Exposure Process 2

- About alignment
- How high-precision alignment is required
- Basic concept of alignment technology
- Structure of exposure equipment
- Types of misalignment that can occur
- Countermeasures for “position misalignment”
and “rotation misalignment”
- Countermeasures for “distortion misalignment”
and “expansion/contraction misalignment”
- Recap

Development

- Position of the development process
within the overall flow
- Purpose of the development process
- Development process flow
- Types of developer solutions
- Key parameters during development
- Recap

Overview of the Photolithography Process

What

Technology for Forming Circuit Patterns on Silicon Wafers Using Light

Etymology

Derived from Greek
λίθος (lithos) + γράφειν (graphein)

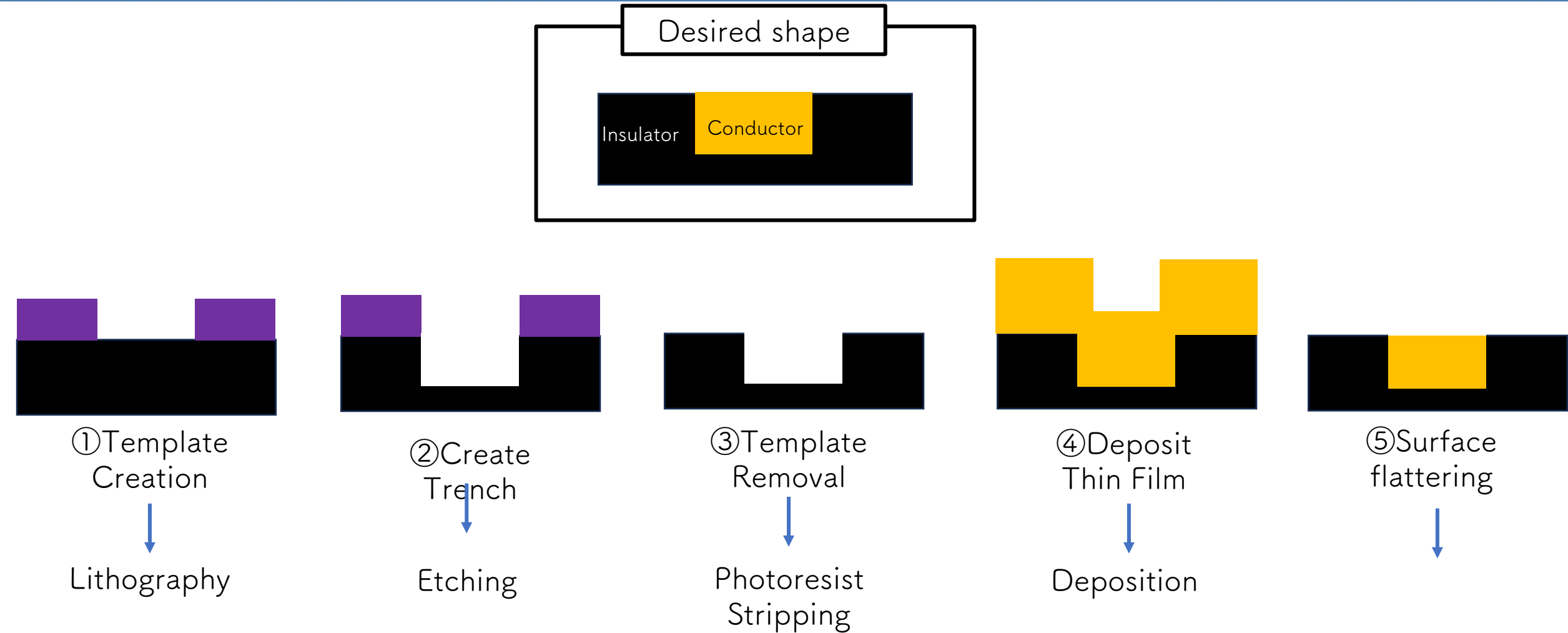
Stone + Draw → Stone Printing → Substrate Transfer

Characteristics

Using light, it is possible to transfer extremely fine patterns at the micro- and nanometer scale with high speed and precision.



The integration of billions of transistors becomes possible, making photolithography a critical technology for creating compact and high-performance devices.

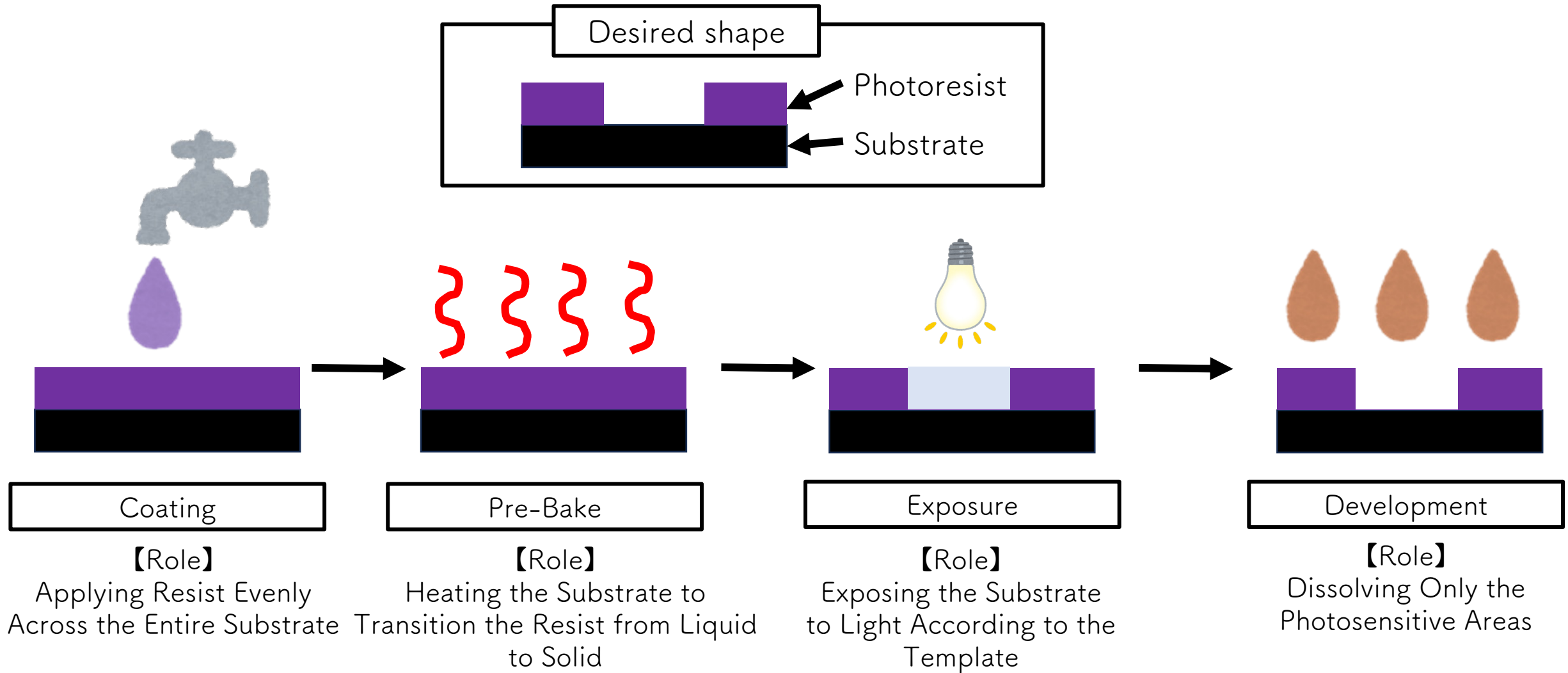


Point

The lithography process is the step in semiconductor manufacturing used to create templates.

Process flow of photolithography and the role of each step

7



Point

The Lithography Process Consists of Coating → Pre-Baking → Exposure → Development in Sequence

The Photolithography Process Is

- A technology that uses light to form circuit patterns on a substrate
- Responsible for creating templates in semiconductor manufacturing
- Consists of four steps:
Coating → Pre-Bake → Exposure → Development

About Photoresist

What is photoresist

A resin that dissolves in specific liquids when exposed to light

Components

A liquid composed of resin and solvent



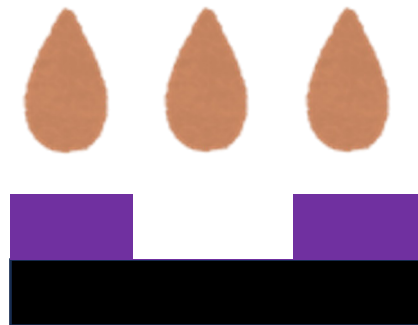
Resin
+
Solvent

Photoresist

Type

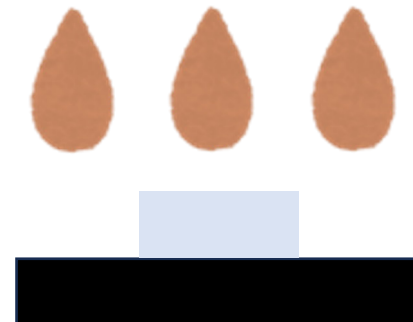
Positive Photoresist

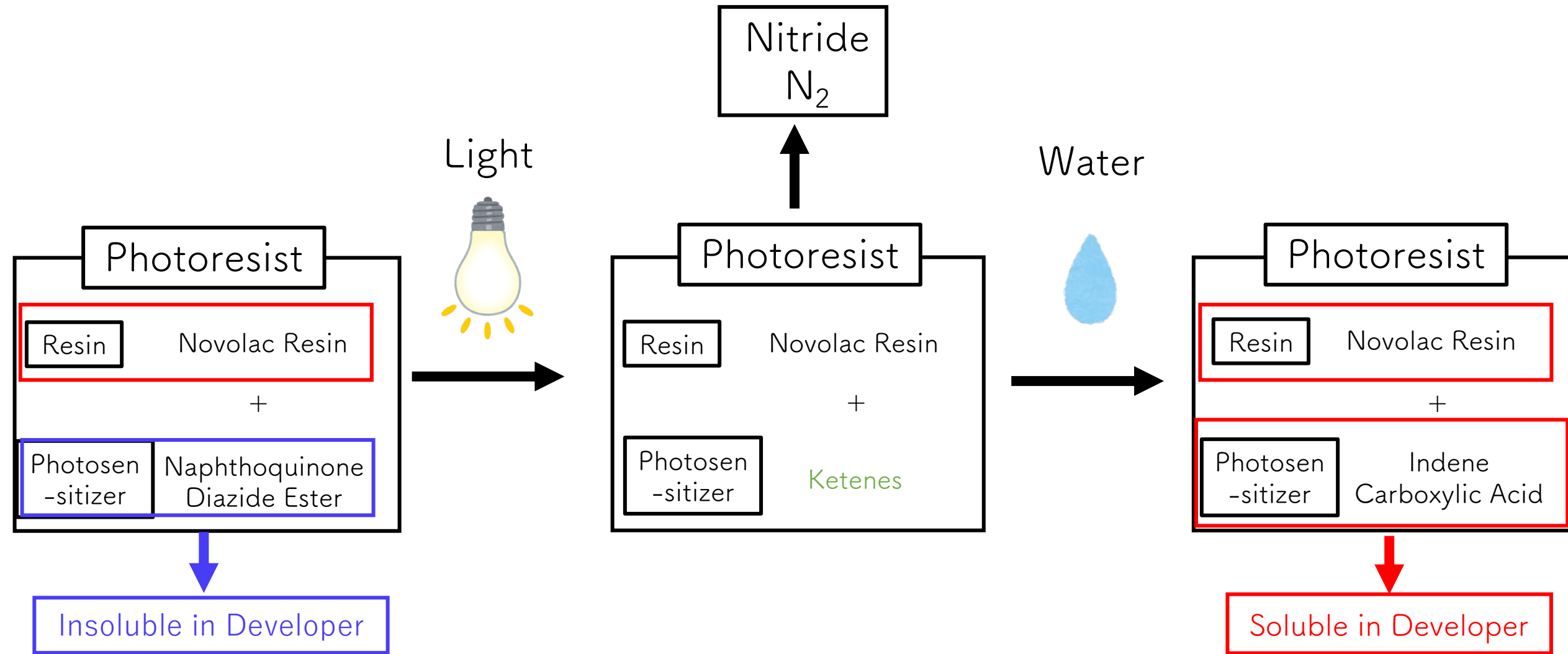
The areas exposed to light dissolve



Negative Photoresist

The areas exposed to light remain





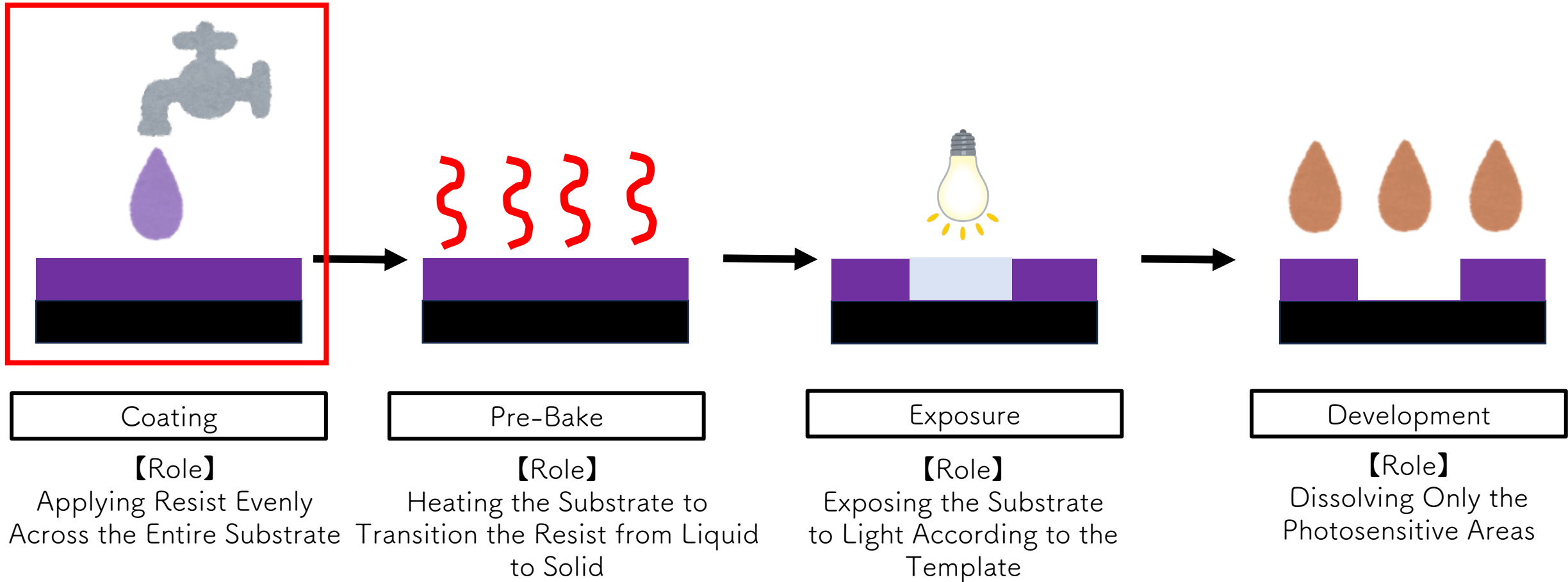
Point

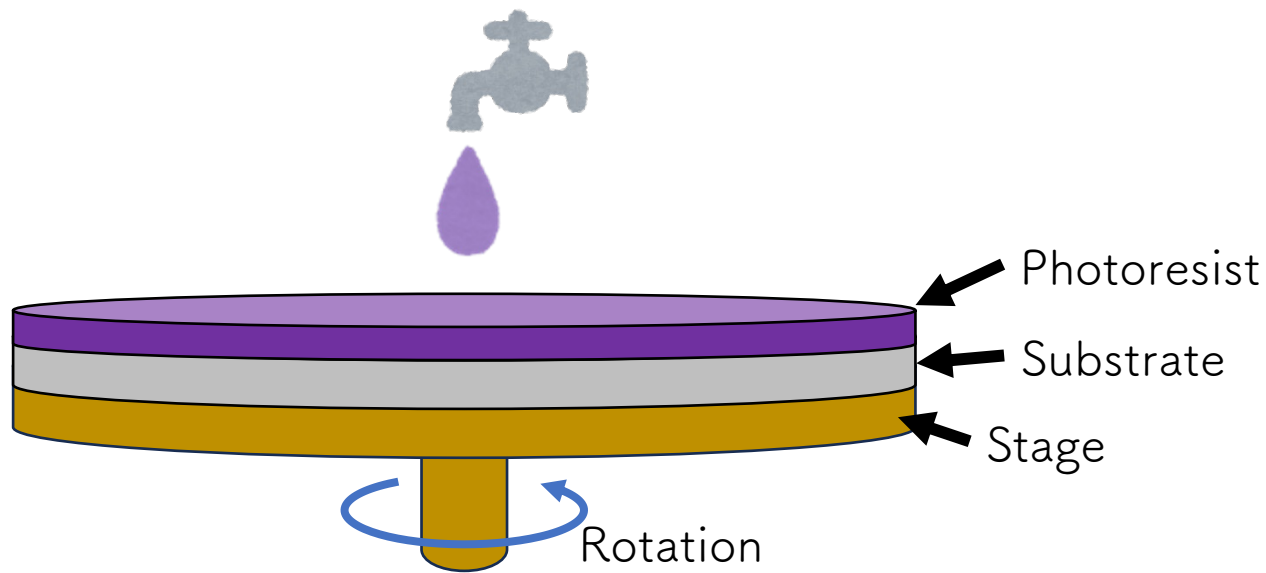
The photosensitizer in the resist changes its composition when exposed to light, making it soluble in the developer.

Photoresist is

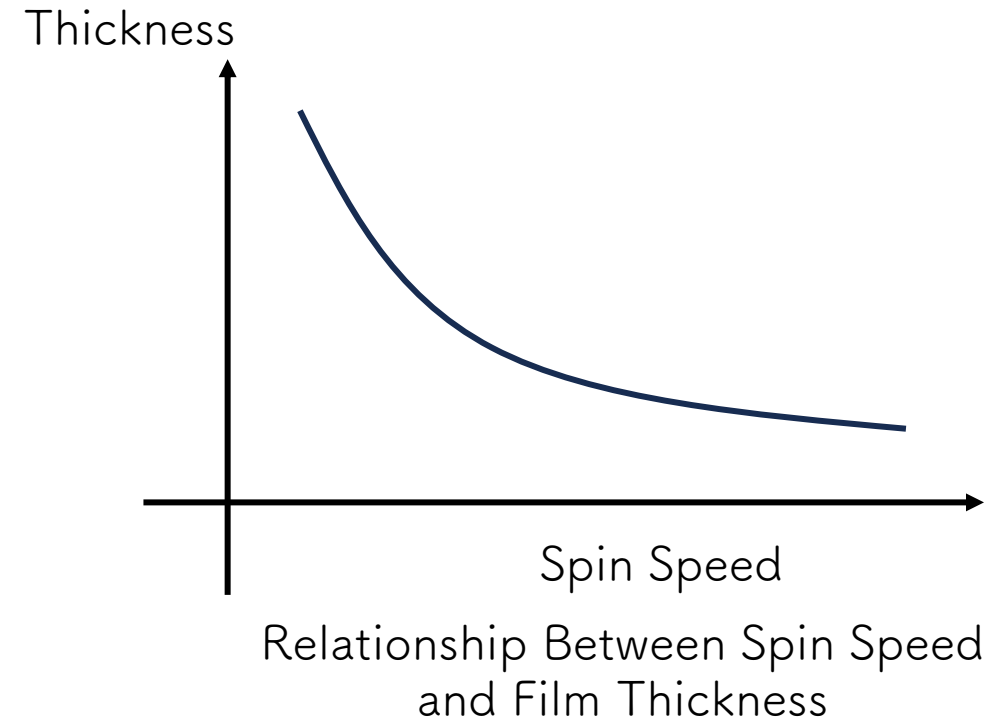
- Before coating, it is in a liquid state composed of resin + solvent.
- A resin in which the photosensitizer changes composition upon exposure to light, becoming soluble in the developer.

Coating Process





Schematic drawing of spin coating

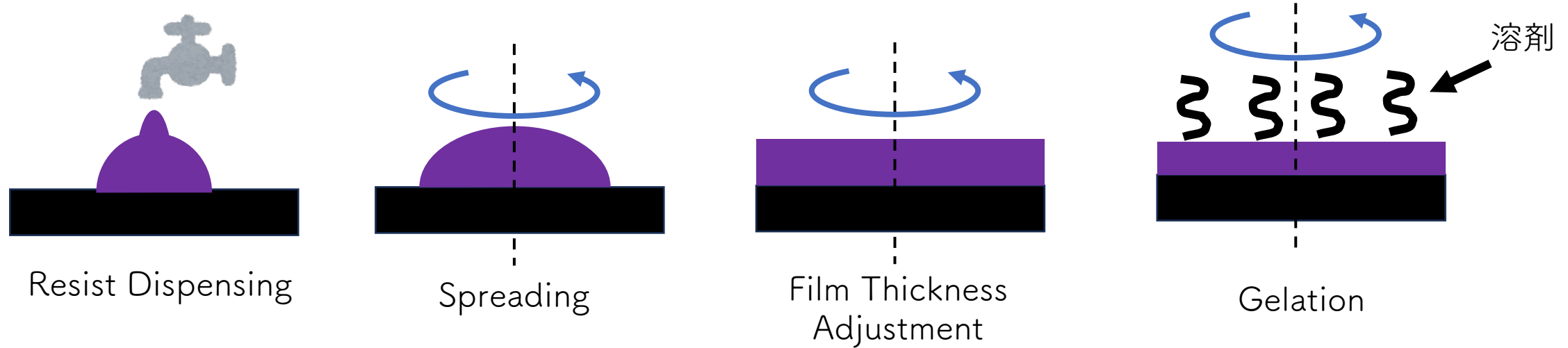


Pros

- The desired resist film thickness can be achieved.
- Optimal film thickness distribution can be achieved depending on the conditions.

Cons

- A large amount of resist is wasted.



Resist coating steps⁽²⁾

Point

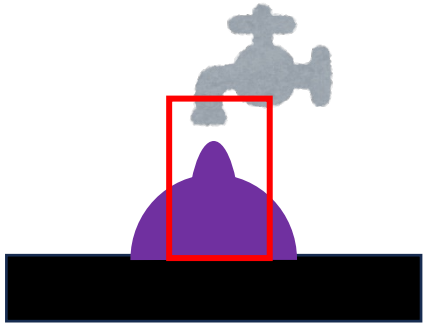
Consists of Four Steps:

Dispensing → Spreading → Film Thickness Adjustment → Gelation

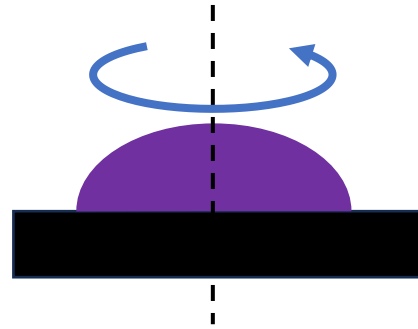
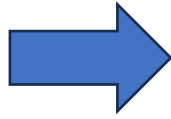
(2)Tyona, M. D. "A theoritical study on spin coating technique." *Advances in materials Research* 2.4 (2013): 195.

Static Coating

Spreading begins after the resist is fully dispensed.



Resist Dispensing



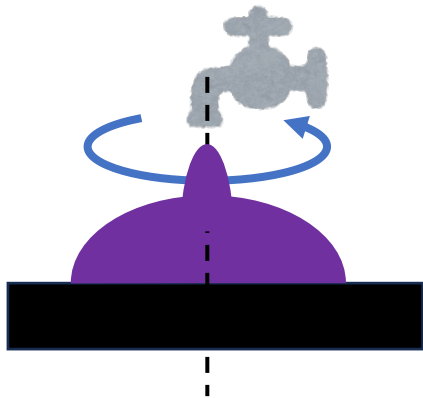
Resist Spreading

Pros : Simple Equipment Design
Affordable Cost

Cons : Poor Coating Uniformity

Dynamic Coating

Dispensing and spreading the resist are performed simultaneously.



Resist Dispensing & Spreading

Pros : Good Coating Uniformity

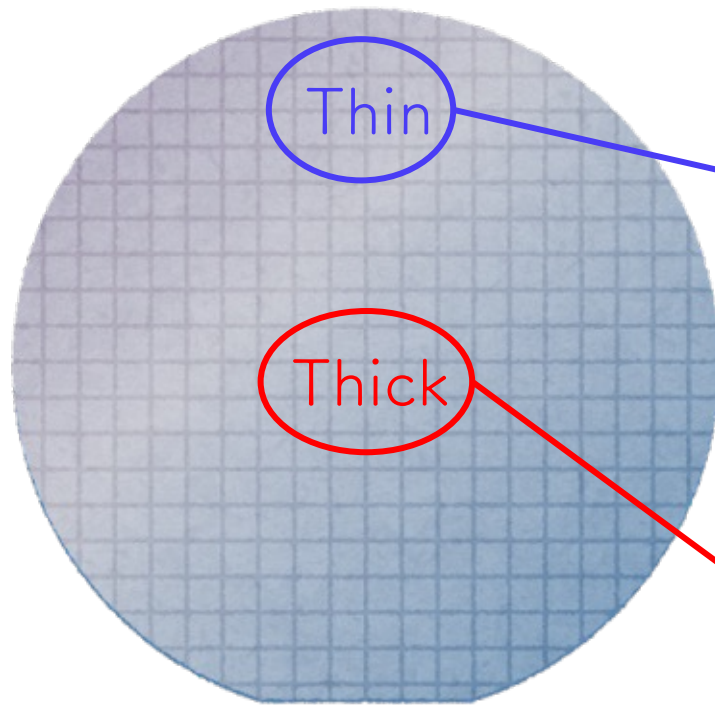
Cons : Complex Equipment Design with High Cost
Challenging Condition Optimization

Requirements

The resist's film thickness distribution must be uniform across the entire substrate.

Reason

Variations in the resist dimensions lead to inconsistencies in the wiring dimensions of the final product.



基板

Edge

Photoresist



Wire



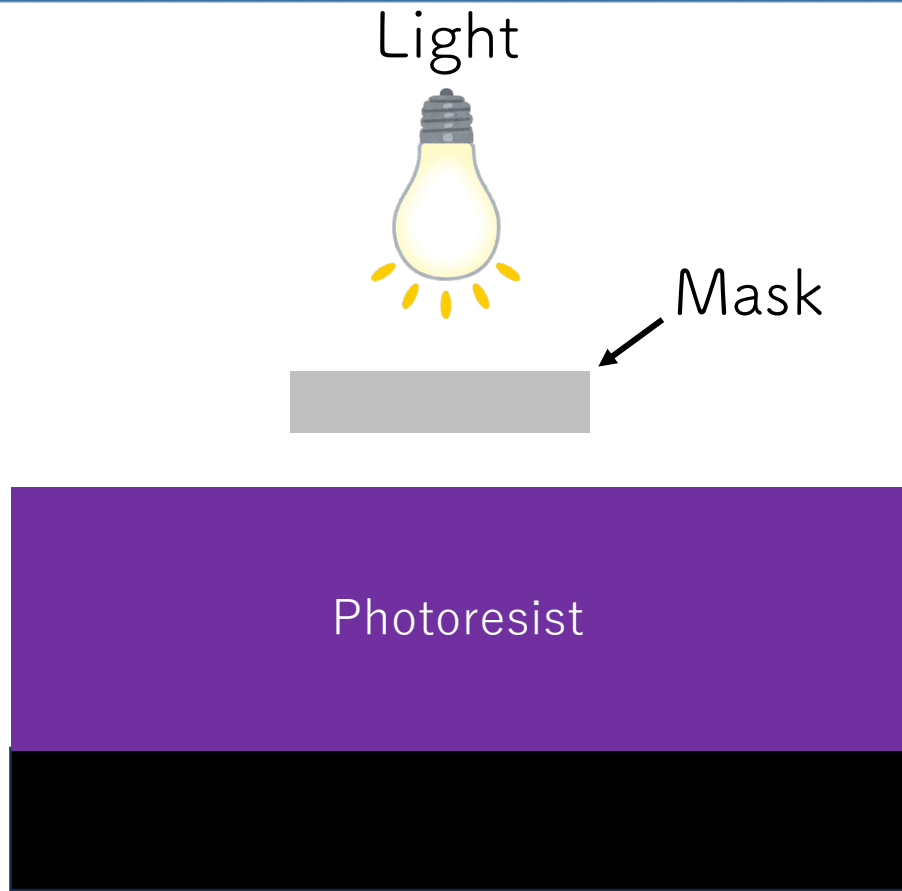
※Top-View Diagram

Center



※Top-View Diagram

1. Non-Uniformity of Exposure Dose
2. Focus Misalignment During Exposure
3. Variations in Development Process

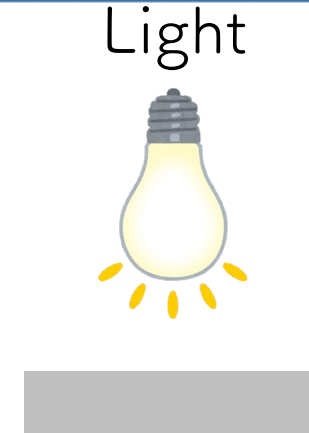


(1) When thick

As the light travels towards the bottom, it is absorbed by the photosensitizer, causing the light intensity to weaken.



The resist width increases.

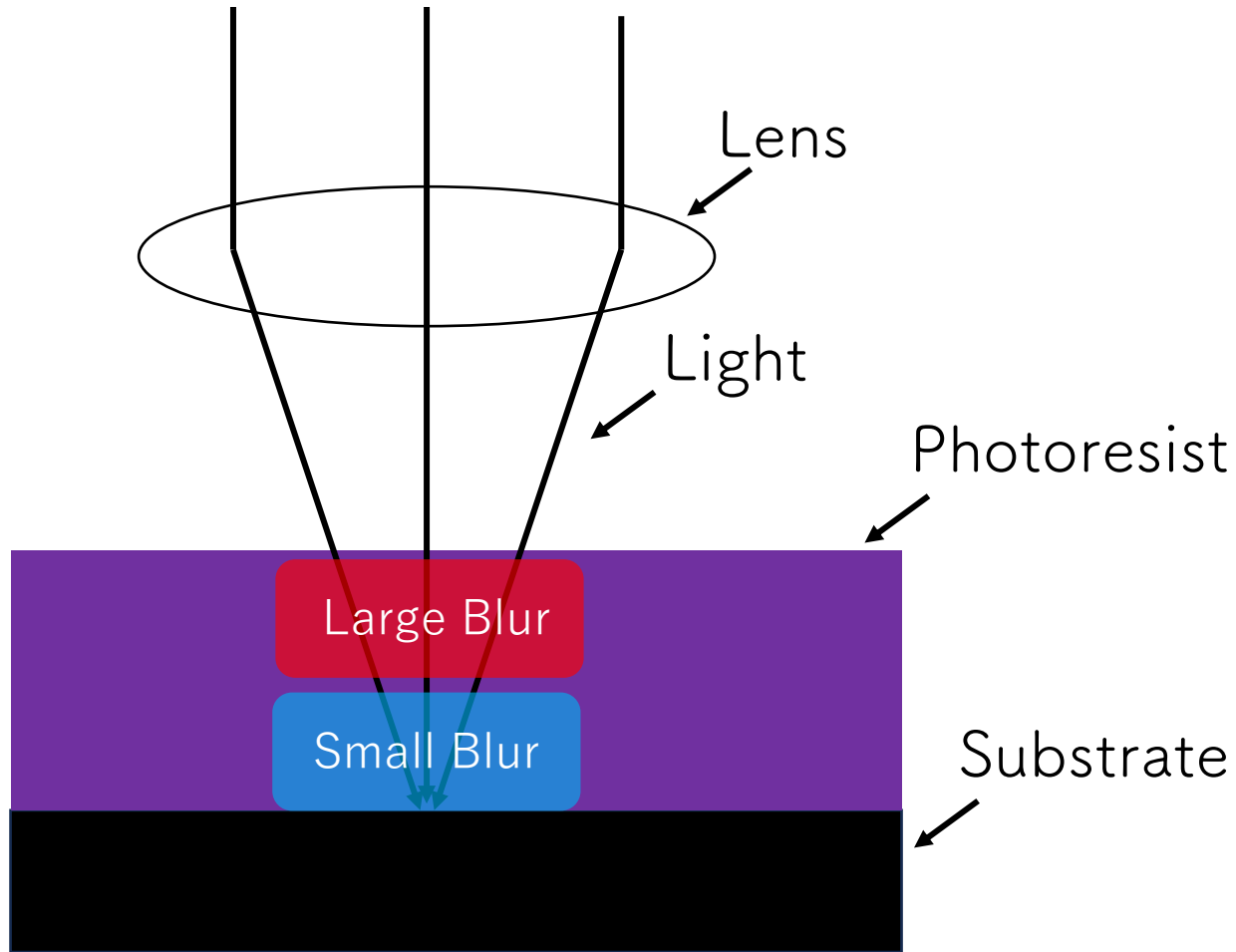


(2) When thin

Excess light is absorbed by the resist, causing light to penetrate into the unexposed areas.



The resist width decreases.

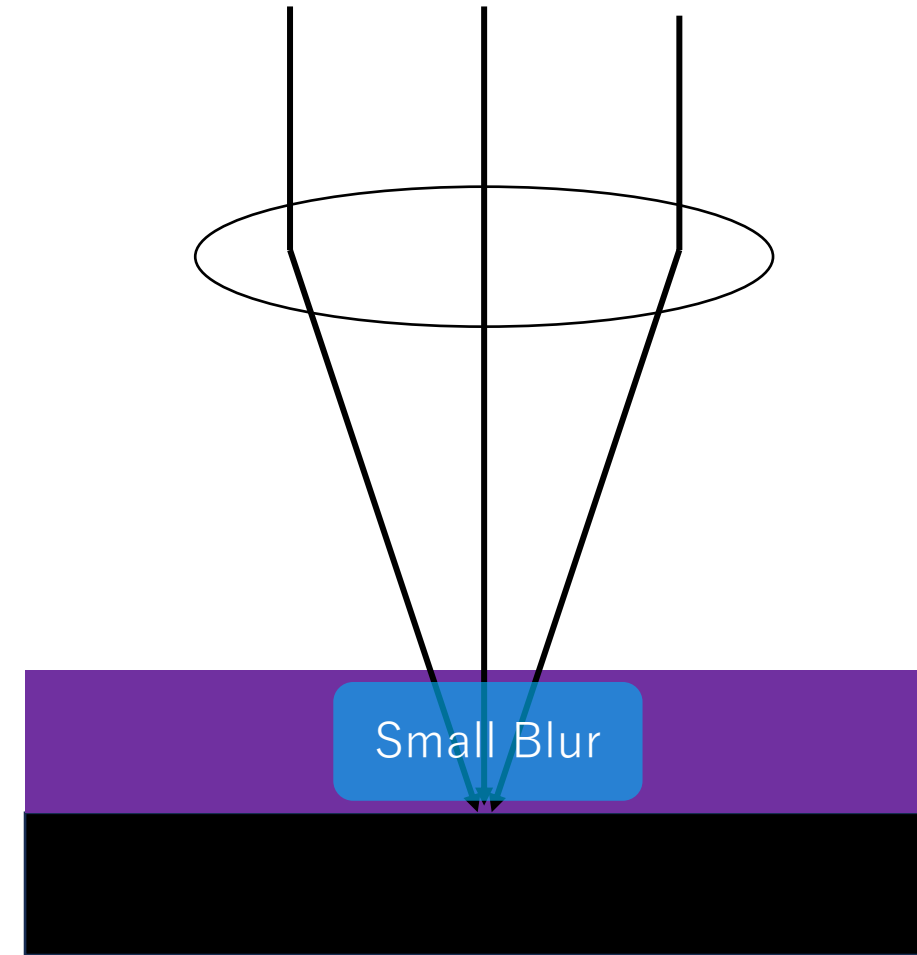


(1) When thick

Areas with both large and small blur exist.



Deviates from the design dimensions

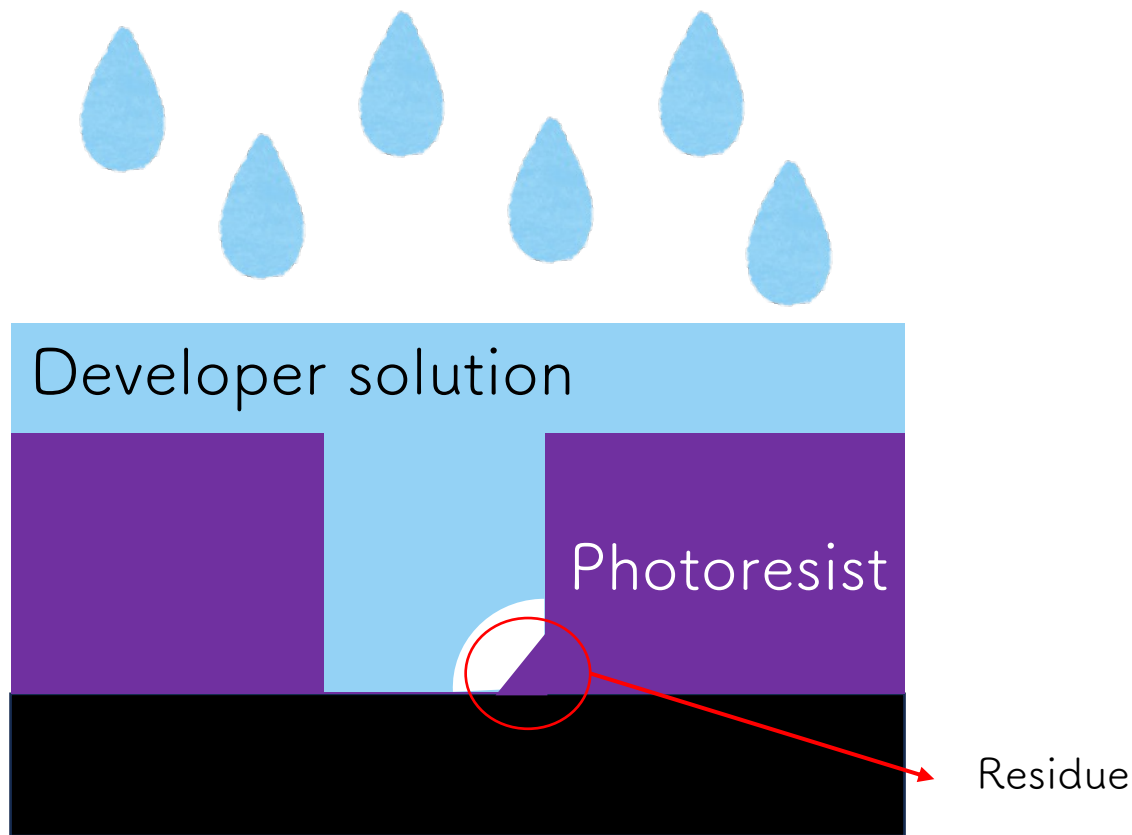


(2) When thin

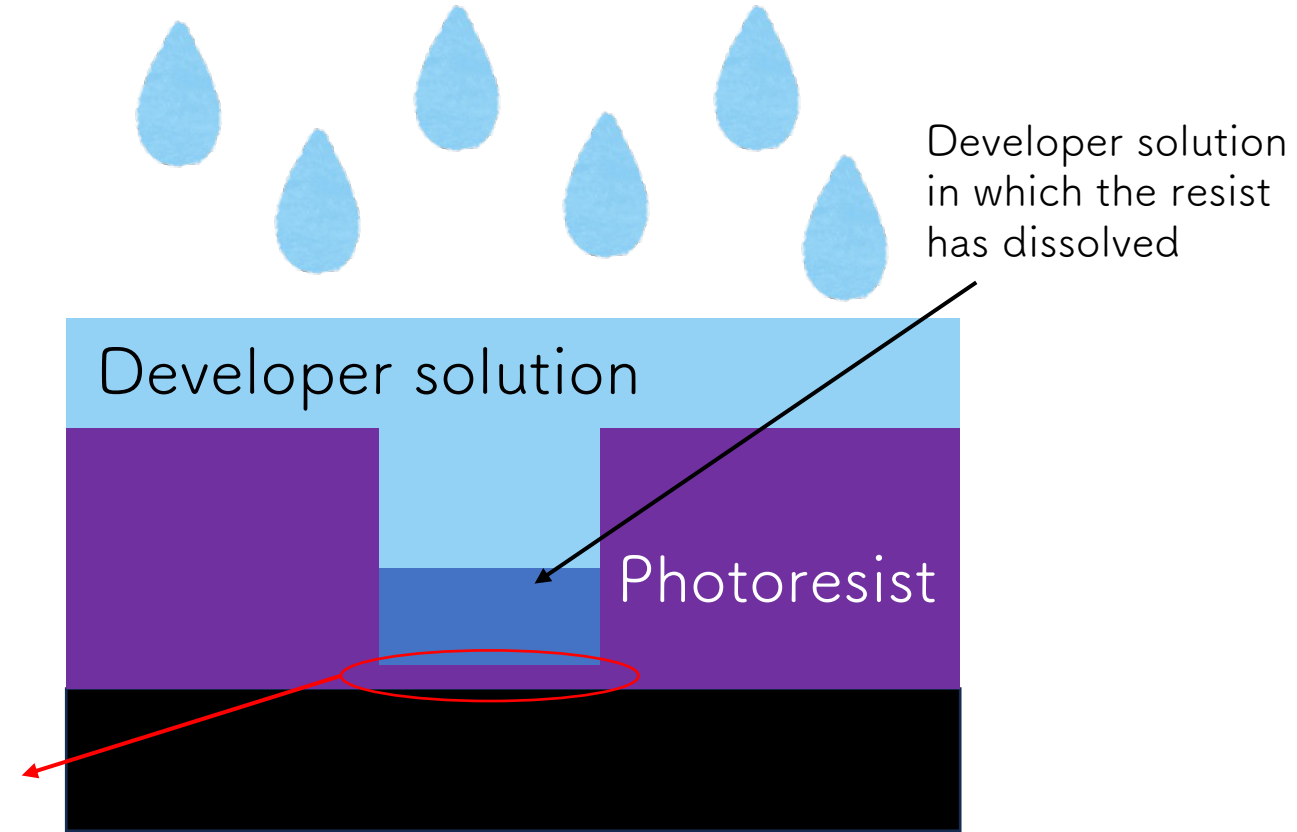
Only small blur exists



Matches the design dimensions

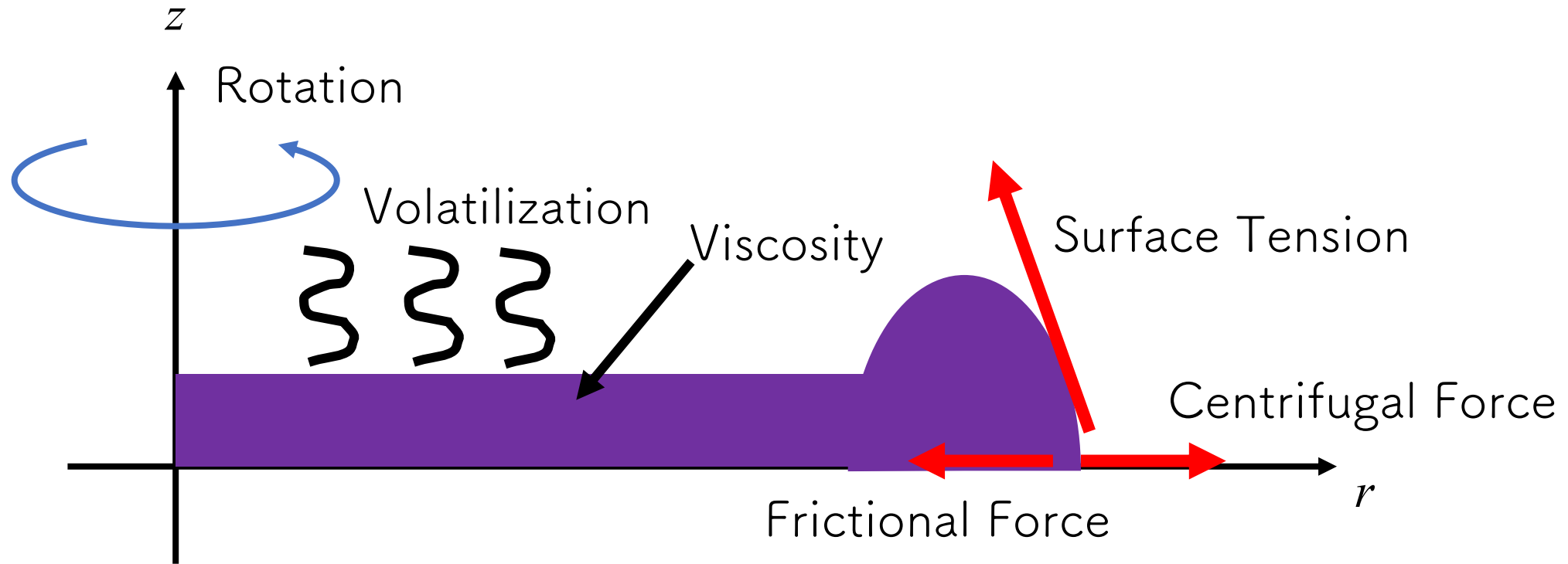


(1) Changes in developer penetration can cause incomplete coverage, bubble formation, and development residue.



(2) Developer solution containing dissolved resist may stagnate at the bottom, potentially causing development residue.

1. Non-Uniformity of Exposure Dose
2. Focus Misalignment During Exposure
3. Variations in Development Process



Physical Forces Acting During Coating ⁽³⁾

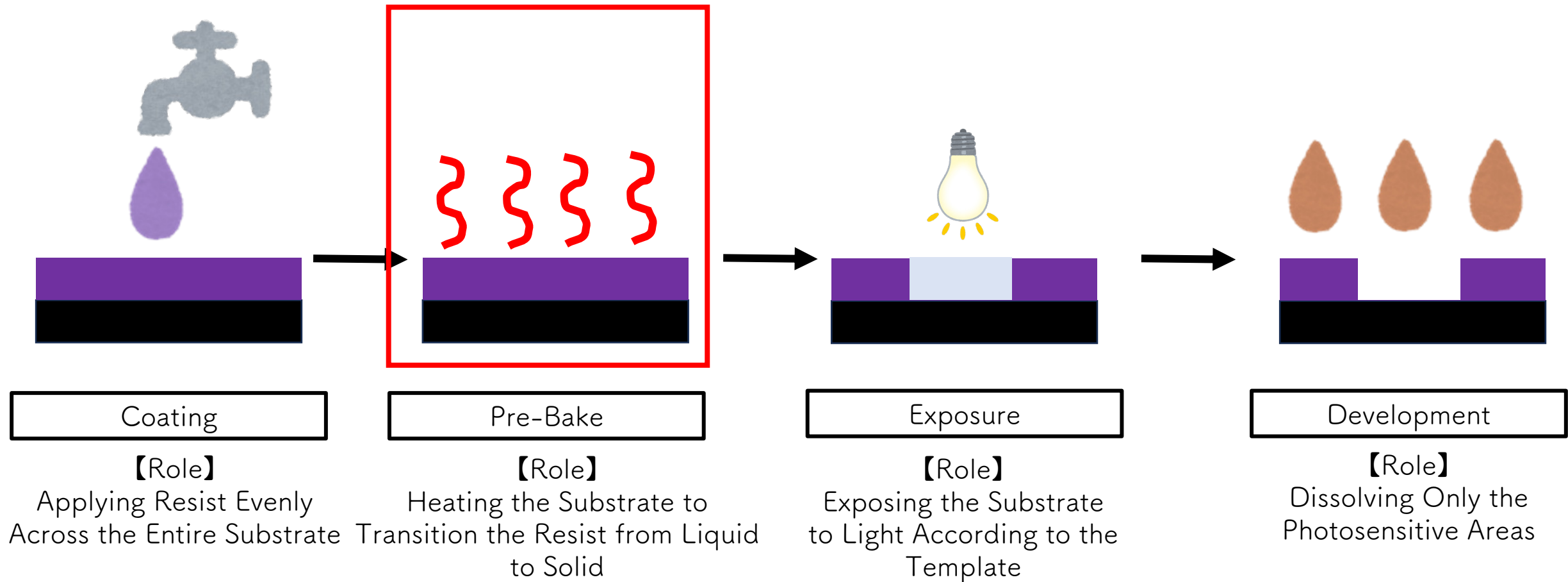
Point

Key parameters are "frictional force between substrate and resist," "surface tension," "centrifugal force," and "viscosity change due to solvent volatilization."

Coating Process is

- Resist is applied using the “spin coating” method.
- The resist film thickness is inversely proportional to the stage’s spin speed.
- When the film thickness distribution varies, the resist dimensions also vary.
- Set conditions such as spin speed while being aware of the physical key parameters.

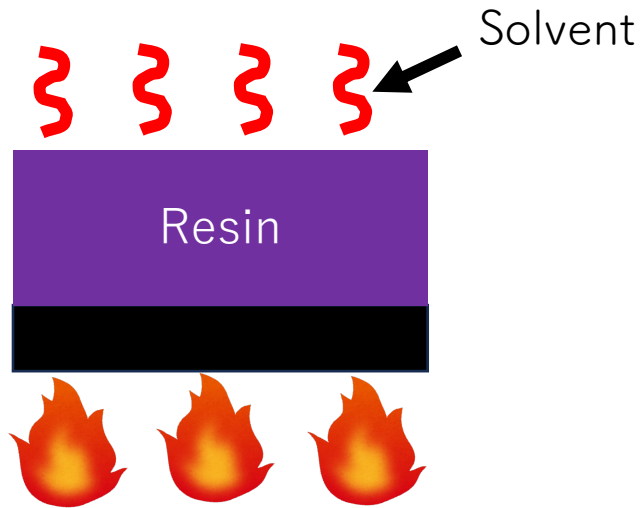
Pre-bake



What

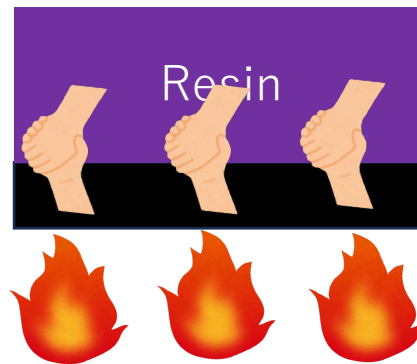
Pre + Bake → The process of applying heat to the substrate before exposure

Purpose



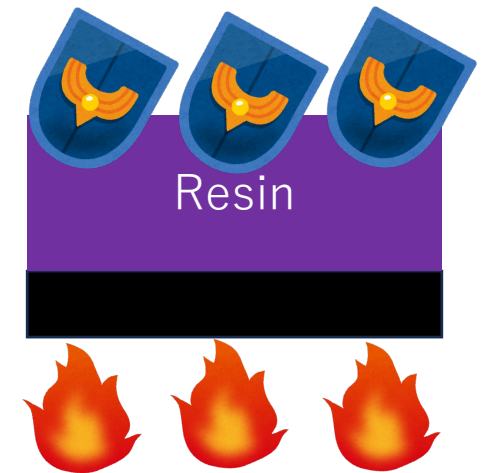
1. Solvent Removal

Evaporate the solvent to solidify the resist, changing it from a liquid to a solid state.



2. Improved Adhesion

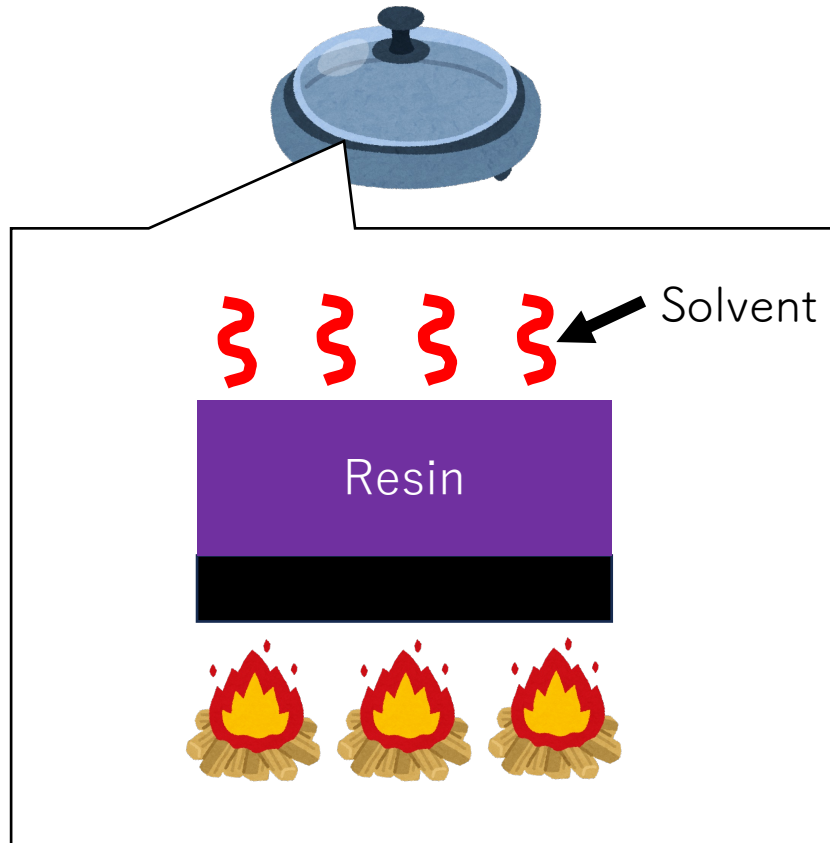
As the solvent evaporates, the resist contracts, becomes denser, and improves adhesion strength.



3. Increased Strength

By solidifying, the resist becomes harder, increasing its mechanical strength.

(1) Hot plate-type



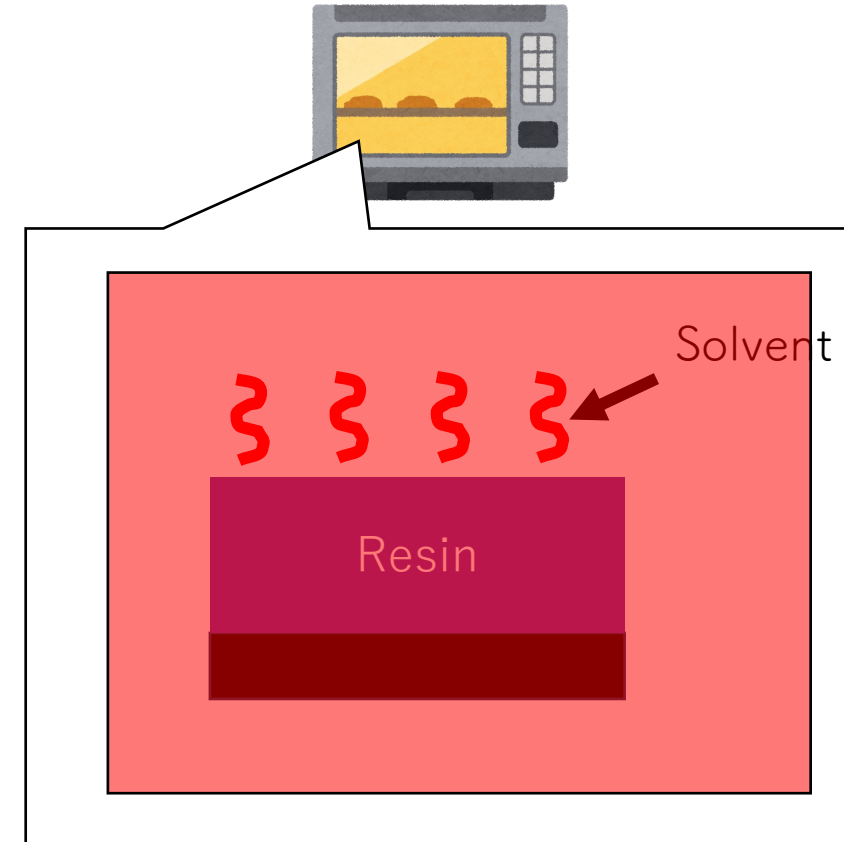
Pros

Minimal temperature variation across substrates

Cons

Limited number of substrates that can be processed at once

(2) Oven-type

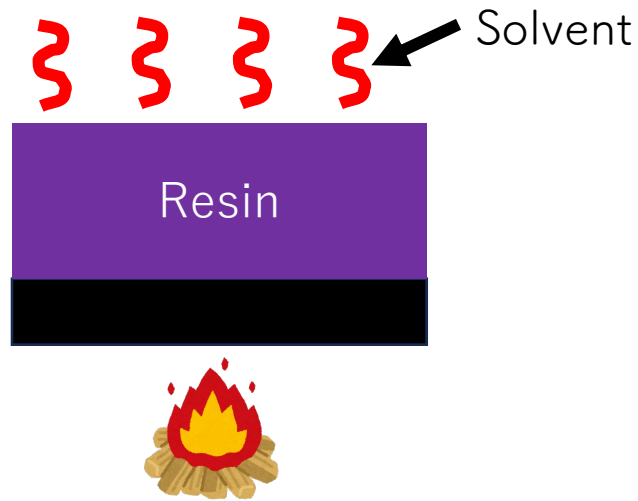


Pros

Multiple wafers can be processed simultaneously

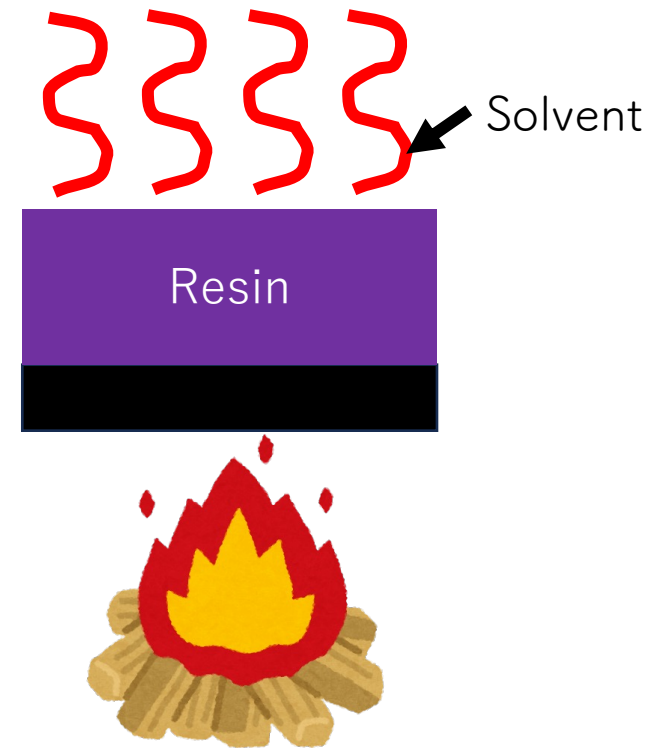
Cons

Prone to temperature variation across the substrates



(1) Insufficient Heating

- Solvent Residue
→ Poor exposure and development results
- Decreased Adhesion
→ Peeling off



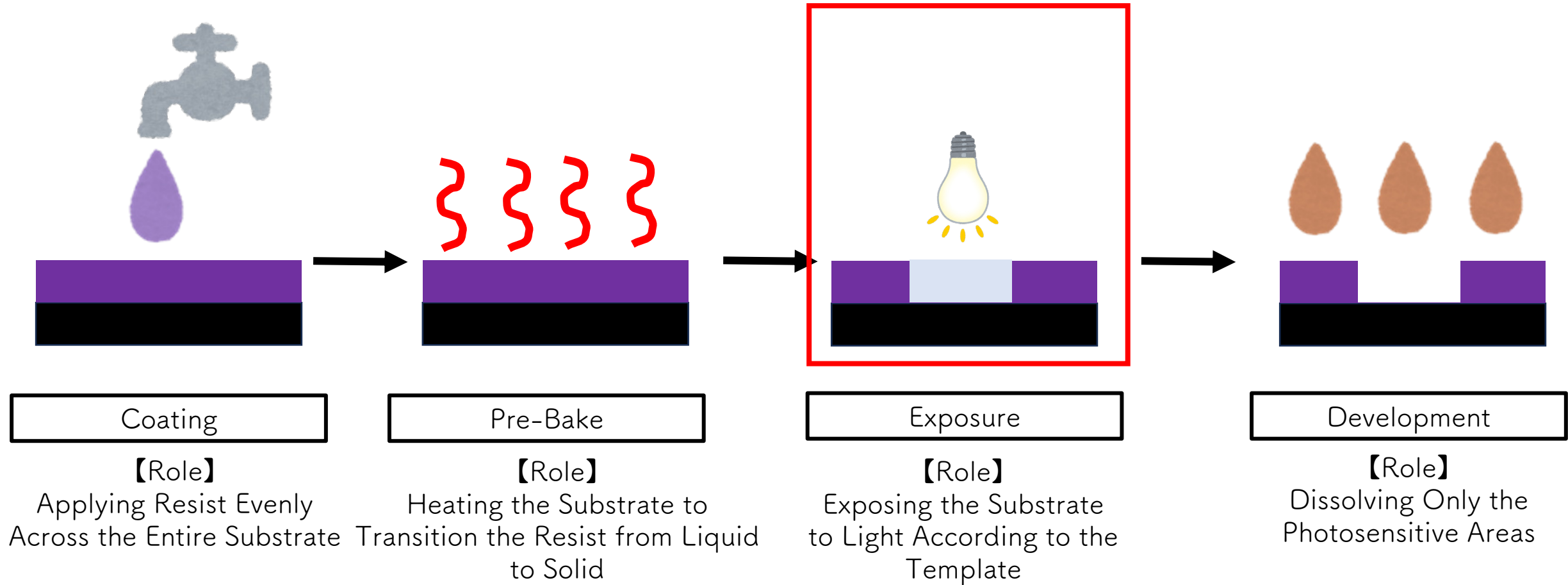
(2) Overheating

- Resist Hardening
→ Resist residue
- Decreased Sensitivity
→ Poor exposure and development
- Cracking of the Resist
→ The resist no longer functions properly

Pre-bake is

- The process of applying heat to the substrate before exposure
- Performed for “solvent removal,” “improved adhesion,” and “increased strength.”
- There are two types: hot plate type and oven type.
- The issues are insufficient heating and overheating.

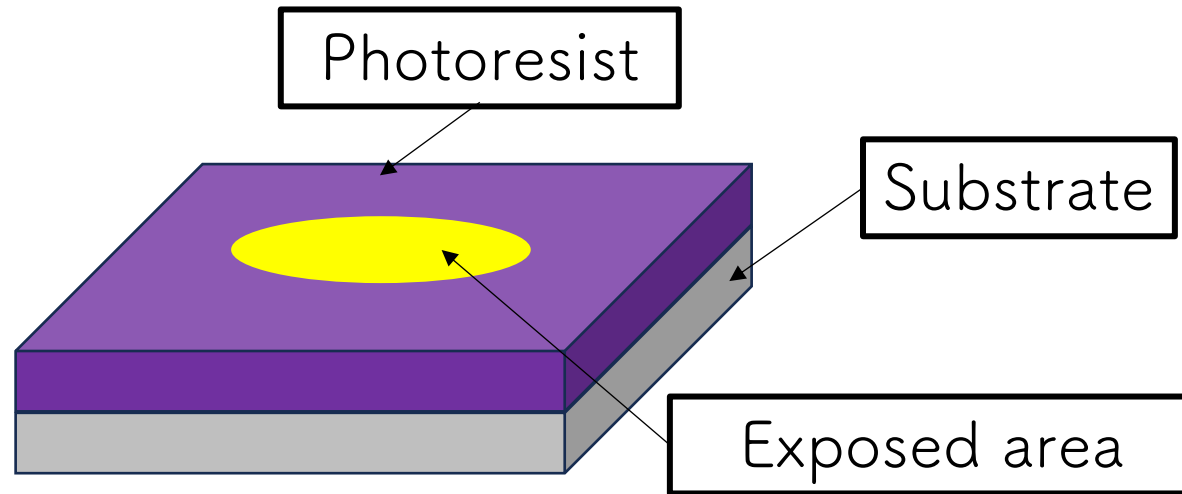
Exposure



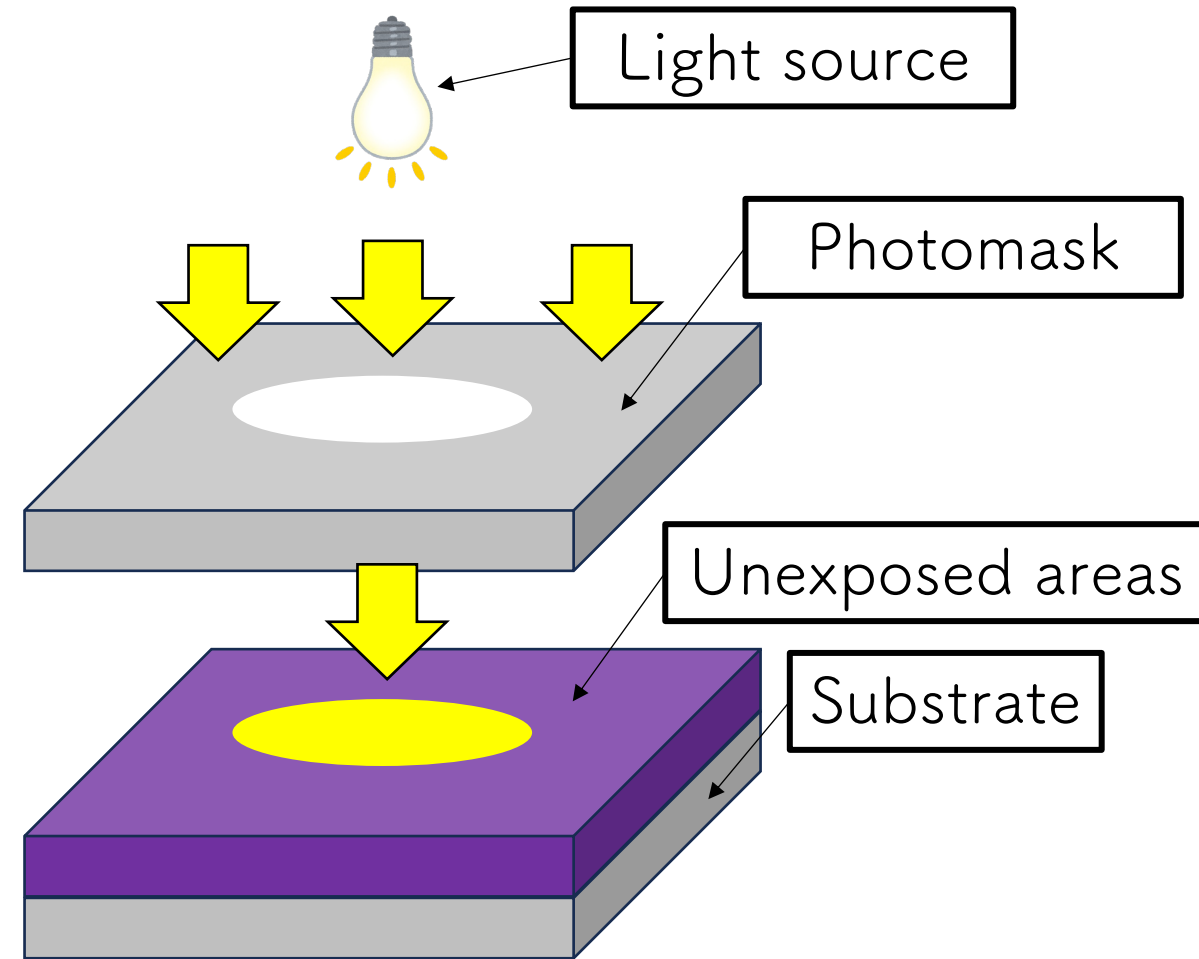
Purpose

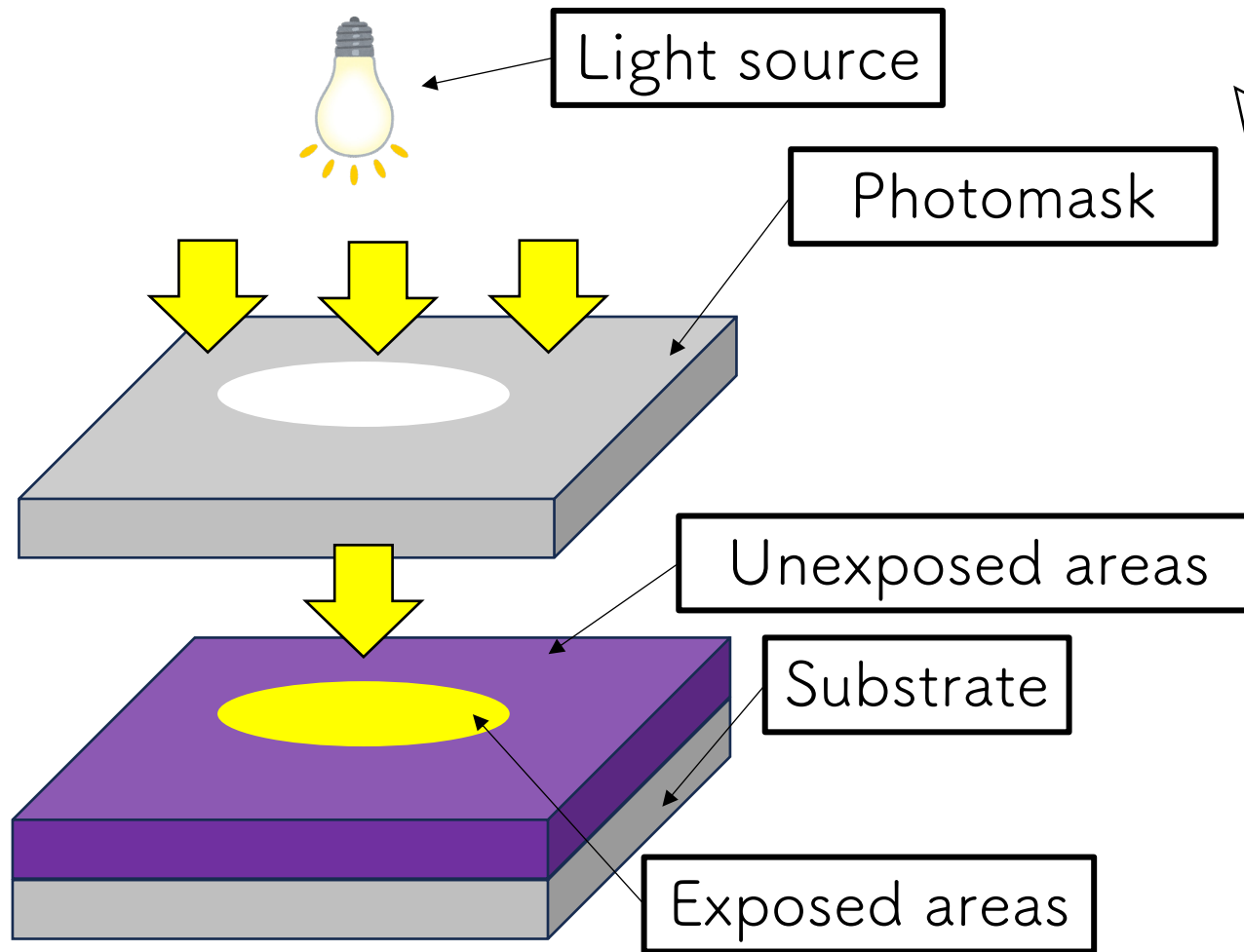
Apply light only to the areas of the substrate that need to be exposed.

→ Forming the circuit pattern



Substrate after exposure



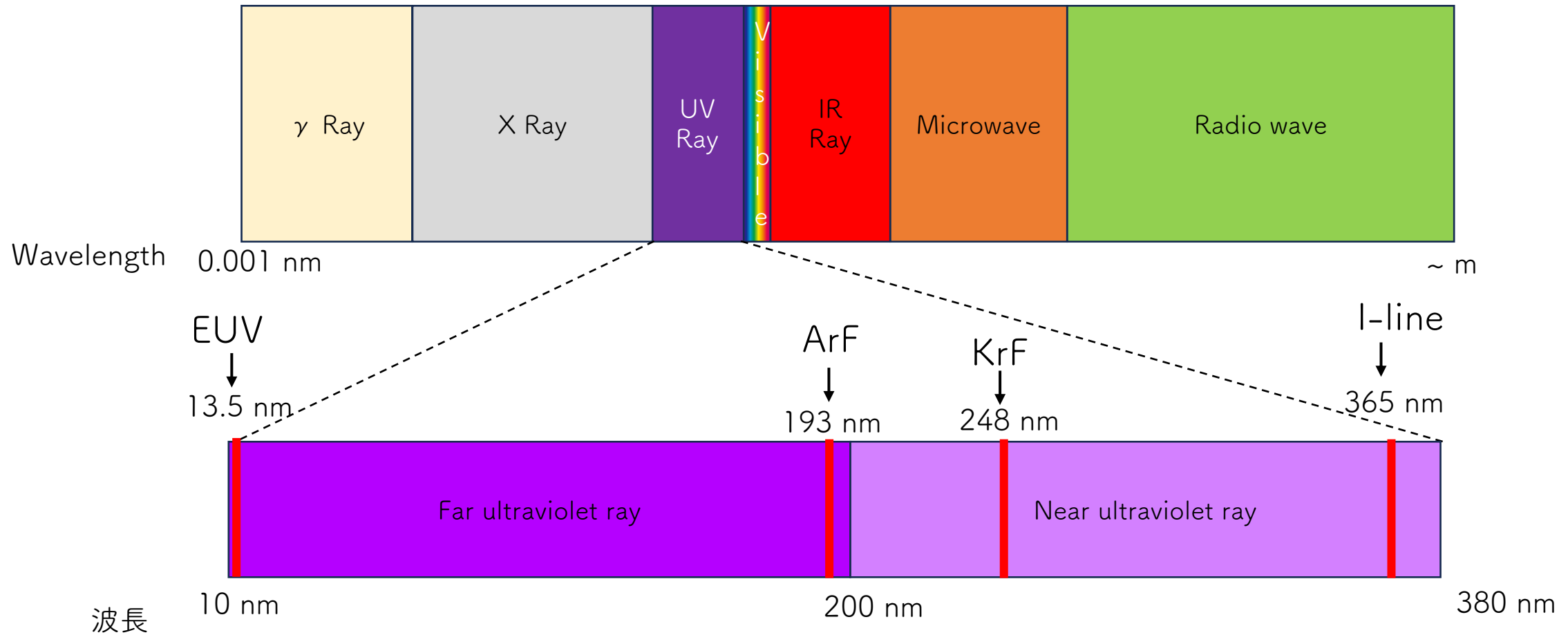


Used materials

Cr

Reasons⁽⁴⁾

- Scratch-resistant
→ Mask Longevity
- High Light Blocking Capability
→ Capable of accurately transferring patterns
- Relatively affordable
→ Reduce manufacturing costs



Point

- Ultraviolet (UV) light (10nm–380nm) is used for exposure.
- Within the ultraviolet spectrum, a specific wavelength is used for exposure.

Wavelength
Change

Pattern Resolution

$$R = k_1 \frac{\lambda}{NA}$$

R : Pattern Resolution
 λ : Wavelength of Light
NA : Numerical Aperture (NA)
 k_1 : Process Constant

The shorter the wavelength,
the finer the pattern that can be formed.

Depth of Focus

$$DOF = k_2 \frac{\lambda}{NA^2}$$

DOF : Depth of Focus
 λ : Wavelength of Light
NA : Numerical Aperture (NA)
 k_2 : Process Constant

The shorter the wavelength, the smaller the depth of focus,
making it more susceptible to the effects of surface irregularities.

Point

There is a trade-off relationship between pattern resolution and DOF

Definition

Two holes are created through the photoresist to the substrate⁽⁵⁾, with the width between the resist areas referred to as the resolution.

Photoresist

Substrate

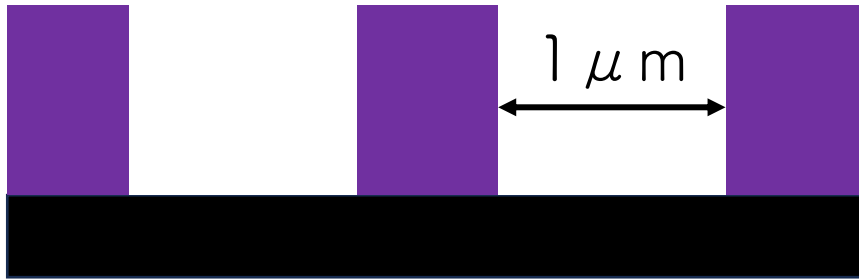
Resolution

Width

① Resist is completely removed.

② The resist in the middle remains.

(5) 田中秀治. 回折限界を超える近接場光リソグラフィに関する研究. Diss. 東京大学, 1999.



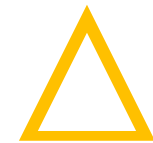
Resolution = $1\ \mu\text{m}$



Resist Residue NG



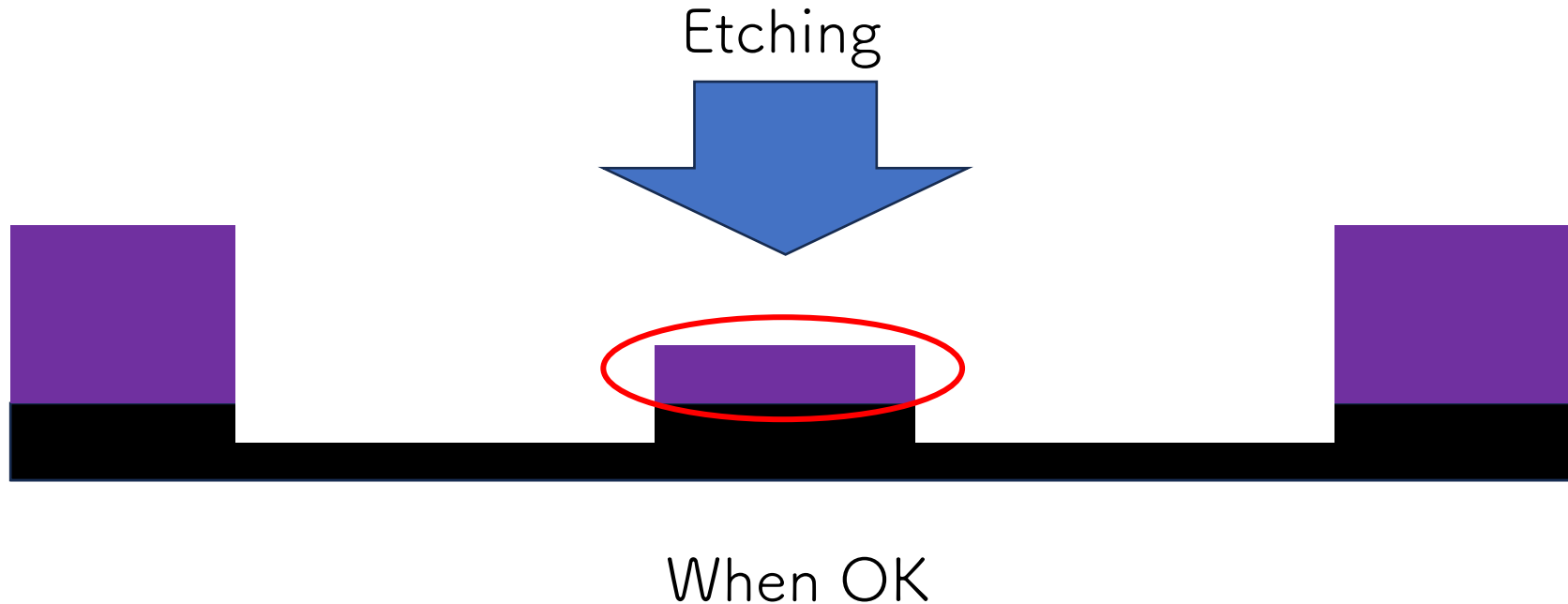
Pattern Loss NG



Depends on next
process

What

The resist has served its purpose after the next process.

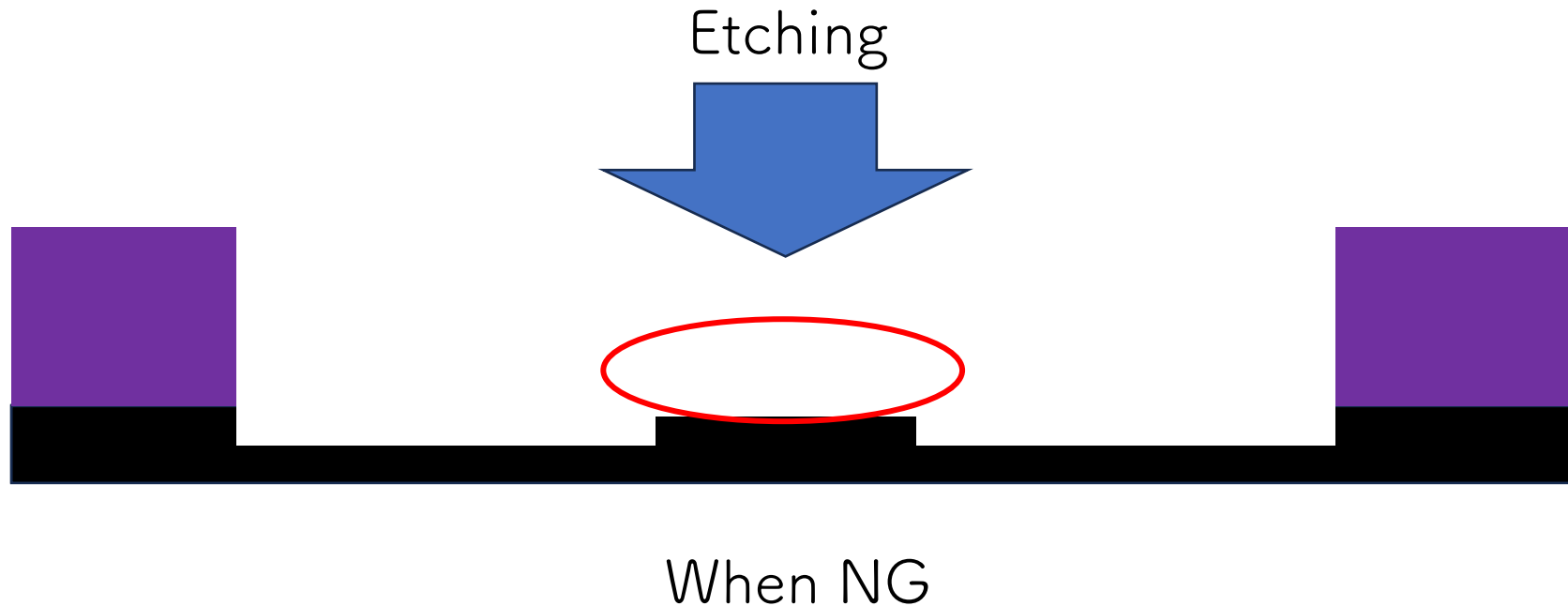


Point

If the resist remains after etching and the central pattern is not etched away, it is considered OK.

What

The resist has served its purpose after the next process.



Point

After etching, the resist is completely removed, and the central pattern, along with the substrate, has been etched away.

NA Change

Pattern resolution

$$R = k_1 \frac{\lambda}{NA}$$

R : Pattern Resolution
 λ : Wavelength of Light
NA : Numerical Aperture (NA)
 k_1 : Process Constant

The higher the Numerical Aperture (NA),
the better the resolution (smaller the resolution).

Depth of Focus

$$DOF = k_2 \frac{\lambda}{NA^2}$$

DOF : Depth of Focus
 λ : Wavelength of Light
NA : Numerical Aperture (NA)
 k_2 : Process Constant

The larger the Numerical Aperture (NA), the smaller the depth of focus,
making it more susceptible to the effects of surface irregularities.

Point

NA has an inverse relationship with both pattern resolution and DOF

What is NA (Numerical Aperture)?

43

What

Indicates the “light-gathering ability” determined by the lens and the surrounding environment during exposure.

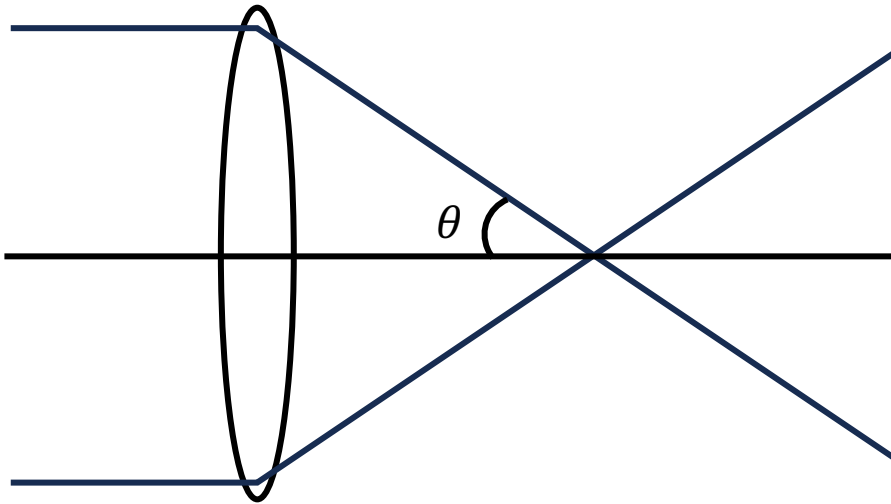
Definition

$$NA = n \sin\theta$$

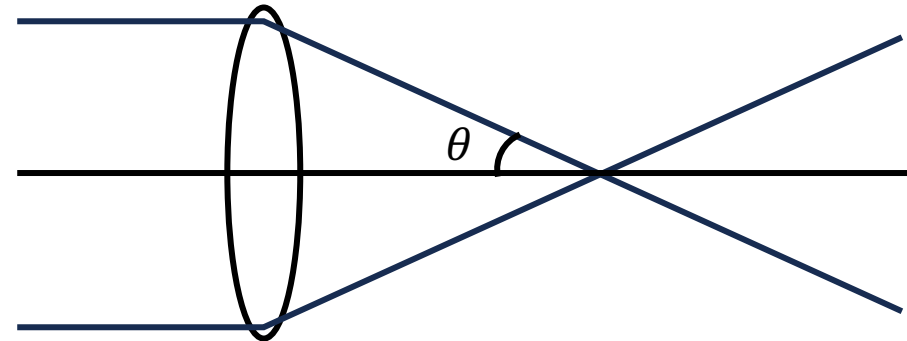
NA : Numerical Aperture

n : Refractive index of the medium in which the lens is placed

θ : Angle



Large NA

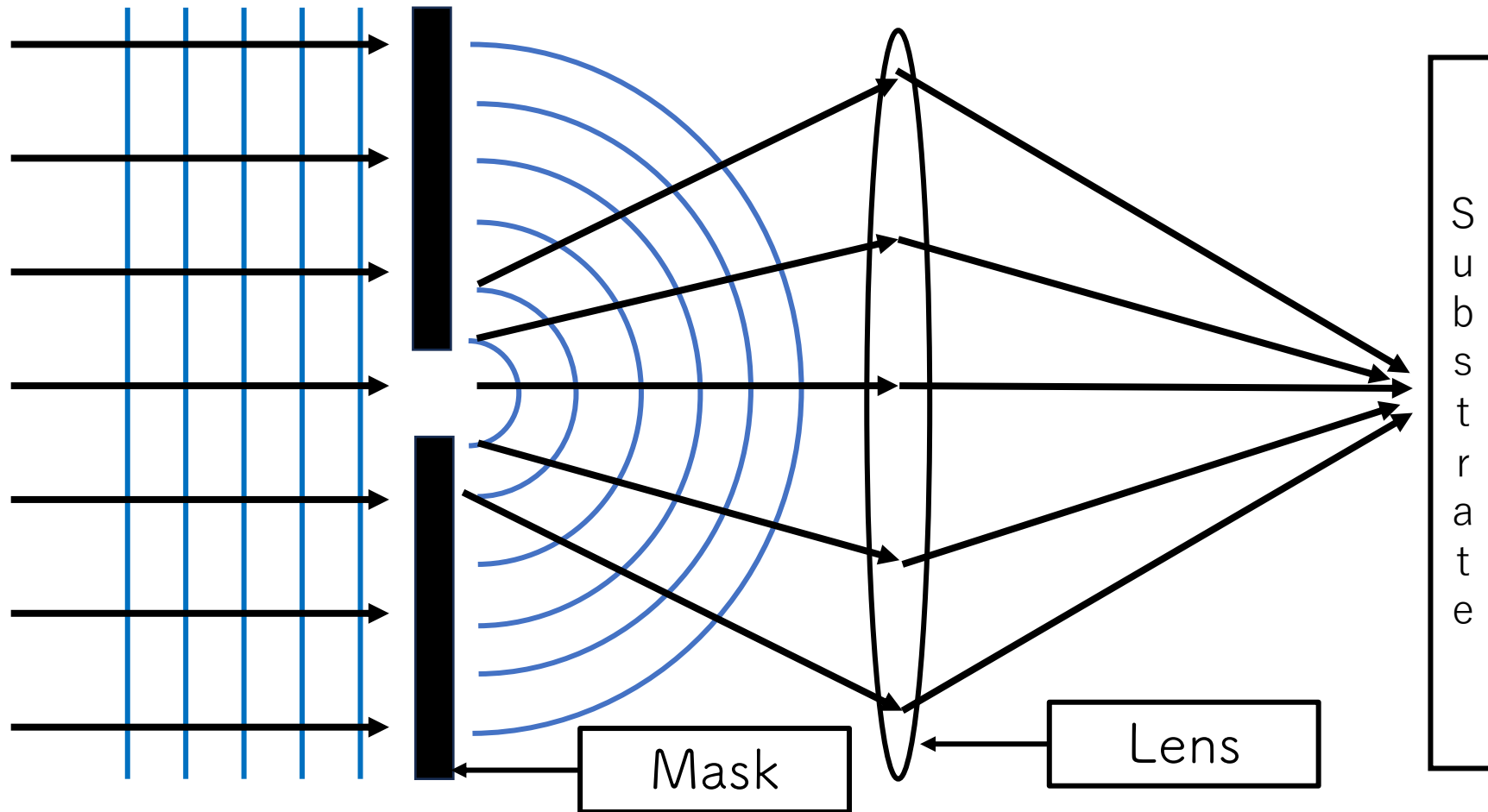


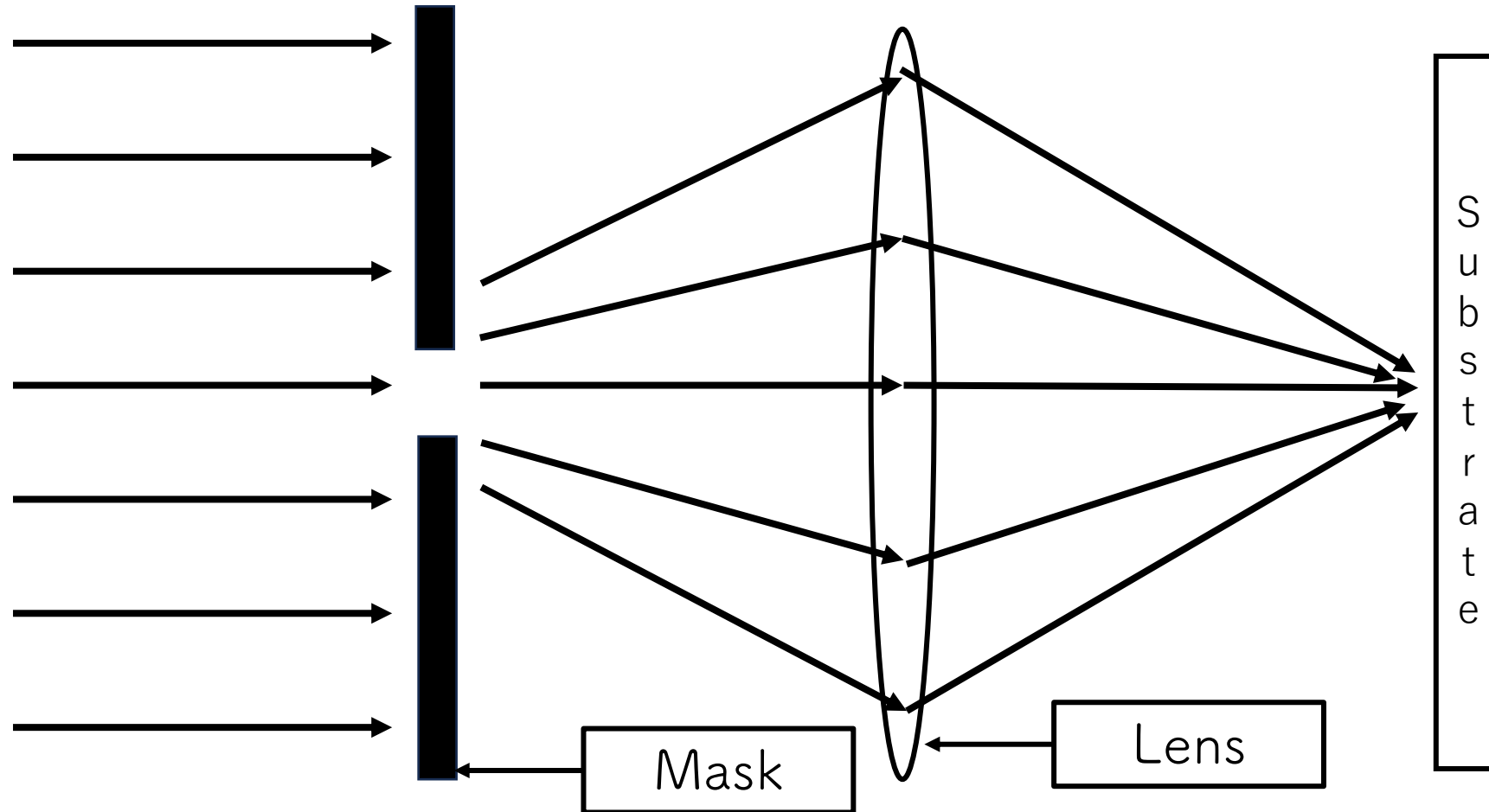
Small NA

$$R = k_1 \frac{\lambda}{NA}$$

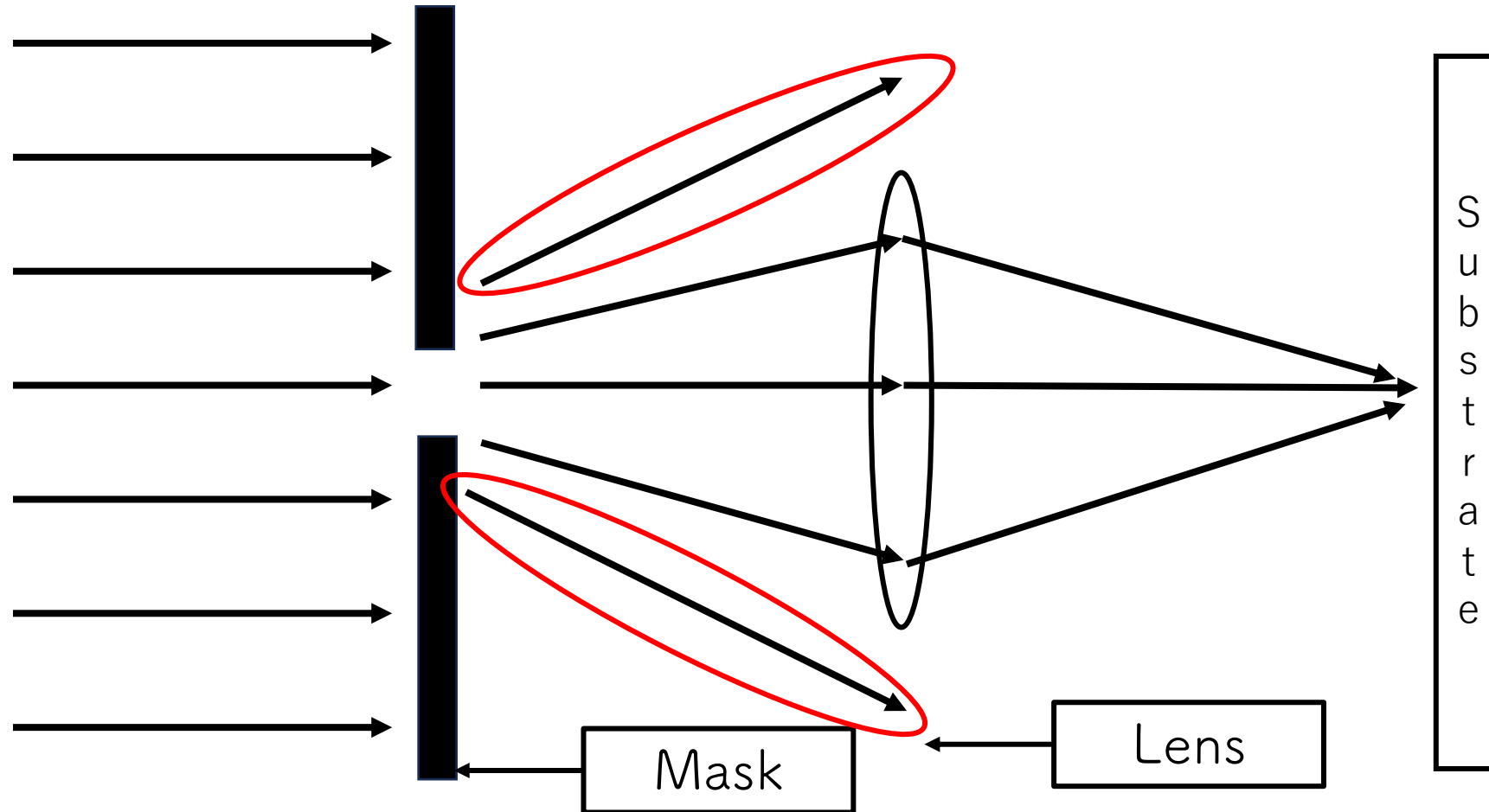
R : Pattern Resolution
 λ : Wavelength of Light
NA : Numerical Aperture (NA)
 k_1 : Process Constant

The higher the Numerical Aperture (NA), the better the resolution (smaller the resolution).





When NA is large, diffracted light can be gathered by the lens,
→ Focus light on the substrate even for fine patterns, resulting in better resolution.



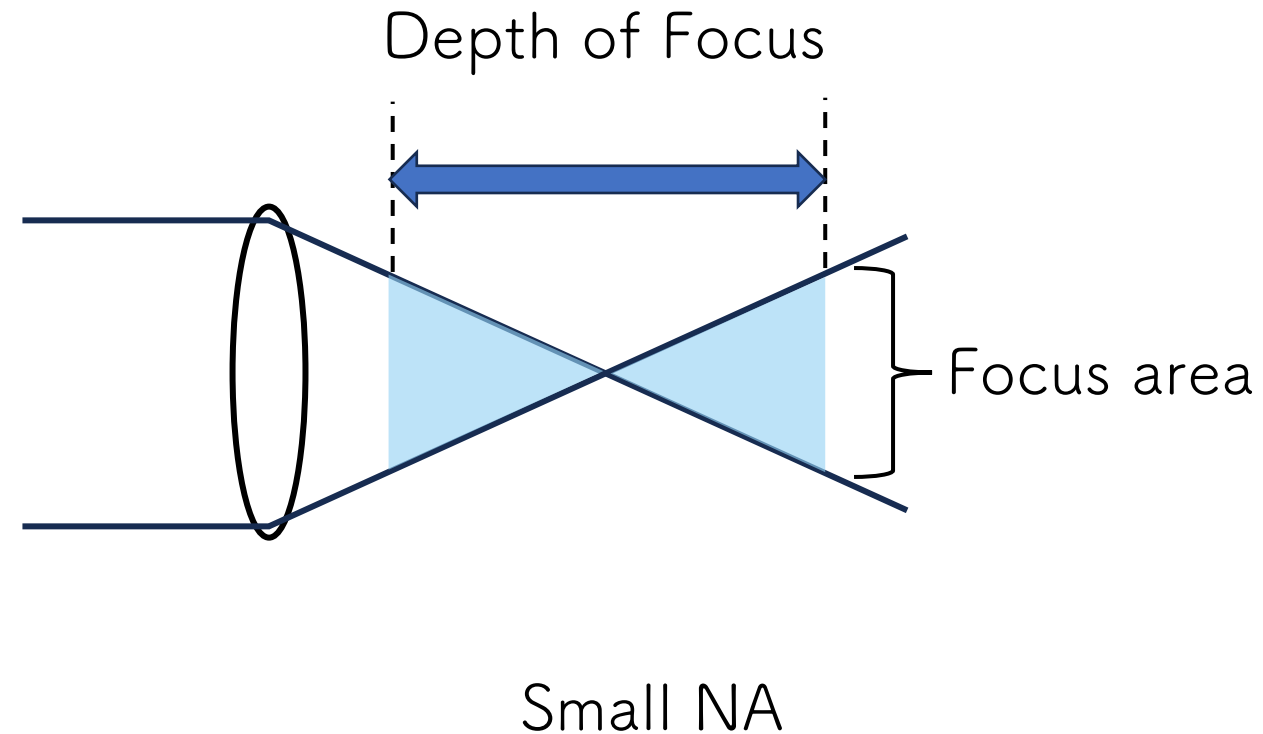
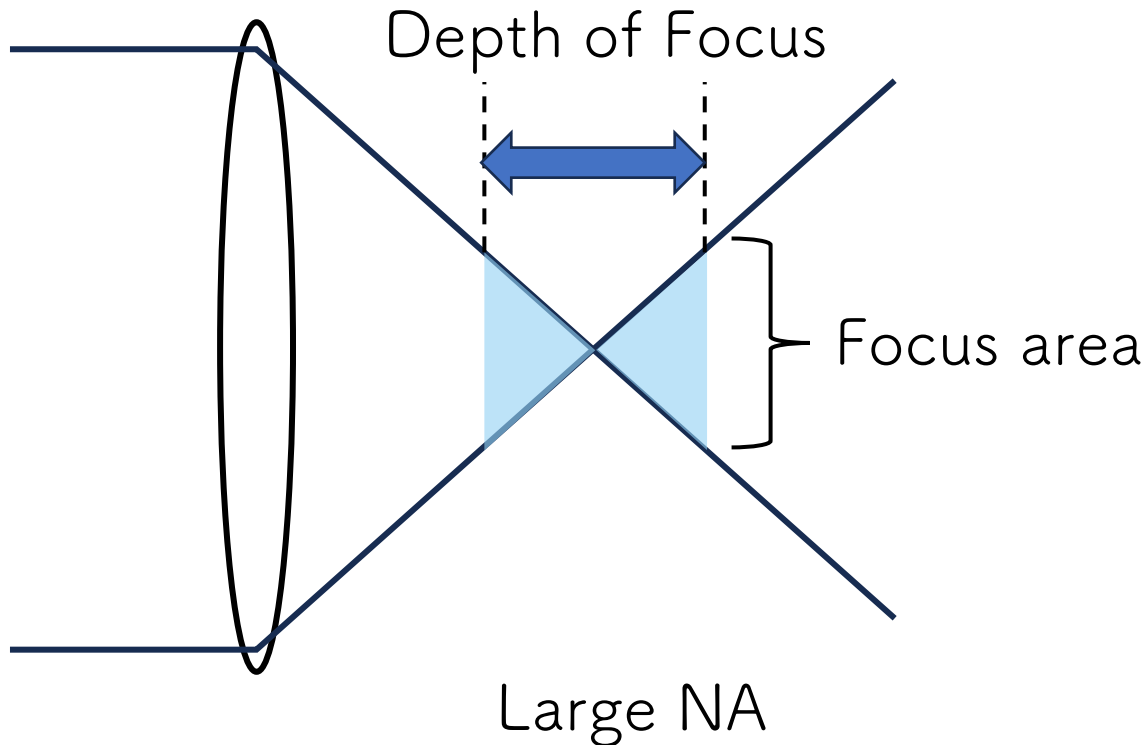
When NA is small, some of the diffracted light cannot be collected by the lens
→ inability to obtain the pattern, leading to a decrease in resolution.

What is DOF

The range where the focus can be considered sharp.

$$\text{DOF} = k_2 \frac{\lambda}{NA^2}$$

DOF : Depth of Focus
 λ : Wavelength of Light
NA : Numerical Aperture (NA)
 k_2 : Process Constant



What is alignment?

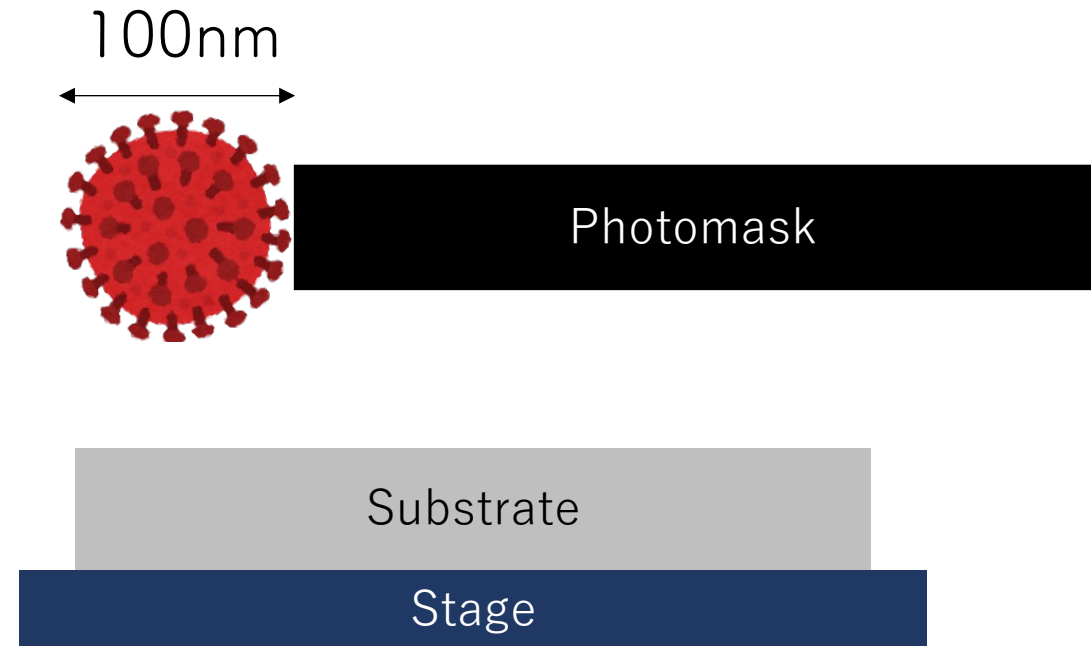
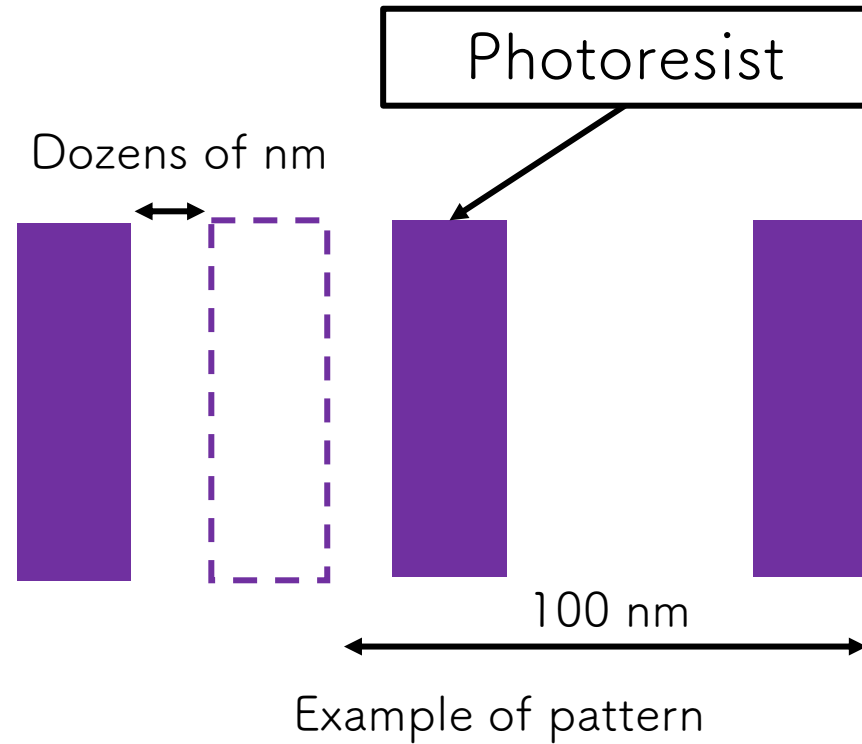
It refers to the alignment technology between the “substrate” and the “mask.”

Why important?

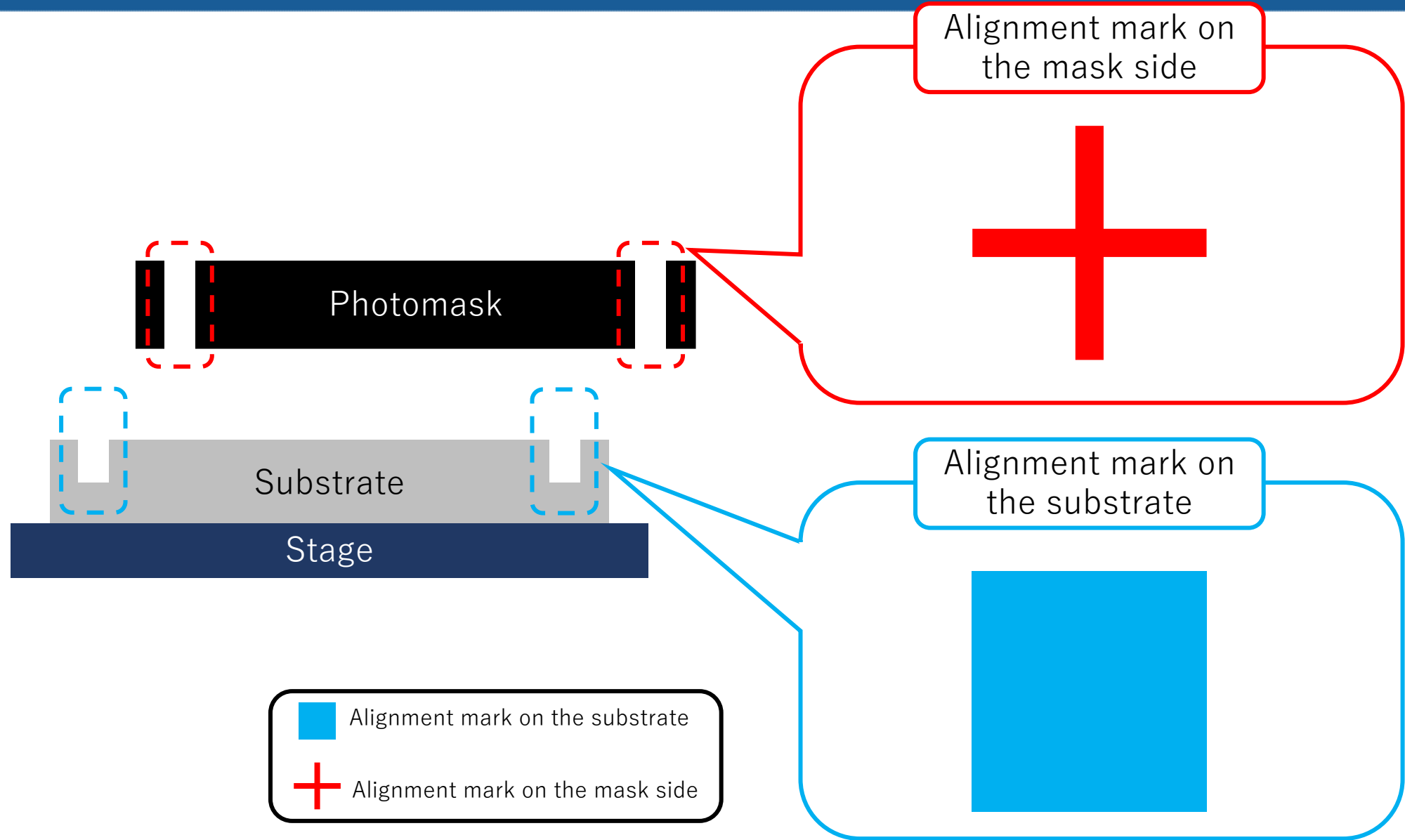
In semiconductor processes, the scale of sizes handled ranges from micrometers (μm) to nanometers (nm).

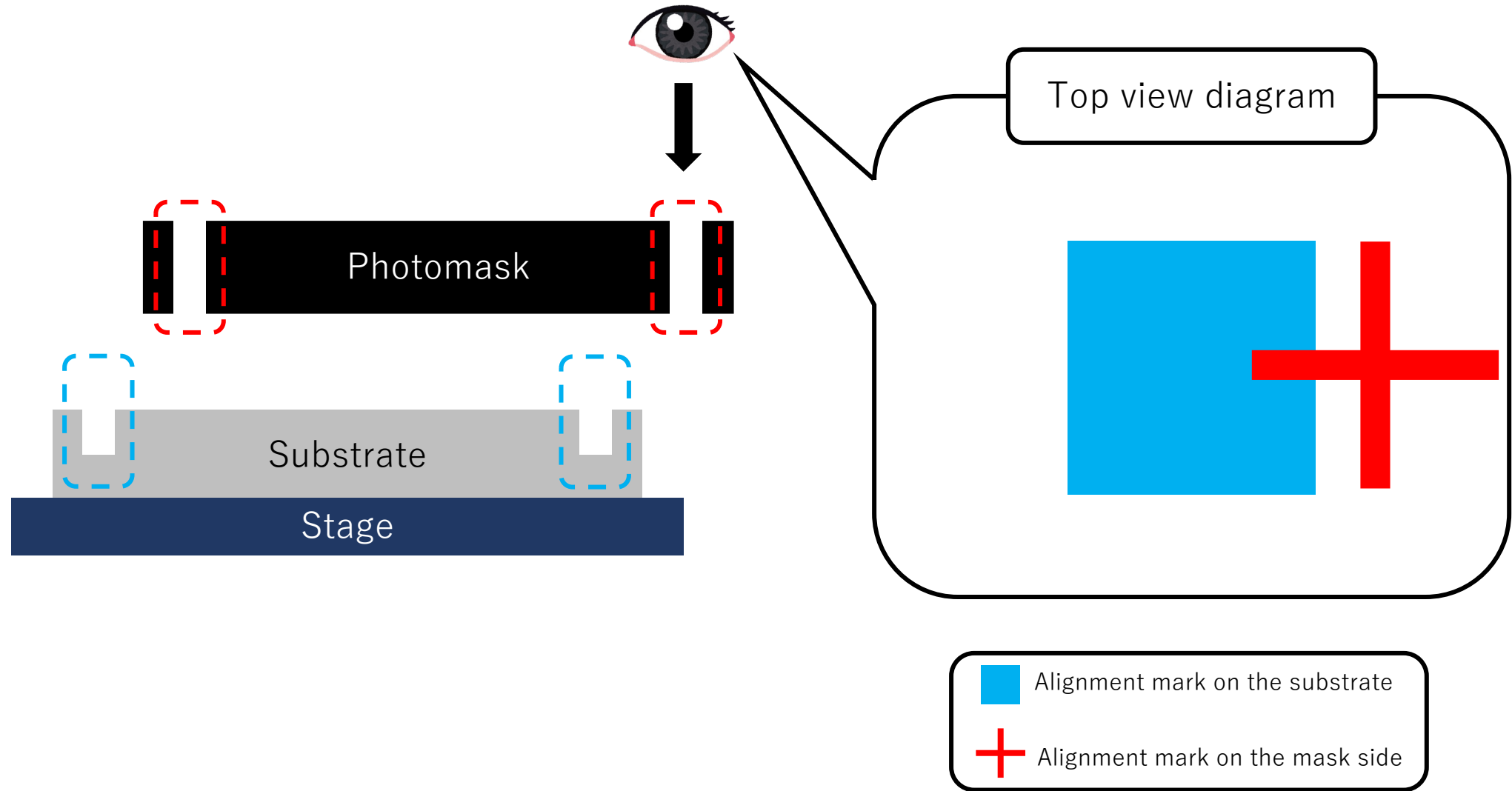


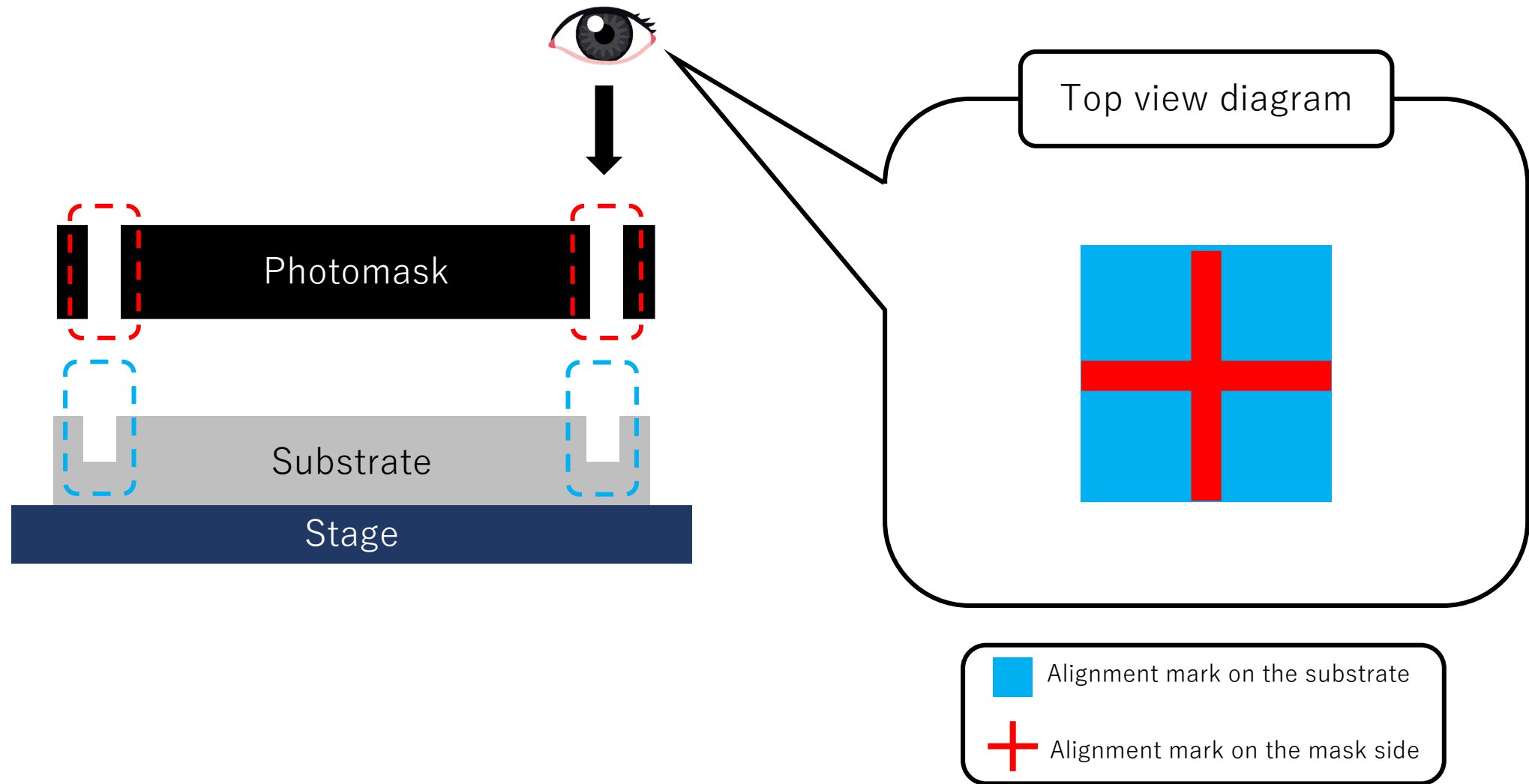
To achieve accurate patterning,
extremely precise alignment technology is required.



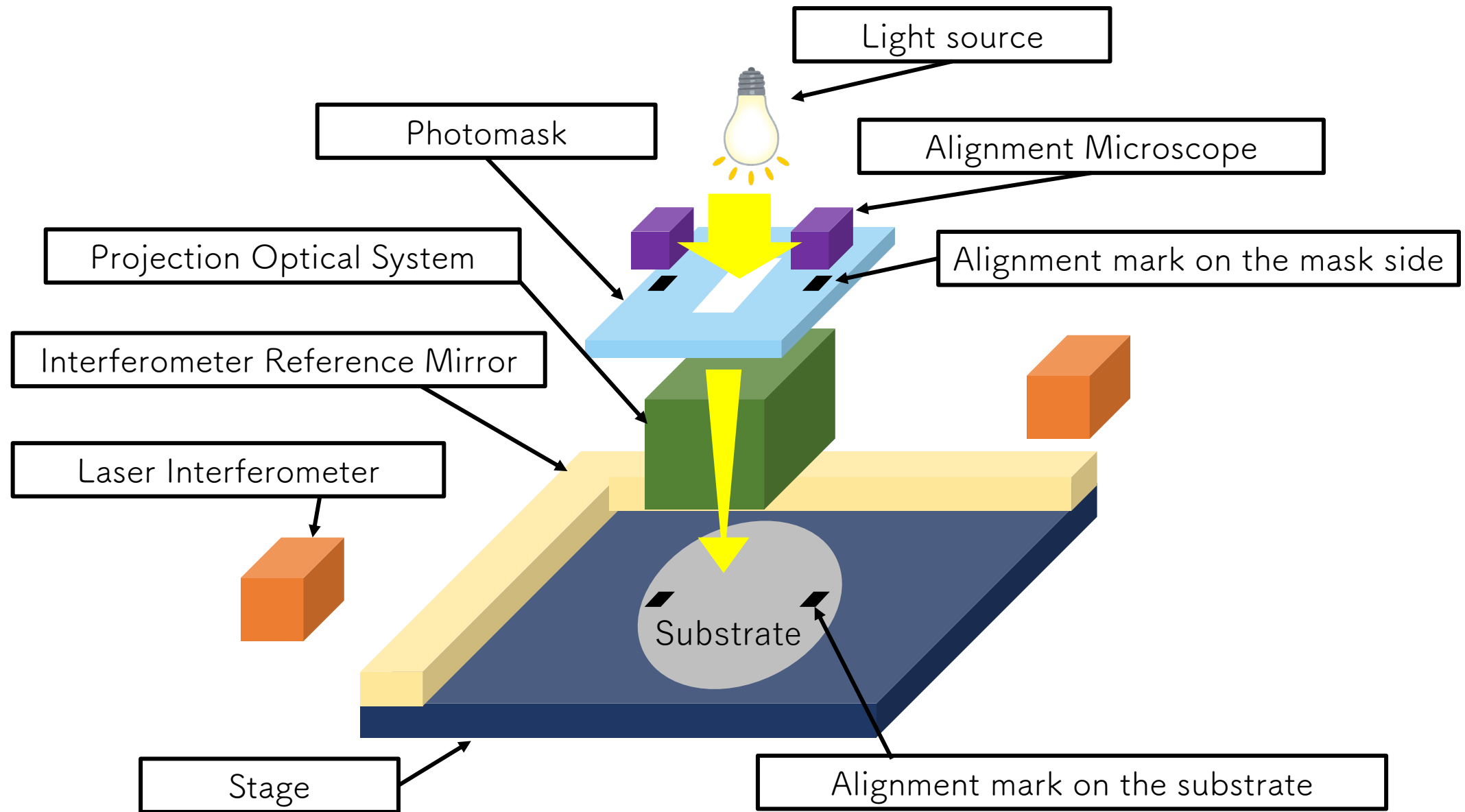
The size of a single virus. Even a small misalignment between the mask and substrate can prevent accurate manufacturing according to the design.

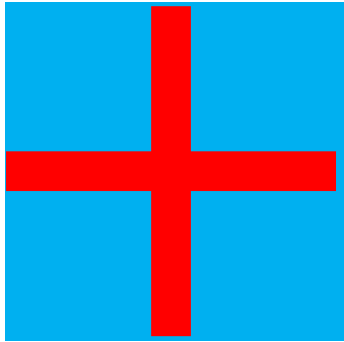




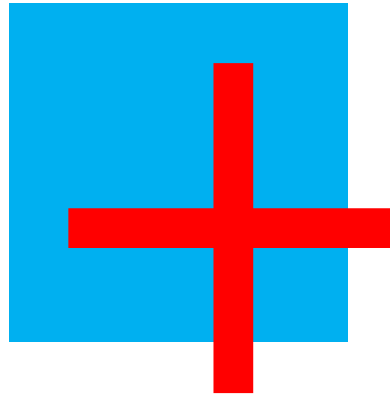


Adjust the alignment marks on the mask side and substrate side to match.

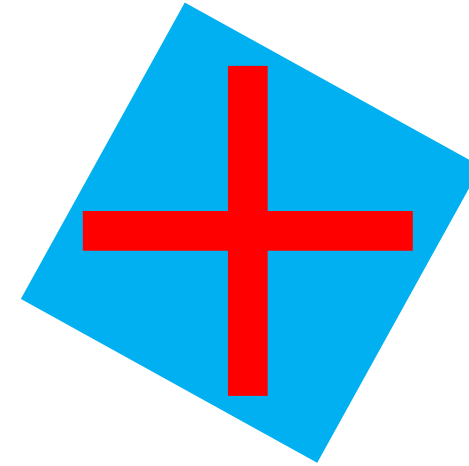




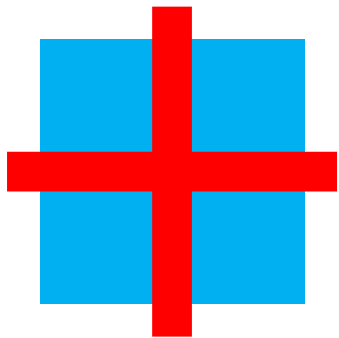
Good alignment



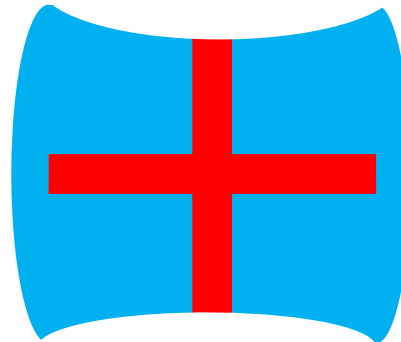
Positional shift



Rotational shift



Distortion shift



Expansion/Contraction
shift



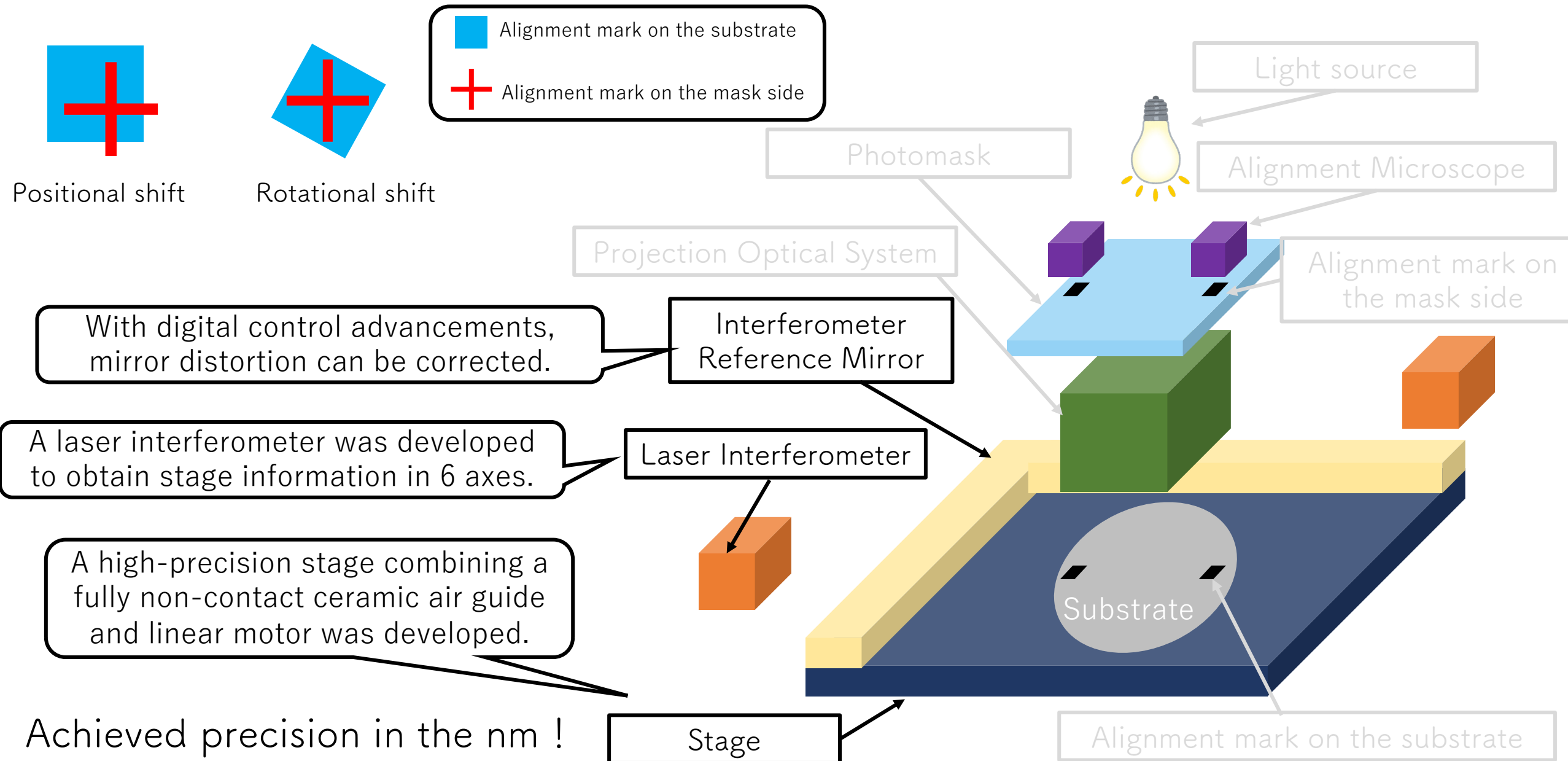
Alignment mark on the substrate



Alignment mark on the mask side

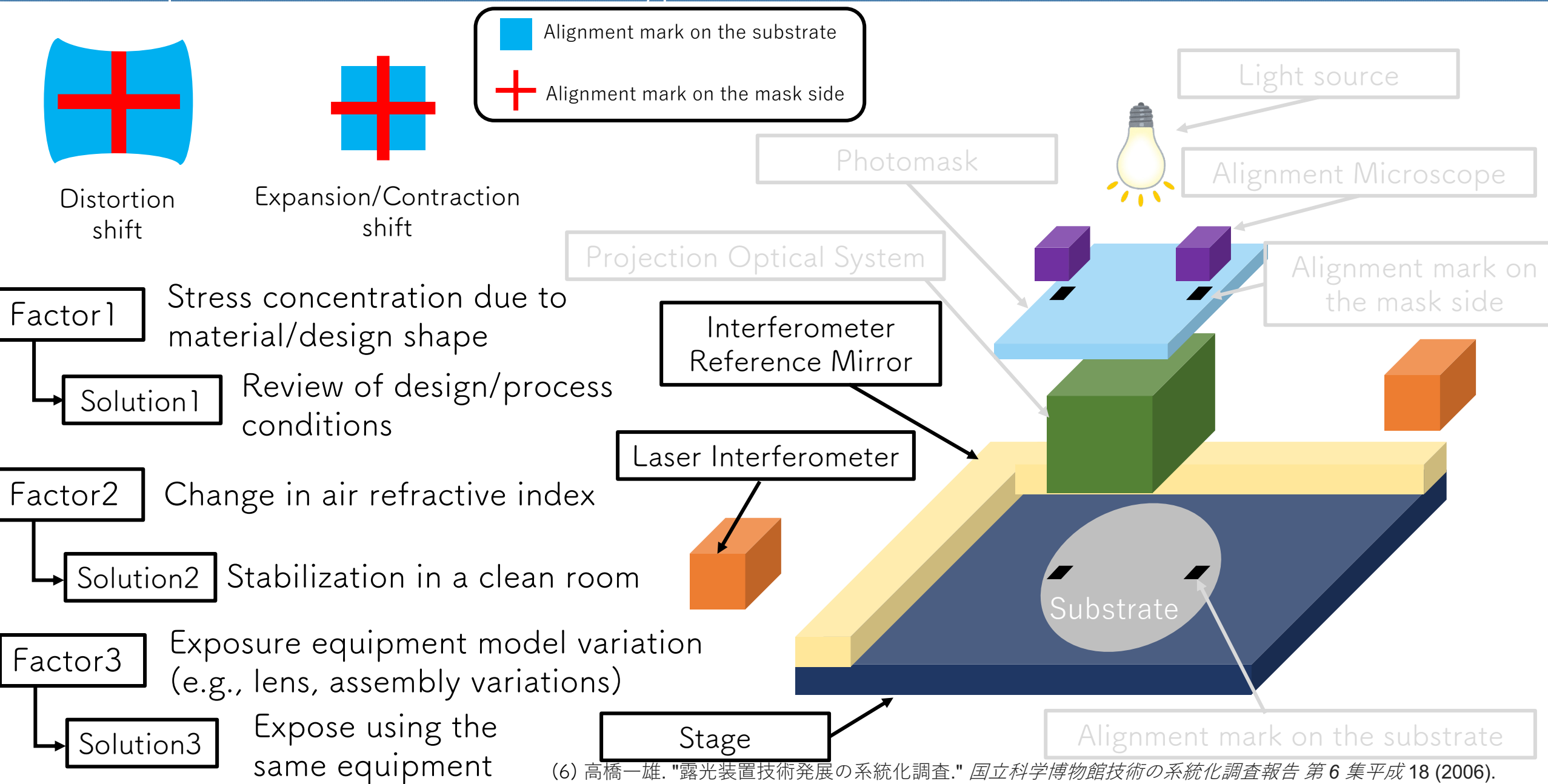
(6) 高橋一雄. "露光装置技術発展の系統化調査." 国立科学博物館技術の系統化調査報告 第6集平成18 (2006).

Countermeasures for “position misalignment” and “rotation misalignment”⁽⁶⁾ 55



Countermeasures for “distortion misalignment” and “expansion/contraction misalignment”⁽⁶⁾

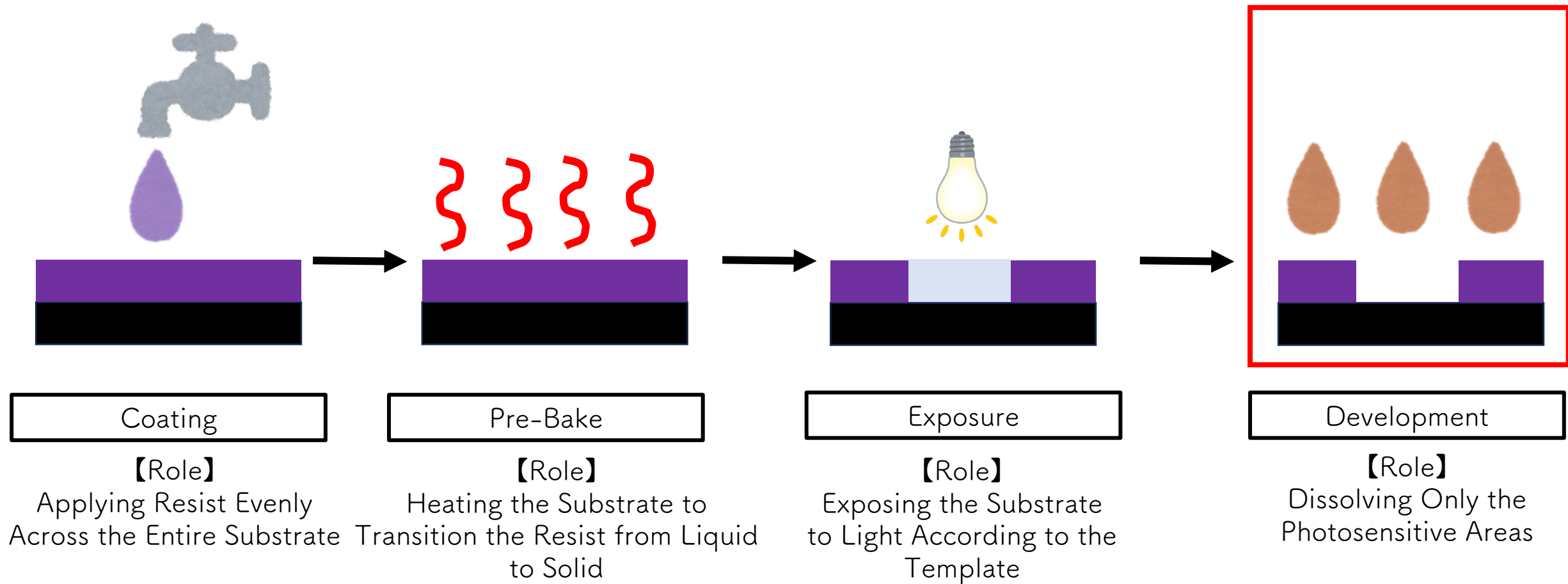
56



Exposure is

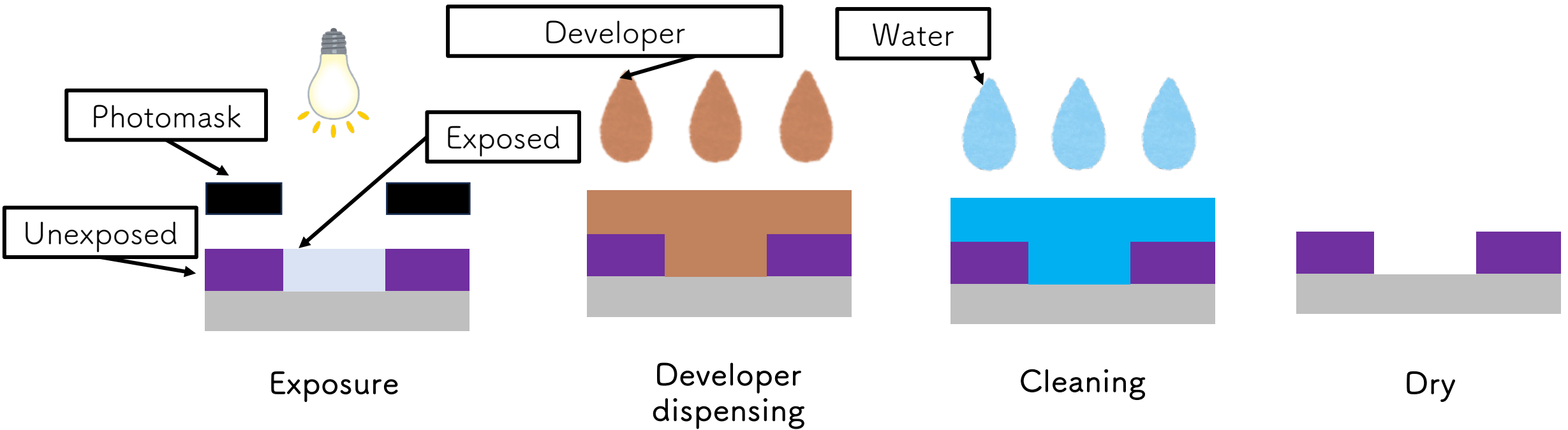
- Expose light only to the desired area on the substrate to form the circuit pattern.
- Ultraviolet light (10nm~380nm) is used for exposure.
- The resolution of the pattern is proportional to the wavelength of the light and inversely proportional to NA.
- The depth of focus is proportional to the wavelength and inversely proportional to the square of NA.

Development



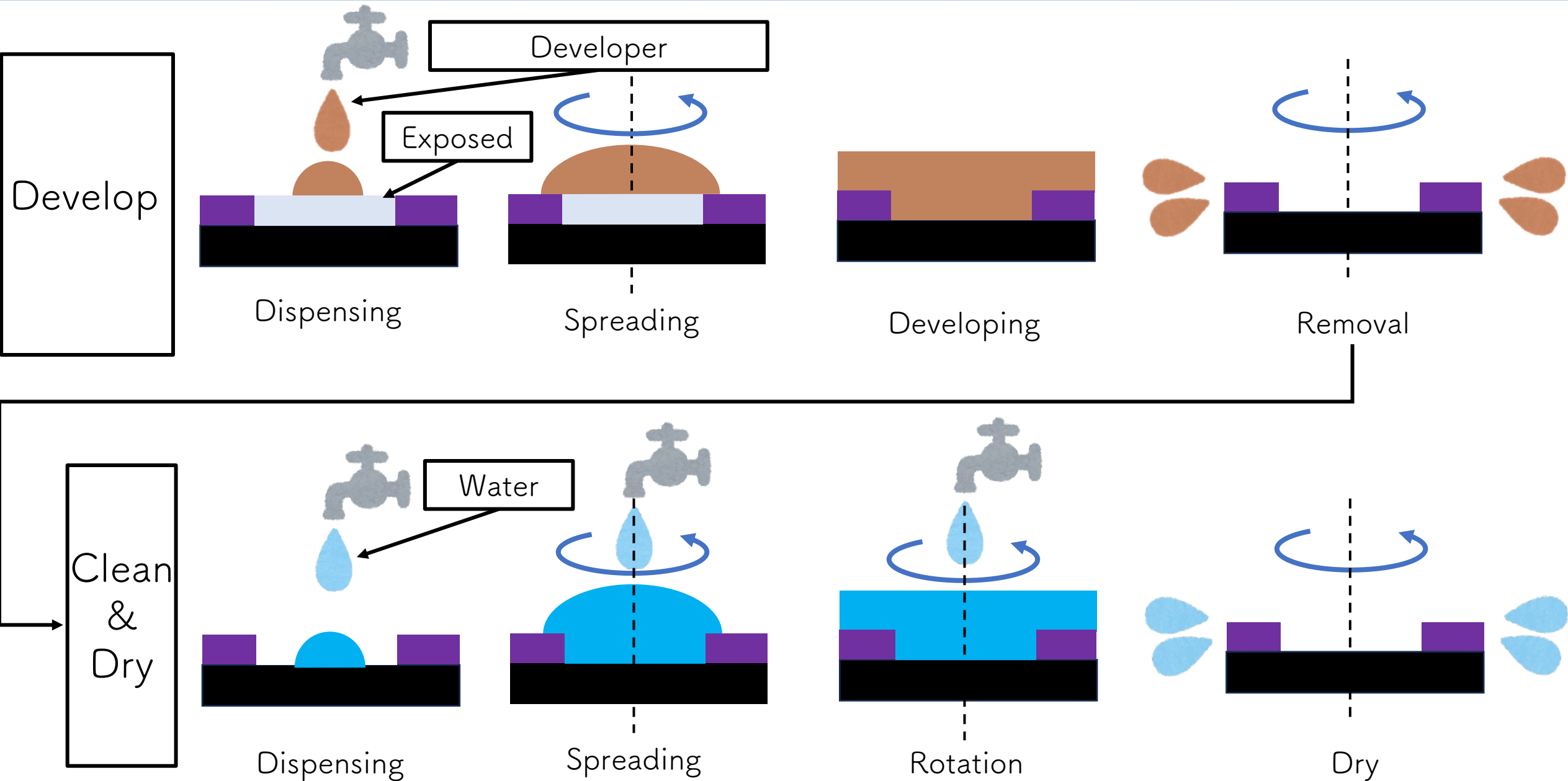
Purpose

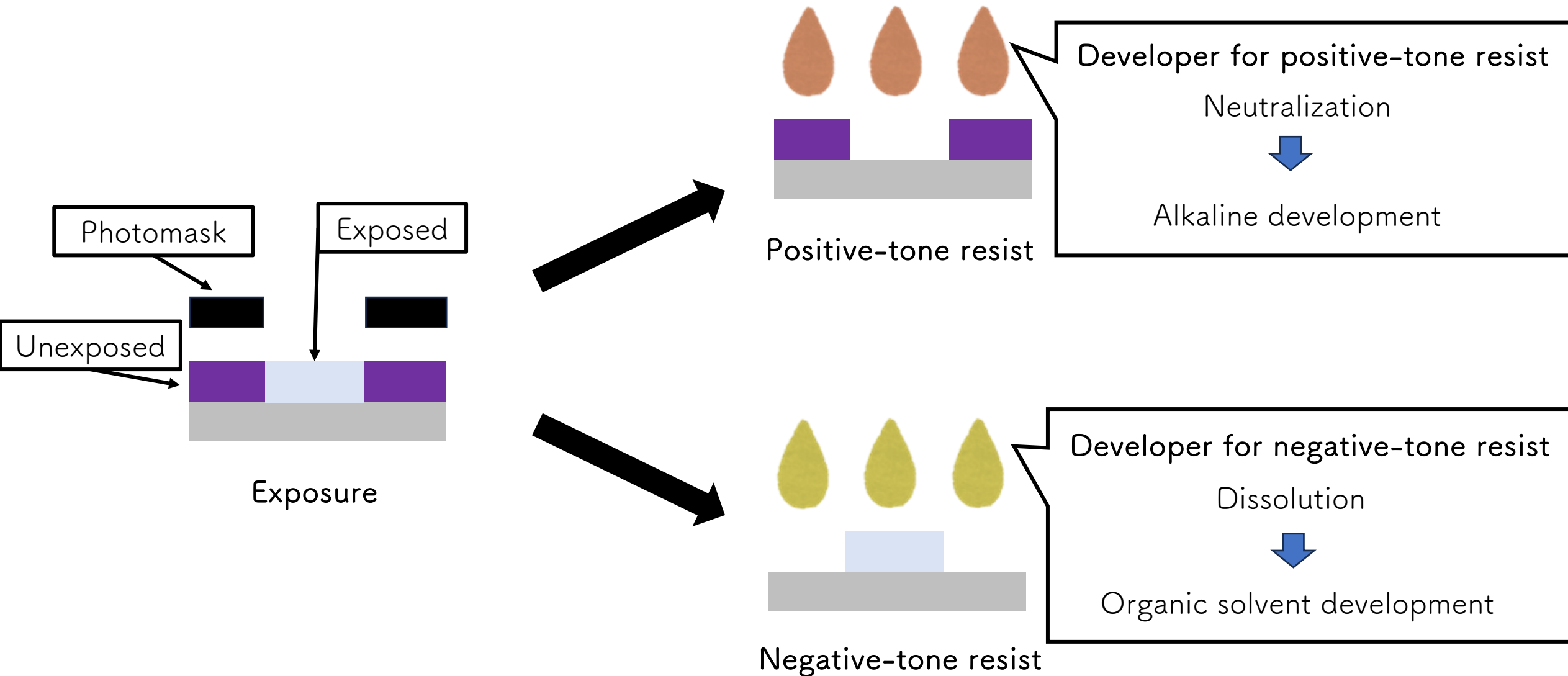
The selective removal of the unwanted portions of the exposed resist, forming the desired pattern on the substrate



Development process flow

61



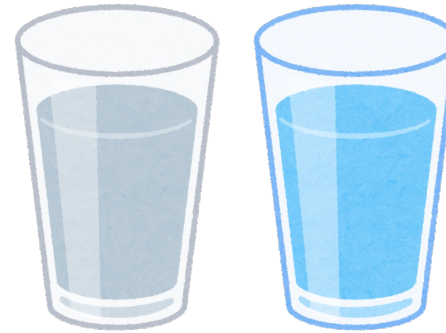




Temperature



Time



Concentration



Agitation

Influence	Development rate	Development amount	Development rate	• Unevenness • Pattern loss
Relation	High → Fast Low → Slow	Long → Many Short → Little	High → Fast Low → Slow	Strong → Pattern loss Weak → Unevenness
Reason	Activate the reaction	Change reaction time	Amount of reactants increases or decreases	Too Strong → Resist peeling occurs Too weak → Unstable reaction

Development process is

- Selective dissolution of the exposed resist to form the desired pattern on the substrate.
- Divided into development, cleaning, and drying steps.
- Positive resist uses alkaline development.
Negative resist uses organic solvent development.
- “Temperature,” “time,” “concentration,” and “agitation” are key parameters that determine quality.

Advanced Process Control in Semiconductor Manufacturing

Tom Sonderman
Costas Spanos

Outline

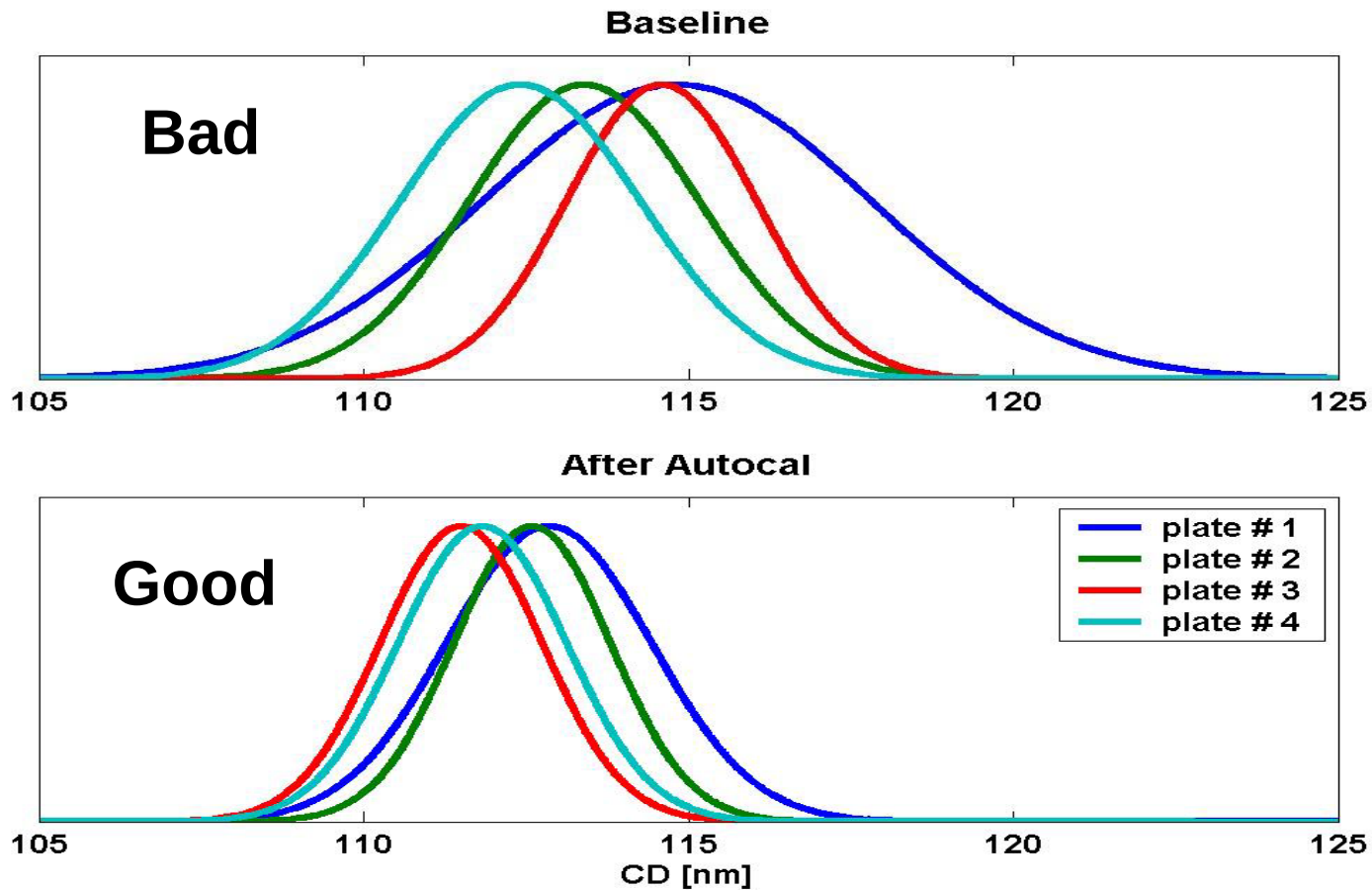
► The Challenge

- Historical Perspective
- Measuring Variability
 - The Importance of Metrology
- Controlling Variability
 - Applied Concepts – APC at AMD
- Modeling Variability
 - Interaction of Design and Manufacturing
 - Design for Controllability
- Future Perspectives

Introduction

- Error budgets cannot keep up with shrinking dimensions.
- In the sub-100nm generations, Critical Dimensions (CDs) are hard to control.
 - 65nm features, have an error budget of +/- ~5nm.
 - Often metrology offers about +/- 2nm precision!
 - Processing and metrology equipment are not being built to facilitate integration and control

The objective is to maintain both Wafer to Wafer and Across Wafer CD Uniformity



Key Manufacturing Objectives

Maximize Revenue Potential of Every Wafer

1

Rock Solid Consistency and Discipline

“Zero Uncertainty” in the minds of our customers through consistent on-time delivery of the highest quality product.

2

Maximum Product Performance Through Aggressive Technology Migration

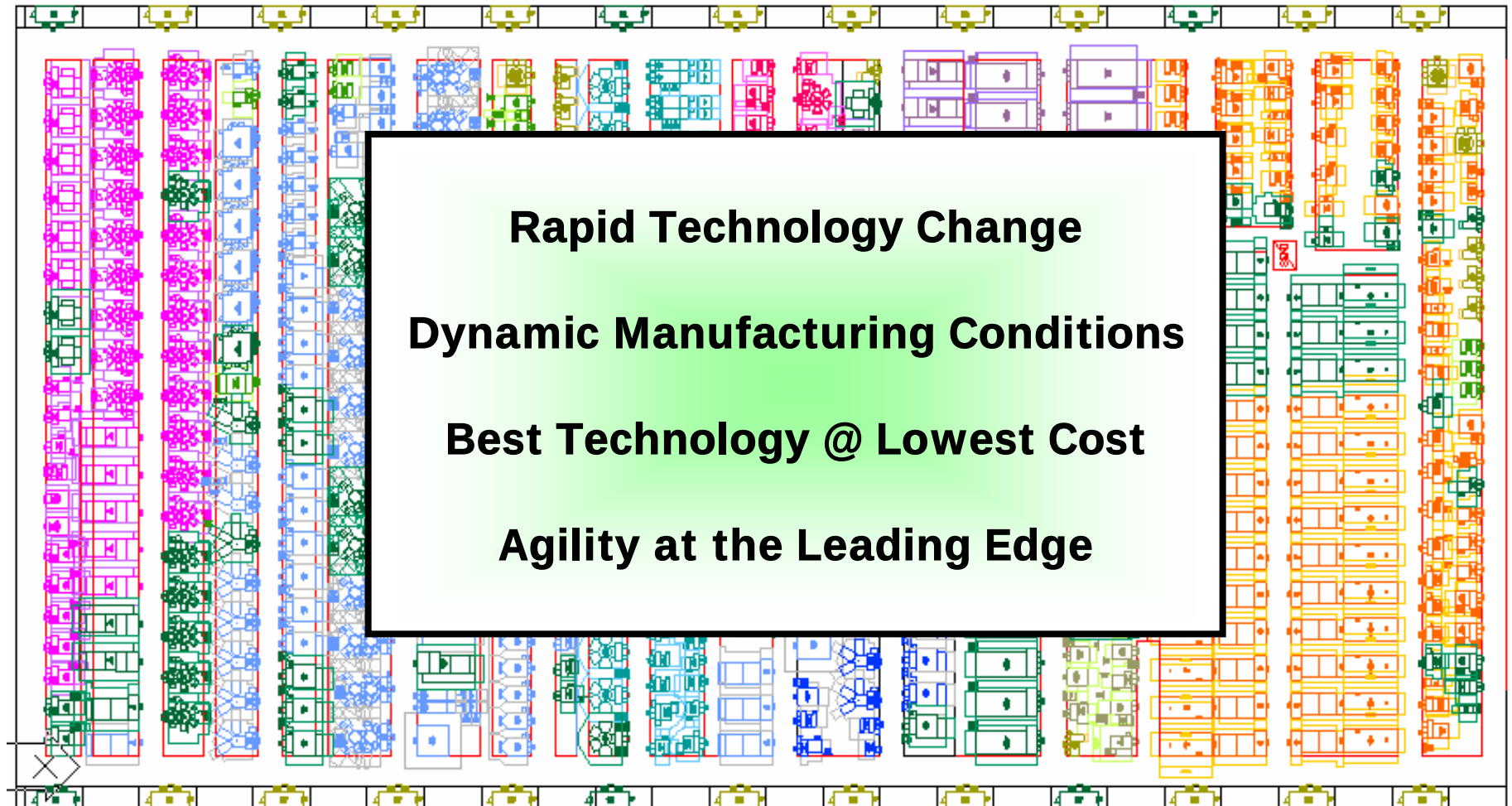
Rapid and efficient 90nm transition. Routine transistor improvements to increase product performance without slowing fab output.

3

Minimum Cost Through Industry-Leading Efficiency and Automation

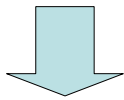
Automatic detection and correction of errors in fabs to lower per-die costs. Integrated fab-wide tool communication and control to accelerate time to mature yields on new generations.

Constant Push for Performance ...

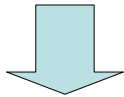


...in a Non-Stop, High Volume Environment

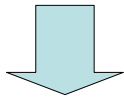
**Mixed mode
manufacturing**



**In-fab technology
development**

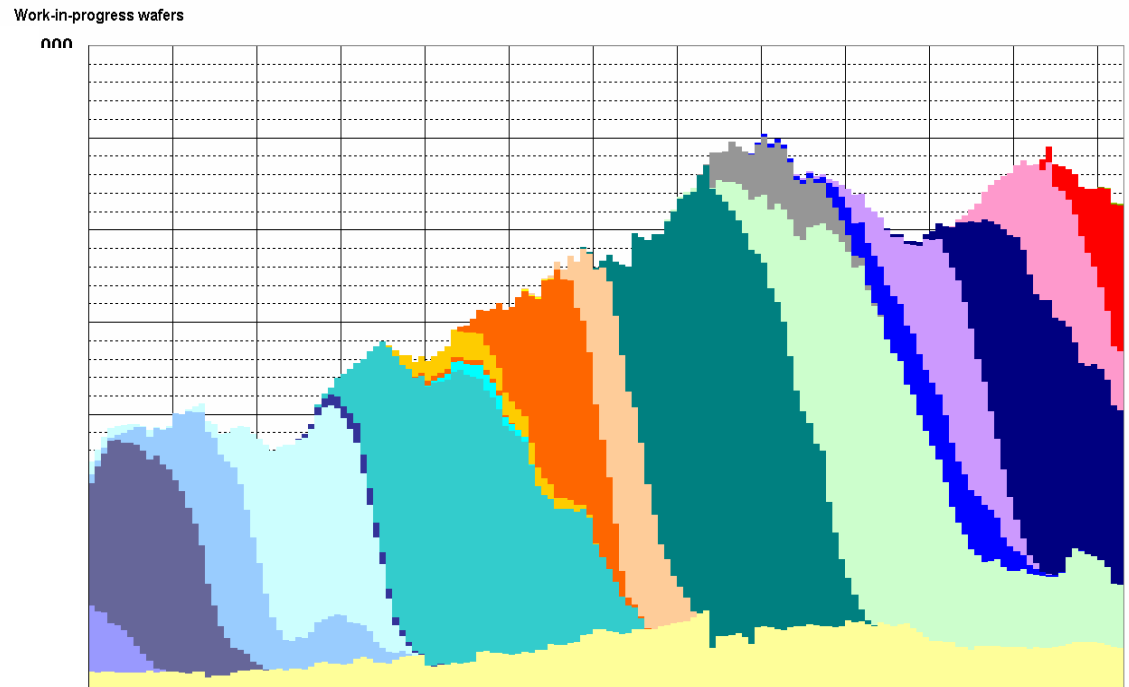


**Enhanced, accelerated
cycles of learning**



**Incremental continuous
performance
improvement**

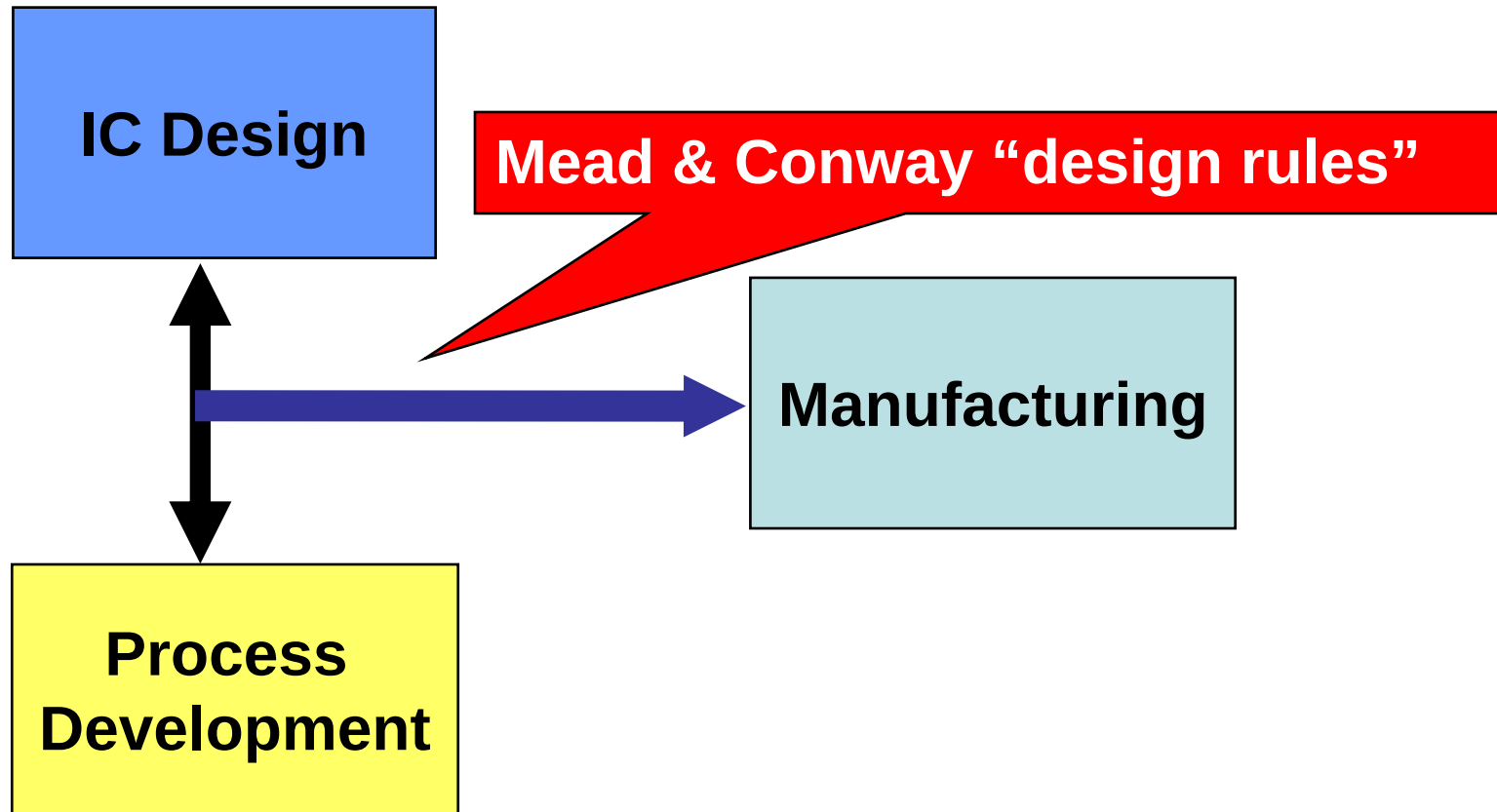
Fab 30 WIP by Transistor Node



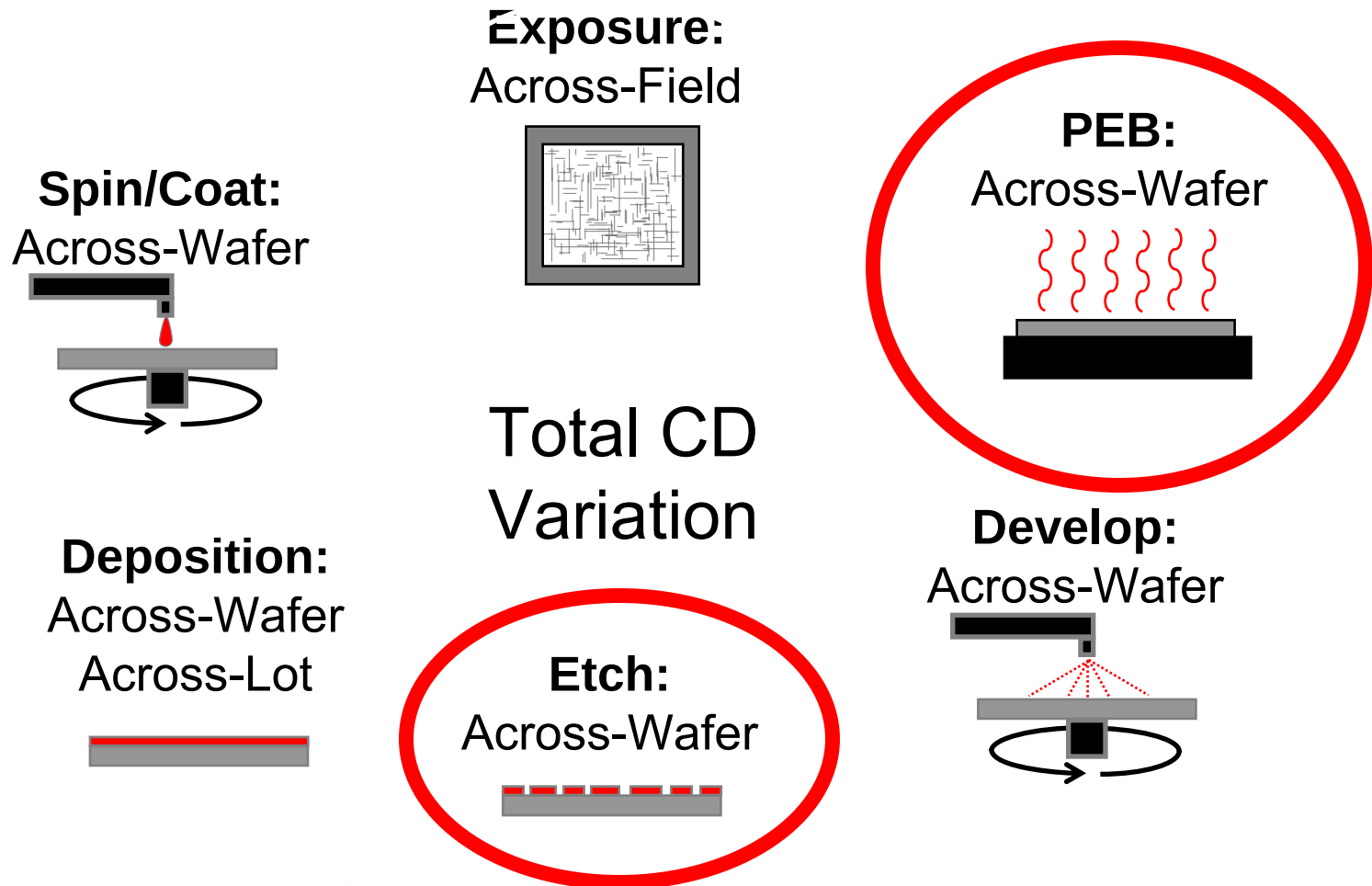
Outline

- The Challenge
- ▶ Historical Perspective
- Measuring Variability
 - The Importance of Metrology
- Controlling Variability
 - Applied Concepts – APC at AMD
- Modeling Variability
 - Interaction of Design and Manufacturing
 - Design for Controllability
- Future Perspectives

The Traditional Semiconductor Manufacturing Environment

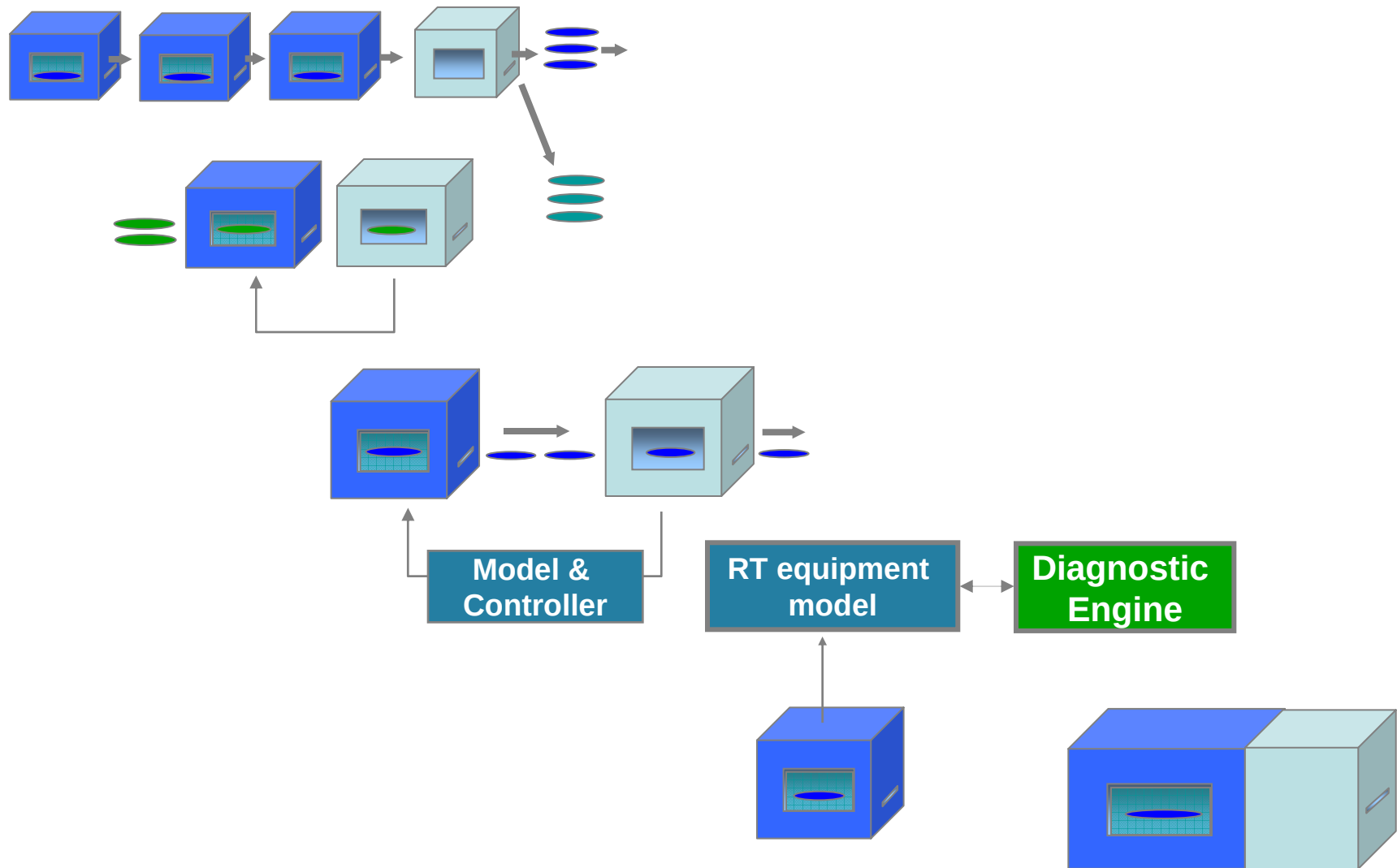


Sources of CD Variation



Systematic CD variation components with different frequencies are combined.

Manufacturing Evolution...



Outline

- The Challenge
- Historical Perspective
- ▶ Measuring Variability
 - The Importance of Metrology
- Controlling Variability
 - Applied Concepts – APC at AMD
- Modeling Variability
 - Interaction of Design and Manufacturing
 - Design for Controllability
- Future Perspectives

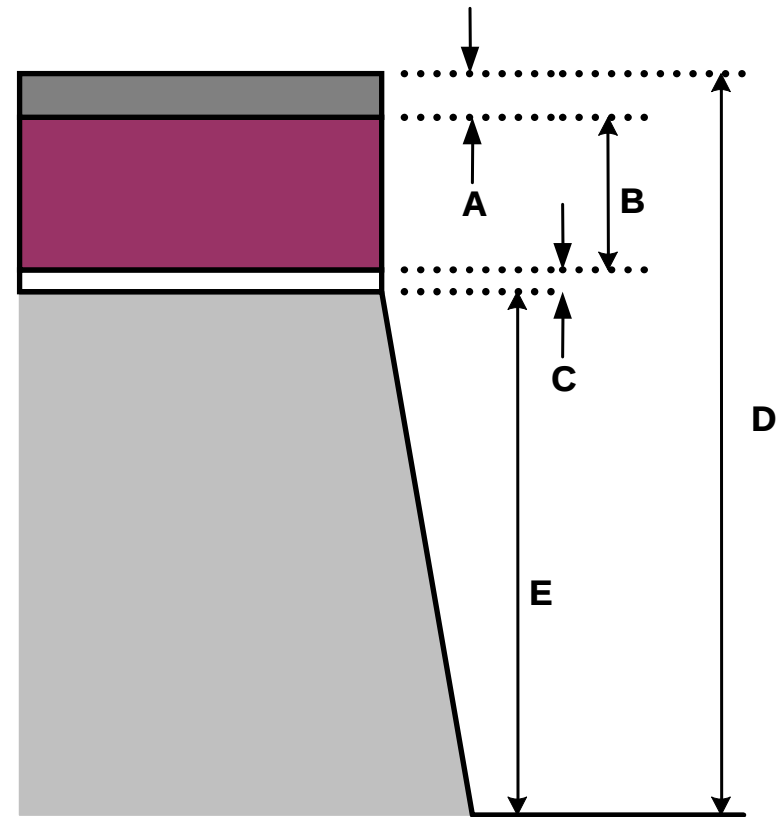
Metrology Outline

- Metrology tools have been created for inspection and quality control – not for APC.
- CD-SEMs, Optical Metrology, in-situ metrology are advancing rapidly towards an APC/FDC model.
 - Direct metrology error budget remains large
 - The expanding use of indirect metrology requires the introduction of advanced modeling tools in the production line.

Shallow Trench Isolation Etch Control

An AMT-APC Success Story

- One of the key metrics for STI Etch control is the depth of the resulting trench into the silicon, but common metrology techniques are unable to directly measure this value.
- Instead, a number of different metrology operations were combined to get an indirect measurement of silicon trench depth.
- The noise from each of these measurements was added to the signal being used for automated control.
- AMT and APC decided to explore scatterometry as an alternative metrology

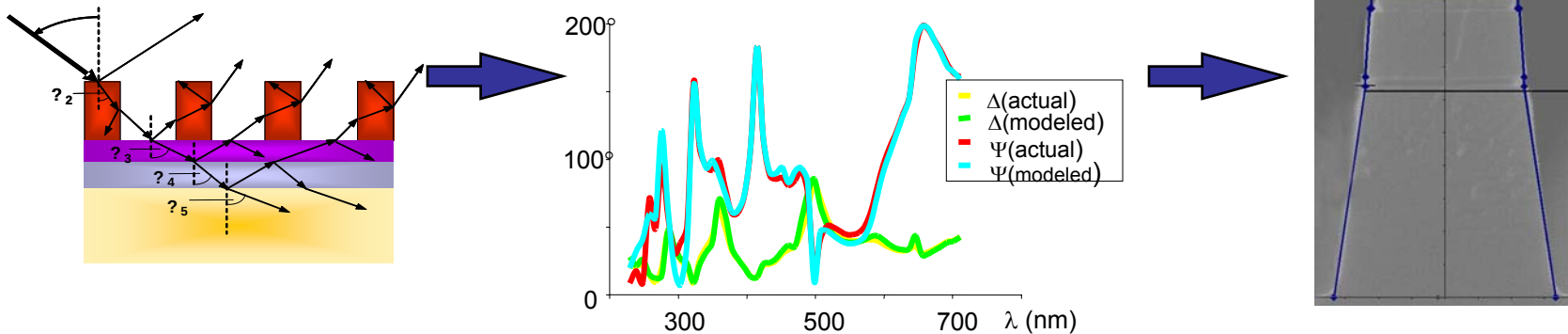


A = ARC Thickness (Optical)
B = Nitride Thickness (Optical)
C = Barrier Oxide (Optical)
D = Trench Depth (Profilometer)
E = Derived Silicon Depth

Advanced Measurement Technology

Scatterometry for STI Etch Production

- Efforts continue to evaluate and implement advanced measurement technology for production solutions
 - New metrology (e.g. scatterometry)
 - Sensor solutions
 - Etc...
- The implementation of scatterometry for STI etch control in FASL Fab 25 was one of the first large-scale uses of scatterometry in a production semiconductor manufacturing facility.

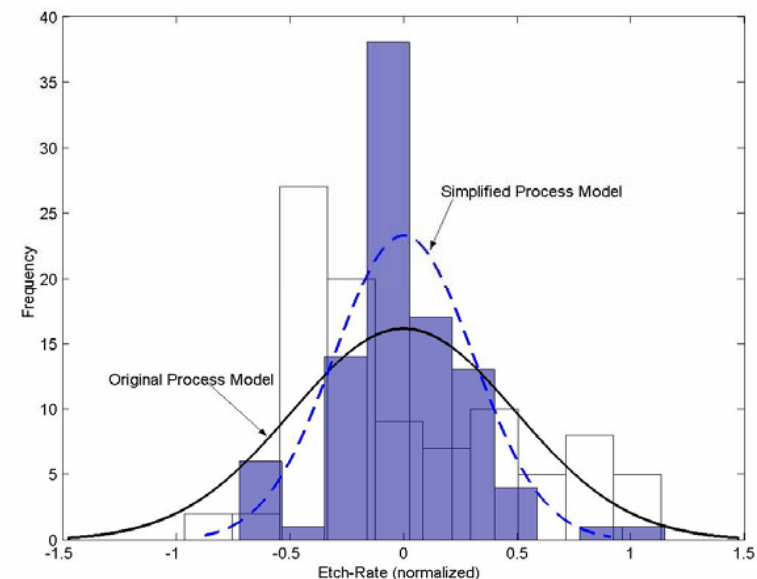


Shallow Trench Isolation Etch Control

An AMT-APC Success Story

- Reduced Uncertainty
 - **Precision**
Scatterometry represents a 5x improvement in precision compared to the profiler
 - **Accuracy**
Scatterometry measures at design rule spacing instead of a large isolated test pad
 - **Stability**
Scatterometry hardware does not drift like the contact profiler
- Created New Capabilities
 - Direct measurement of silicon depth (no feed-forward derivation)
 - Combined CD, profile, and thickness metrology

- Results
 - Improved Control
 - Cpk improvement of 90%
 - Reduced Cost
 - Eliminated feed-forward metrology
 - Improved Cycle Time
 - 3x faster metrology system

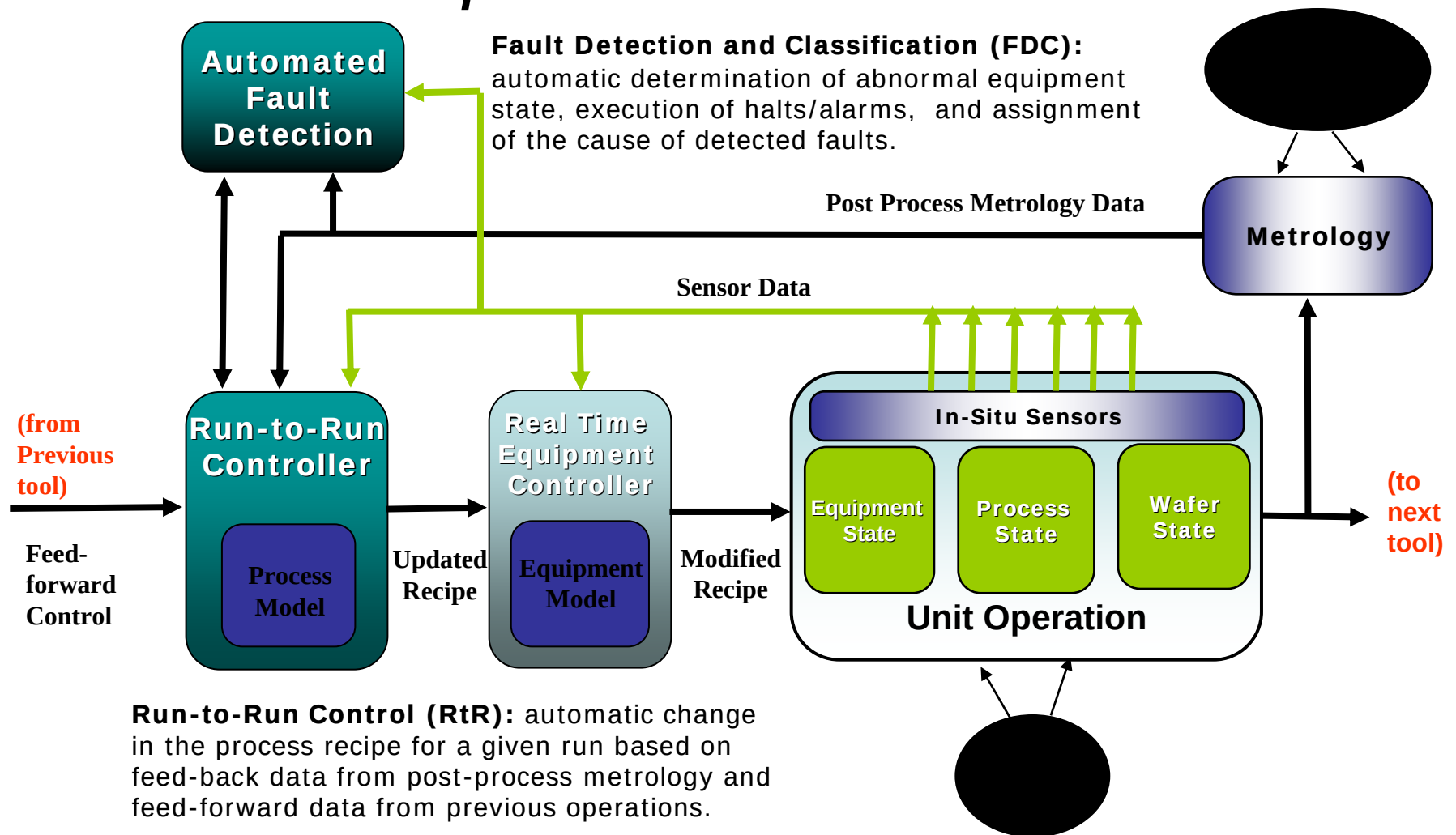


Outline

- The Challenge
- Historical Perspective
- Measuring Variability
 - The Importance of Metrology
- ▶ Controlling Variability
 - Applied Concepts – APC at AMD
- Modeling Variability
 - Interaction of Design and Manufacturing
 - Design for Controllability
- Future Perspectives

Advanced Process Control

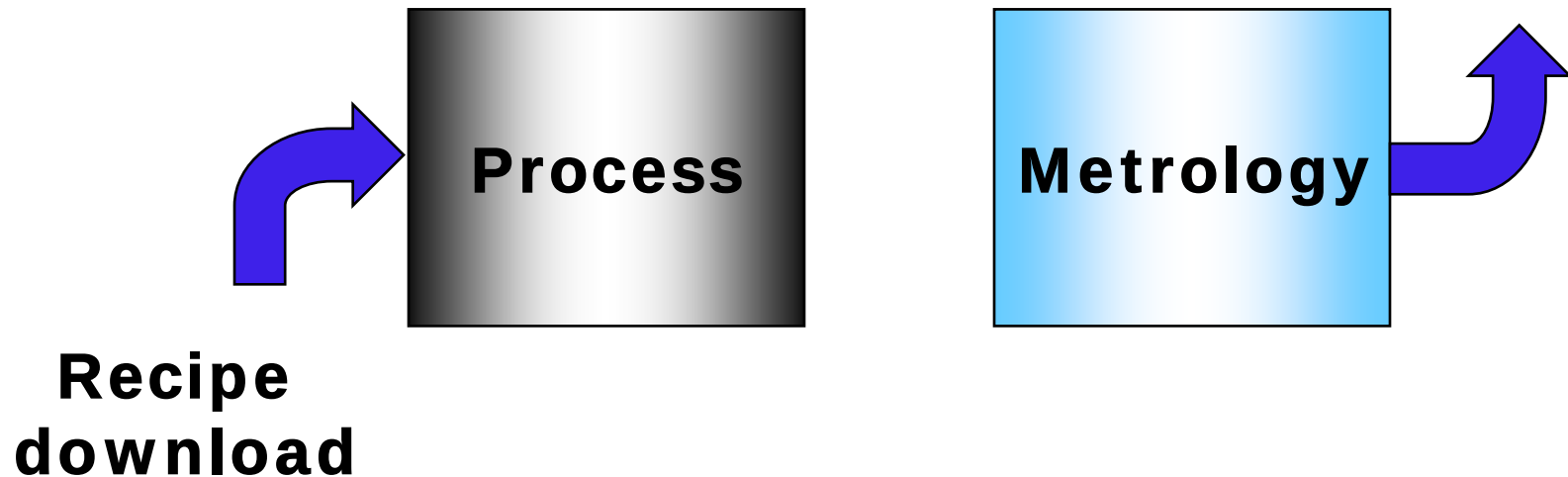
APC incorporates RtRPC and FDC



Automated Precision Manufacturing *Control Evolution*

APM 1.5: Stability

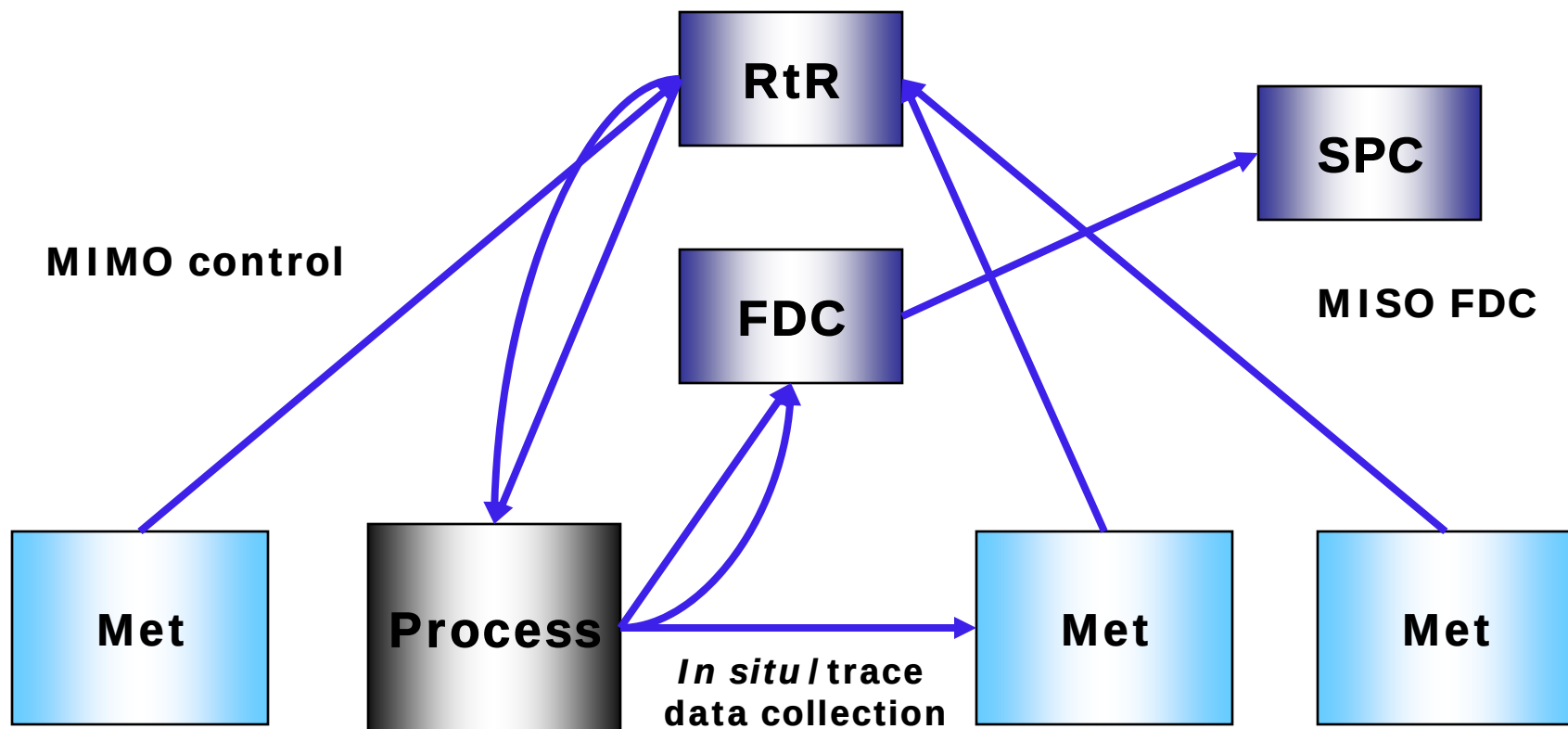
SPC



- Automated recipe download through an Equipment Interface minimized process setup errors and improved productivity.
- Statistical Process Control provided visibility into and control of run-to-run stability of each process.

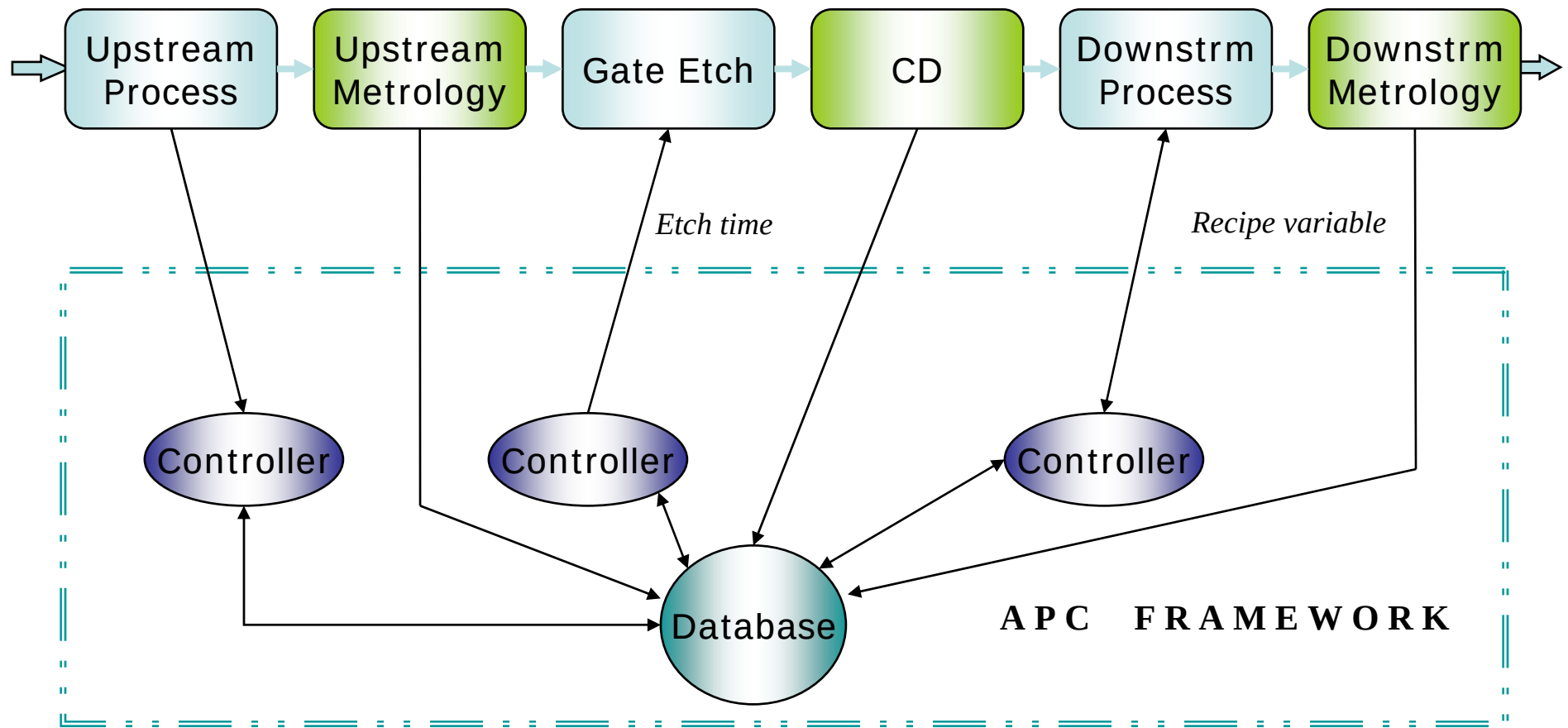
Automated Precision Manufacturing *Control Evolution*

APM 2.0: Controllability



Transistor Control

Gate Etch Control with FF/FB Control



- Resist trim etch control, along with controls at upstream and downstream operations, effectively control drive current according to our understanding of device physics.

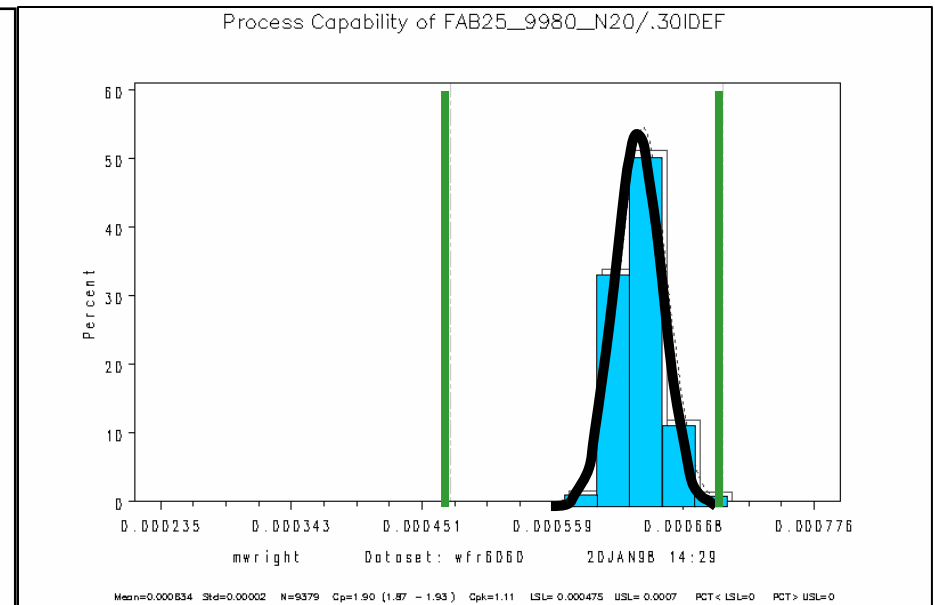
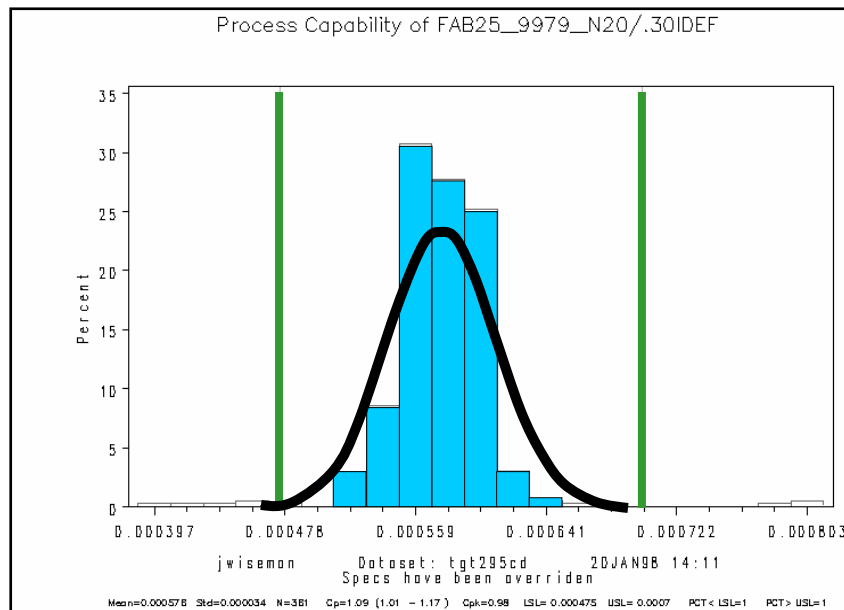
Transistor Control

Effect on Microprocessor Speed Distributions

Initial Impact of APC on MPU Performance – 1998 – Narrower transistor drive current distribution allows AMD to push our process to the edge of the spec without fallout for high power or slow parts

AMD-K6® drive current
distribution before Id,sat control

AMD-K6® drive current
distribution after Id,sat control

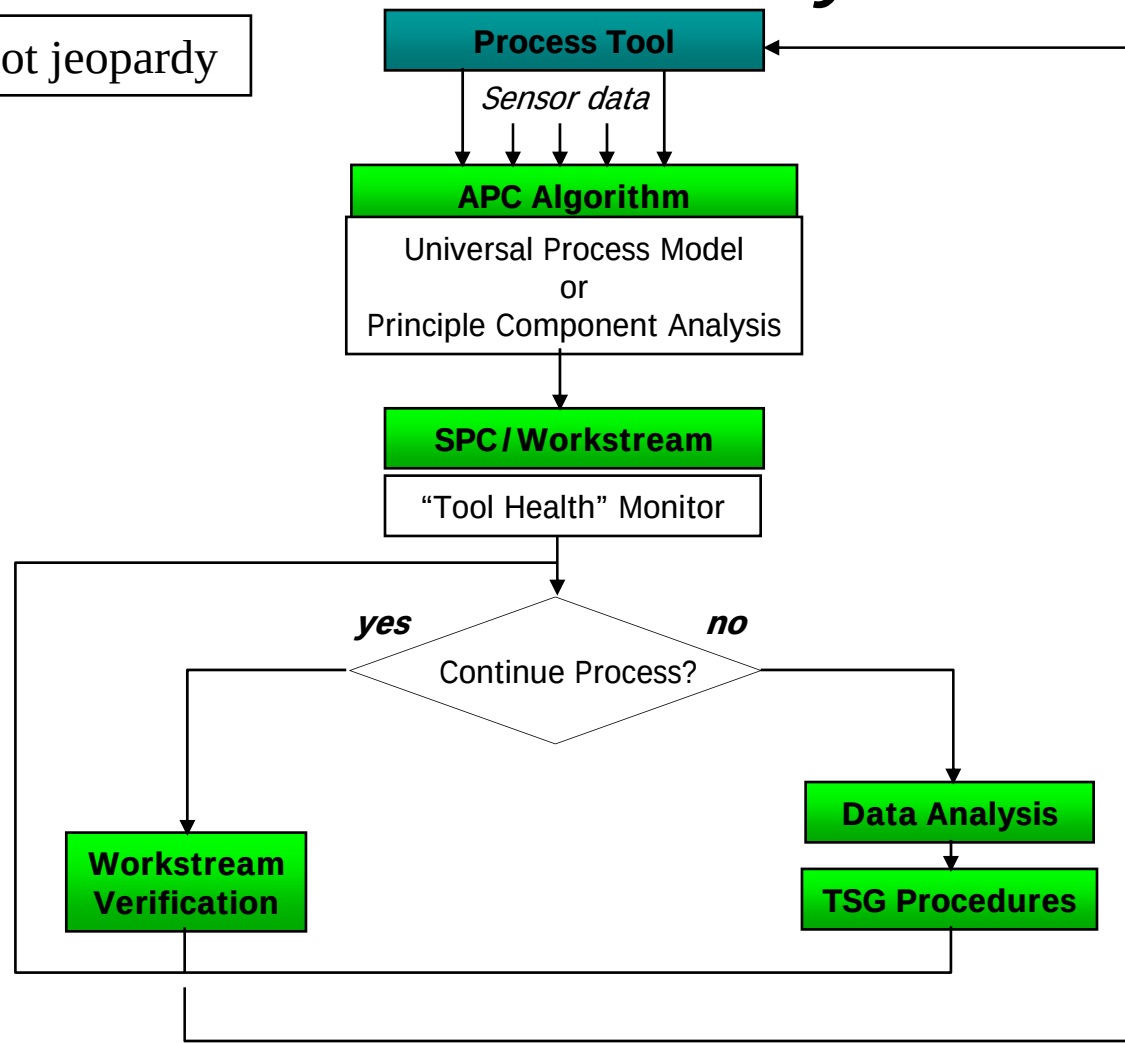


Initial side-by-side comparison: Left APC used ever since in F25 and from start-up in F30

Fault Detection and Classification

MISO FDC Analysis

Goal: single lot jeopardy

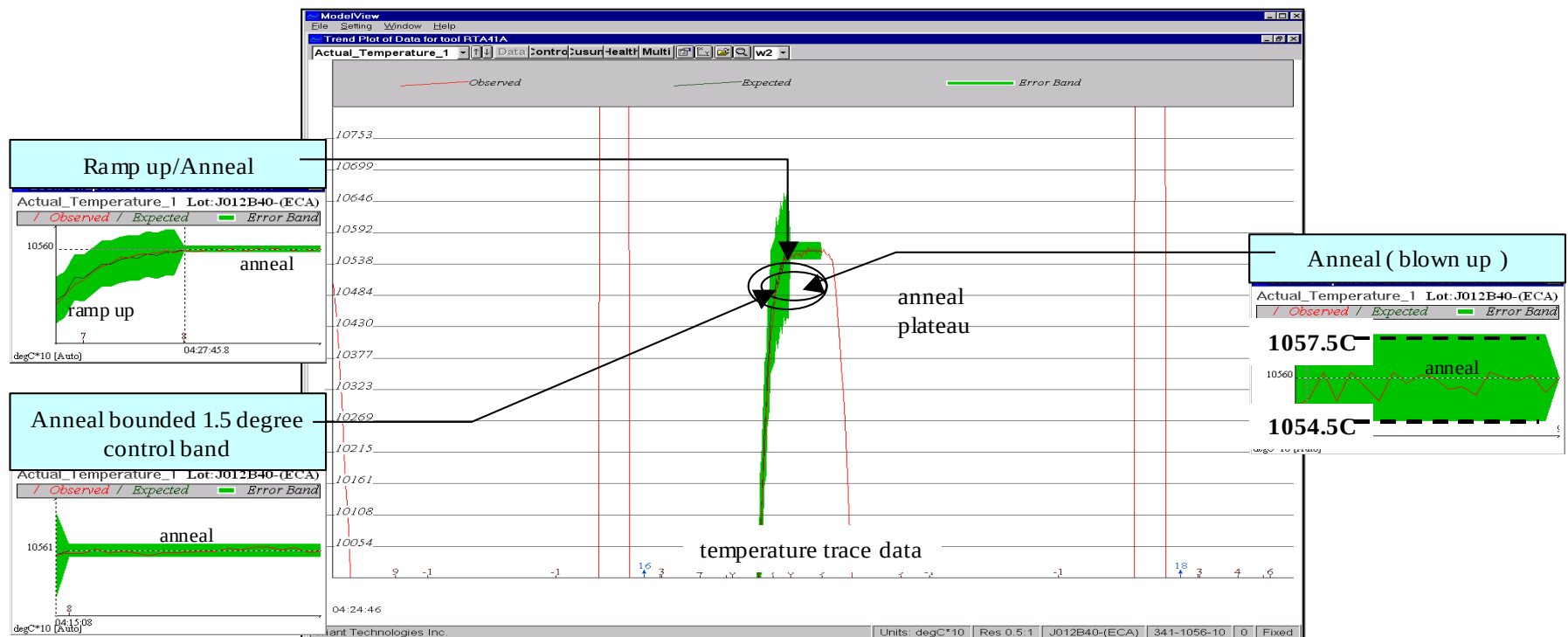


Fault Detection and Classification

Rapid Thermal Anneal FDC

FDC model tightened control beyond manufacturer's specifications to + 1.5 °C alarm limits (3- σ)

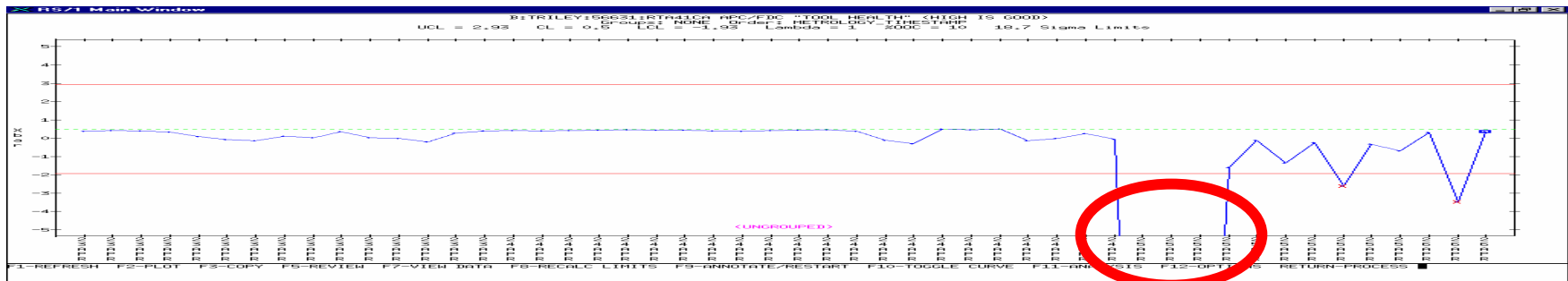
"Temp_probe_1" trace is bounded by a +/- 1.5 degree control-band (green) in real-time during last half of the anneal at target temperature.



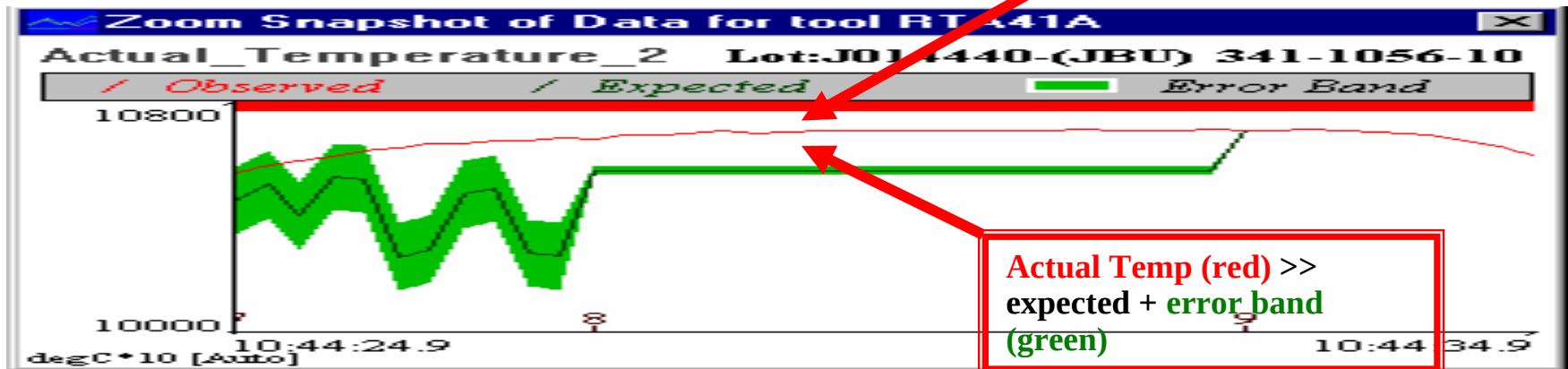
Fault Detection and Classification

RTA process monitoring and interdiction

AMAT RTA Fault Detection & Classification
Chart Caught RTA41A Zone 2 Temp Problem (14° C Too High)



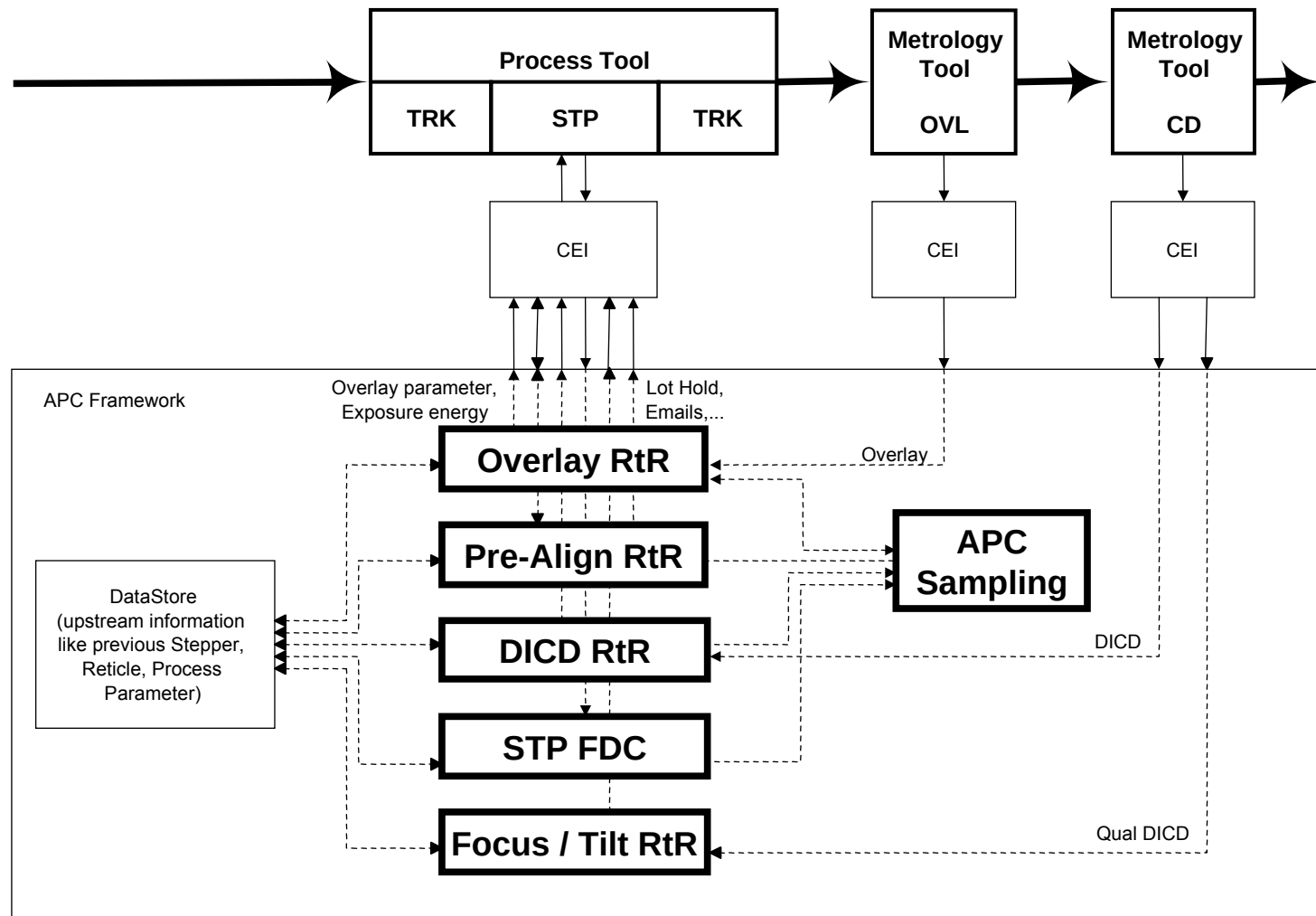
Workstream chart reconstructed to reflect current, proper, LCL to reduce false fails



Bottom Line: Achieved a 86% reduction in product lot risk

Lithography Control

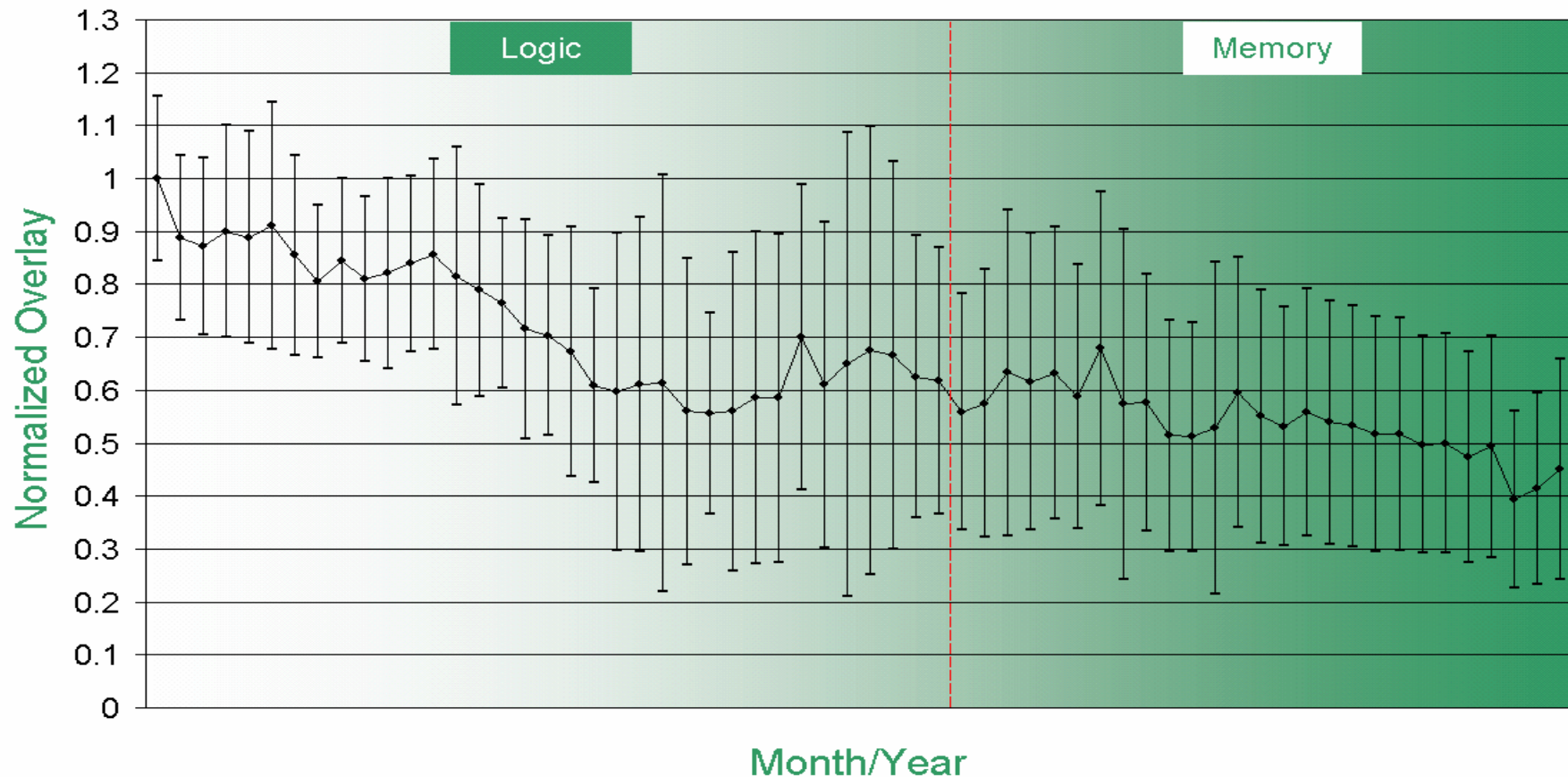
Schematic of Integrated APC Applications



Lithography Overlay Control

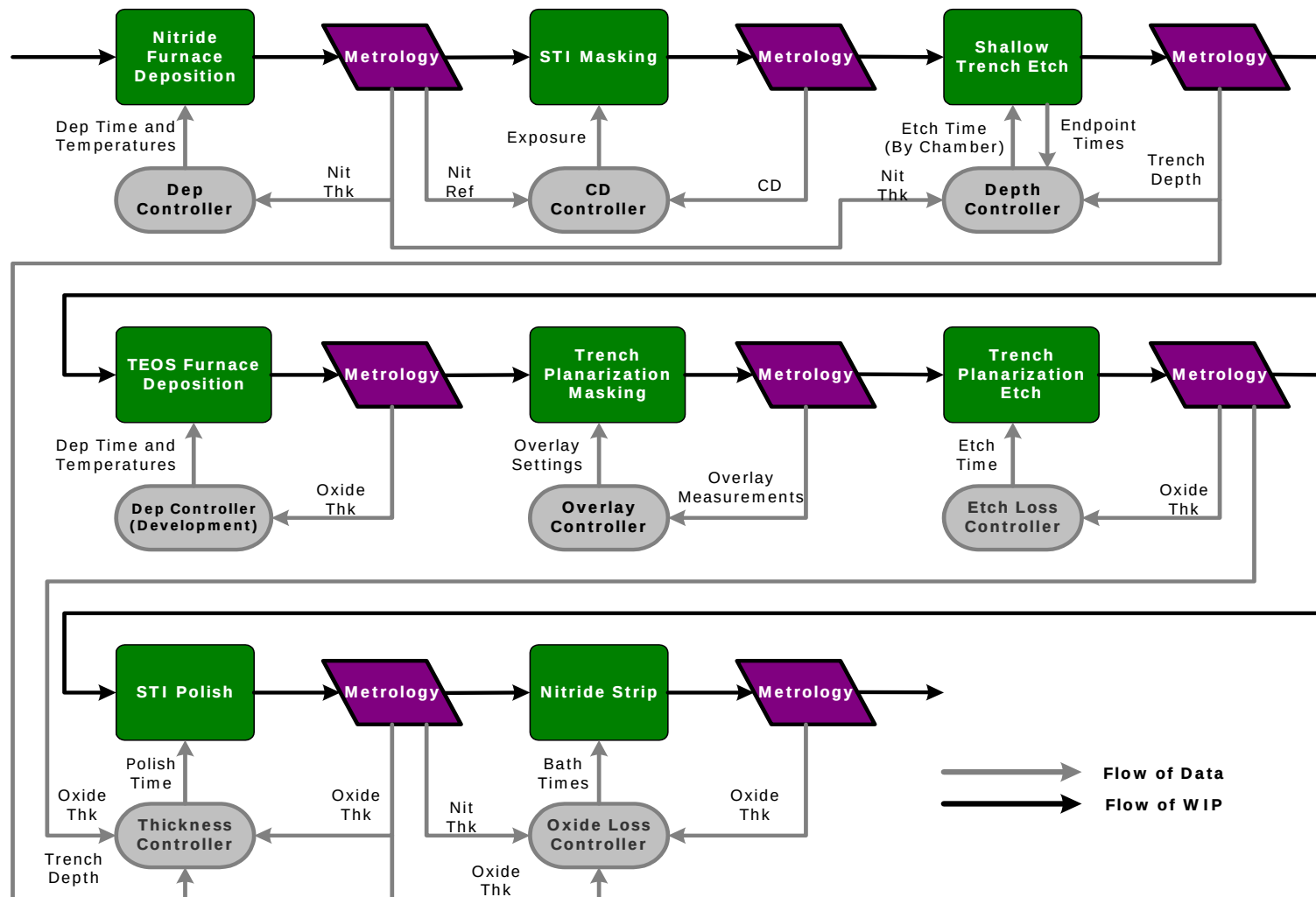
Performance improvement over 6+ years

Fab 25 Average Max Overlay Errors For All Layers



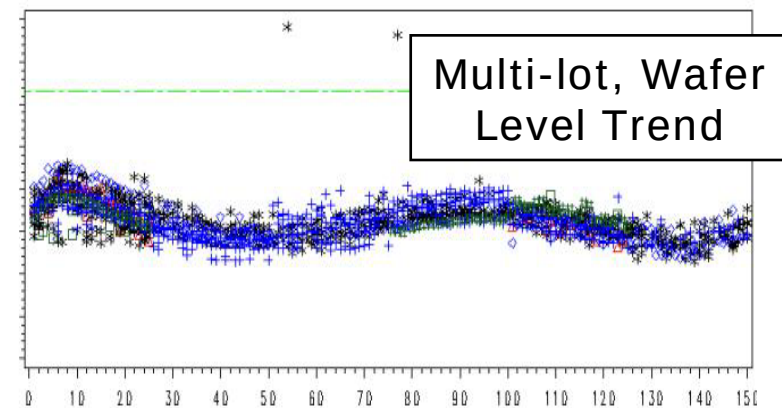
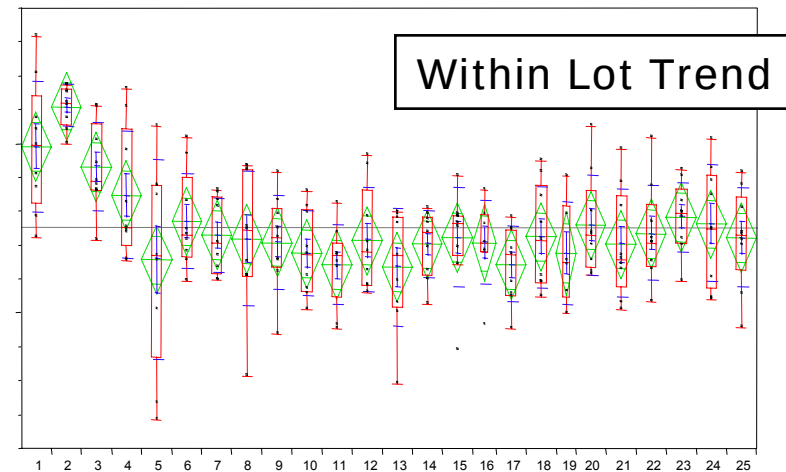
Islands of Control

FASL Fab 25: Shallow Trench Isolation



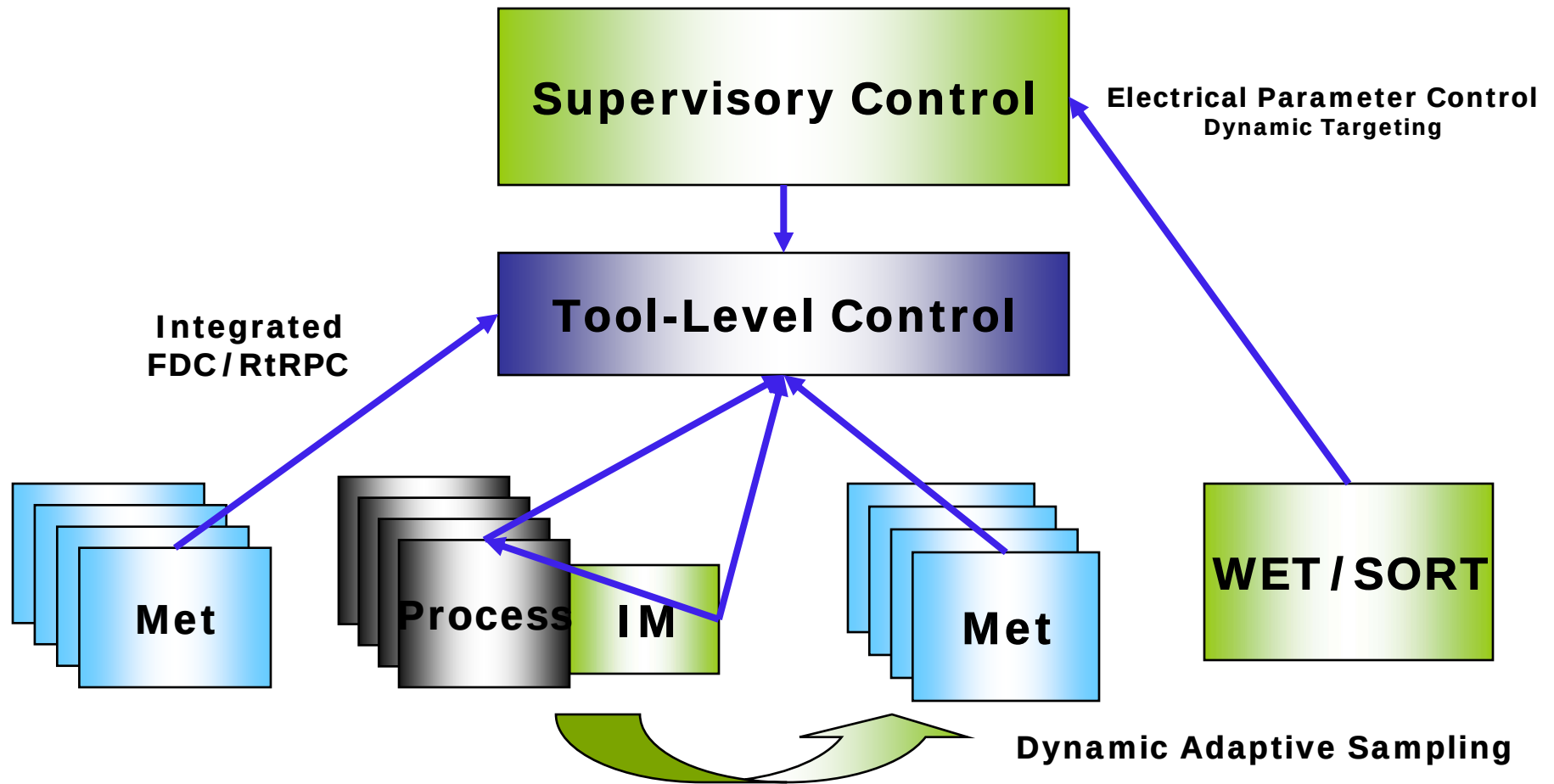
The Need for Wafer Level Control

- Processes within the factory exhibit drift that show repeatable signals within the lot or over larger periods.
- With the higher costs of 300mm wafers and processing, the economic impact of this variation is not acceptable.
- Wafer level control applications can be used to eliminate much of this variation and thereby improve the economics of the factory.



Automated Precision Manufacturing *Control Evolution*

Phase 3: Predictability

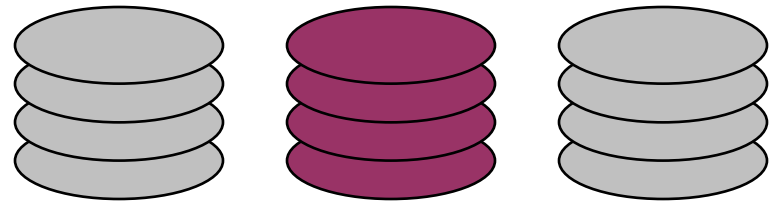


Fab-Wide Control Technology

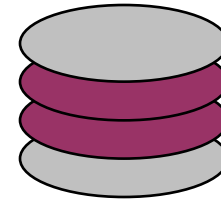
Dynamic Adaptive Sampling

- Like dispatching, the decisions on what type of metrology sampling also could benefit from the inputs of an APC application.
- This input could impact all levels of sampling:
 - Do you measure a particular lot?
 - Do you measure a particular wafer in a lot?
 - What is the pattern of sites on the wafer?
 - What structures should be measured at each site?

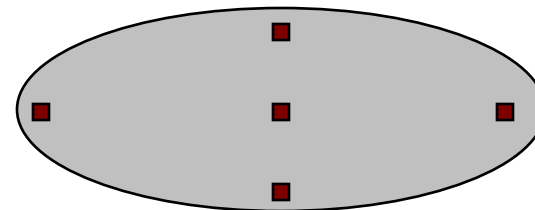
Which Lots?



Which Wafers?

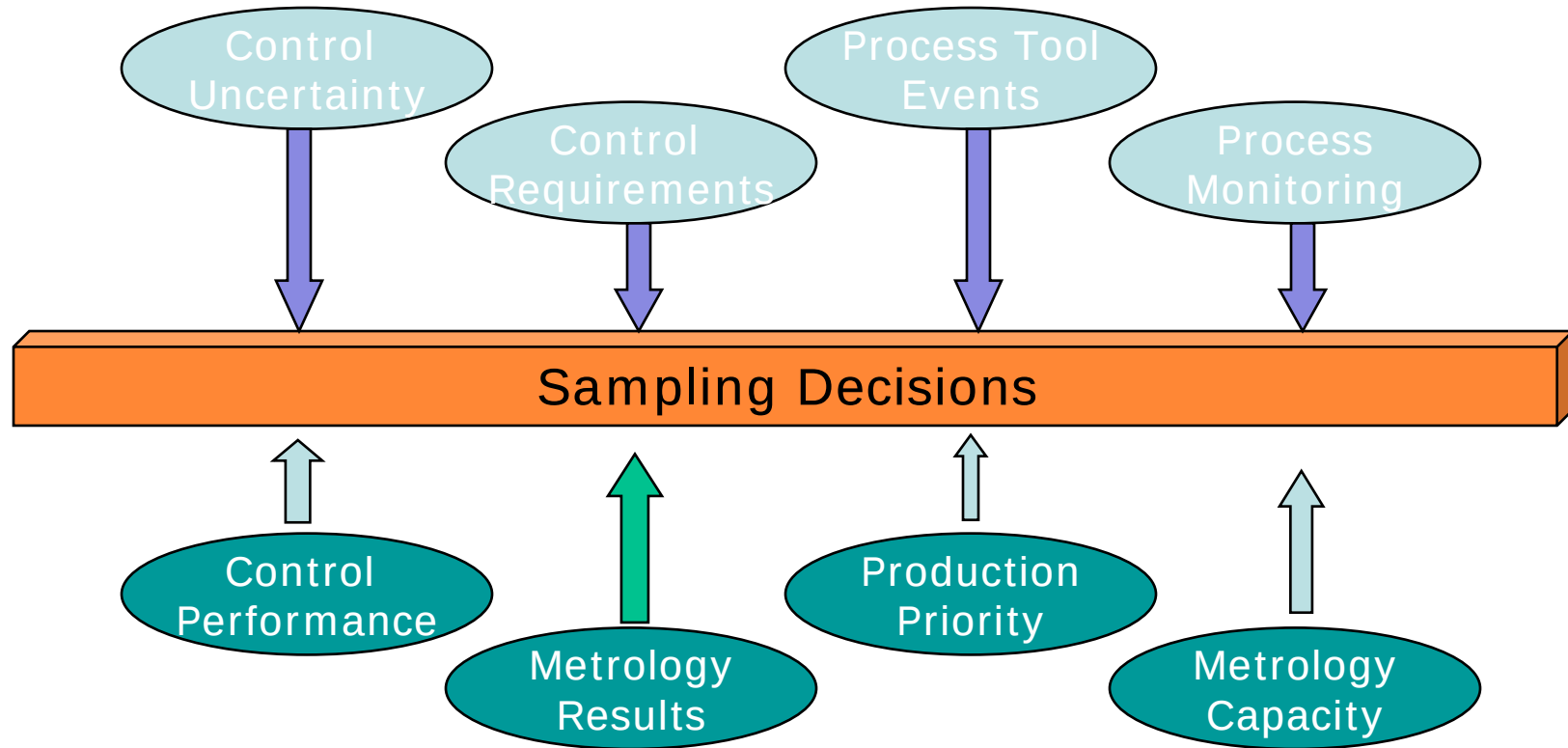


Which Sites?



Fab-Wide Control Technology

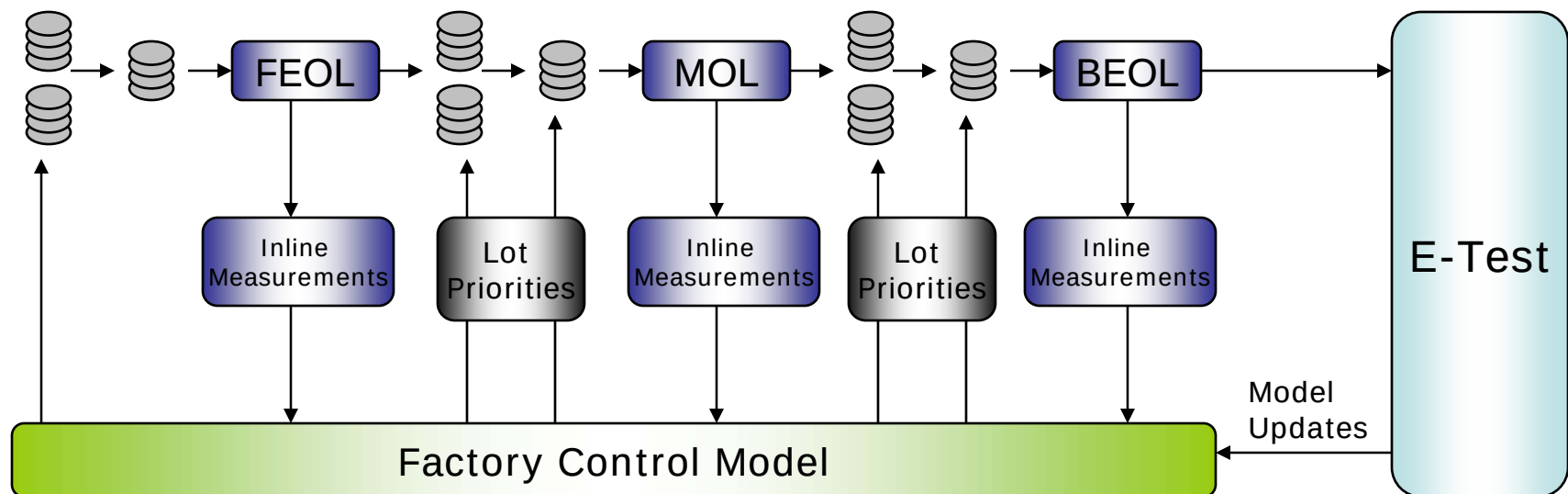
Factors Influencing Sampling



Incorporation of control related inputs allows for “smarter” sampling based on factory performance.

Fab-Wide Control Technology

Dynamic Production Control



Determine speed distribution for each lot based on in-line measurements and transistor model

Change priority of lot based on estimate of # of parts in speed bins and production outs requirements (based on user input of outs@speed / week)

Change equipment set for each lot based on equipment performance (defectivity, speed impact) and requirements for outs

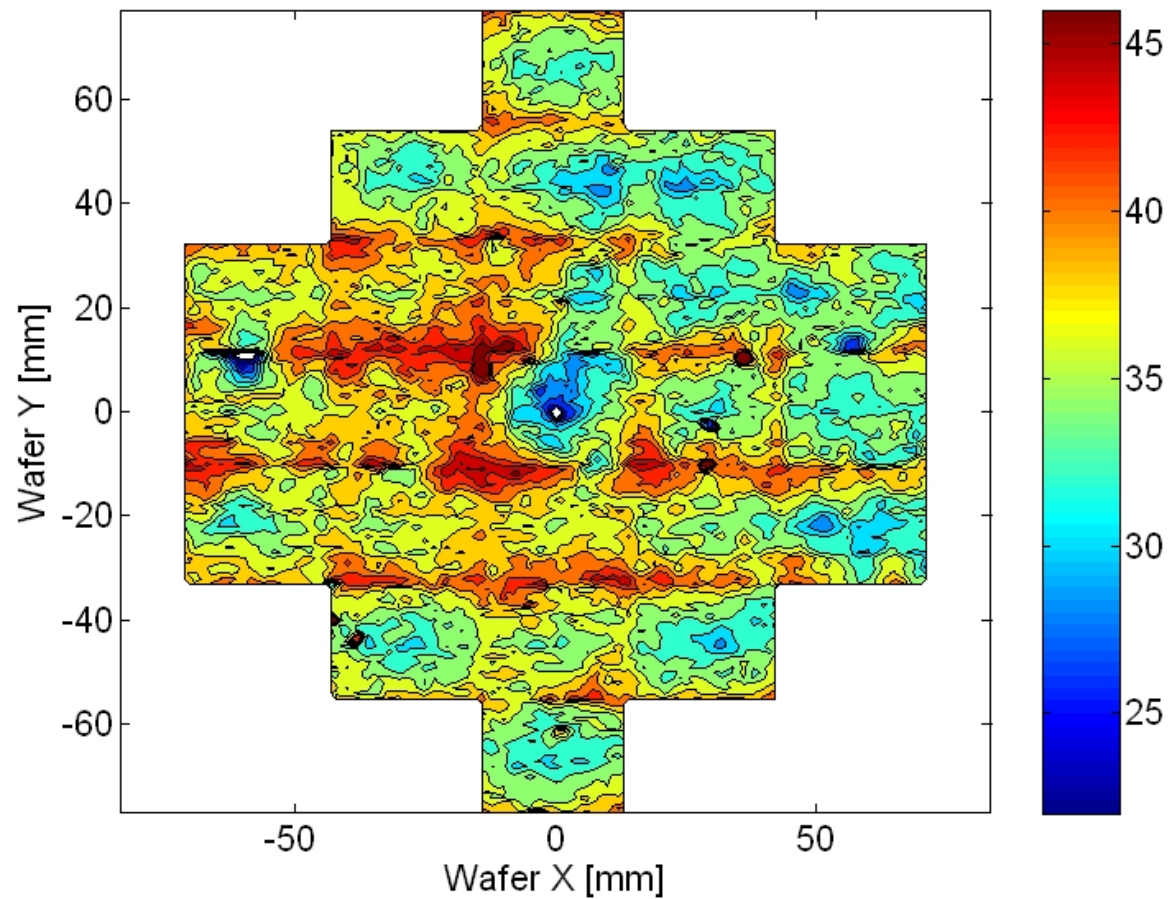
Change starts based on current estimations of in-line yield@speed and requirements for outs

Outline

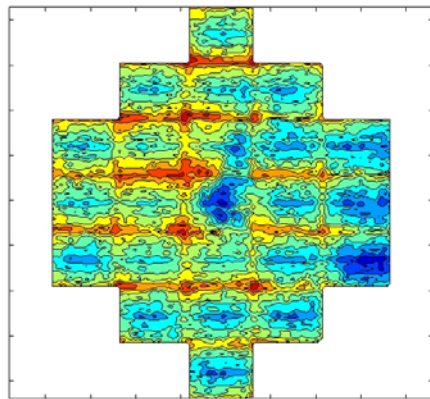
- The Challenge
- Historical Perspective
- Measuring Variability
 - The Importance of Metrology
- Controlling Variability
 - Applied Concepts – APC at AMD
- ▶ Modeling Variability
 - Interaction of Design and Manufacturing
 - Design for Controllability
- Future Perspectives

Spatial Correlation Analysis

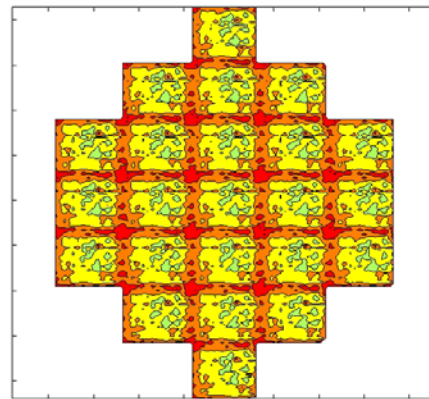
Exhaustive poly-CD measurements (280/field):



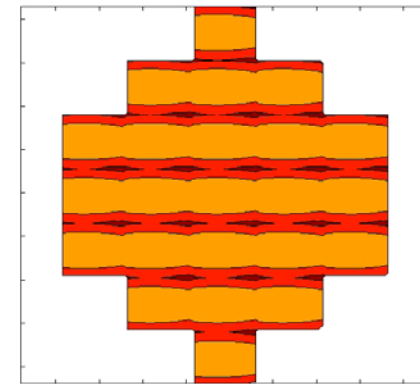
Origin of Spatial Correlation Dependence



Average Wafer

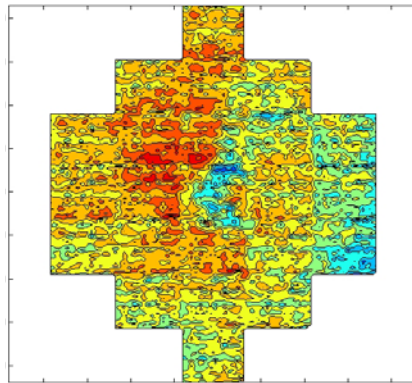


Scaled Mask Errors

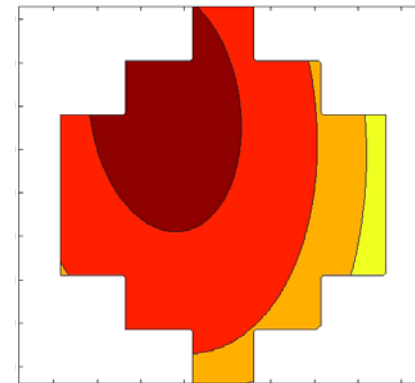


Across-Field Systematic Variation

=



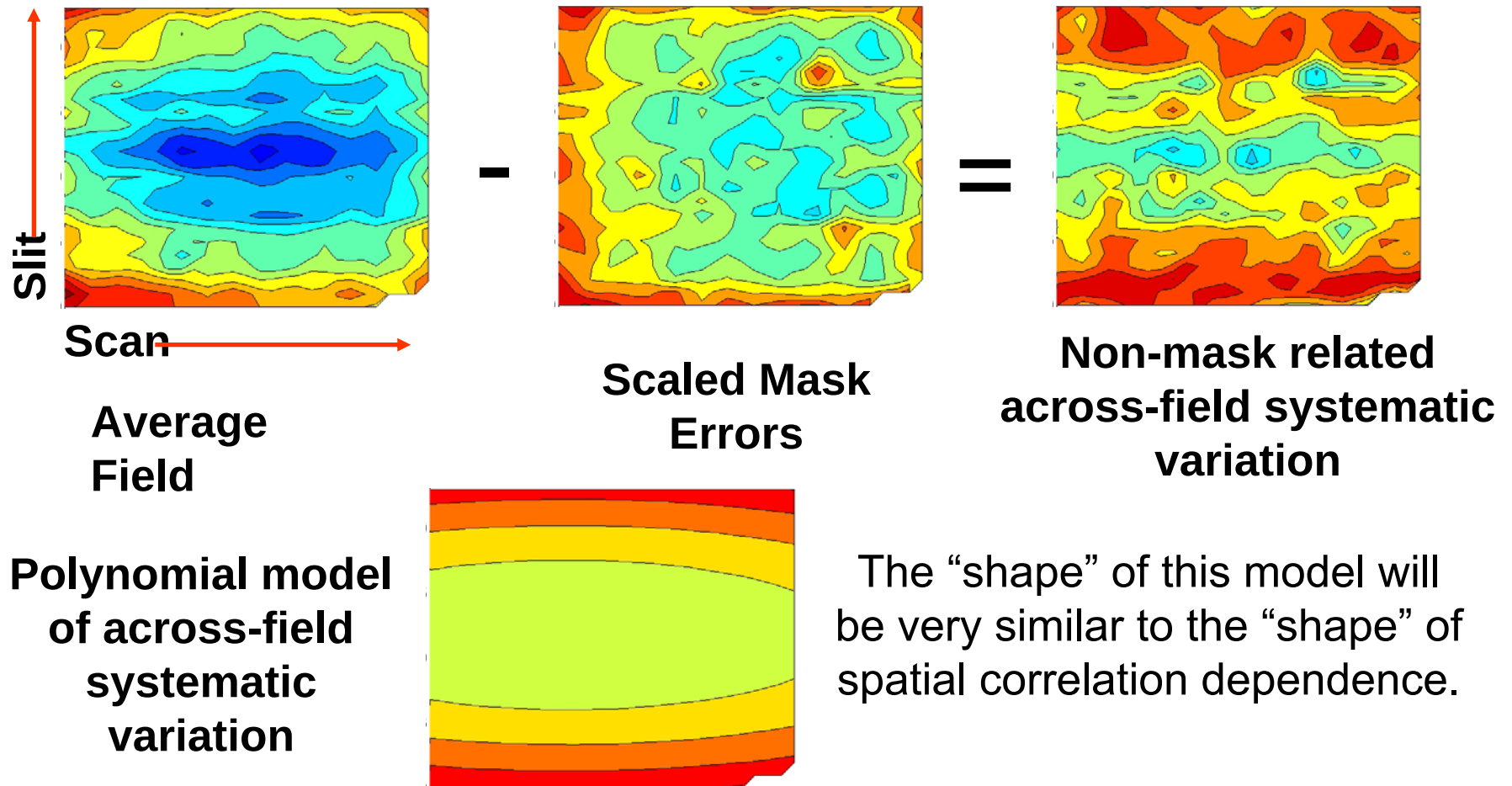
Across-Wafer Systematic Variation



Polynomial Model

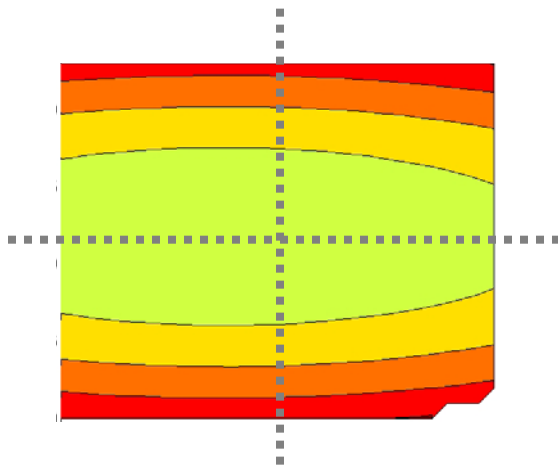
Origin of Spatial Correlation Dependence

- Within-die variation:

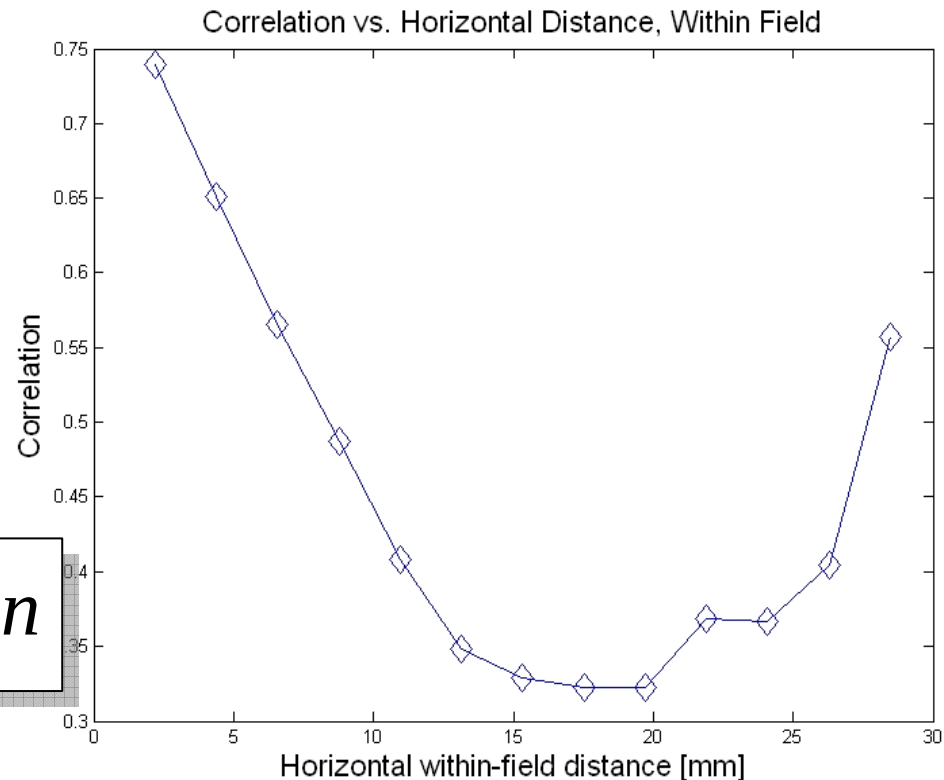


Origin of Spatial Correlation Dependence

- By slicing the within-die variation, we can see where the origin of the spatial correlation plots



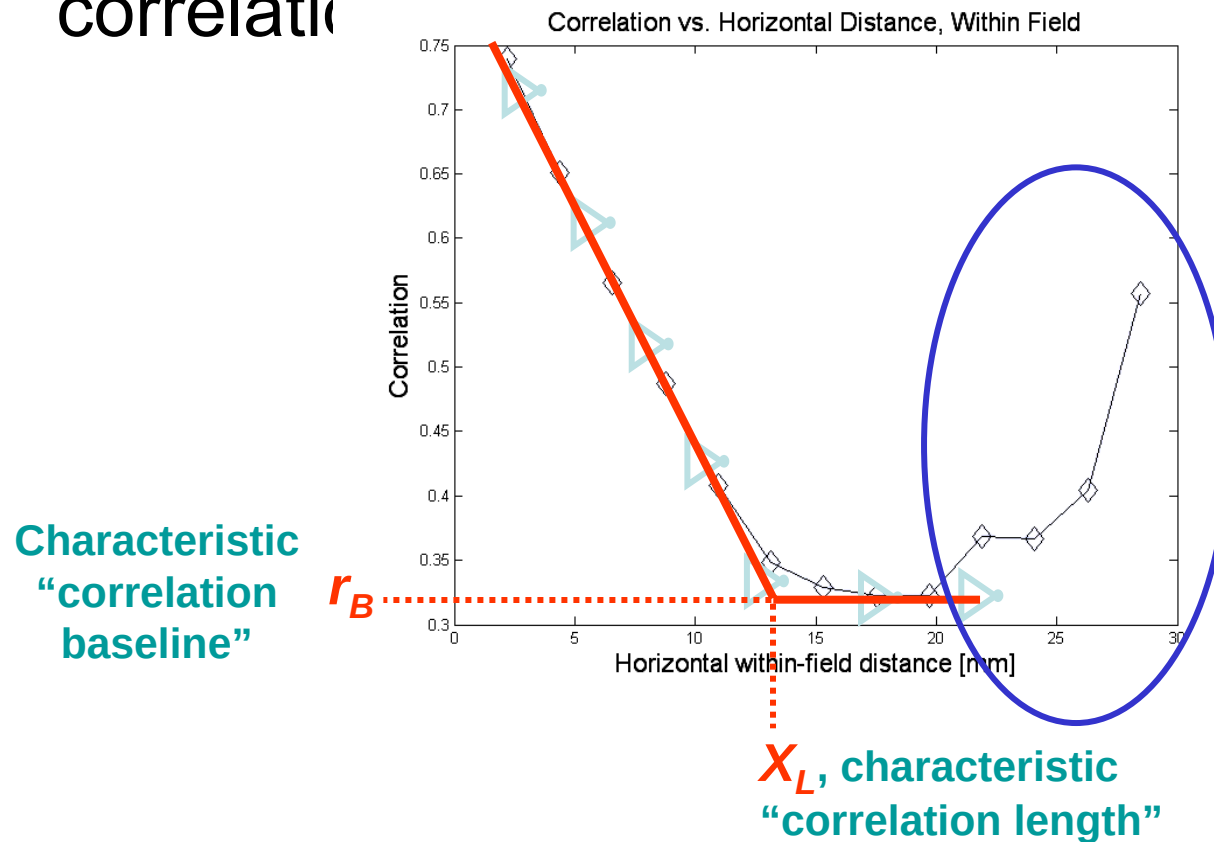
$$\rho_{jk} = \left(\sum z_{ij} * z_{ik} \right) / n$$



Spatial Correlation Model

- Fit rudimentary linear model to spatial correlation

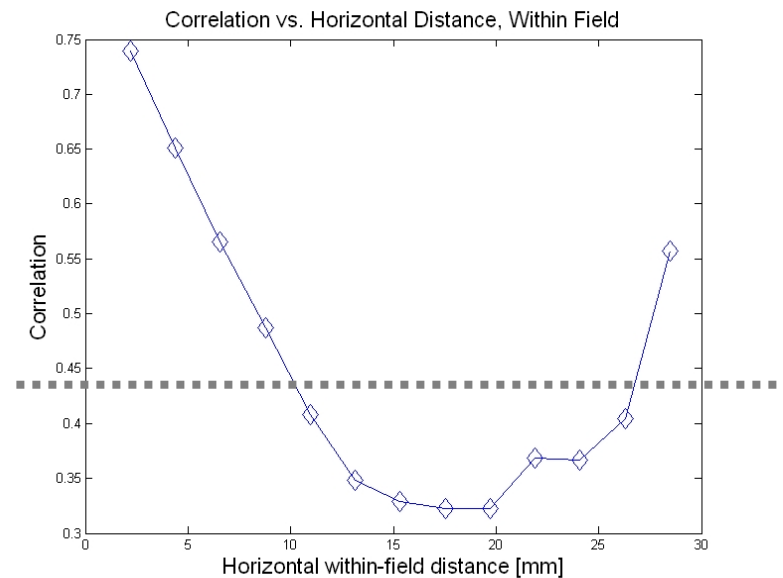
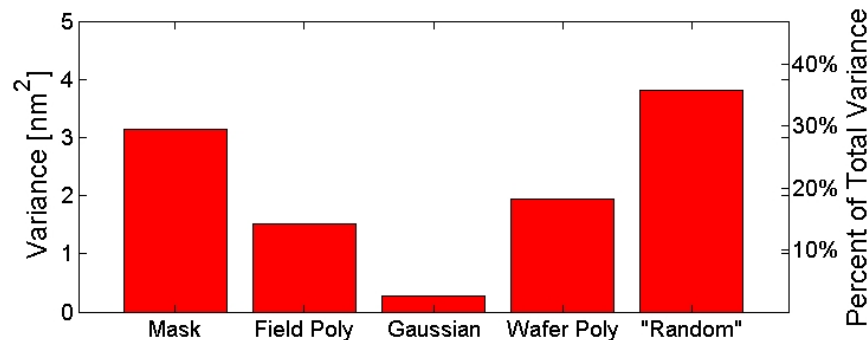
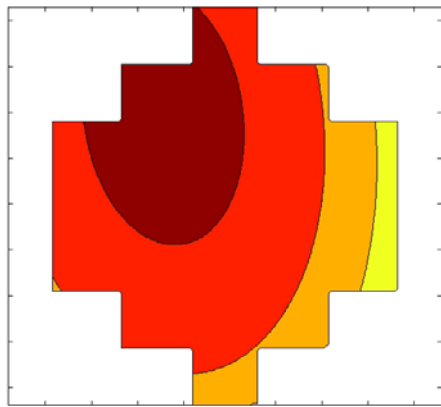
empirical data:



Ignore this part - distances too large for typical IC sub-circuit

Origin of Spatial Correlation Dependence

- The across-wafer variation impacts ρ_B :



Relative “weights” of field & wafer variation determine overall shape of spatial correlation curve

Calculation of Expected Effect

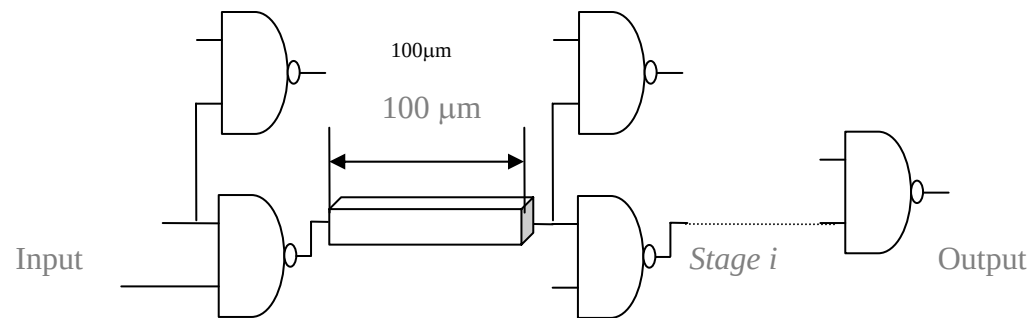
- Assuming n independent random variables with equal mean and variance, the variance of the sum of the n variables is:

$$\sigma_{tot}^2 = n\sigma_{indv}^2 + 2\rho\binom{n}{2}\sigma_{indv}^2 = [n + \rho(n)(n-1)]\sigma_{indv}^2$$

- For $\rho = 1$, $\sigma_{tot} = n\sigma_{indv}$
- For $\rho = 0$, $\sigma_{tot} = n^{1/2}\sigma_{indv}$
- Total potential improvement is factor of $n^{1/2}$ (i.e., for a 16-stage path, maximum total delay variation reduction is ~4x...)

Monte Carlo Simulations

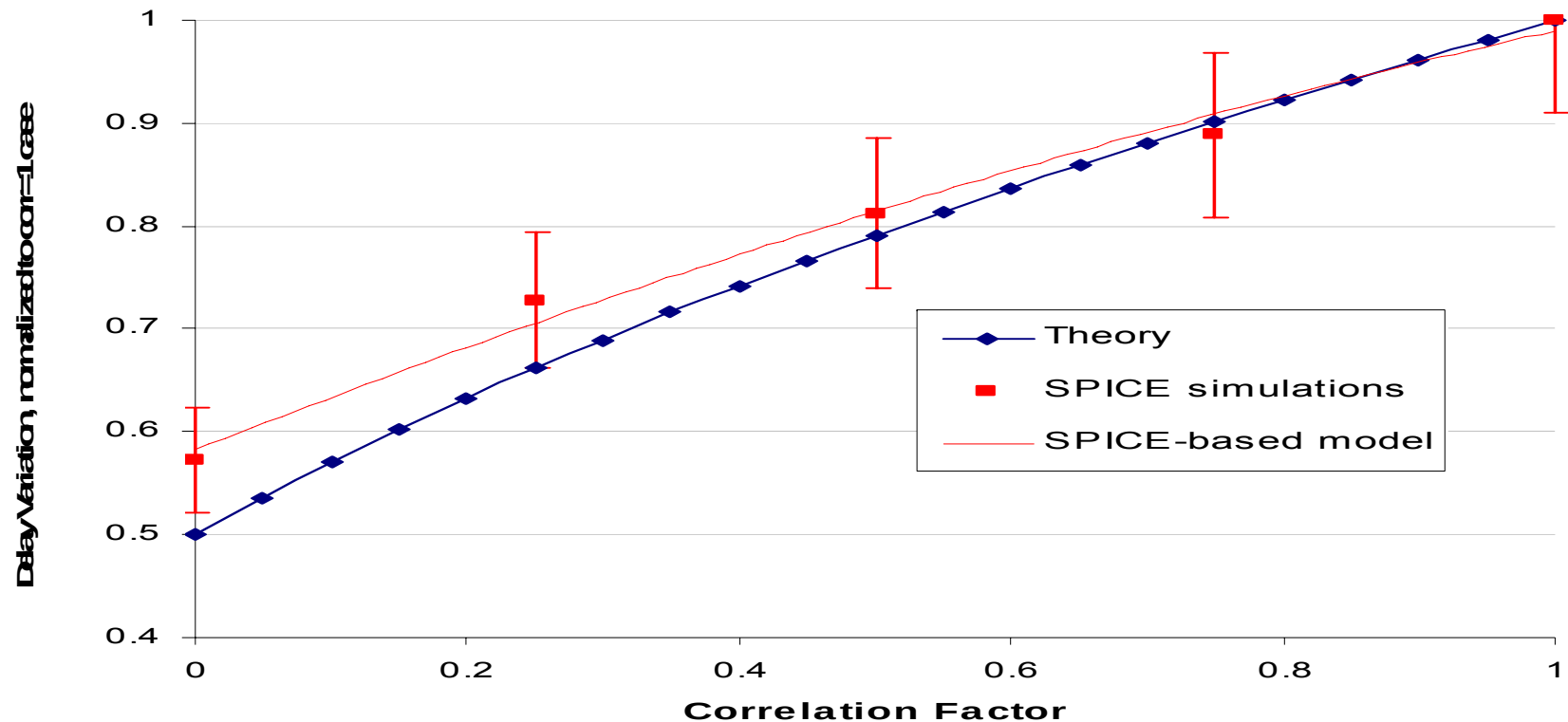
- Use canonical circuit of FO2 NAND-chain w/ stages separated by $100\mu\text{m}$ local interconnect, ST 90nm model:



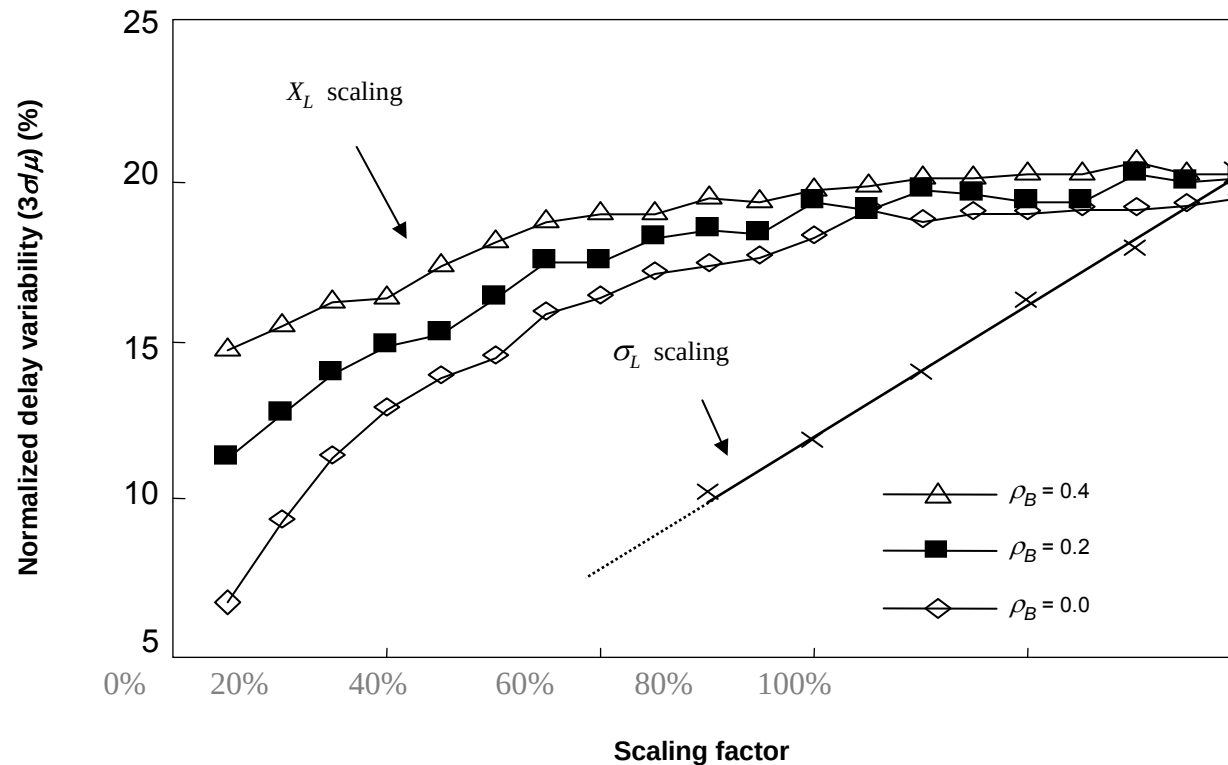
- Perform Monte Carlo simulations for various combinations of X_L , ρ_B , and σ/μ (gate length variation)
- Measure resulting circuit delays, extract normalized delay variation ($3\sigma/\mu$)

Statistical Theory vs. Circuit Simulation

- Prediction of potential impact of spatial correlation using SPICE simulations on simple

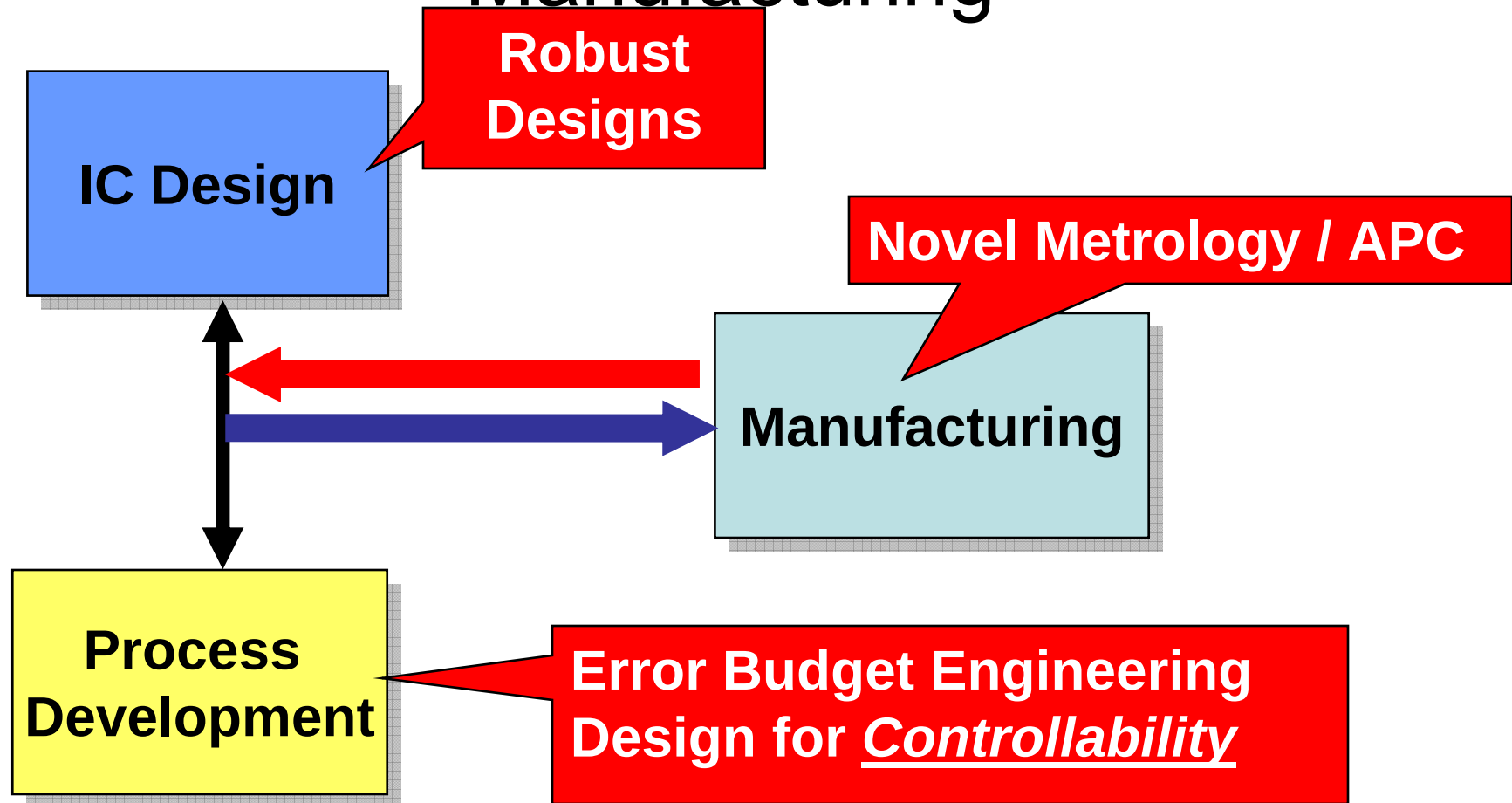


Delay Variability vs. X_L , ρ_B , σ/μ



- Scaling gate length variation directly has most impact
- Reducing spatial correlation also reduces variability

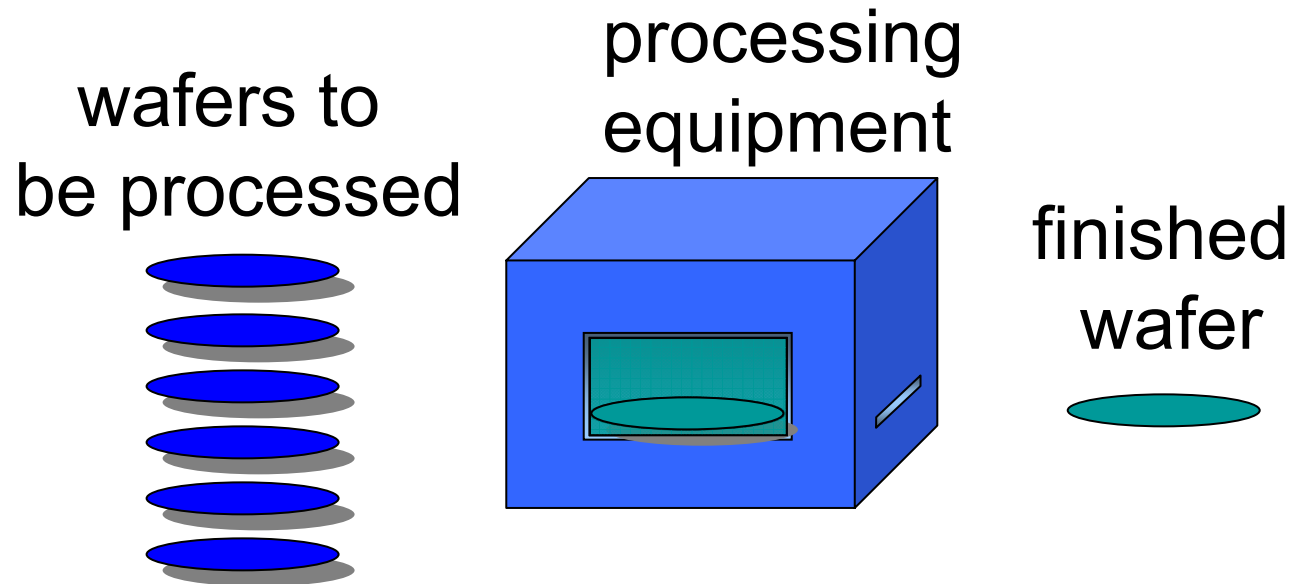
Coping with Variability in Semiconductor Manufacturing



Outline

- The Challenge
- Historical Perspective
- Measuring Variability
 - The Importance of Metrology
- Controlling Variability
 - Applied Concepts – APC at AMD
- Modeling Variability
 - Interaction of Design and Manufacturing
 - Design for Controllability
- Future Perspectives

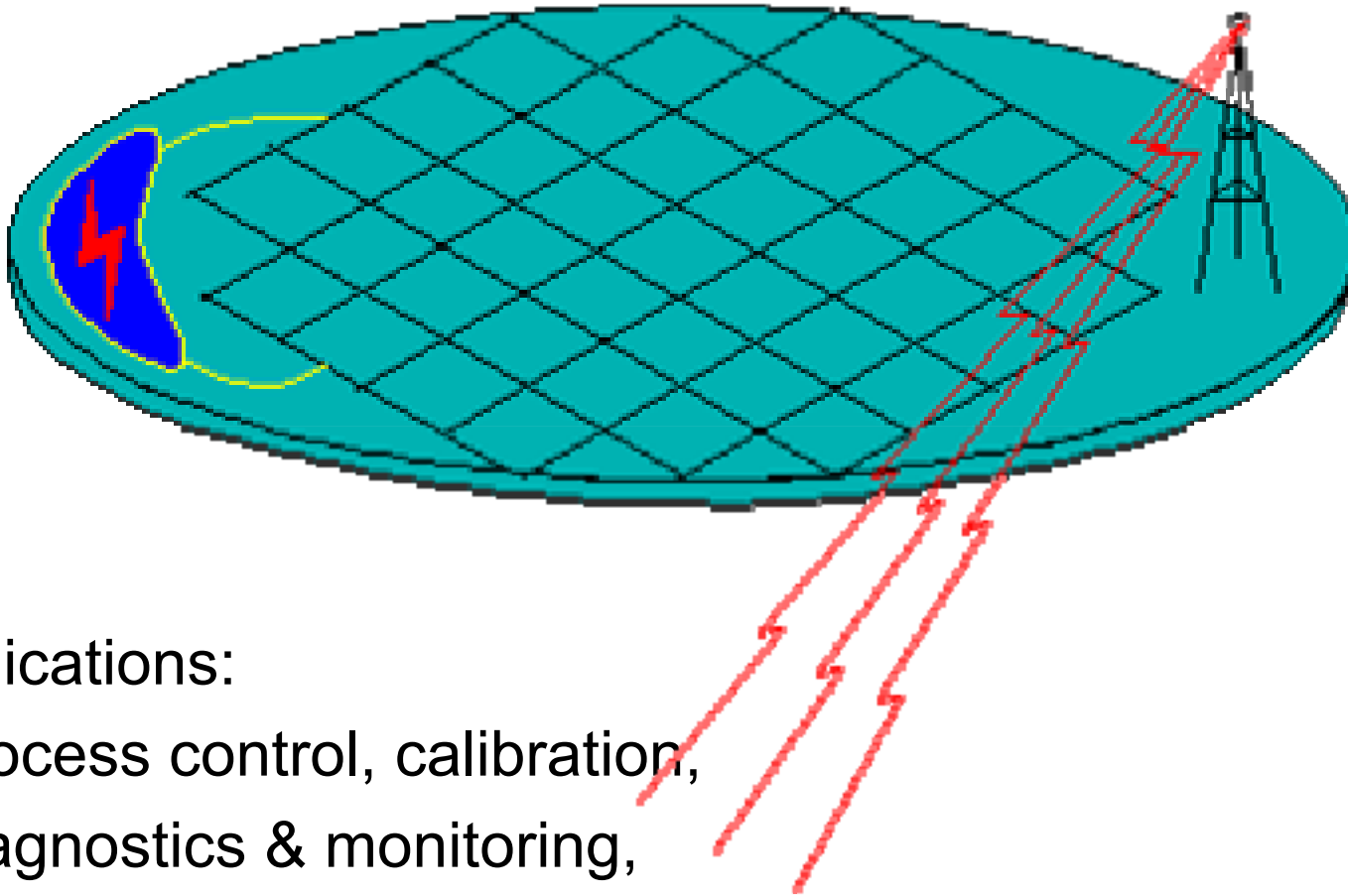
Processing Sequence



What was the state of the wafer during processing?

Smart Sensor Wafers

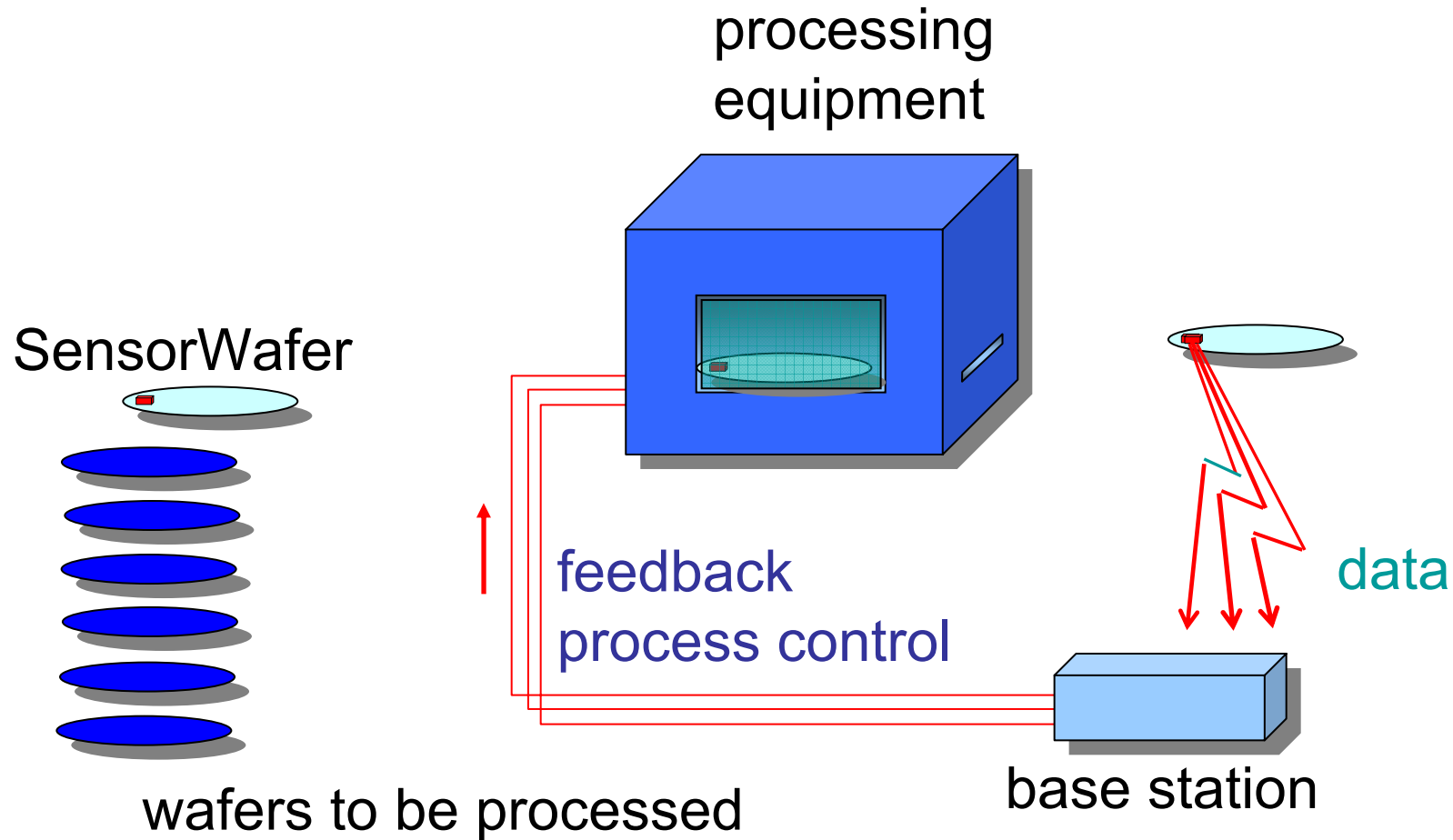
In situ sensor array, with integrated power and telemetry



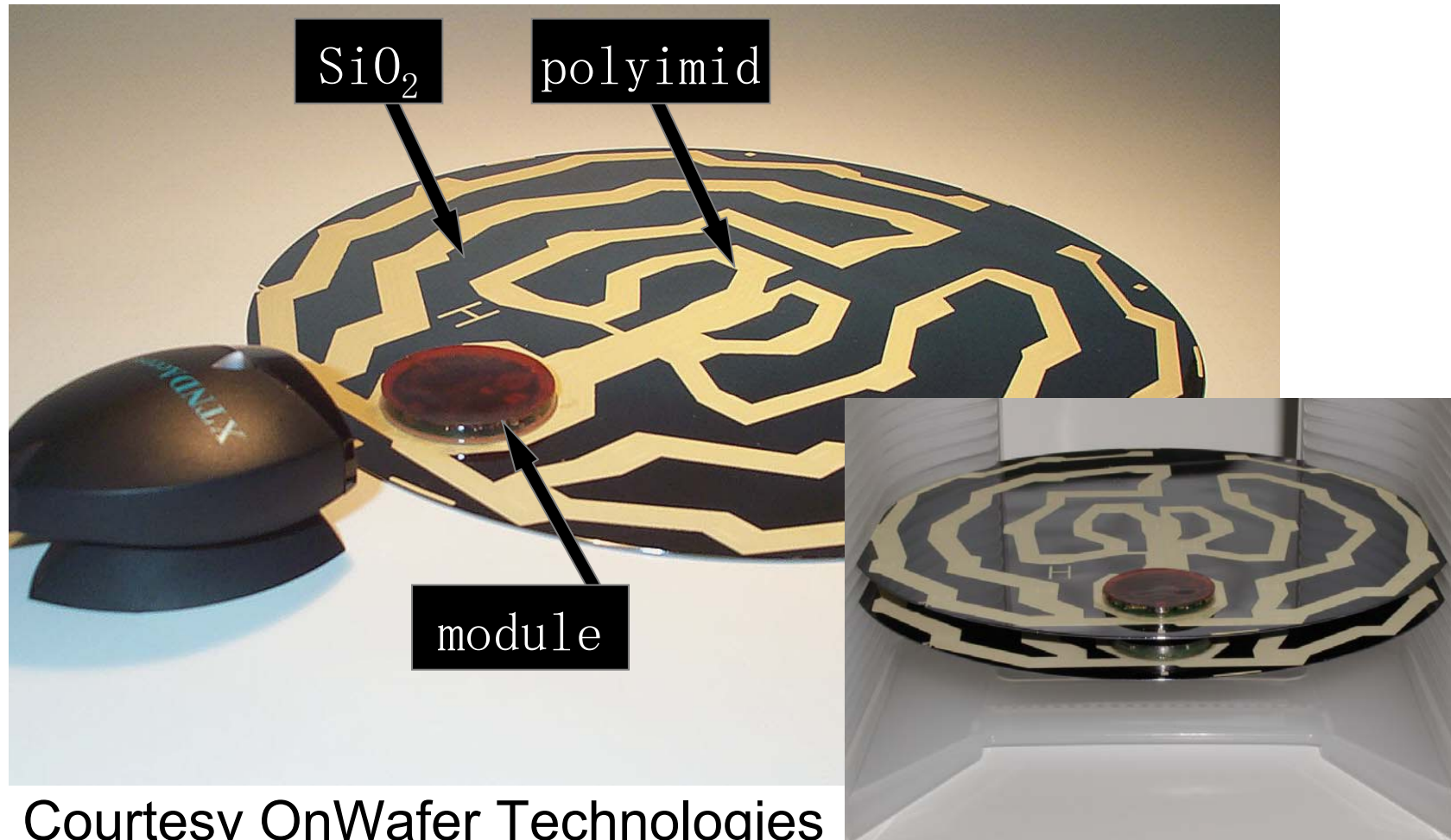
Applications:

process control, calibration,
diagnostics & monitoring,
process design

The Approach

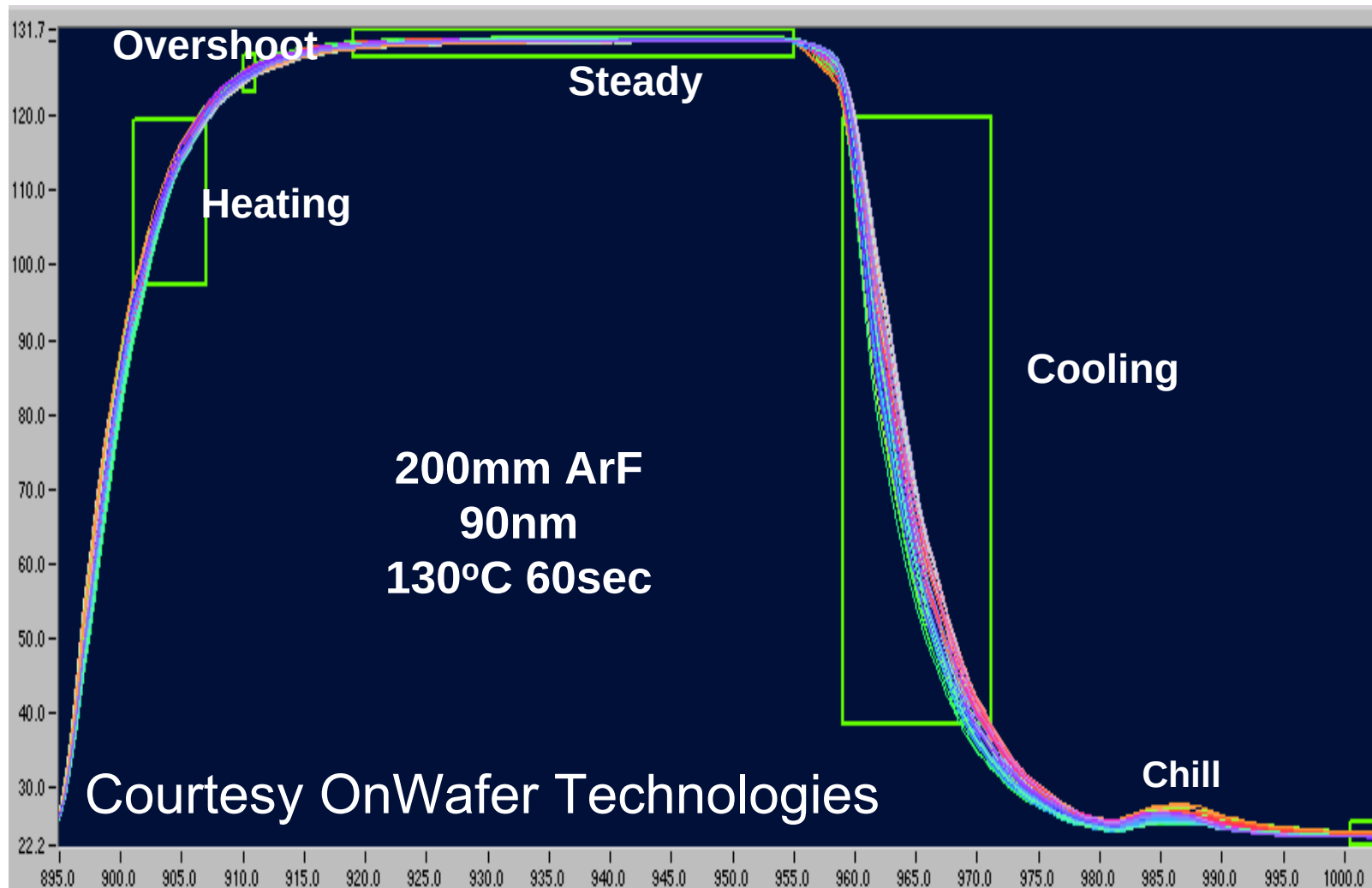


Today: The Sensor Wafer

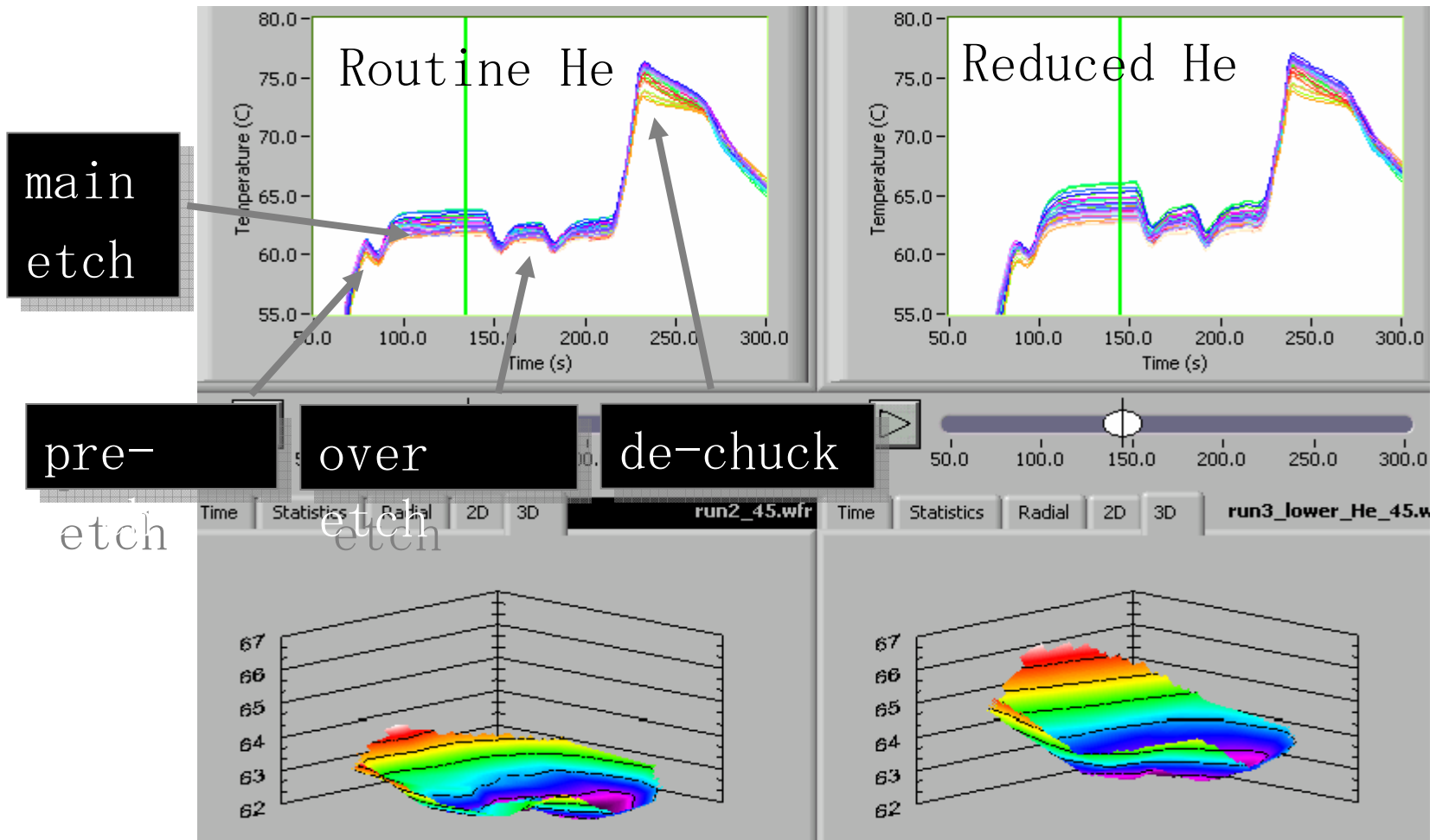


Courtesy OnWafer Technologies

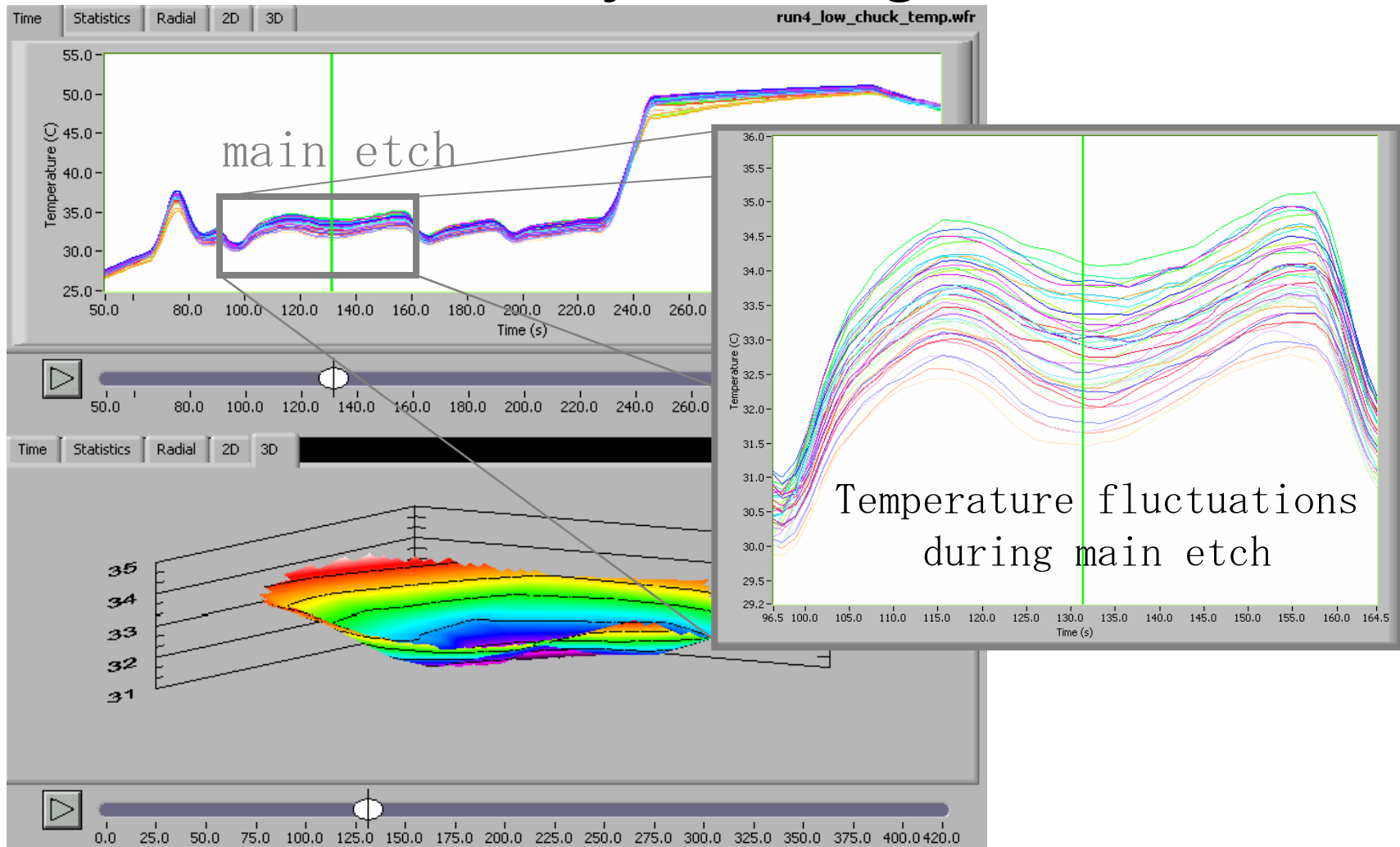
Much more than you ever wanted to know about Post Exposure Bake



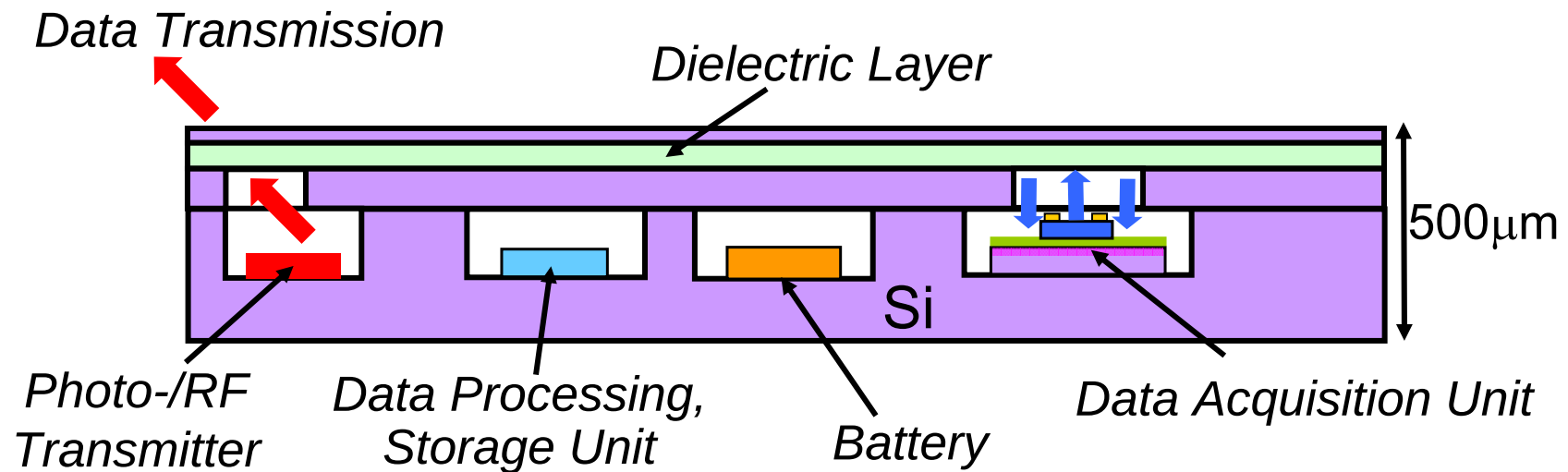
On-Wafer Plasma Monitoring 200mm Poly Etching



Cool chuck - 200mm Poly Etching

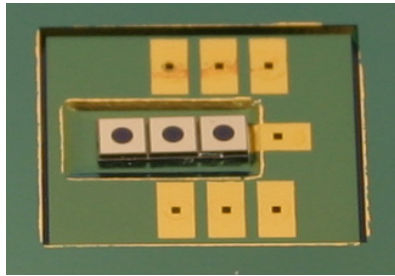
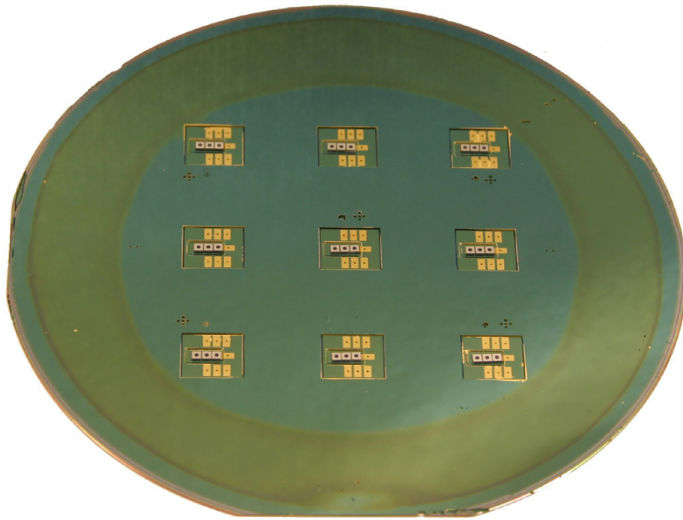


Next Step: Zero-Footprint Metrology Wafer

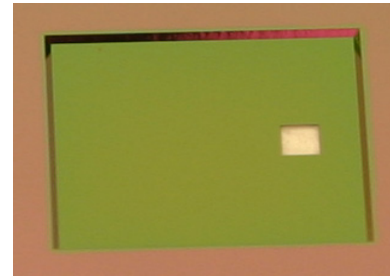
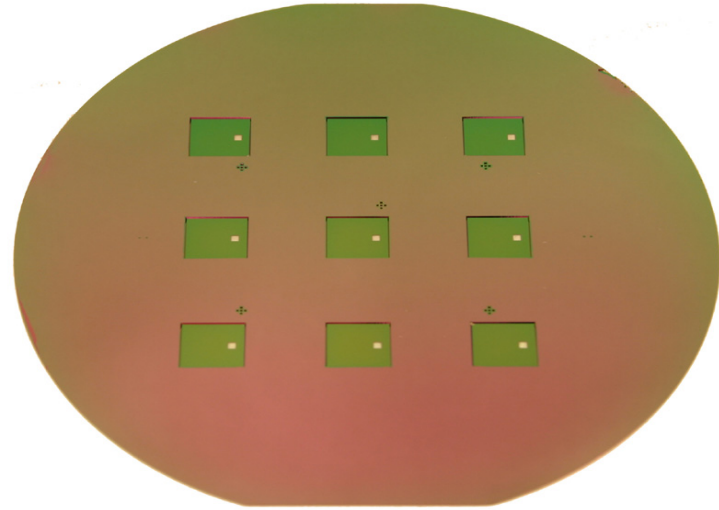


Prototyping a zero-footprint metrology wafer with optical detection unit and encapsulated power source.

3 x 3 Pixels Optical Metrology Prototype

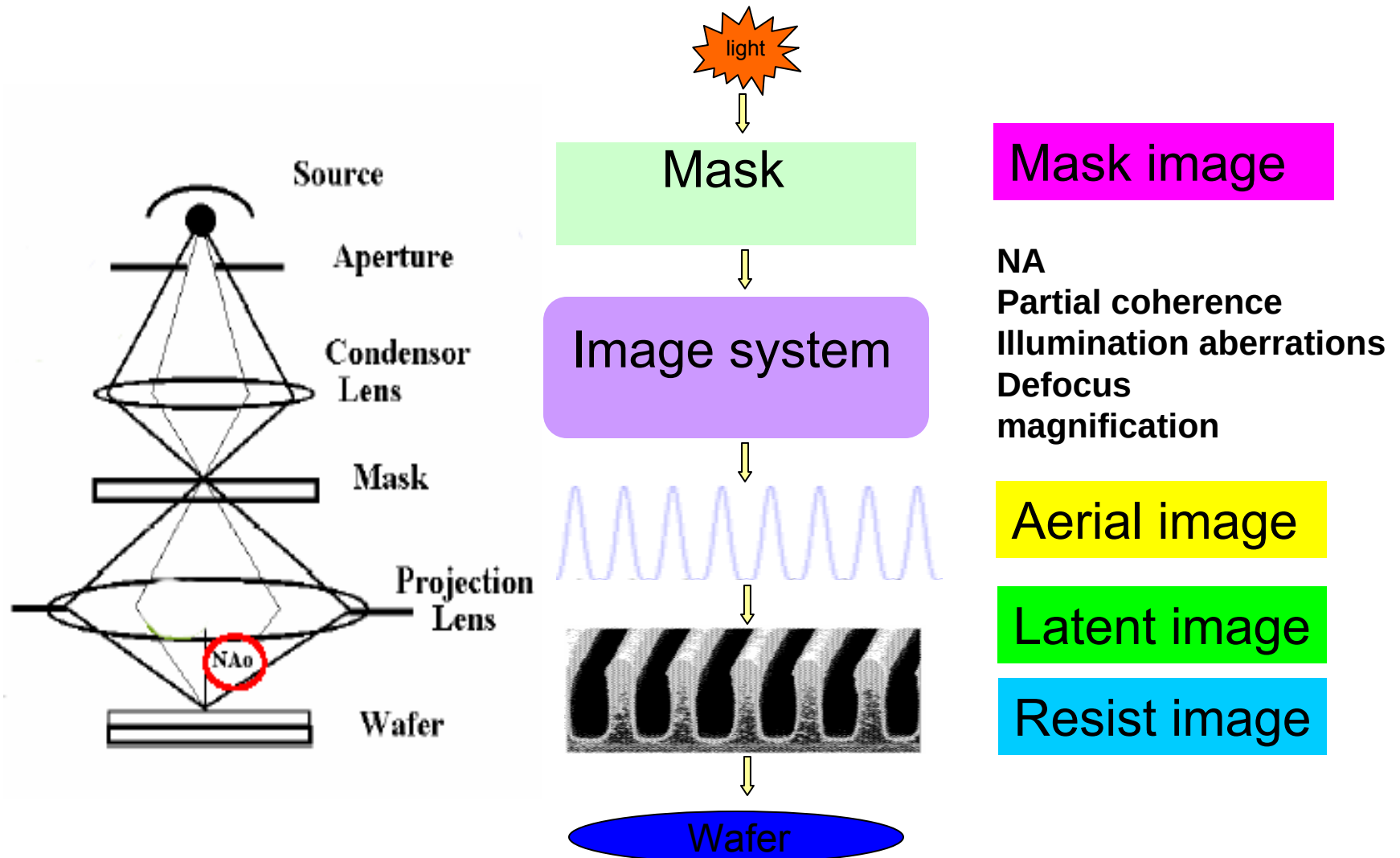


Bottom Wafer with LED
Photodetector integrated



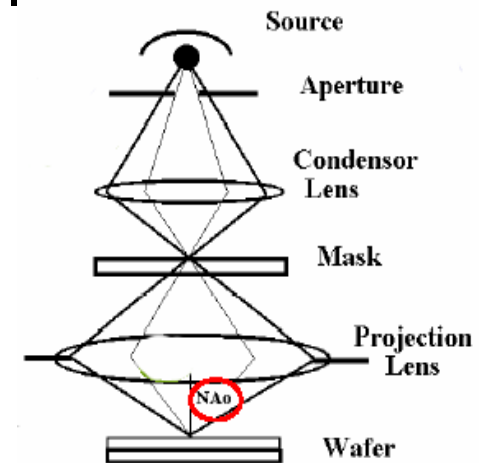
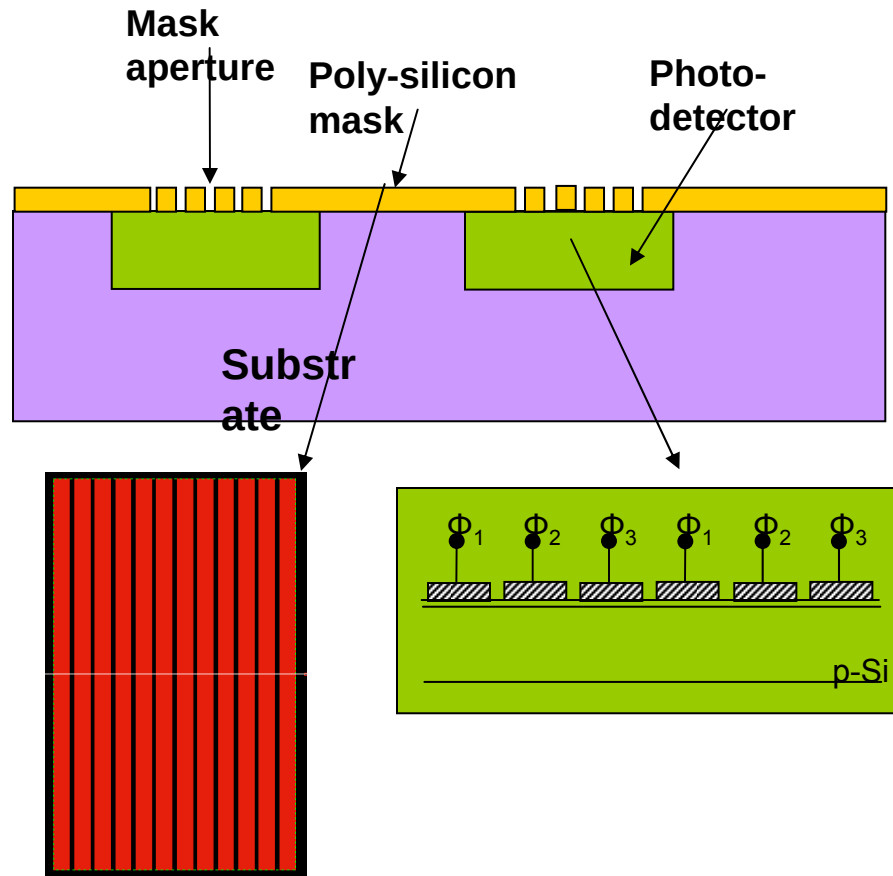
Top Wafer

Wireless Aerial Image Metrology



An Integrated Aerial Image Sensor

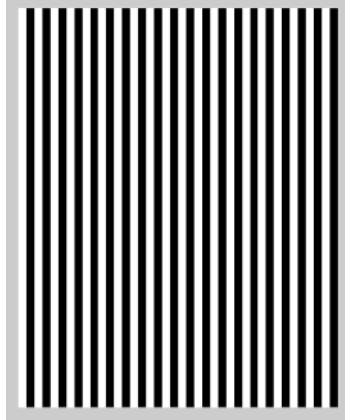
Dark contact mask forms a “moving” aperture to capture incident electromagnetic field.



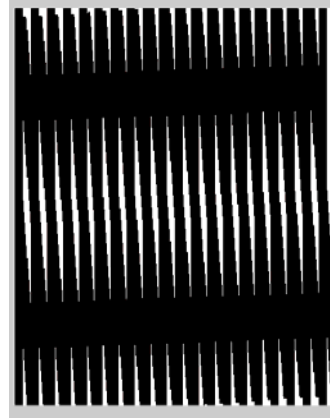
How can a μm detector retrieve nanometer-scale resolution of the aerial image?

Moiré Patterns for Spatial Frequency Shift

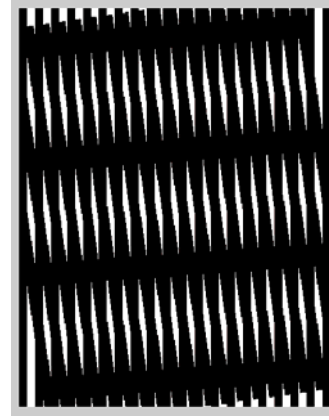
Narrow CD



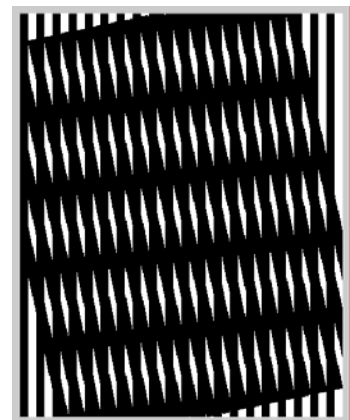
Patterns Overlap



Pattern rotates 4°

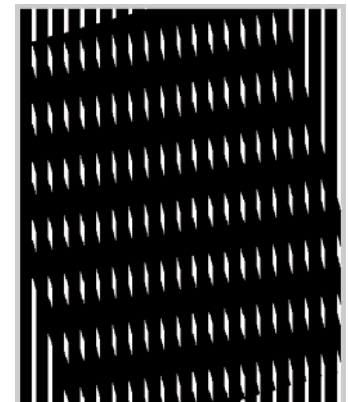
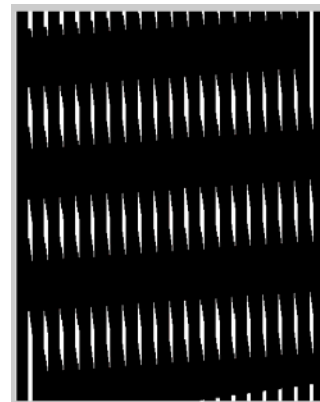
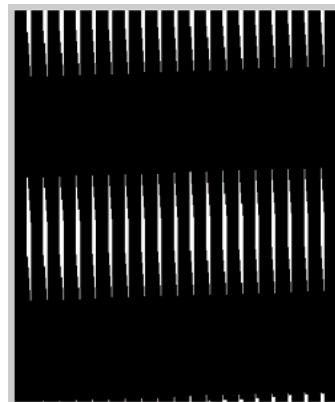
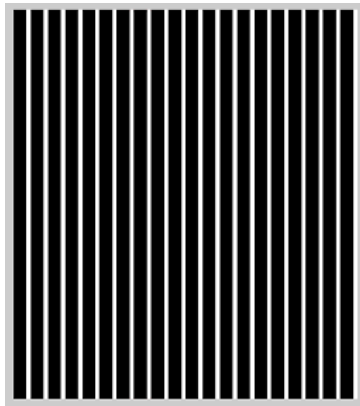


Pattern rotates 8°

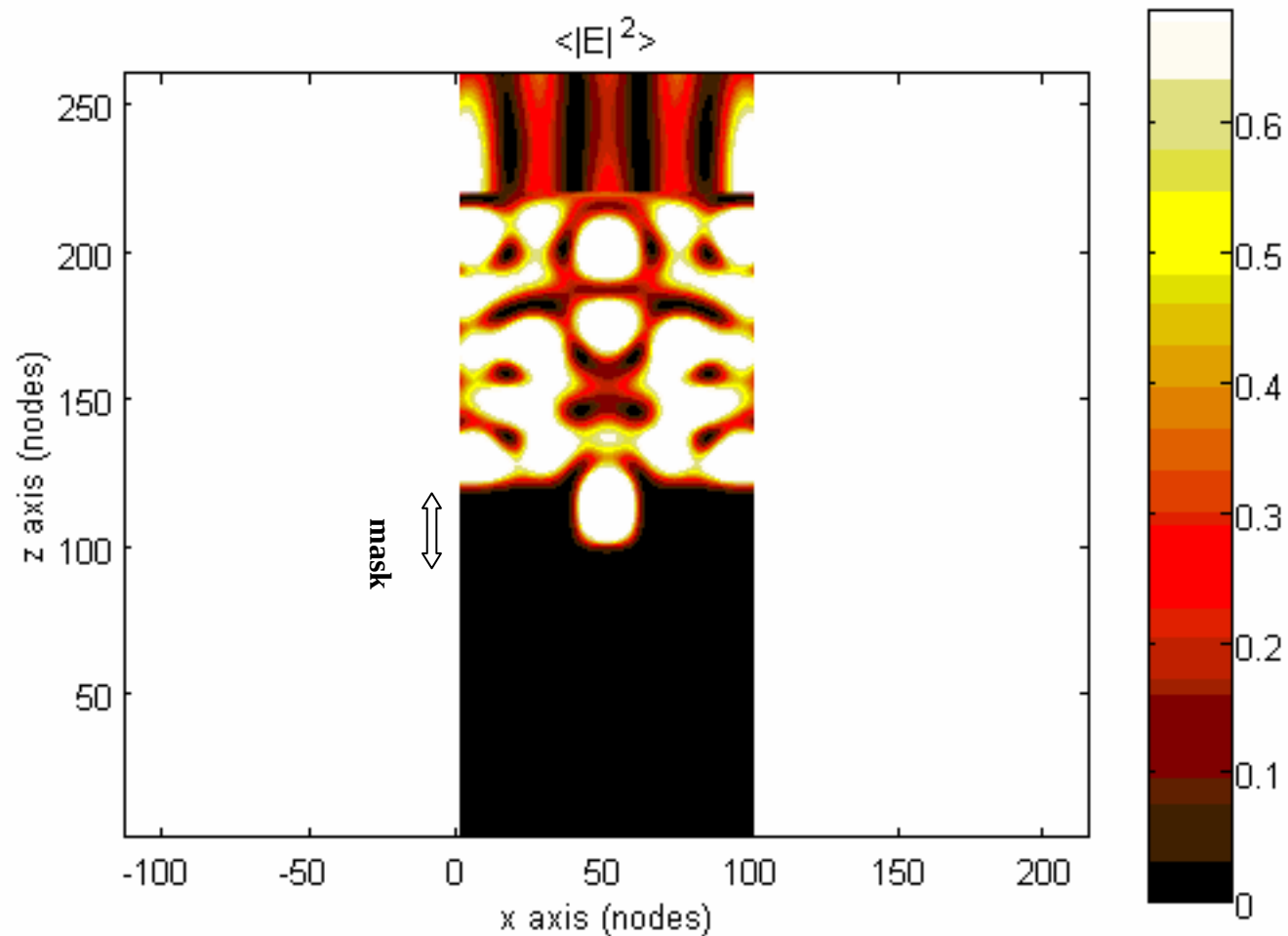


Pattern rotates 16°

Wide CD



Near-Field Optical Simulation

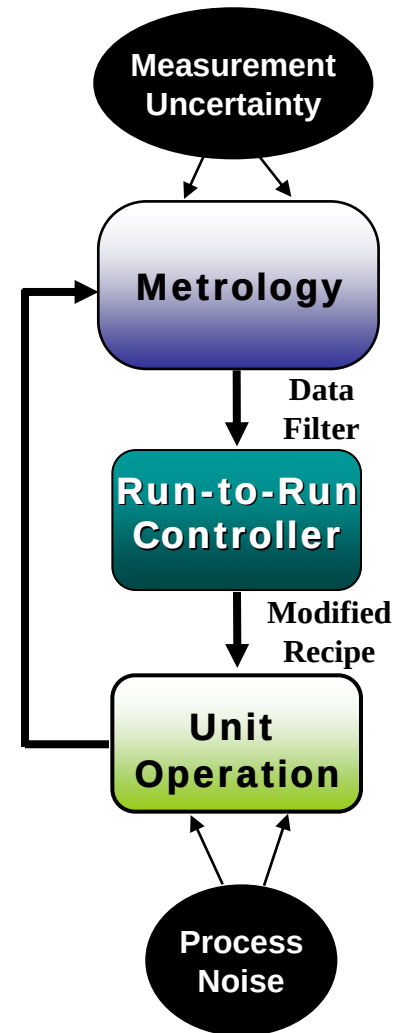


Intensity at the center of the simulation domain

Metrology for APC

Why does APC care about metrology?

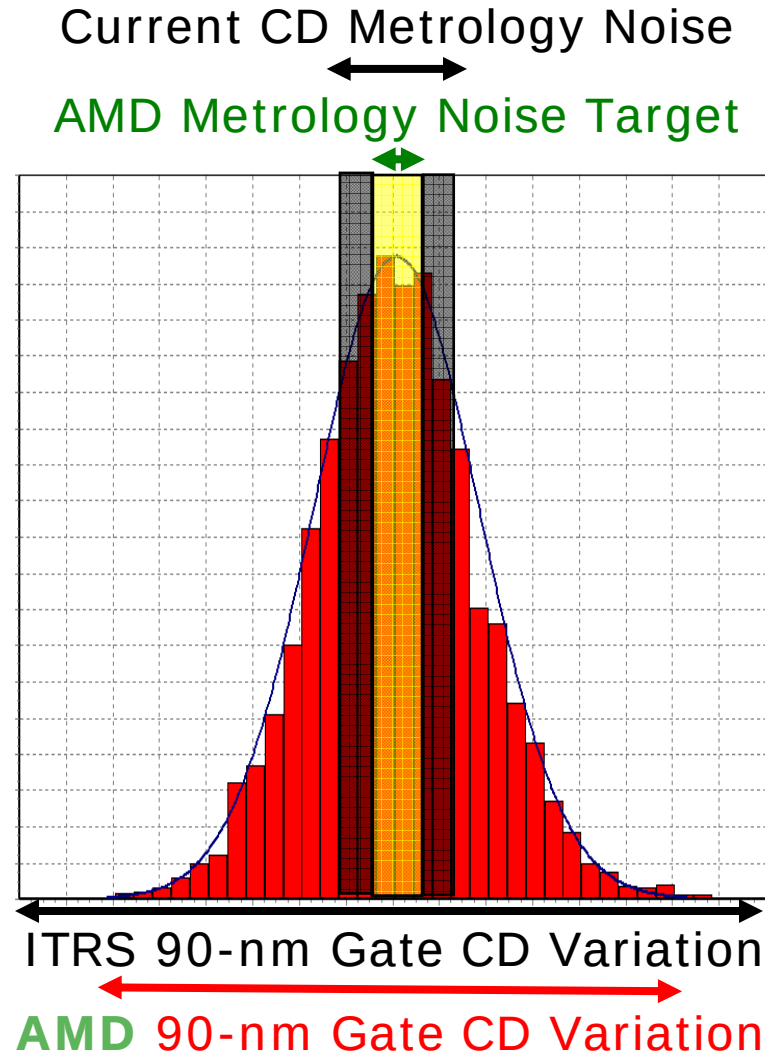
- Garbage-in, Garbage-out (GIGO)
 - Noisy metrology limits controller effectiveness
 - Controllers must filter, dampen and average the signal to account for noise
 - These actions may compromise control objectives or constrain the factory
 - Metrology errors become process errors
 - RtR controllers can not distinguish between metrology and process errors
 - The controller will adjust for a metrology trend or shift as if it were a process deviation
 - The result is a mis-targeted process
- No metrology means no APC
 - New control needs are emerging with no robust measurement solution
 - Metrology is the gating item to controller development



Advanced Measurement Technology (AMT)

- The AMD APC roadmap can not be achieved without directly addressing metrology challenges
 - **Reducing Uncertainty**
 - **Improve existing metrology metrics to meet APC needs**
 - Metrology noise must be reduced (precision)
 - Metrology data must reflect real process variation (accuracy)
 - Metrology trends and shifts must be eliminated (stability)
 - **Creating new capabilities**
 - **Develop product metrics to enable new APC applications**
 - Pattern height and shape
 - Line roughness
 - Design-based Metrology
- The AMT team is dedicated to working with APC to destroy metrology roadblocks

Gate APC and Metrology Precision

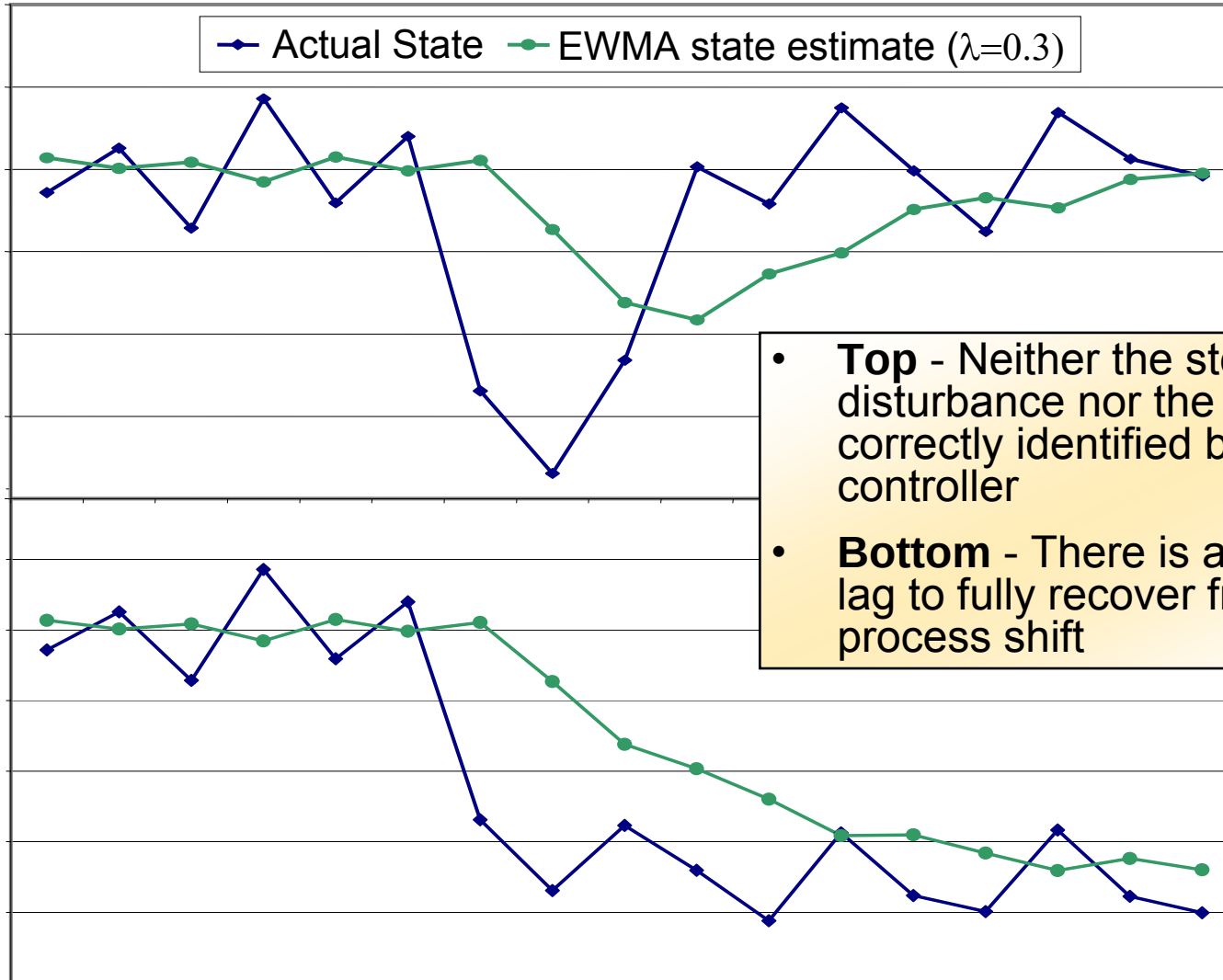


- CD-SEM metrology tools are meeting the ITRS spec for precision at the 90-nm node
- But gate etch RtR control has reduced the overall process range well below the ITRS benchmark
- Metrology precision is not adequate for these tighter process windows
- Metrology precision must improve by more than 100%!

Managing the Precision Problem *with Algorithms*

- Mathematical methods to mitigate input noise:
 - Exponentially Weighted Moving Average (EWMA) state estimation
 - Averages the process state over time to ensure that the controller responds to a “real” signal
 - May hide some signals or delay detection of others
 - Statistical filtering (e.g. box filters) of incoming metrology data
 - Eliminates statistically erroneous measurements
 - May incorrectly ignore real process faults
 - Increasing the metrology sample size
 - Additional measurements increase confidence in statistics
 - Also burdens the factory with metrology cycle time

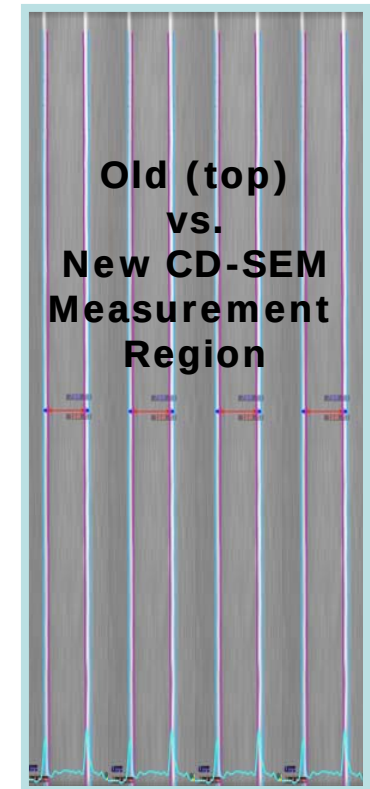
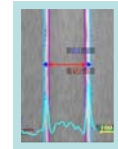
Disturbances and EWMA Filtering



- **Top** - Neither the step disturbance nor the recovery is correctly identified by the controller
- **Bottom** - There is an eight run lag to fully recover from this process shift

Managing the Precision Problem *with Advanced Measurement Technology*

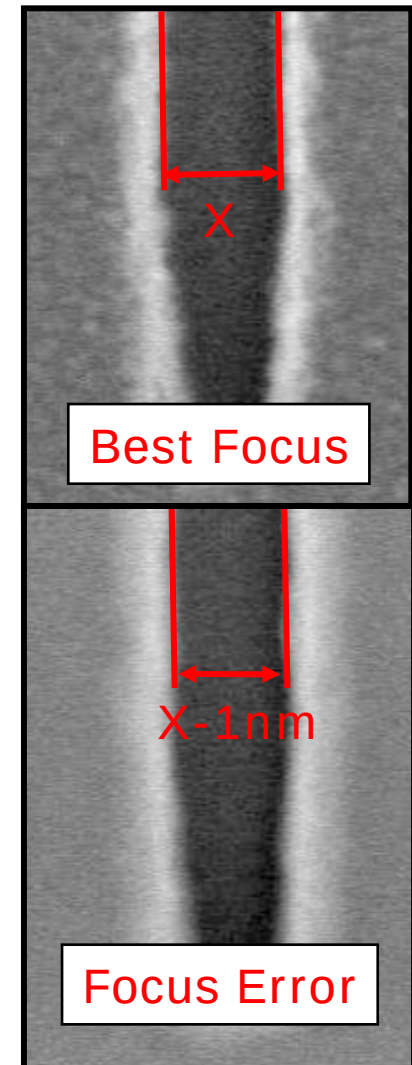
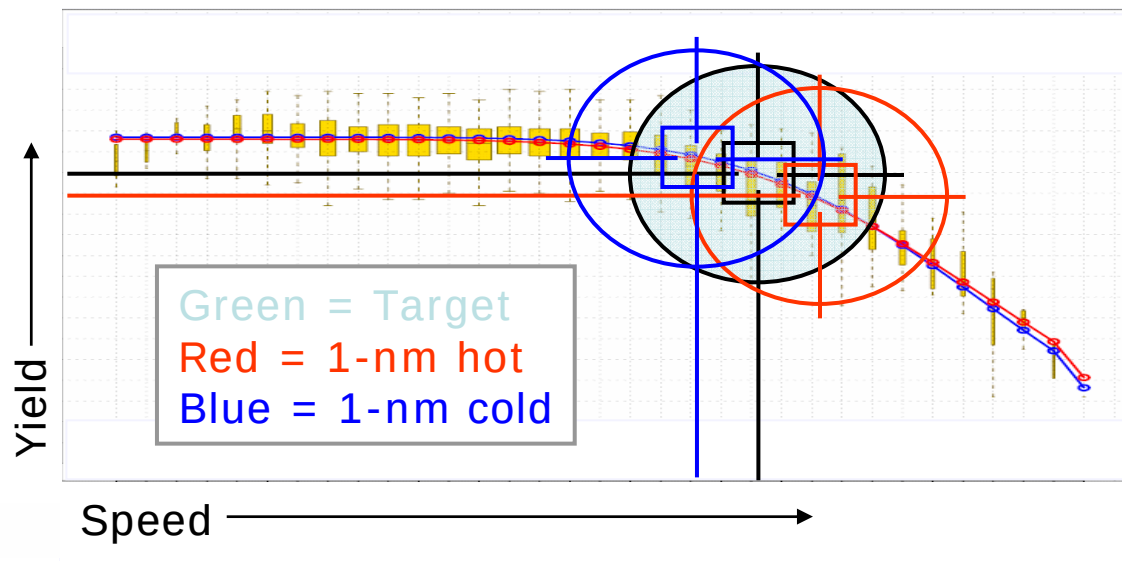
- Identify key areas where control improvement is limited by measurement noise
 - CD metrology
 - Step height metrology
- Focus on improving the metrology rather than manipulating control algorithms
 - Implement rigorous metrology tool controls
 - With engineering focus and advanced equipment monitoring, CD-SEM precision can reach 2x of manufacturer's specifications
 - Drive supplier roadmaps for improved hardware and software
 - New SEM columns and measurement algorithms allow 100x more signal averaging than previous generation of tools
 - Investigate paradigm-shifting emerging technologies
 - Scatterometry has displaced the profilometer for STI step-height metrology and is challenging the CD-SEM on other fronts



Improving Measurement Stability

Gate CD Control

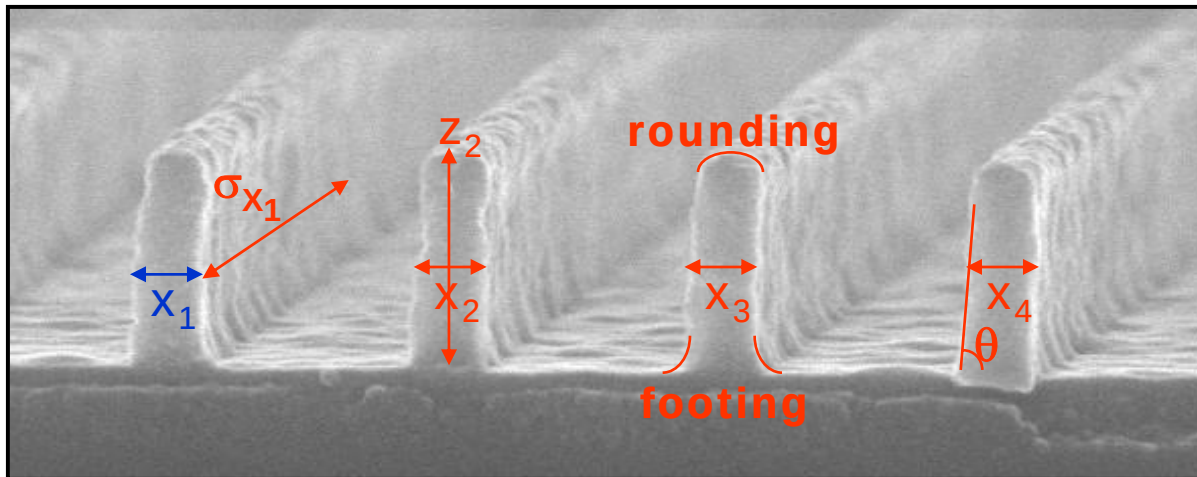
- Slight variations in SEM imaging can cause metrology errors (like focus variation in these SEM images)
- A shift in CD distributions as little as 1-nm can lead to dramatic yield or speed degradation
- With APC control active and tolerances so severe, metrology error must be eliminated



Creating New Capabilities

Pattern Fidelity Metrology

- For 90-nm and beyond, it is not enough to know the width of a single printed line
- Robust litho metrology must be able to answer the question:
“How accurately did I reproduce the mask pattern in all three dimensions?”
- **Old Paradigm:** What is the CD of this array?
 - Metrology: point solution CD (X_1)
- **Beyond 90-nm:** What is the pattern fidelity of this array?
 - Metrology: 3-dimensional details (X_{1-4} , X_{bar} , σ_x , θ , Z , footing, rounding)



Cross-section SEM image of 90-nm gate resist lines.

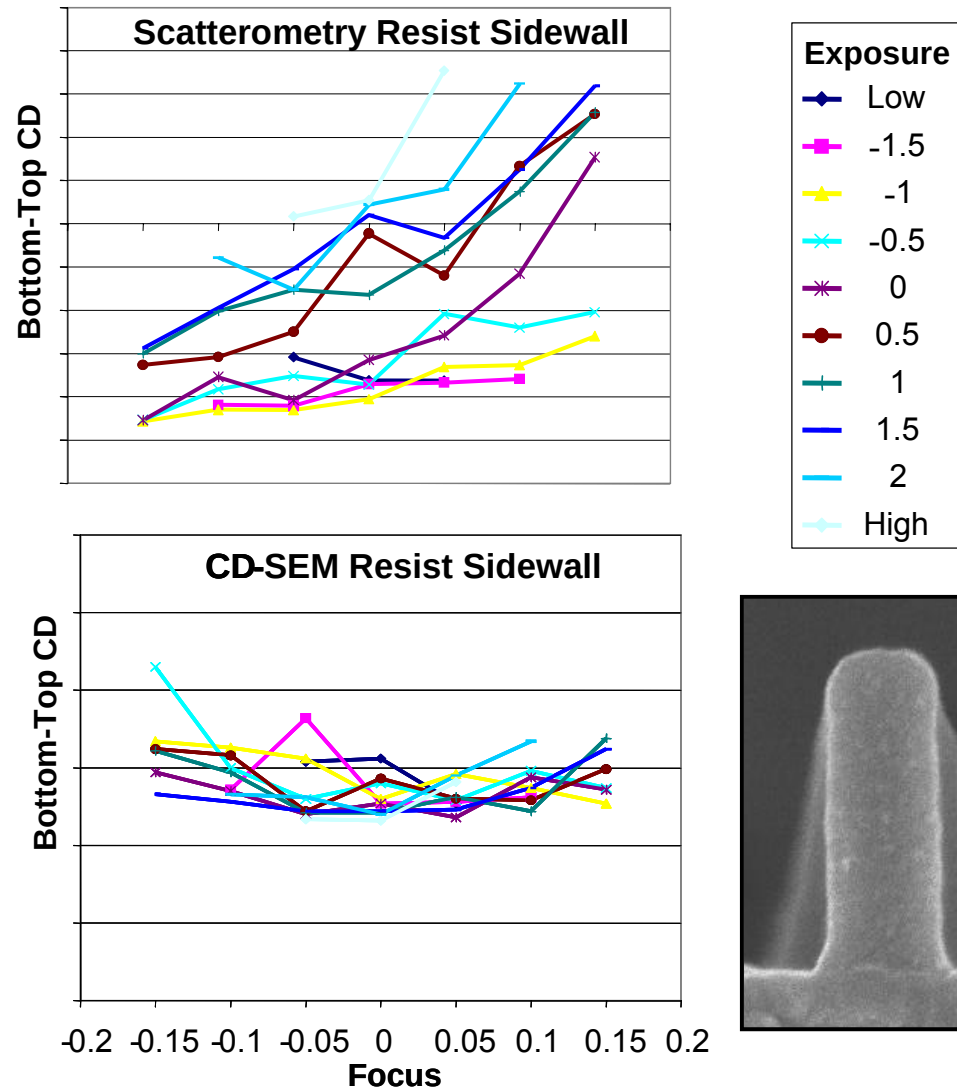
Pattern Fidelity Tools

- Advanced SEM
 - X_1, X_2, X_3, X_4 or X_{bar}
 - σ_{X1} (roughness)
- Scatterometry
 - $\theta, Z, X_{\text{bar}}$
 - Rounding/footing
- AFM
 - $X, Z, \theta(?)$
 - Reference metrology

Creating New Capabilities

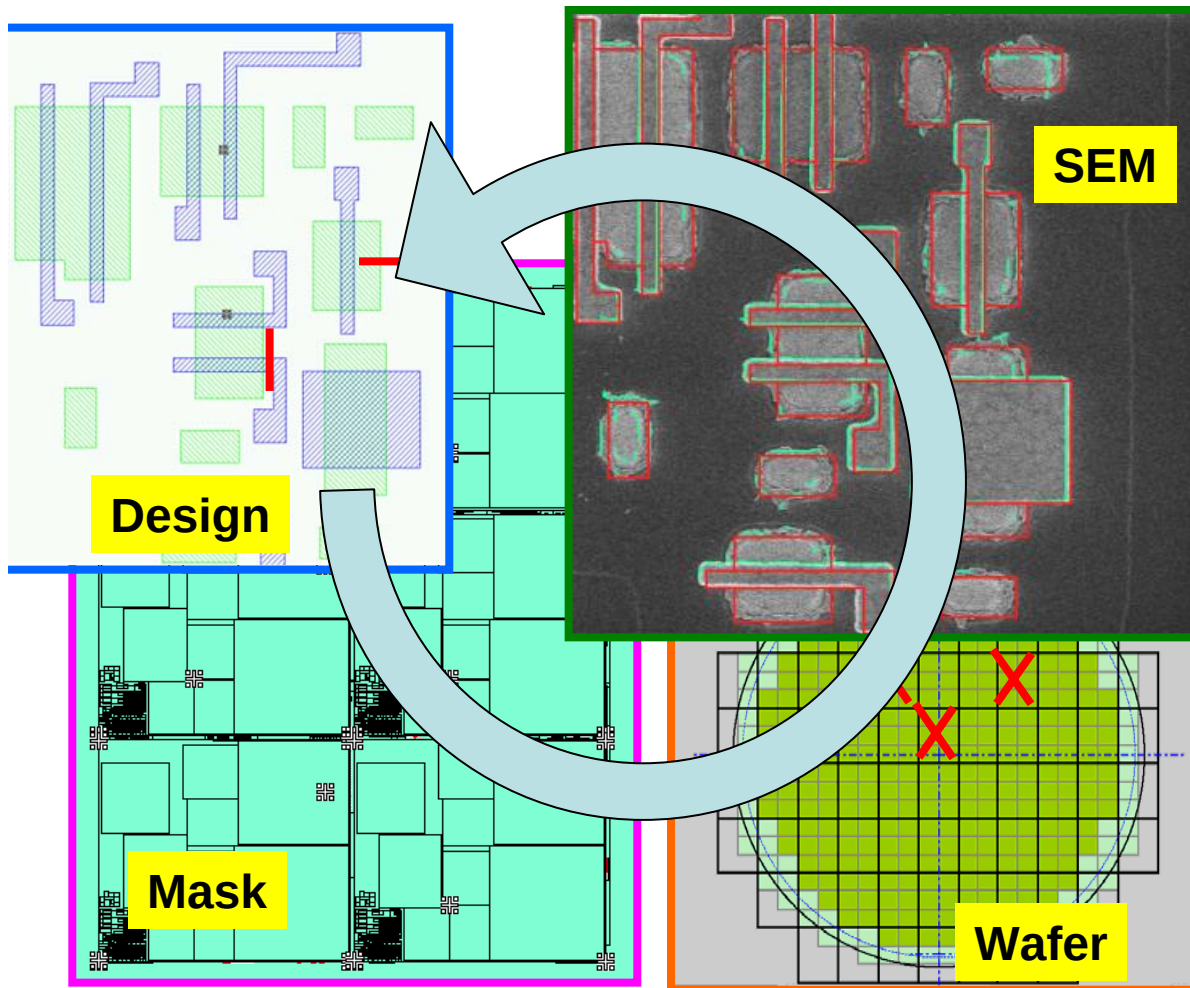
Sidewall Angle Metrology (θ)

- Most scatterometry models return the sidewall angle of the feature
- The CD-SEM is incapable of measuring sidewall variation, especially for vertical profiles
- Sidewall data is a critical early indicator of focus error
- AMD APM is using scatterometry to develop integrated dose and focus RtR control



Creating New Capabilities

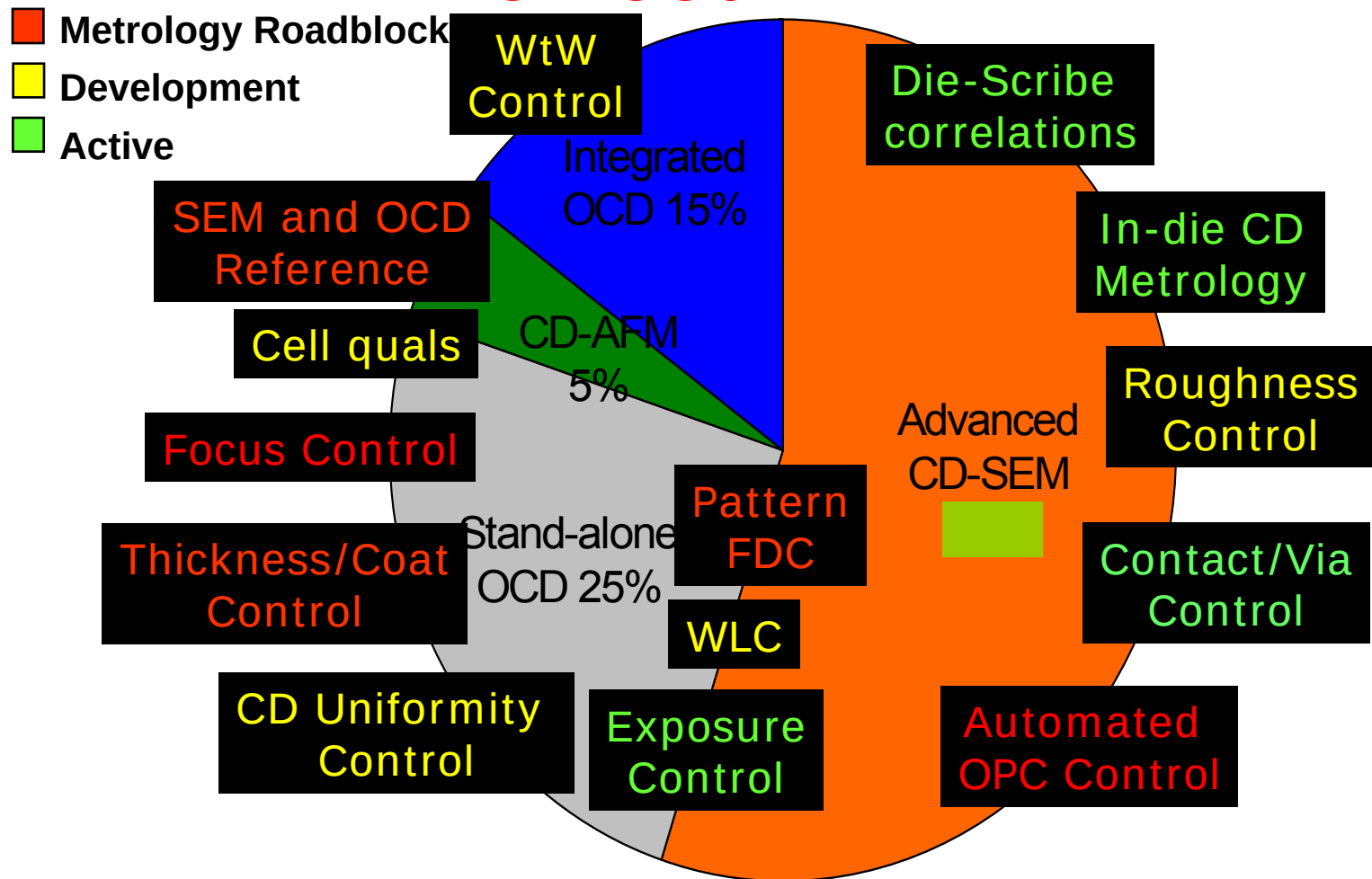
Design-based Metrology



- Shrinking pitch and CD targets have forced complex lithography strategies (RET)
- One implication of RET is large proximity effects
- Optical Proximity Corrections (OPC) are designed into the mask to make the wafer image match design intent
- OPC model building and validation requires thousands of wafer measurements
- AMT has collaborated with the OPC community to create SEM recipes automatically from design data
- The SEM will soon be able to score the agreement between wafer and design and give numerical feedback
- The next frontier for APC???

APC Lithography Applications

We need AMT!



Conclusion

- APC technology relies heavily on metrology development
 - Reducing the uncertainty of existing RtR input data streams
 - Developing new metrics to enable next-generation APC solutions
- It is naïve to believe that APC can address metrology deficiencies with advanced algorithms alone
- APC and the metrology community must work hand-in-hand to develop, test and implement new advanced measurement technology (AMT)
- The coordination of APC and AMT is a central element in AMD's Automated Precision Manufacturing (APM) strategy



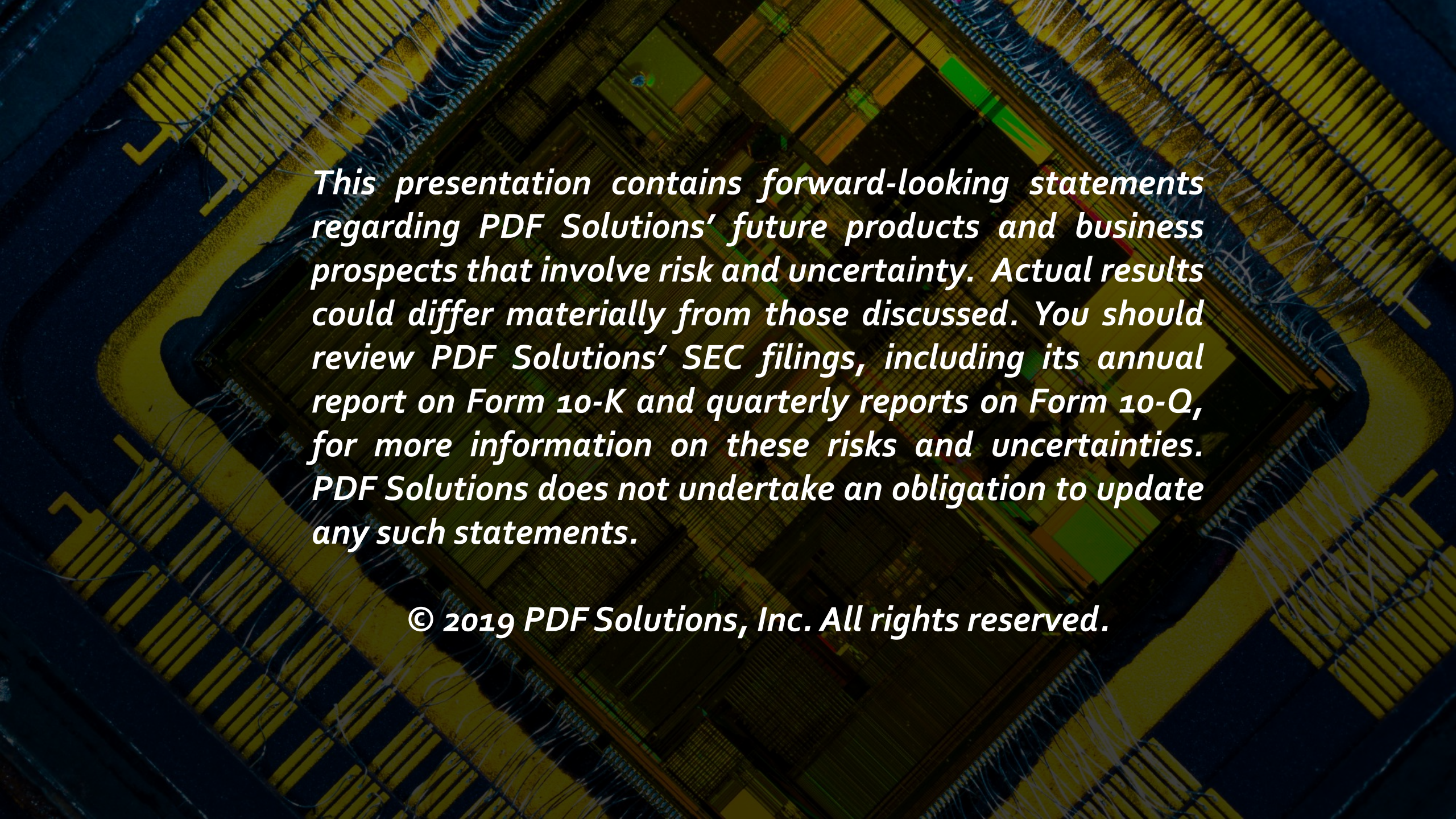
16th Annual PDF Solutions Users Conference

PDF/SOLUTIONS™

S1.3 - Machine Learning in Exensio®

October 15, 2019

Jeff David, VP of AI Solutions

A high-magnification, top-down view of a complex electronic circuit board. The image shows a dense network of gold-colored conductive traces and pads. Numerous small, rectangular components, likely integrated circuits or capacitors, are soldered onto the board. The overall color palette is dominated by the yellow-gold of the metal, with some darker, possibly black or dark green, areas representing the substrate or other components. The lighting creates a sense of depth and highlights the intricate patterns of the circuitry.

This presentation contains forward-looking statements regarding PDF Solutions' future products and business prospects that involve risk and uncertainty. Actual results could differ materially from those discussed. You should review PDF Solutions' SEC filings, including its annual report on Form 10-K and quarterly reports on Form 10-Q, for more information on these risks and uncertainties. PDF Solutions does not undertake an obligation to update any such statements.

© 2019 PDF Solutions, Inc. All rights reserved.

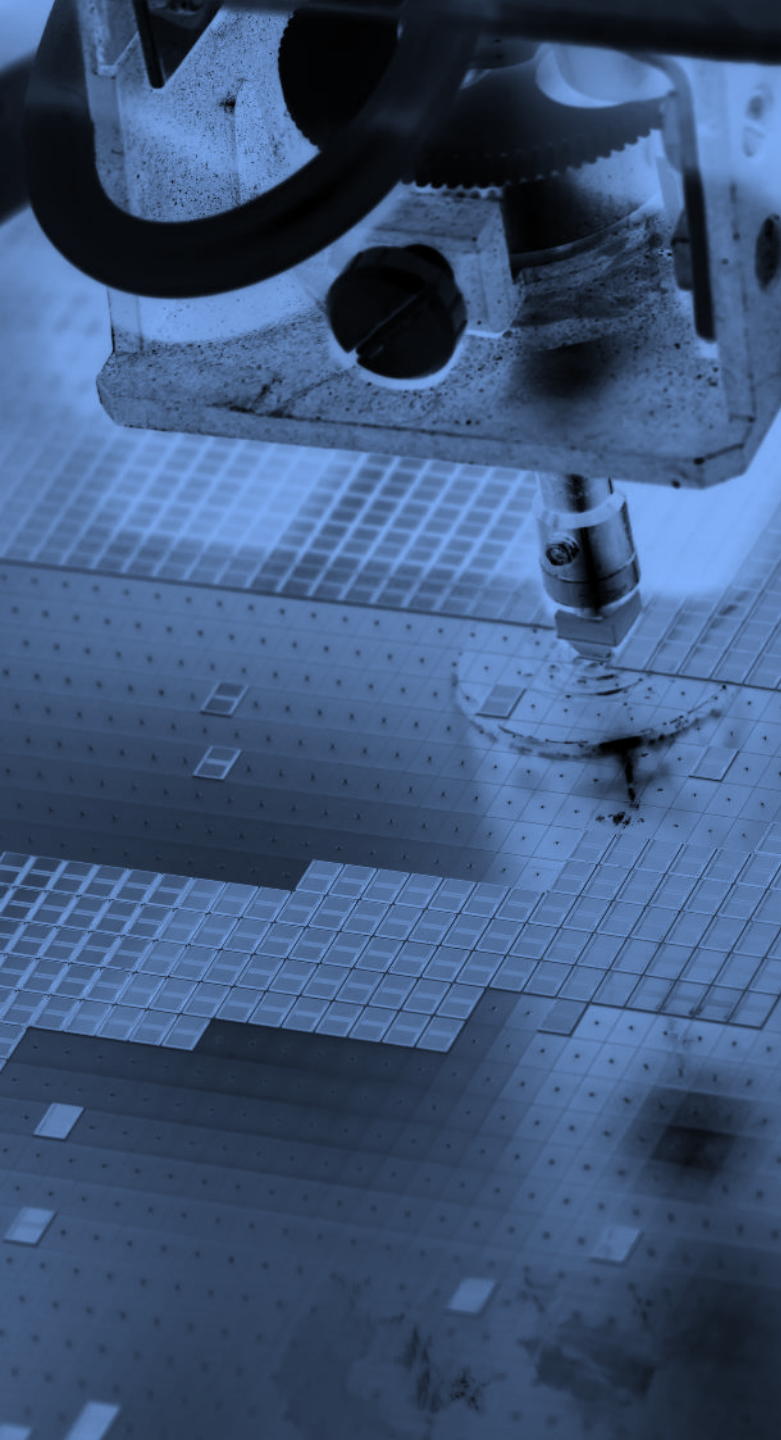
What is Machine Learning??

- *Algorithms that iteratively learn from data to find hidden insights*
- *Allows for automatic generation of models that can analyze massive amounts of data to deliver faster, more accurate predictions*
- *"Learns" complex correlations between incoming data and outputs.*

Everyday Examples of ML



- ☐ *Optical character recognition*
- ☐ *Face detection: Facebook*
- ☐ *Banking: Mobile Check deposits / Fraud detection*
- ☐ *Traffic Prediction: Waze / Google Maps*
- ☐ *Recommendation Engines: Netflix / YouTube / Google*
- ☐ *Spam filtering: identify emails as spam or non-spam*
- ☐ *Self Driving Cars: Google, Tesla*



Agenda

- o Why is implementing machine learning so difficult in the semiconductor industry??*
- o PDF is UNIQUELY positioned to deliver the promise of AI to the industry*
- o Augmenting analytics with AI/ML to improve results and make it easier to use*

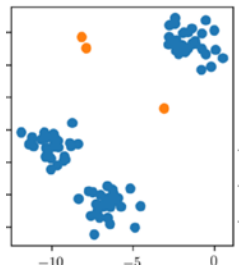


Other industries have more sophisticated AI than the semiconductor industry

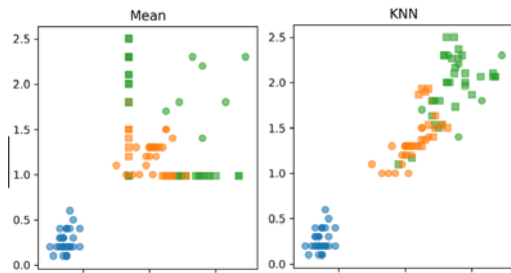
Why is that?

Model Pipeline: a Collection of Steps, Methods and Best-Known Practices that Solve the Problem

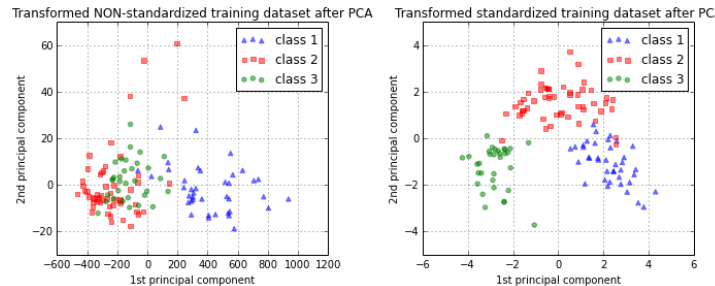
Outlier detection



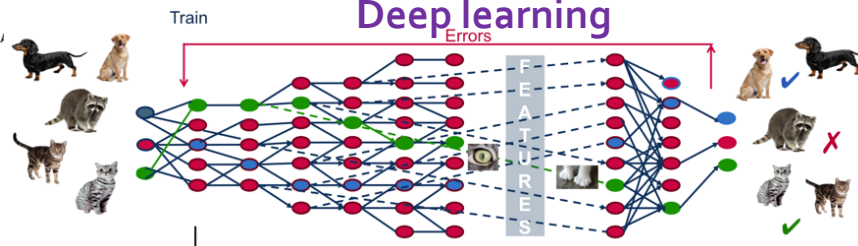
Imputation



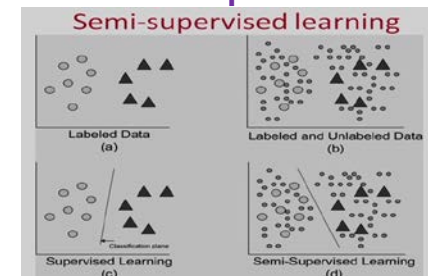
Scaling / Normalization



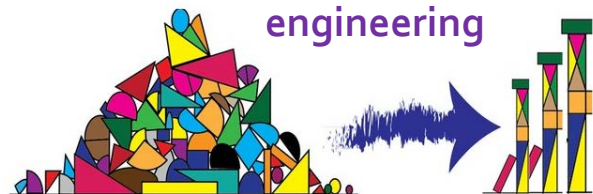
Deep learning



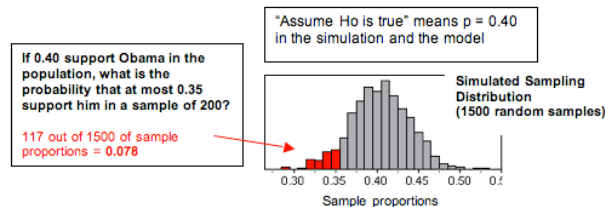
Semi-Supervised



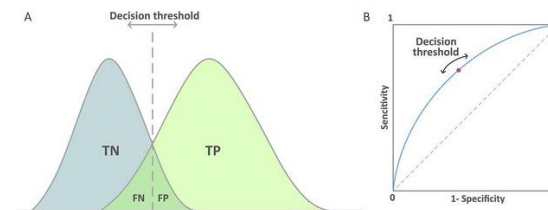
Feature selection & engineering



Statistical tests & uncertainty estimation

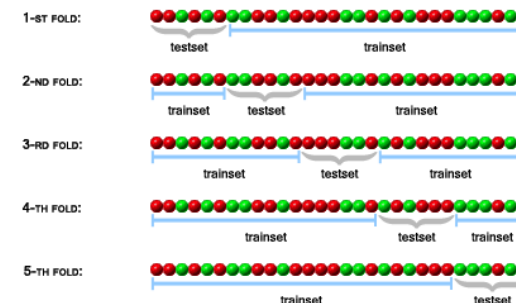


Threshold selection

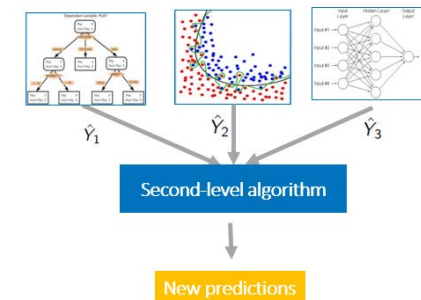


Hyper-parameter tuning
Stratified K-fold cross validation

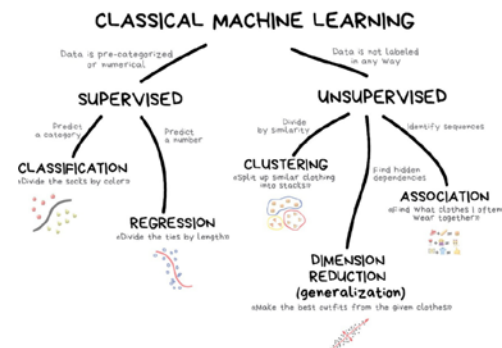
ONE ITERATION OF A 5-FOLD CROSS-VALIDATION:



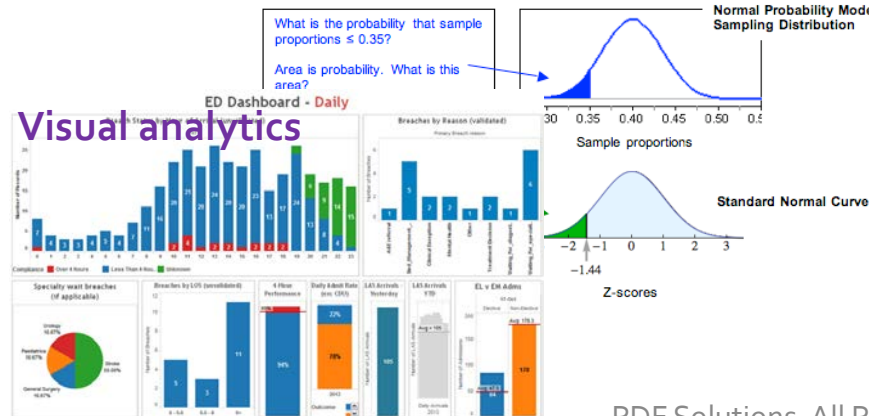
Ensemble

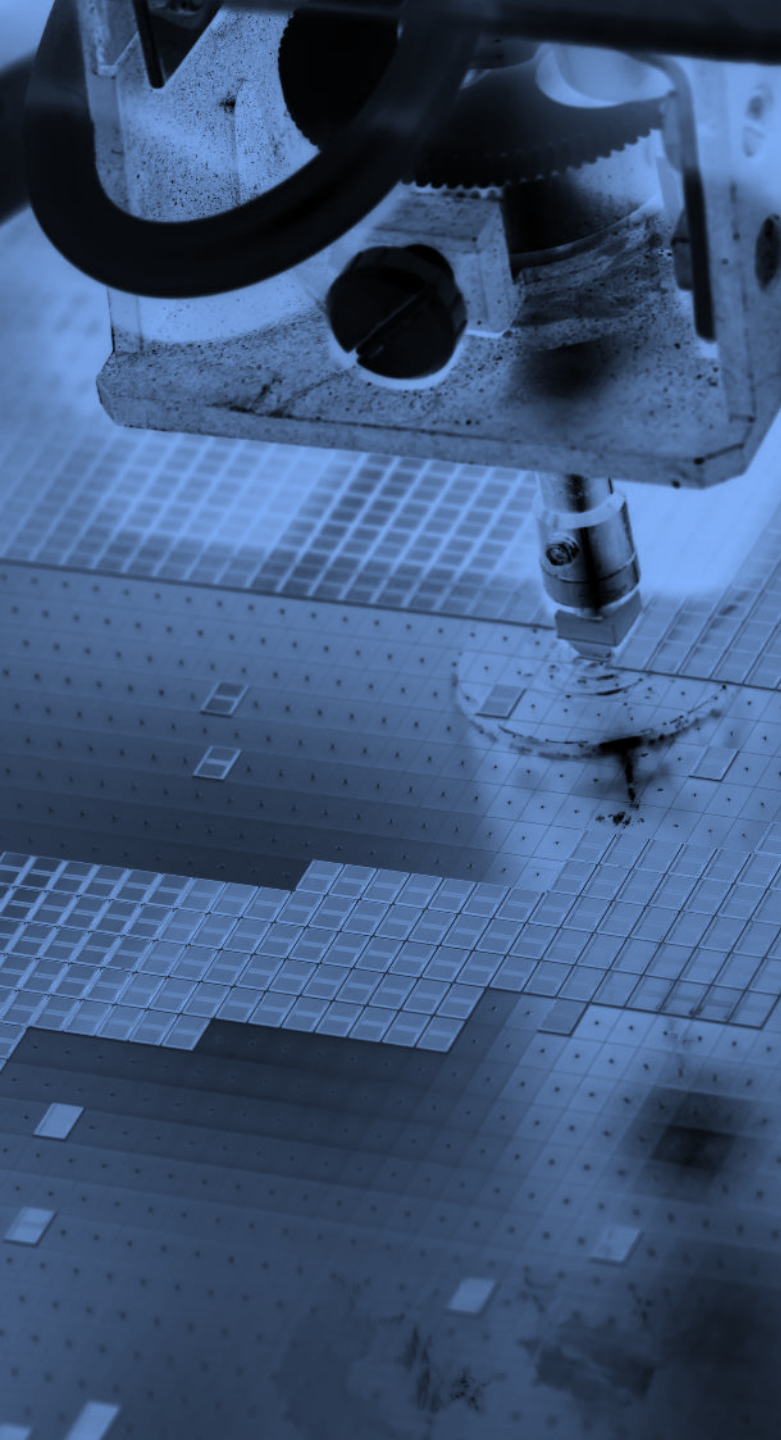


Algorithms / ML



Visual analytics

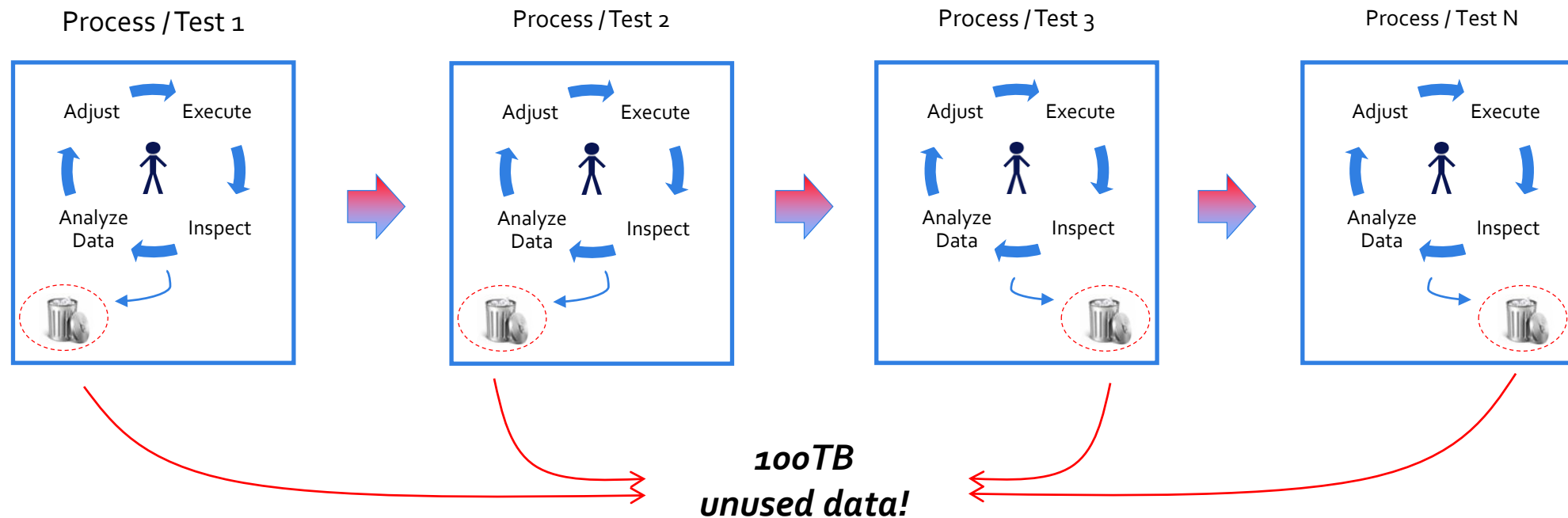




Some Challenges for ML in the Semiconductor Industry

- o Multimodal batch trajectories due to product mix
 - o Test program changes
 - o Process drift and shift, tool recalibration
 - o Changing failure modes
 - o Small amount of training data
 - o Lack of labels
 - o Lack of data for emerging technologies
 - o Lack of traceability for root cause
-
- *What happens when your data doesn't arrive?*
 - *What do you do if it is corrupt?*
 - *What kind of prediction do you make?*

Today's Struggle: Silos and Local Optimization



Our customers want to be able to do something better than humans stuck in silos

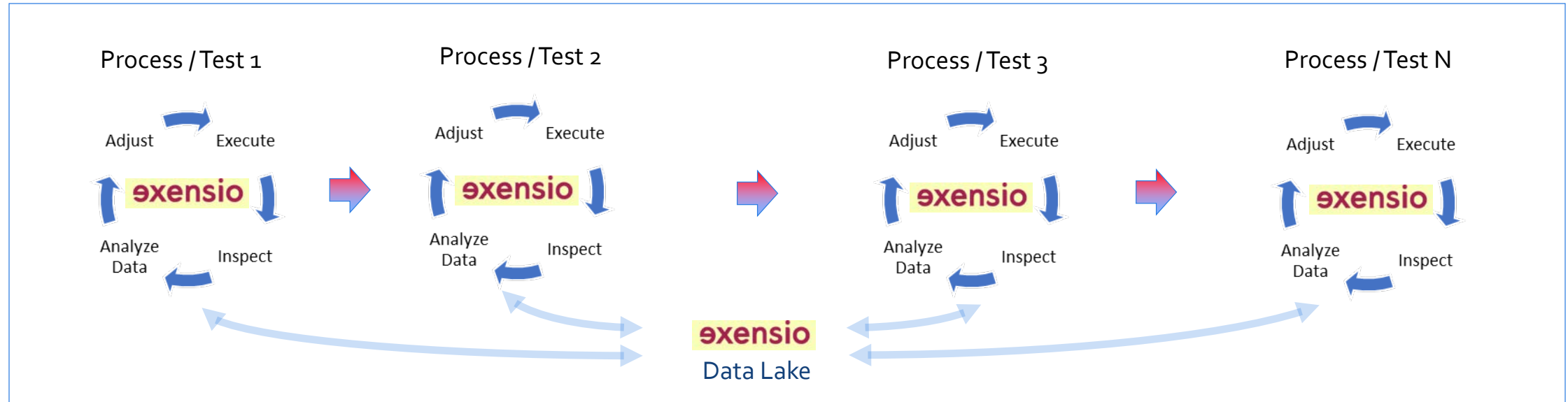
- One step learning
- One step adjustment
- Data used once
- Human heavy



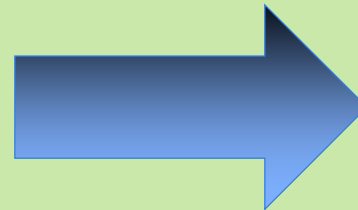
**Higher cost
Lower quality
Falling yields**

By integrating data and applying ML, our customers can achieve better results

Exensio®: Global and Local Optimization



- Multi-step learning
- Multi-step adjustment
- Uses in-situ, upstream & historical data
- Human light, machine learning

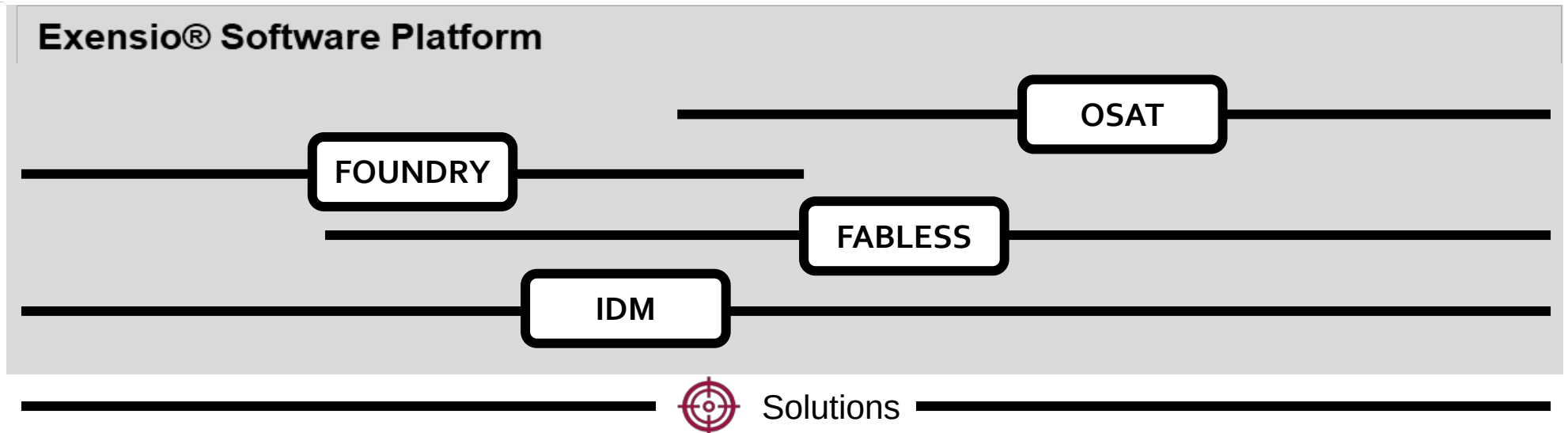
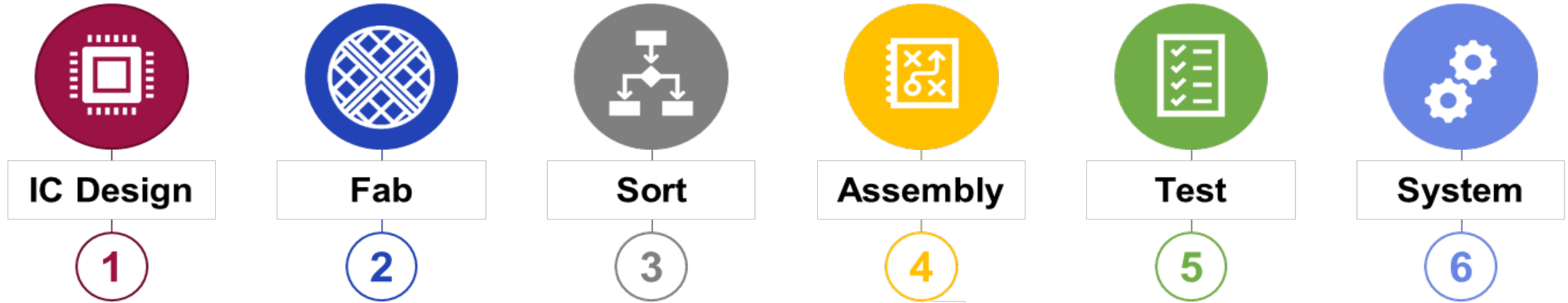


Lower cost
Improved quality
Higher Yield



PDF is UNIQUELY Positioned to Deliver the Promise of AI to the Industry

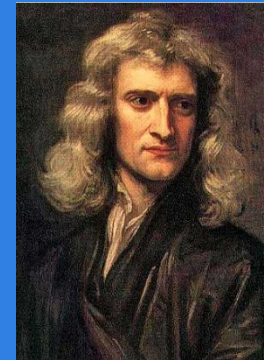
PDF Solutions: Spanning the Supply Chain



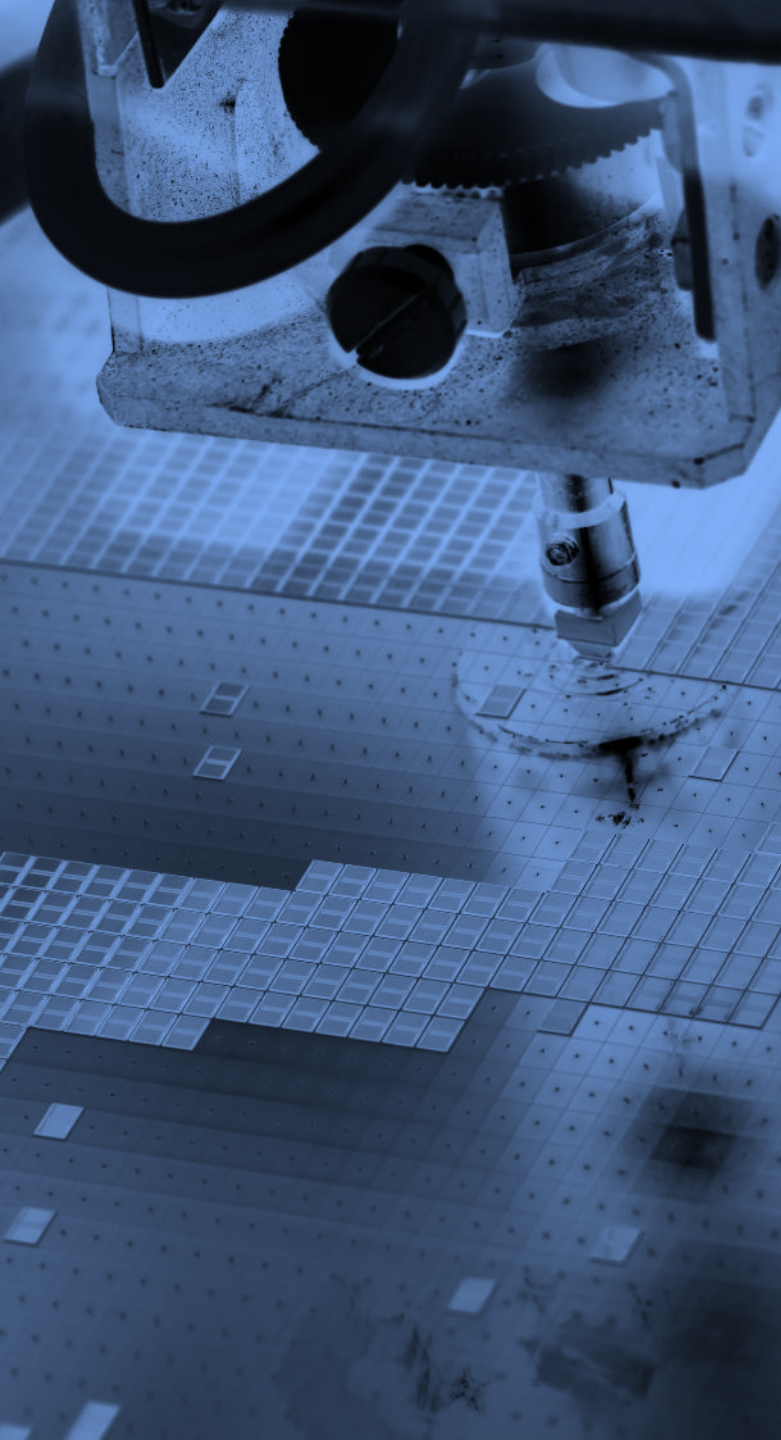
PDF Solutions: Spanning the Supply Chain

- ✓ PDF has access to the RIGHT data, including new types of data
- ✓ PDF has access to a breadth of data across the supply chain. PDF aligns this data to make it relevant.
- ✓ PDF has the infrastructure built up to integrate and leverage all of this data to deliver on the promise of AI
- ✓ “If I have seen further, it is by standing upon the shoulders of giants”

–Sir Isaac Newton



https://en.wikipedia.org/wiki/Isaac_Newton#/media/File:GodfreyKneller-IsaacNewton-1689.jpg



Semantic Models: A Key Element for Applying Machine Learning

- o The semantic data model is a method of structuring data in order to represent it in a specific logical way. It is a conceptual data model that includes semantic information that adds a basic meaning to the data and the relationships that lie between them.

<https://www.techopedia.com/definition/30489/semantic-data-model>

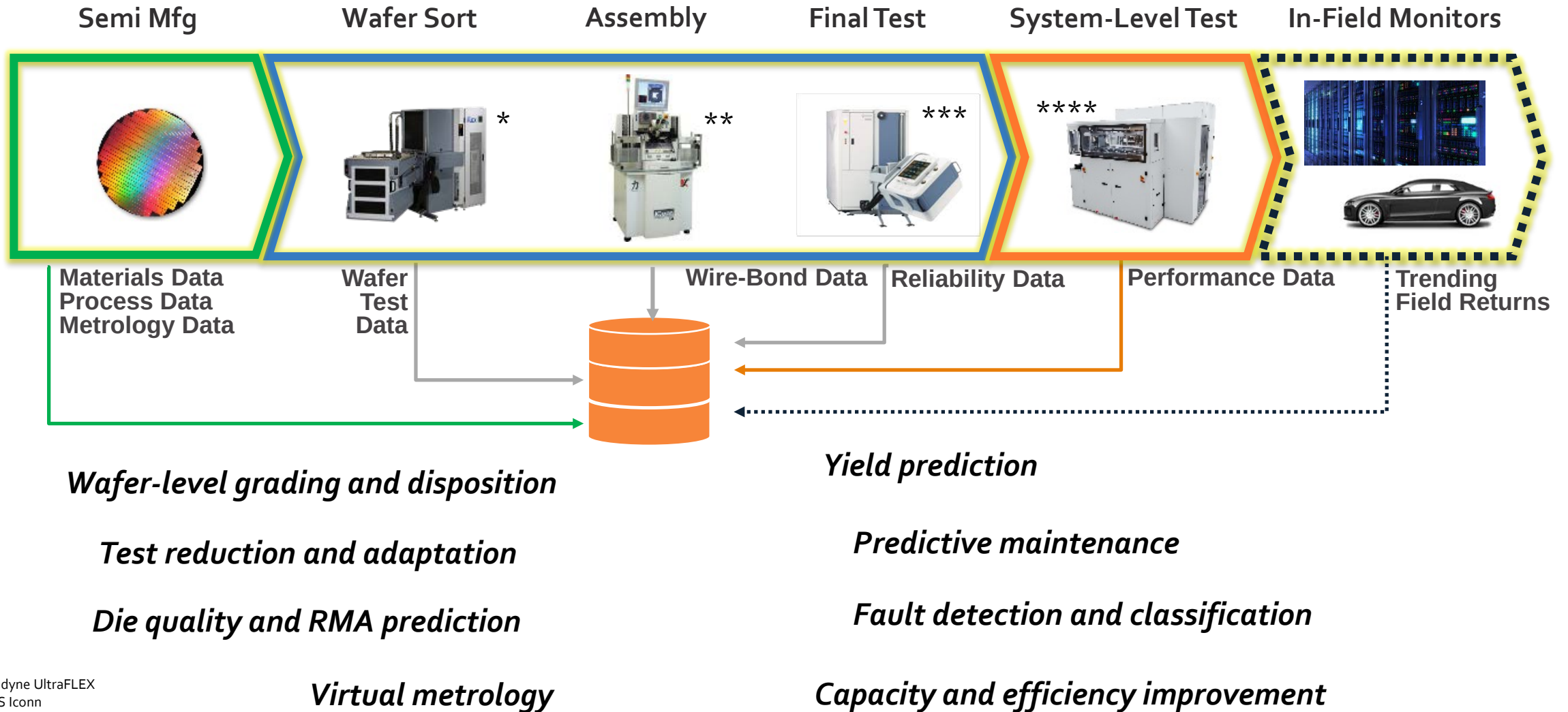
- o Examples:
 - Aligning events in a fab with wafer data to answer question like “which wafers were processed with the new batch of resist”?
 - Mapping equipment signals across a fleet of tools to account for configuration differences
 - Meaningful merging of chip data as the chips flow through wafer sort, assembly, and final test

**Semantic models allow our customers to
deploy machine learning to production**



Augmenting Analytics with ML to Improve Results and Make it Easier to Use

A Unified View of Semiconductor Data is Needed



* Teradyne UltraFLEX
** KnS Iconn
*** Advantest V93000
**** Astronics ATS 5034

Lots of Great Modules in Exensio® that Incorporate 20 Years of PDF Expertise, but Requires all the Right Skills to get the Needed Results

EXENSIO® Modules

Rule Ensemble Engine (REE)

Spatial Signature Analysis (SSA)

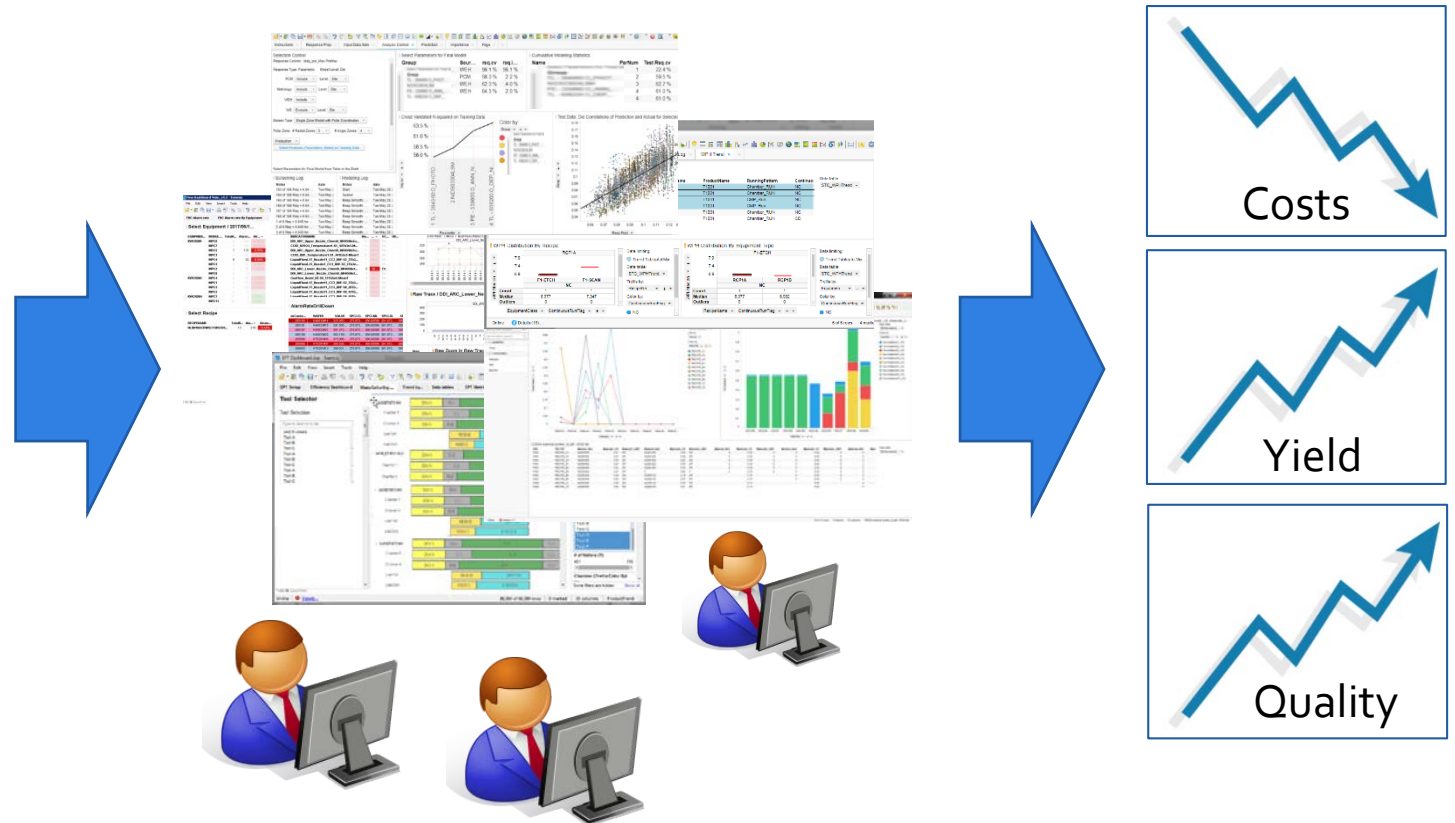
Fail Signature Detection and Analysis (FSDA)

Product Sensitivity Analysis (PSA)

Parameter Screening Report (PSR)

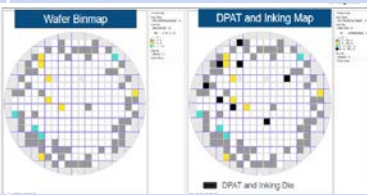
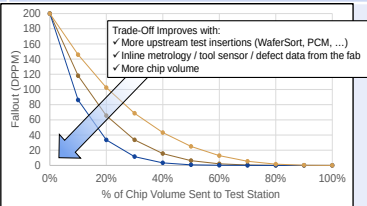
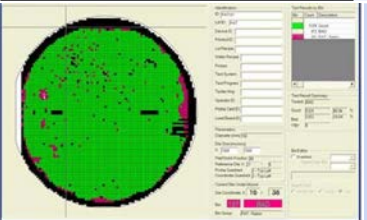
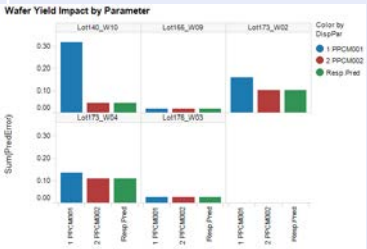
Indicator Screening Report (ISR)

Automatic Spatial Classification (ASC)

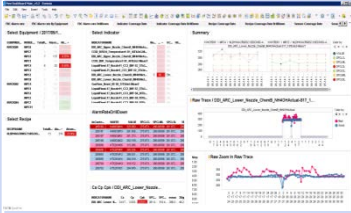
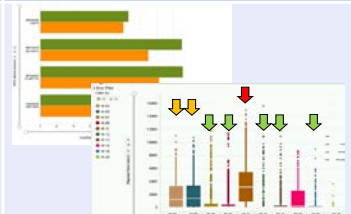
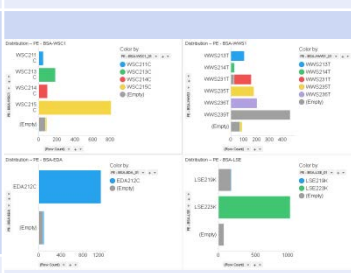


Results require Data Science expertise and understanding of semiconductor manufacturing

AIM Solutions Overview (Fabless/IDM)

Solution		Description	ROI
1. IMD – Intelligent Material Disposition (aka MRB)		Wafer level grading & disposition with near real time execution	✓ Reduce engineering disposition time by > 50% ✓ Improves quality of disposition
2. ST - Smart Testing – Predict Final Test & Burn-in		Prediction of test requirements based on electrical wafer Sort parametrics	✓ Reduce test or burn-in requirements by 20-60%
3. ELF – Early Life Failure detection (die level MRB)		Comprehensive Die Quality Grading - Classify risk based on Sort parametrics	✓ Prevent quality and reliability escapes by detecting high risk die at Sort
4. FPM – Fab & Final Test Predictive Modeling		Predict wafer and die level yield & parametric prior to Sort & Final Test operations (or other responses).	✓ Improved inventory management ✓ Reduce yield loss & excursions ✓ Reduces engineering investigative resource

AIM Solutions Overview (Foundry/IDM)

Solution	Description	ROI
5. FDC – Fault Detection and Classification	 FDC to YMS AI modeling & prediction for yield variability reduction & control plan upgrade	✓ 8% Yield Improvement ✓ 40% excursion reduction ✓ 7% Faster NPI Ramp Learning Rate
6. CEI – Capacity & Efficiency Improvement (aka OEE)	 Improve OEE, Fab Capacity, & wafer through-put by matching tools & chamber operations	✓ 10% improvement in bottleneck tool capacity ✓ >20% improvement in efficiency (thru-put)
7. ASD – Adaptive Signature Diagnostics (Smart Analysis)	 Uses spatial signature analysis, ML, to classify Sort failures and auto-diagnose likely root cause of yield loss	✓ Identify sources of yield loss immediately after Sort. ✓ 5x Reduction in engineering investigative resources.
8. TEP – Trace based Excursion Prevention	 Use AI to identify abnormal tool operation via text log & raw FDC sensor trace data	✓ Prevent large scale excursions at the tool through trouble prevention

AIM Solutions with AI Makes it Easier to Obtain Optimal Results

EXENSIO® AIM SOLUTIONS

Intelligent Material Disposition (IMD)

Smart Testing (ST)

Early Life Failure Prediction (ELF)

Fab Predictive Modeling (FPM)

Fault Detection and Classification (FDC)

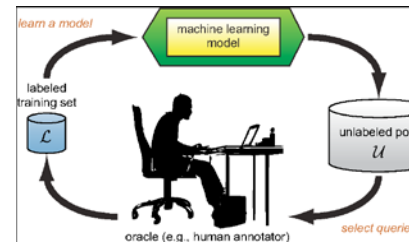
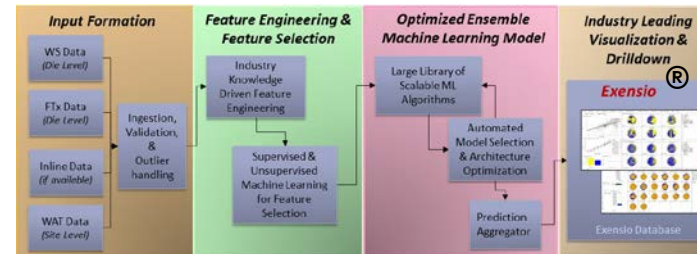
Capacity & Efficiency Improvement (CEI)

Adaptive Signature Diagnostics (ASD)

Trace-based Excursion Prevention (TEP)



ML Pipeline



Collaborative Learning



Costs

Yield

Quality

AIM Solutions with AI Makes it Easier to Obtain Optimal Results

EXENSIO® AIM SOLUTIONS

Intelligent Material Disposition (IMD)

Smart Testing (ST)

Early Life Failure Prediction (ELF)

Fab Predictive Modeling (FPM)

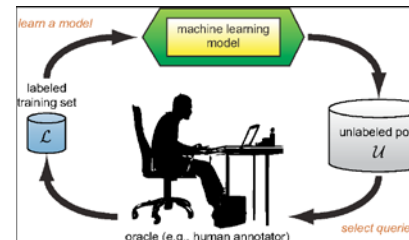
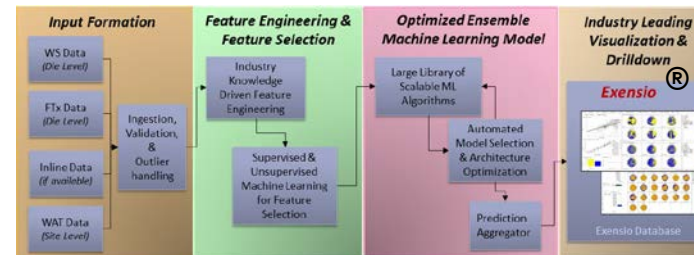
Fault Detection and Classification (FDC)

Capacity & Efficiency Improvement (CEI)

Adaptive Signature Diagnostics (ASD)

Trace-based Excursion Prevention (TEP)

ML Pipeline



Collaborative Learning

- ✓ **Human light, Better results**
- ✓ **Data Science enabled in the tool – you don't have to be a data scientist to set it up or maintain it**
- ✓ **1000's of virtual experts – enabled by AI**

Costs

Yield

Quality

With AI, 20 years of PDF expertise is applied in more productive ways

Augmenting Analytics with ML to Make it Easier to Use and Deploy

○ *Our systems make it easy for our customers to incorporate their knowledge to improve performance*

- Allow customer to deploy their own models
- Allow user to implement their own feature engineering and transformation
- Collaborative Learning – scalable across different use cases and datasets
 - Provides capability such that you don't need to be a semi expert and data scientist
 - Includes our customer knowledge in the training and refinement of the models.

○ *Using ML, we automate the work performed by our customers for their use cases*

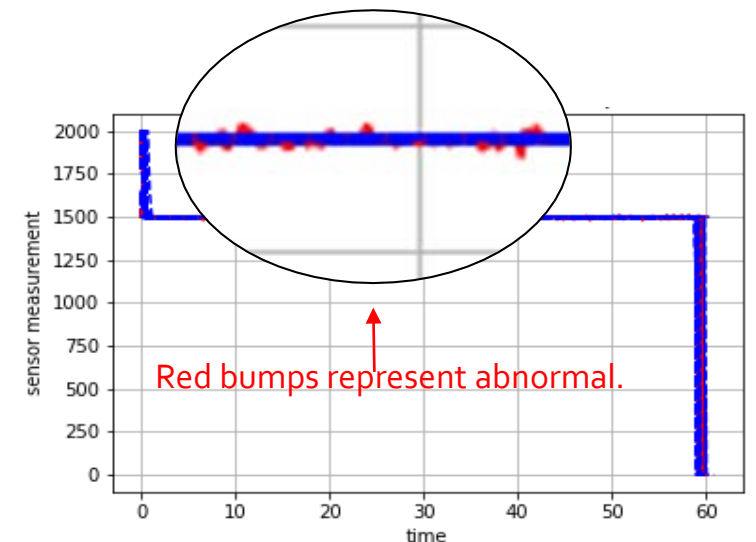
- ETL – Dynamic Schema
- Automatic model training designed to be insensitive to product/tool type
- Allow users to choose models to deploy based on confusion matrix response
- *Aside:* we want to be careful how many knobs we add as this dilutes the automated nature of the model training

FDC – RESULTS on Large Volume of Production Wafers Across Multiple Process Steps

Process Step	True Positive Rate (actual = normal, prediction = normal)	True Negative Rate (actual = abnormal, prediction = abnormal)
Process Step 1	100.00%	100.00%
Process Step 2	99.82%	99.90%
Process Step 3	97.12%	99.38%
Process Step 4	99.86%	99.23%

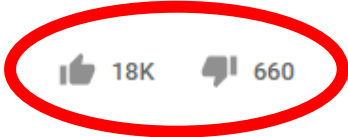
- ❑ Single Machine Learning Pipeline produced accurate model for multiple processes
- ❑ No user modification of algorithm settings from process to process
- ❑ Many tools and recipes per process step
- ❑ Test results on a large volume of advanced device node production wafers

- ✓ Data Science enabled in the tool – you don't have to be a data scientist to set it up or maintain it
- ✓ 1000's of virtual experts – enabled by AI
- ✓ Better IP security – distributed knowledge remains in the tool itself



Collaborative Learning: Captures User Knowledge in the System

- o Final report provides the user a way to assign a new class per wafer, and retrain the model
- o Human interaction via Collaborative Learning allows the user to continually improve and validate wafer classifications, building user confidence in the system.

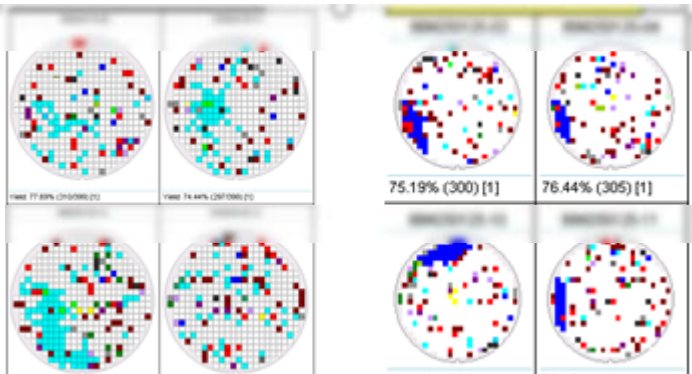


Similar to the "like" button

6 columns from MRBWaferTable

LOT	WAFER	Partial Wafers	RB Class	CL Class	User Class
		True	A	A	A
		True	A	A	A
		True	C	B	A
		True	A	B	Untagged
		True	B	B	B
		True	C	C	B
		False	A	C	C
		True	B	B	Untagged
		True	B	B	Untagged
		True	B	B	Untagged
		True	C	B	Untagged
		True	C	B	Untagged
		True	A	A	A
		True	C	A	A
		True	A	A	Untagged
		True	A	A	Untagged
		True	B	B	B
		True	B	B	A
		True	A	A	A
		True	C	C	A
		True	A	A	A
		True	C	C	A

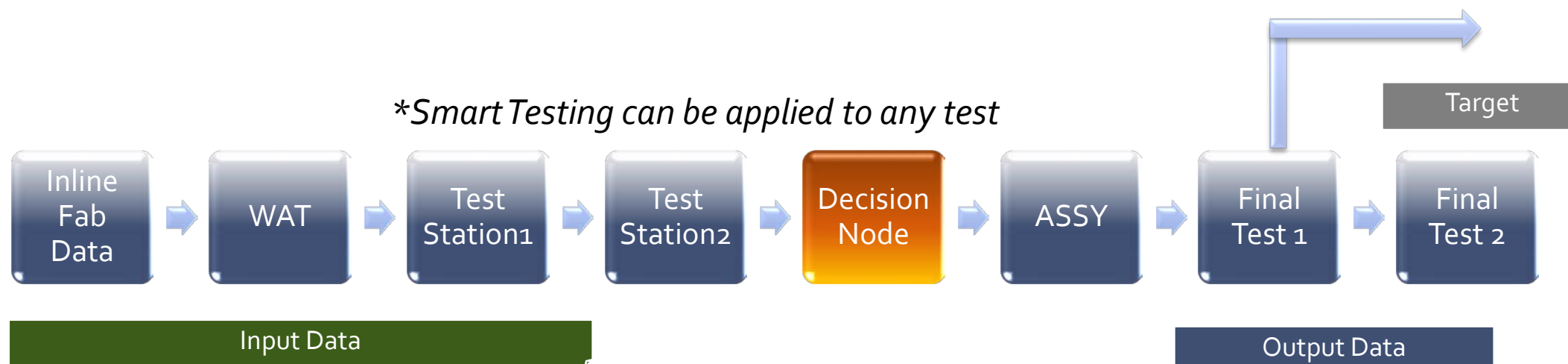
The user can modify the classifications to continuously improve the model



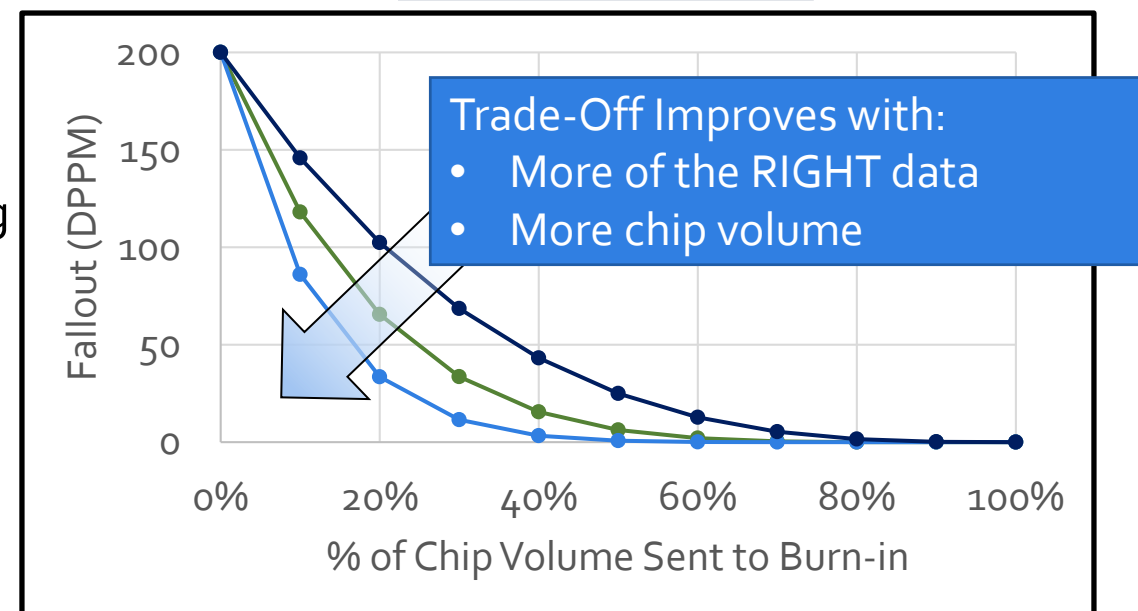
Method	Hit Rate
Standard Statistics	58%
Fab Model (semantic model)	71%
Collaborative Learning	91%

You don't need to be a data scientist or know statistics to use AI

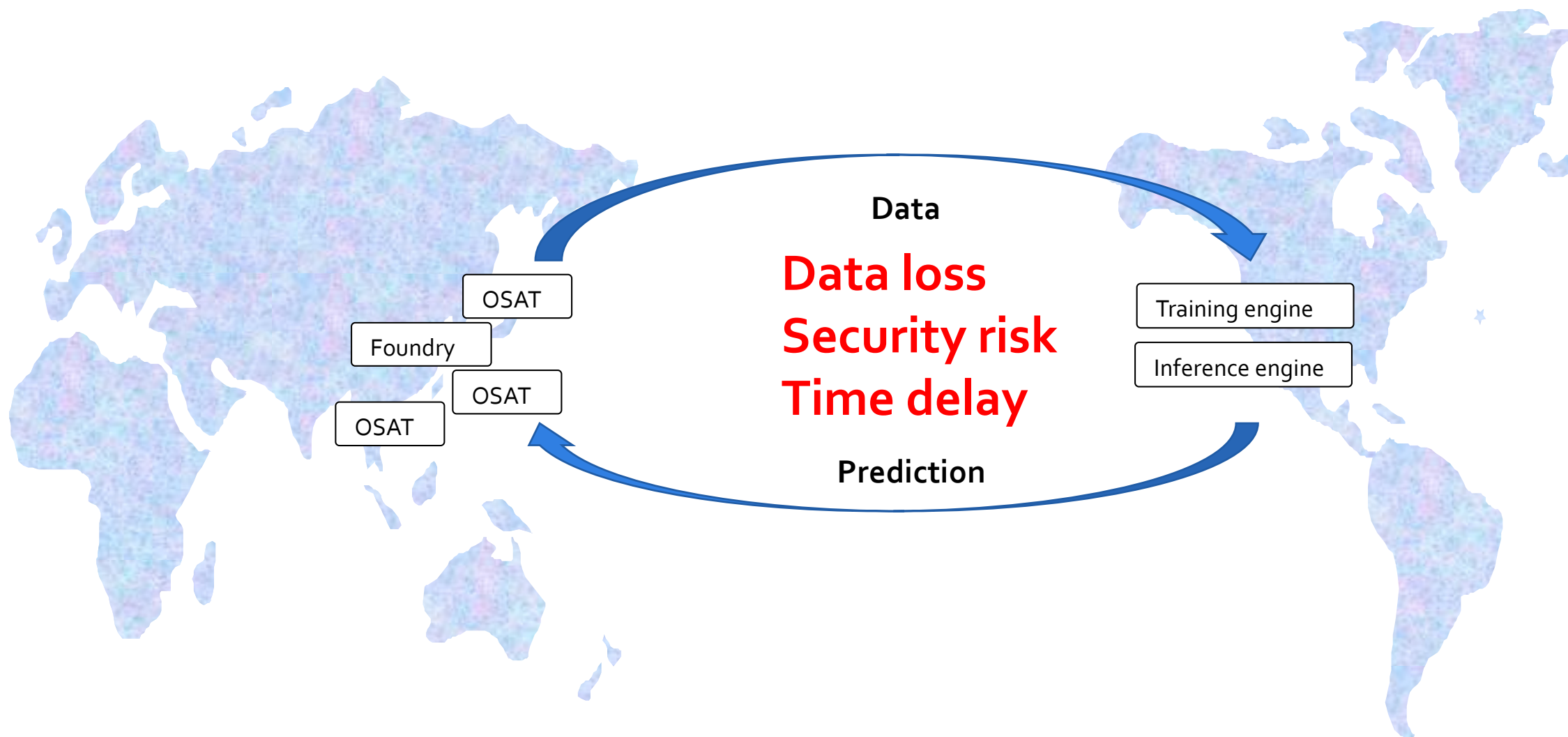
Smart Testing



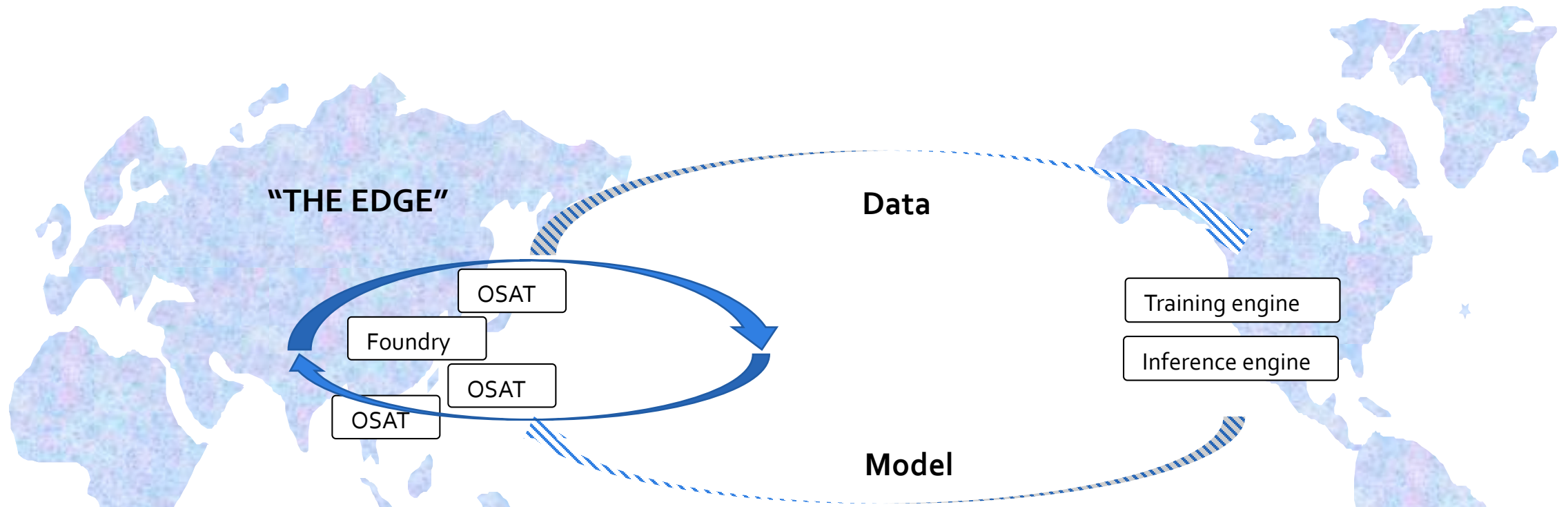
- o Goal: No risky chips to field, while diverting chips from expensive tests
- o Goal: Improve quality and reliability by quickly identifying the root cause of field returns
- o Goal: Focus test resources on product that are at the highest risk for failure
- o Goal: Reduce test cost by diverting low-risk product volume away from expensive tests
- o Goal: Smarter product binning by quality



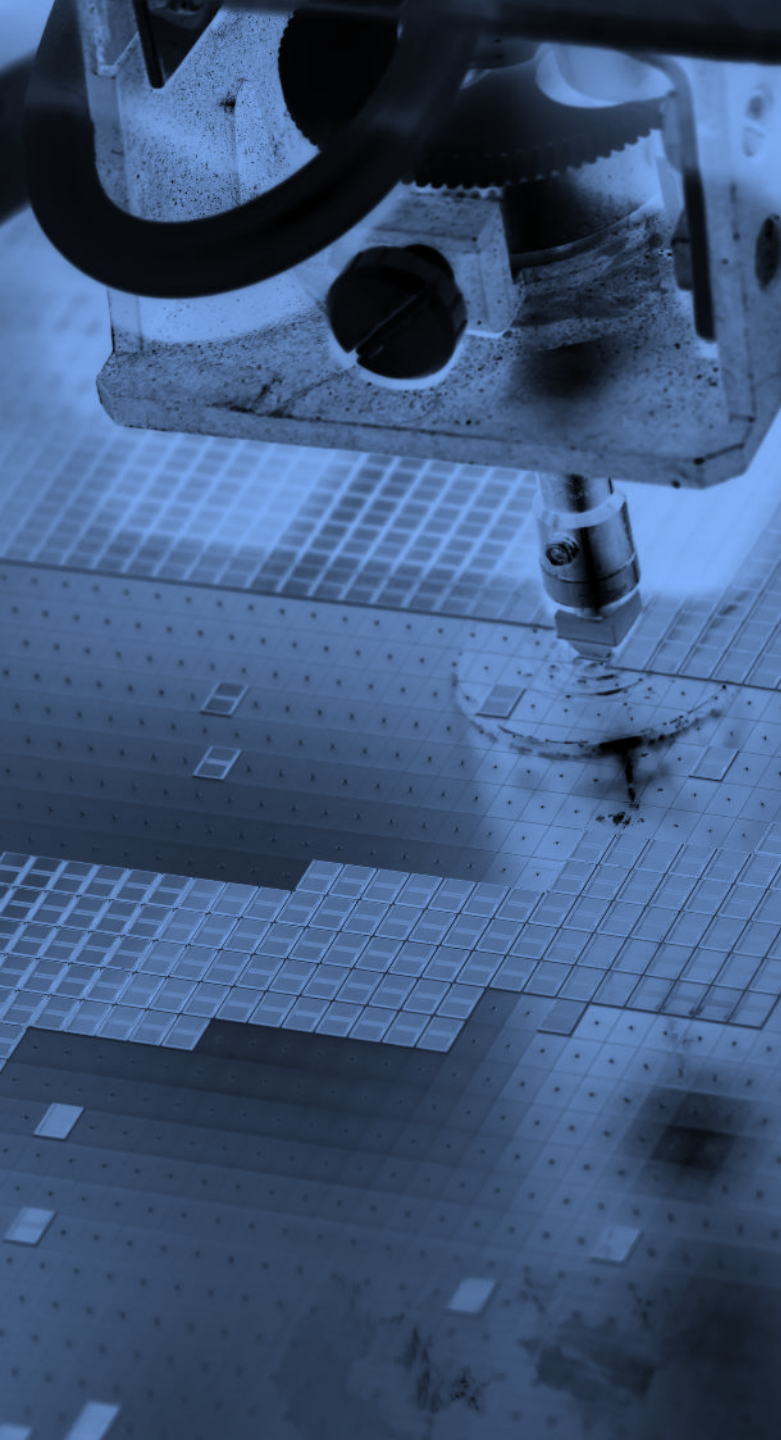
Deployment Challenges



DEX™ Enables Edge Deployment Architecture

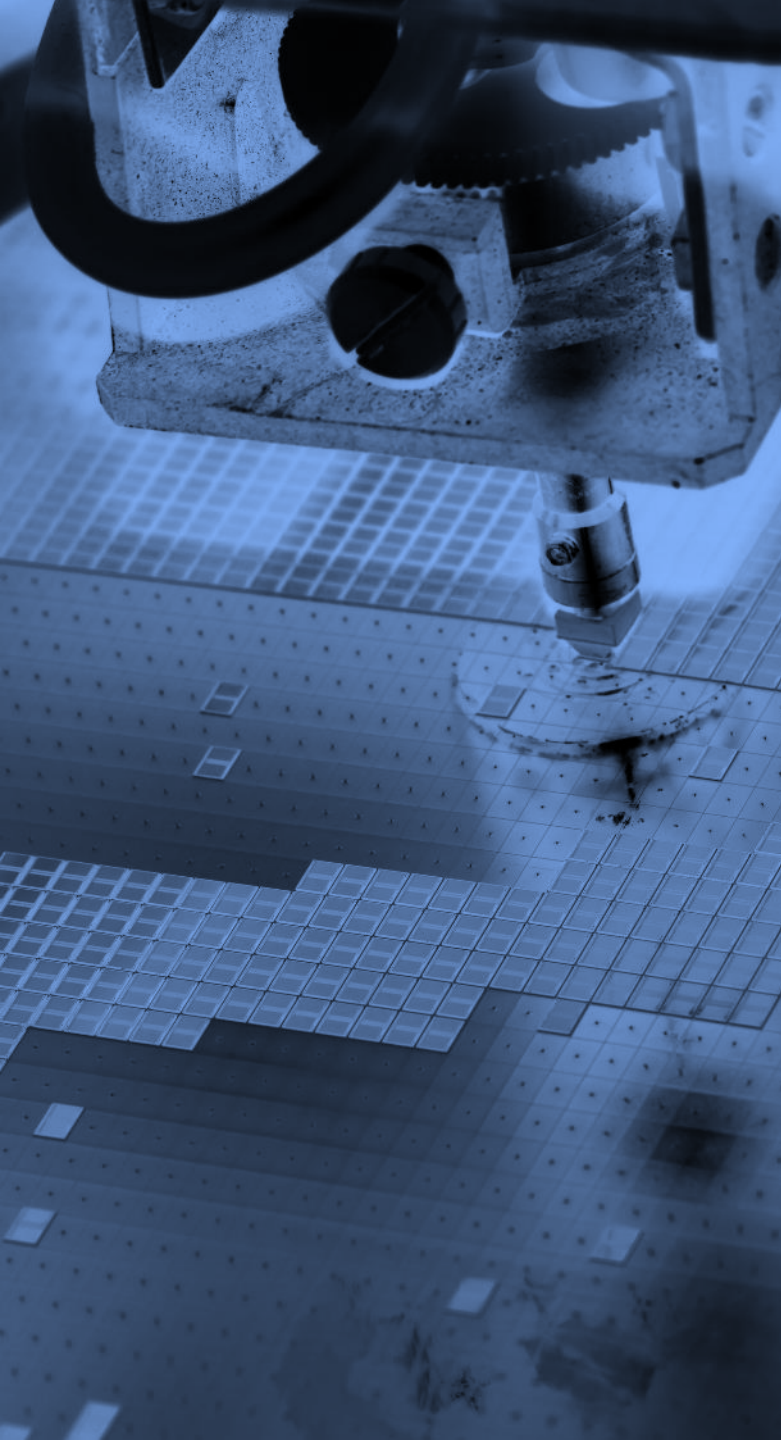


- ✓ Edge Analytics – fast turnaround times on predictions, making ACTIONABLE predictions a reality
- ✓ Reduced data loss
- ✓ Brings advanced ML capabilities to Adaptive Testing
- ✓ Allows user to develop their own models using latest ML technologies



Overall Benefits

- o Access to a breadth of data, and access to the RIGHT data
- o Infrastructure already built up across the supply chain
- o Data Science enabled in the tool – you don't have to be a data scientist to set it up or maintain it
- o Our systems are designed to make it easy for our customers to incorporate their knowledge to improve performance
- o We leverage state-of-the-art AI technologies intended to bring significant ROI to our customers
- o Architected for production-worthiness



In Summation...

- o The semiconductor industry has been slower to adopt AI technologies than other industries
- o Given those challenges, PDF is **UNIQUELY** positioned to deliver the promise of AI to the industry
- o PDF's goal is to make it easier to adopt AI – you don't need to be a data scientist
- o ...the AI enablement provided by PDF is targeted to bring significant value to our customers

Thank You

PDF/SOLUTIONS™



pdf-solutions



pdfsolutionsinc



pdfs.inc



pdf_solutions



pdf_cn

ELEC 7770

Advanced VLSI Design

Spring 2016

Yield, Quality and Moore's Law

Vishwani D. Agrawal

James J. Danaher Professor

ECE Department, Auburn University

Auburn, AL 36849

vagrawal@eng.auburn.edu

http://www.eng.auburn.edu/~vagrawal/COURSE/E7770_Spr16/course.html

VLSI Chip Yield

- A manufacturing defect is a finite chip area with electrically malfunctioning circuitry caused by errors in the fabrication process.
- A chip with no manufacturing defect is called a good chip.
- Fraction (or percentage) of good chips produced in a manufacturing process is called the yield. Yield is denoted by symbol Y .

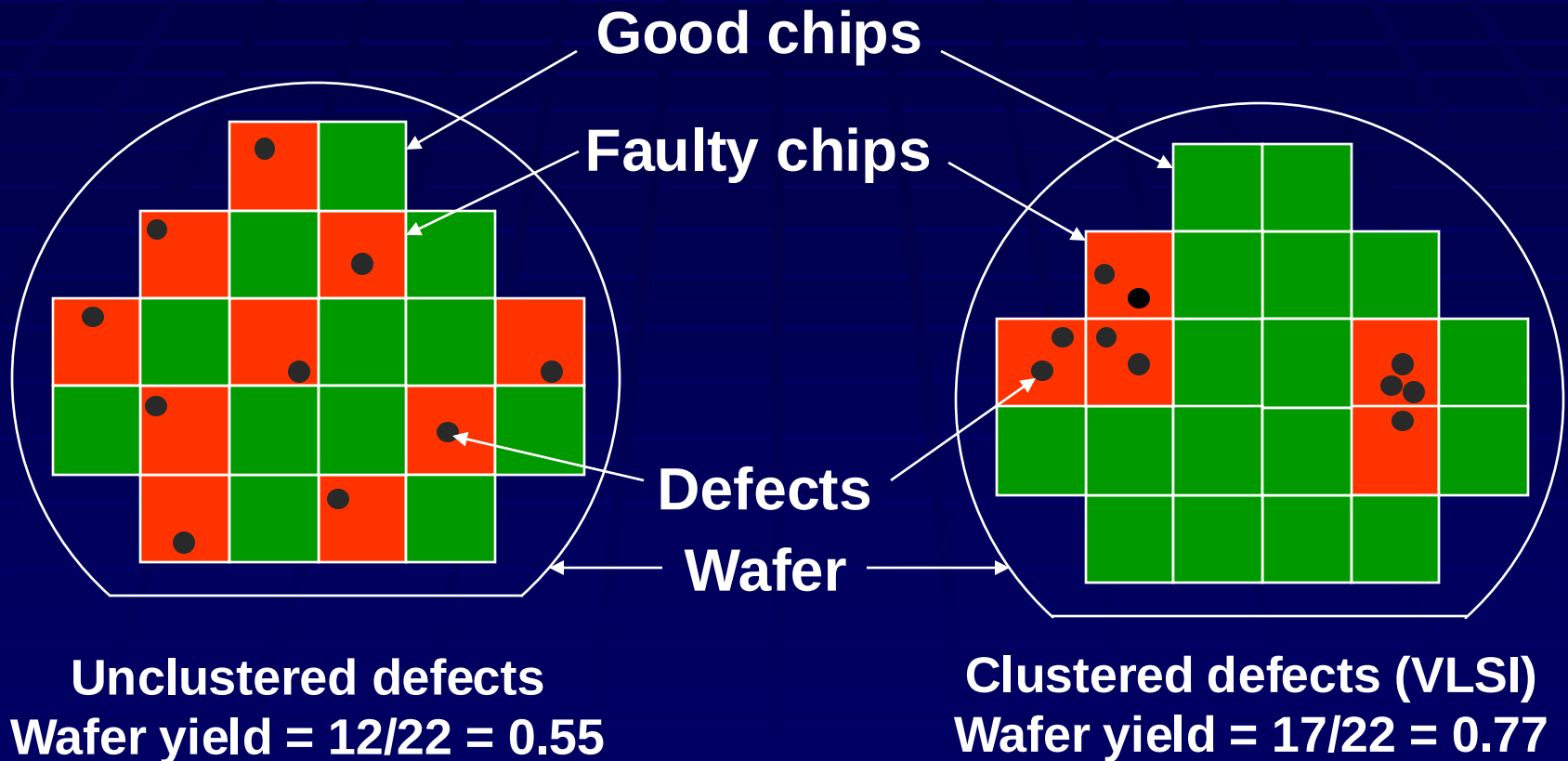
Importance of Yield

- Cost of a chip =

Cost of fabricating and testing a wafer

Yield $\times$ Number of chip sites on the wafer

Clustered VLSI Defects



Yield Parameters

- Defect density (d) = Average number of defects per unit chip area
- Chip area (A)
- Clustering parameter (α)
- Negative binomial distribution of defects,

$$p(x) = \text{Prob}(\text{number of defects on a chip} = x)$$

$$= \frac{\Gamma(\alpha + x)}{x! \Gamma(\alpha)} \cdot \frac{(Ad / \alpha)^x}{(1 + Ad / \alpha)^{\alpha + x}}$$

where Γ is the gamma function

$\alpha = 0$, $p(x)$ is a delta function (max. clustering)

$\alpha = \infty$, $p(x)$ is Poisson distribution (no clustering)

Yield Equation

$Y = \text{Prob}(\text{ zero defect on a chip }) = p(0)$

$$Y = (1 + Ad / \alpha)^{-\alpha}$$

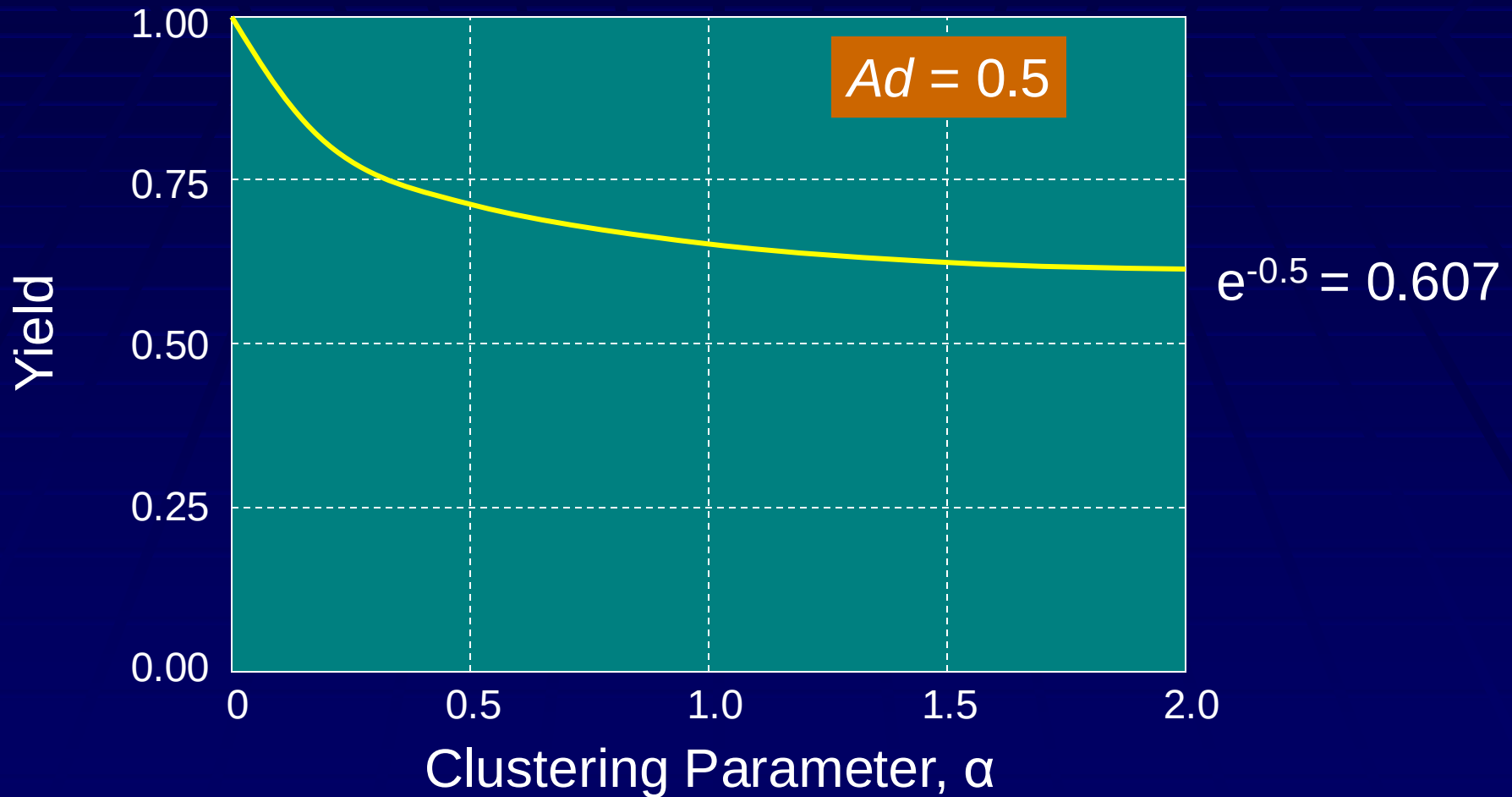
Example: $Ad = 1.0$, $\alpha = 0.5$, $Y = 0.58$

Unclustered defects: $\alpha = \infty$, $Y = e^{-Ad}$

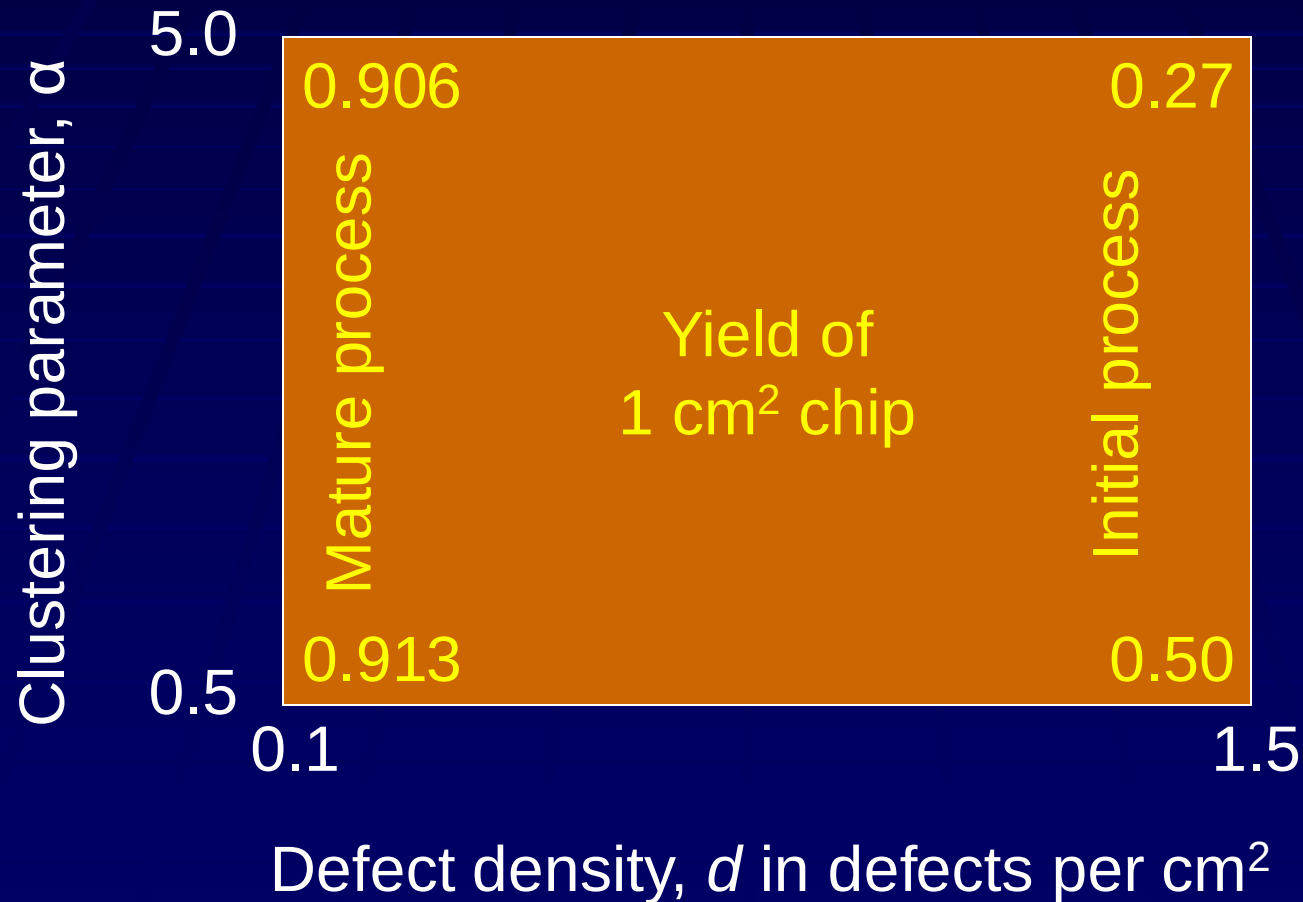
Example: $Ad = 1.0$, $\alpha = \infty$, $Y = 0.37$

too pessimistic !

Effect of Defect Clustering



Ranges of Yield Parameters



References

- Clustered yield model
 - M. L. Bushnell and V. D. Agrawal, *Essentials of Electronic Testing for Digital, Memory and Mixed-Signal VLSI Circuits*, Springer, 2000, Chapter 3.
 - C. H. Stapper, “On Yield, Fault Distributions, and Clustering of Particles,” *IBM Jour. of Res. and Dev.*, vol. 30, no. 3, pp. 326-338, May 1986.
- The unclustered defect model was first described in paper:
 - B. T. Murphy, “Cost-Size Optima of Monolithic Integrated Circuits,” *Proc. IEEE*, vol. 52, no. 12, pp. 1537-1545, December 1964.
- A general reference on clustered distributions:
 - A. Rogers, *Statistical Analysis of Spatial Dispersions*, London, United Kingdom: Pion Limited, 1974.

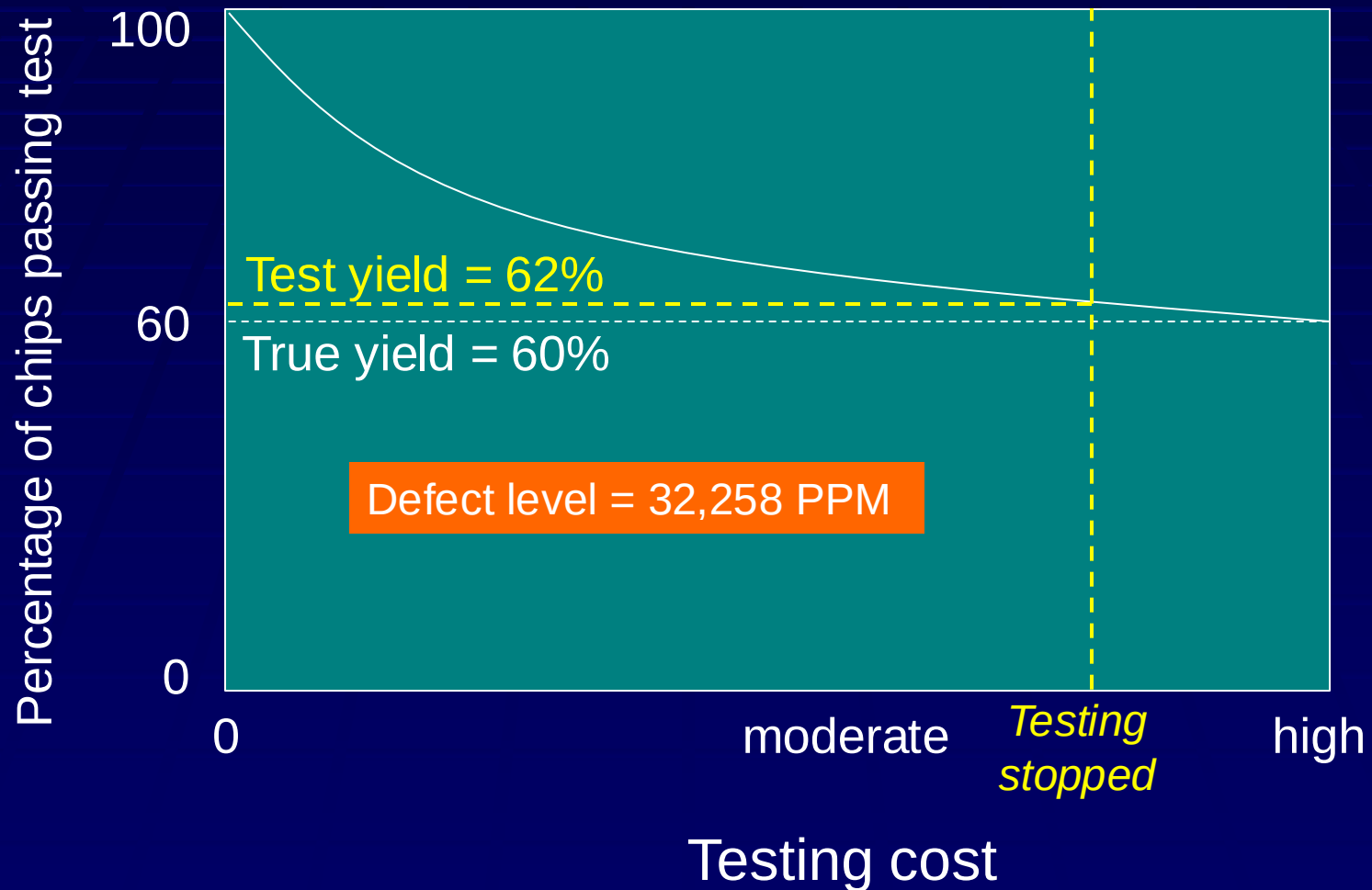
Defect Level or Reject Ratio

- *Defect level* (**DL**) is the ratio of faulty chips among the chips that pass tests.
- **DL** is measured as *parts per million* (ppm).
- **DL** is a measure of the effectiveness of tests.
- **DL** is a quantitative measure of the manufactured product quality. For commercial VLSI chips a **DL** greater than 500 ppm is considered unacceptable.

Determination of DL

- From field return data: Chips failing in the field are returned to the manufacturer. The number of returned chips normalized to one million chips shipped is the **DL**.
- From test data: Fault coverage of tests and chip fallout rate are analyzed. A modified yield model is fitted to the fallout data to estimate the **DL**.

Testing and Defect Level



Test Data Analysis for DL

- Define three parameters:
 - Fault density, f = average number of stuck-at faults per unit chip area
 - Fault clustering parameter, β
 - Stuck-at fault coverage, T
- New yield parameters:
 - $d \rightarrow f$, $Ad \rightarrow TAf$
 - $\alpha \rightarrow \beta$

Fault-Based Yield Equation

- Replace defects with modeled faults:

$$Y(T) = (1 + T Af / \beta)^{-\beta}$$

- Assume that tests with 100% fault coverage ($T = 1.0$) will remove all faulty chips,

$$Y = Y(1) = (1 + Af / \beta)^{-\beta}$$

Defect Level

$$DL(T) = \frac{Y(T) - Y(1)}{Y(T)}$$

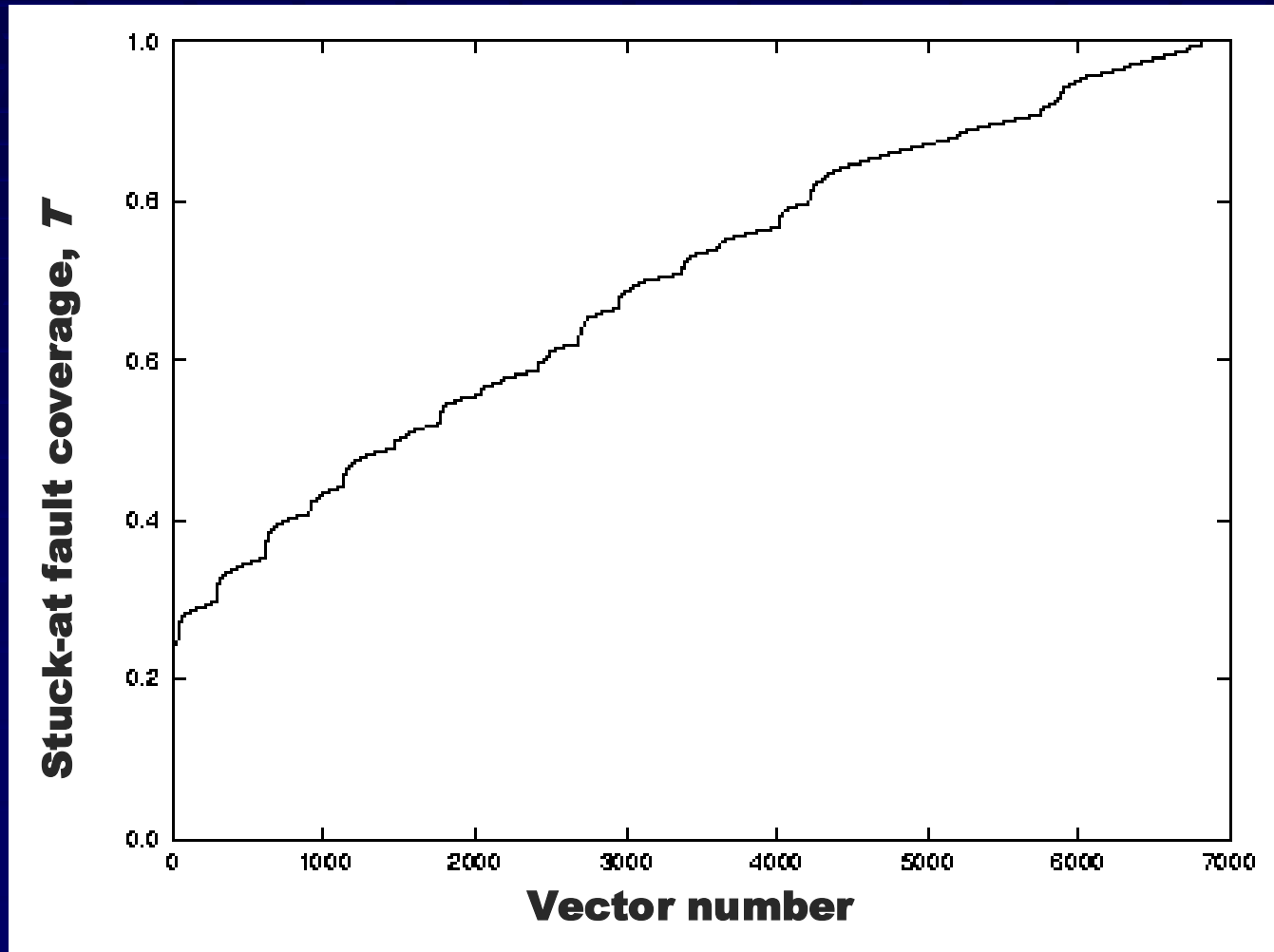
$$= 1 - \frac{(\beta + TAf)^\beta}{(\beta + Af)^\beta}$$

Where T is the fault coverage of tests, Af is the average number of faults on the chip of area A , β is the fault clustering parameter. Af and β are determined by test data analysis.

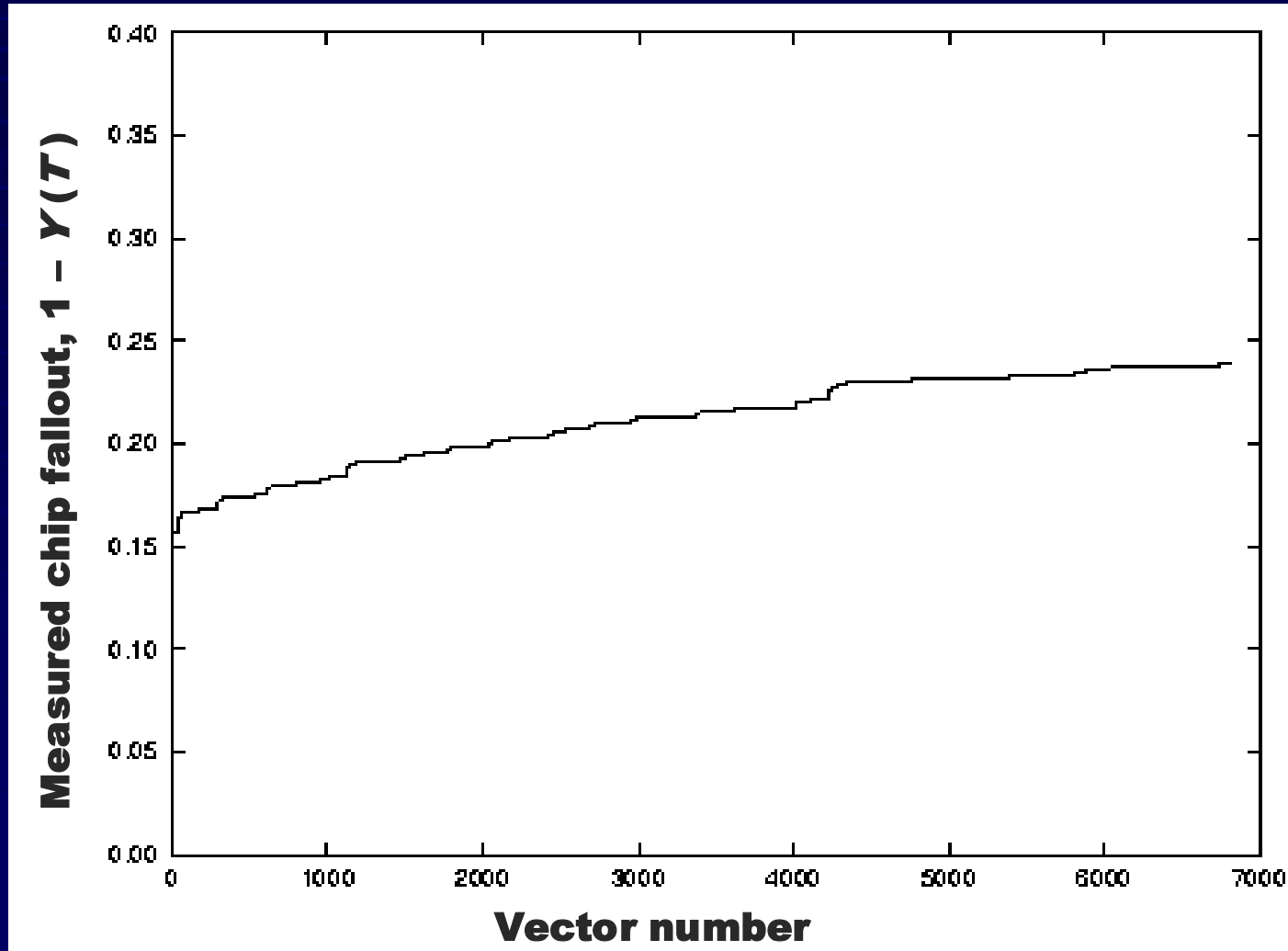
Example: SEMATECH Chip

- Bus interface controller ASIC fabricated and tested at IBM, Burlington, Vermont
- 116,000 equivalent (2-input NAND) gates
- 304-pin package, 249 I/O
- Clock: 40MHz, some parts 50MHz
- 0.8 μ m CMOS, 3.3V, 9.4mm x 8.8mm area
- Full scan, 99.79% fault coverage
- Advantest 3381 ATE, 18,466 chips tested at 2.5MHz test clock
- Data obtained courtesy of Phil Nigh (IBM)

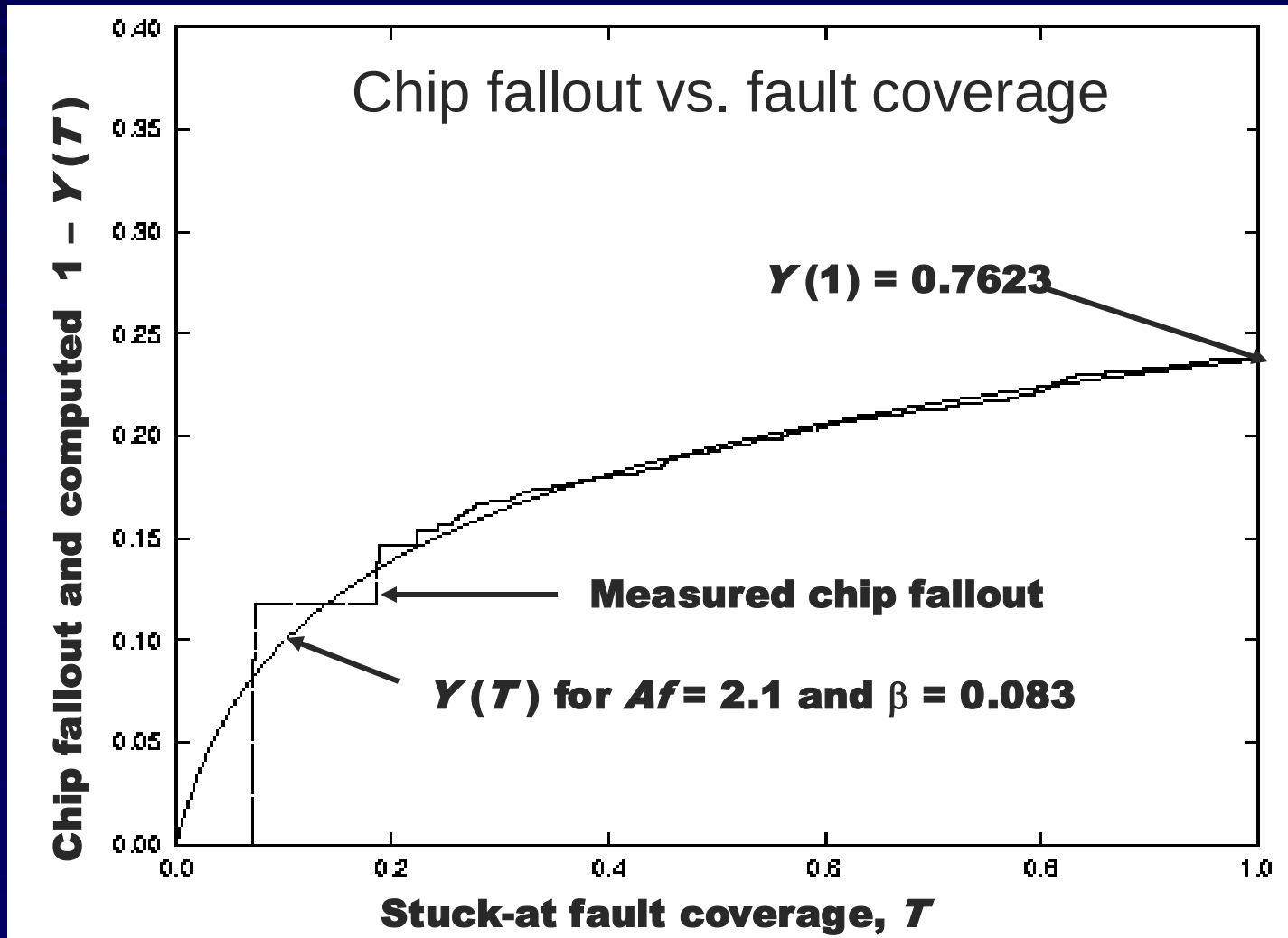
Test Coverage from Fault Simulator



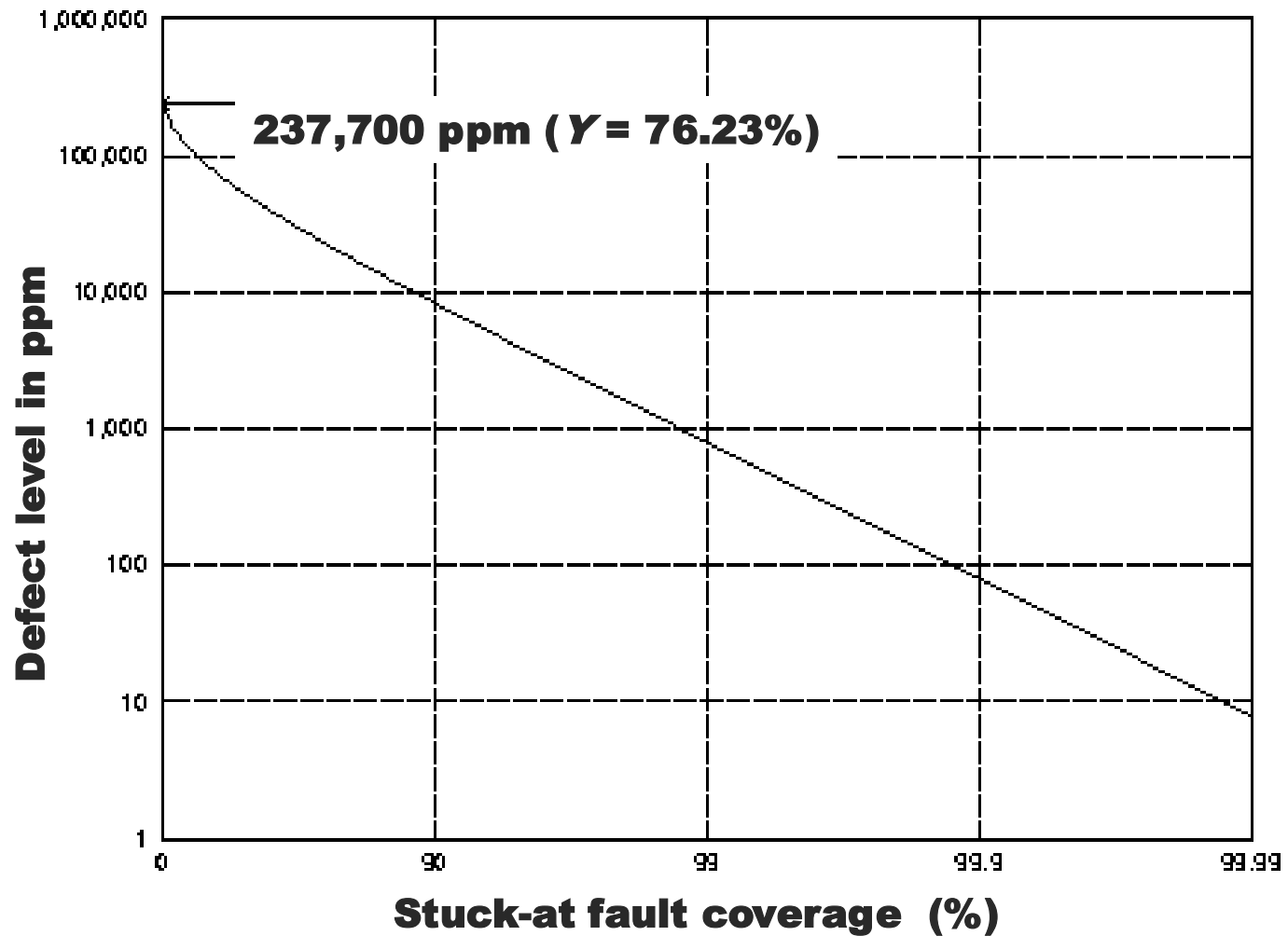
Measured Chip Fallout



Model Fitting



Computed DL



Summary

- VLSI yield depends on two process parameters, defect density (d) and clustering parameter (α).
- Yield drops as chip area increases; low yield means high cost.
- Fault coverage measures the test quality.
- Defect level (DL) or reject ratio is a measure of chip quality.
- DL can be determined by an analysis of test data.
- For high quality: $DL \ll 500$ ppm, fault coverage $\sim 99\%$

Two Problems to Solve

- 1. Using the expression for defect level on Slide 15, derive test coverage (T) as a function of fault clustering parameter (β), defect level (DL), and average number of faults (Af) on a chip.**
- 2. Find the defect level for:**
 - Fault density, $f = 1.45$ faults/sq. cm
 - Fault clustering parameter, $\beta = 0.11$
 - Chip area = 1 cm^2
 - Fault Coverage, $T = 95\%$

Solution to Problem 1

Defect level, DL, is given on Slide 15, as follows:

$$DL = 1 - [(\beta + T Af) / (\beta + Af)]^\beta$$

where T is the fault coverage, Af is the average number of faults on a chip of area A, and β is a fault clustering parameter. Further manipulation of this equation leads to the following result:

$$(1 - DL)^{1/\beta} = (\beta + T Af) / (\beta + Af)$$

or
$$T = [(\beta + Af)(1 - DL)^{1/\beta} - \beta] / (Af) \times 100 \text{ percent}$$

Solution to Problem 2

Defect level, DL, as given on Slide 15, is:

$$DL(T) = 1 - [(\beta + T Af) / (\beta + Af)]^\beta$$

Substituting,

- Fault density, $f = 1.45$ faults/sq. cm
- Fault clustering parameter, $\beta = 0.11$
- Chip area = 1 cm^2
- Fault Coverage, $T = 95\%$

We get,

$$DL(T) = 0.00522 \text{ or } 5,220 \text{ parts per million}$$

Gordon E. Moore

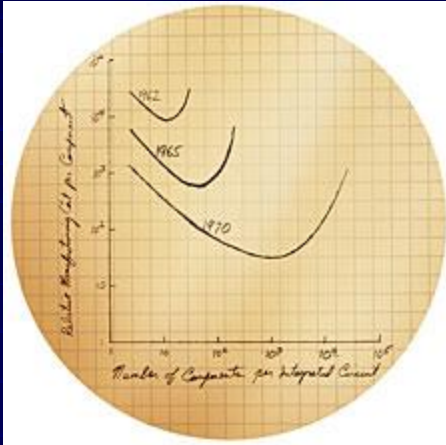


1965

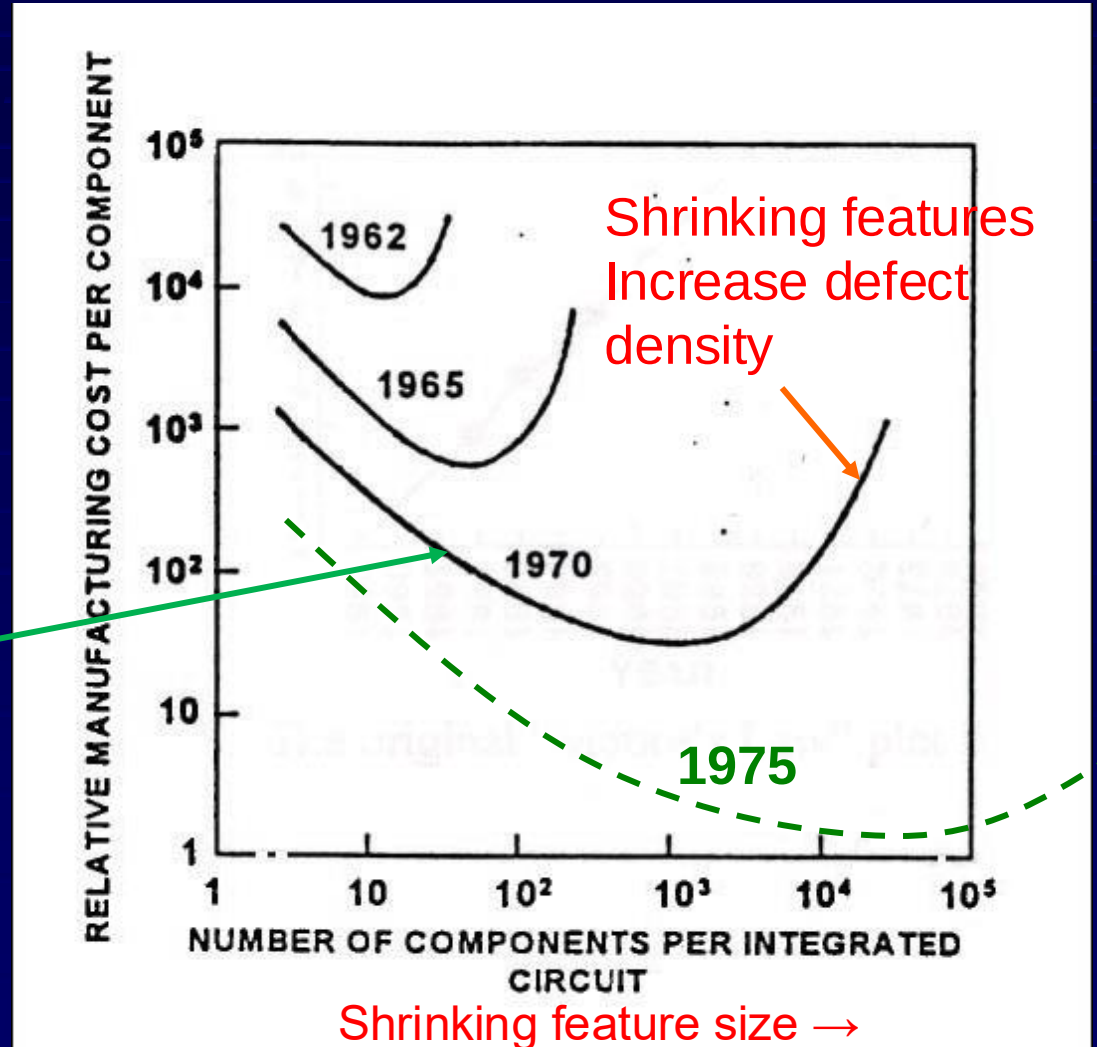
- “Cramming More Components onto Integrated Circuits,” *Electronics*, vol. 38, no. 8, April 19, 1965.
- The complexity for minimum component costs has increased at a rate of roughly a factor of two per year (see graph on next page). Certainly over the short term this rate can be expected to continue, if not to increase. Over the longer term, the rate of increase is a bit more uncertain, although there is no reason to believe it will not remain nearly constant for at least 10 years. That means by 1975, the number of components per integrated circuit for minimum cost will be 65,000.

I believe that such a large circuit can be built on a single wafer.

Moore's 1965 Graph



Components per unit area
Increased by reducing
Feature size



Economics of VLSI

- Assume wafer size and processing cost are fixed.

- Increasing chip area will:

- Reduce number of chips on a wafer
- Reduce yield
- Increase chip cost

$$\text{Chip cost} = \frac{\text{Cost of fabricating and testing a wafer}}{\text{Yield} \times \text{Number of chip sites on the wafer}}$$

- Increase functionality also
- Cost per function should be minimized.

Increasing Functionality/Area

- Increase number of transistors per unit area:
 - Shrink feature size (λ)
 - Each generation of VLSI technology reduces λ by a factor $1/\sqrt{2} = 0.707$, doubling number of transistors per unit area
 - Assume defect density $d = k/\lambda$, where k is a constant of proportionality (technology parameter)
 - Area of a component (transistor) is $k'\lambda$, where k' is a constant (technology parameter)
 - On chip of area A , number of transistors $t = A/(k'\lambda)$

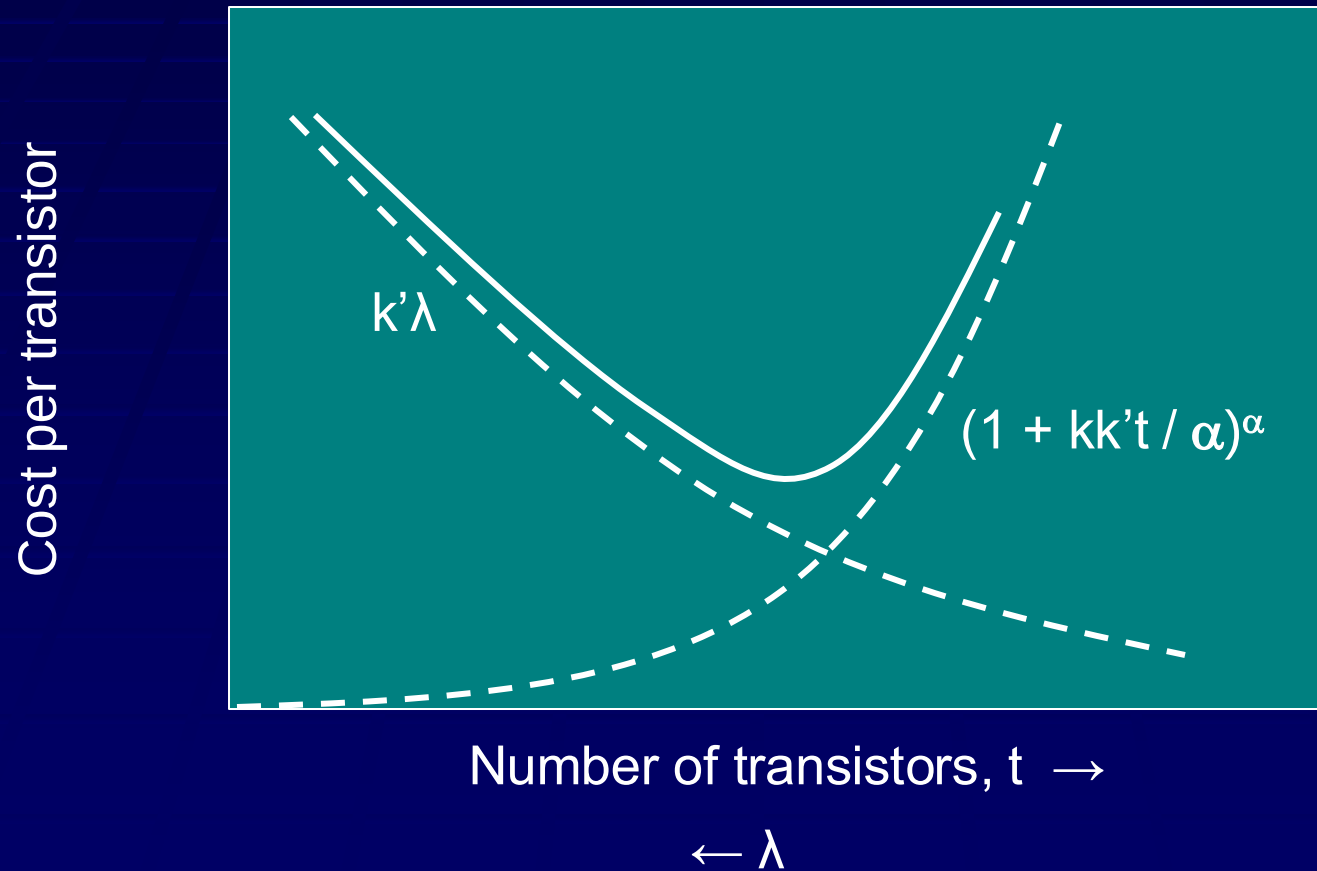
Component (Transistor) Cost, C_t

- C_t = Chip cost/Number of transistors

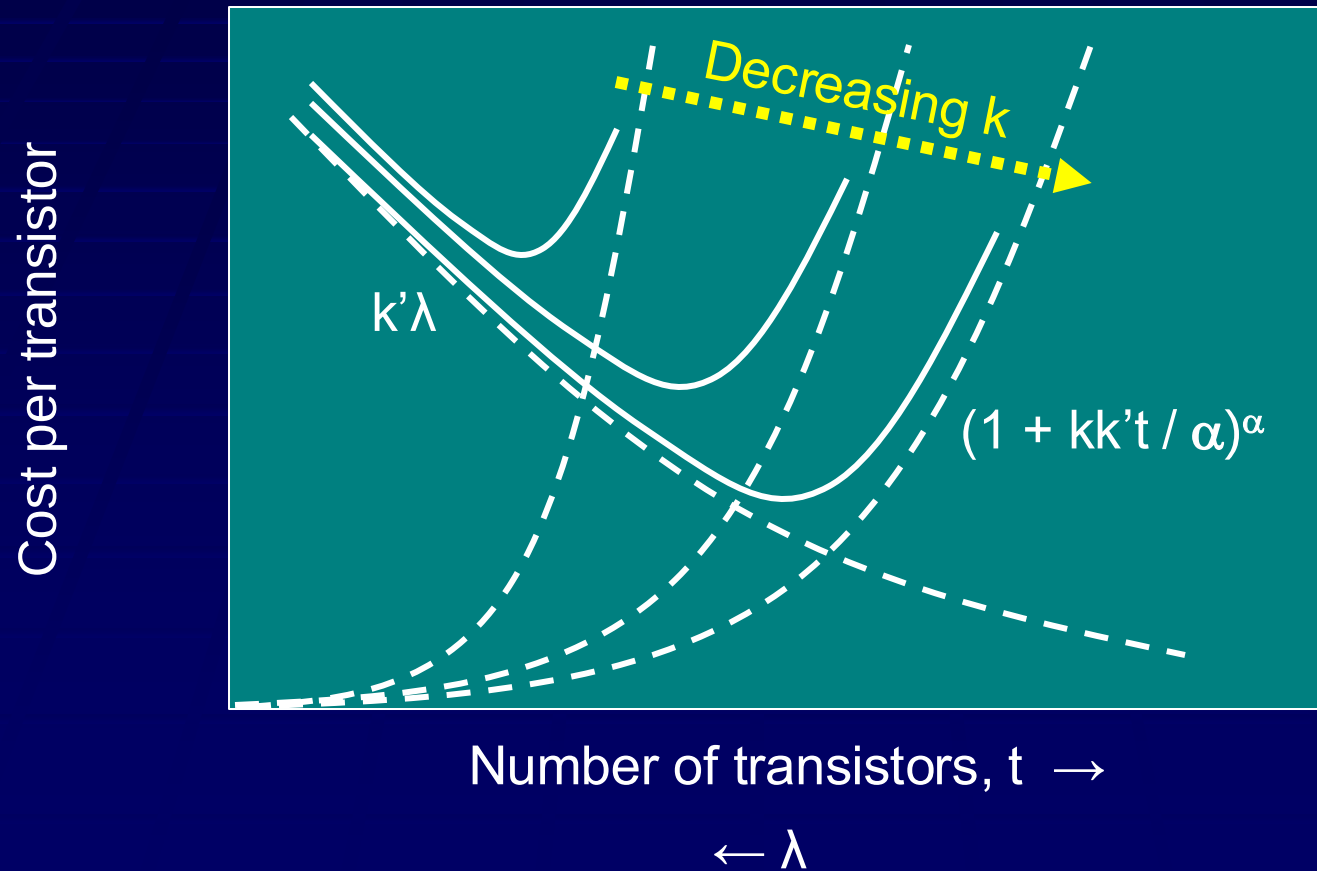
$$= \frac{\text{Wafer cost } (W_c)}{Y \times \text{Wafer area } (W_a)/A} \cdot \frac{k'\lambda}{A}$$

$$= \frac{W_c k'\lambda}{W_a (1 + A_d/\alpha)^{-\alpha}} = \frac{W_c}{W_a} k'\lambda (1 + k k't / \alpha)^\alpha$$

Component Cost



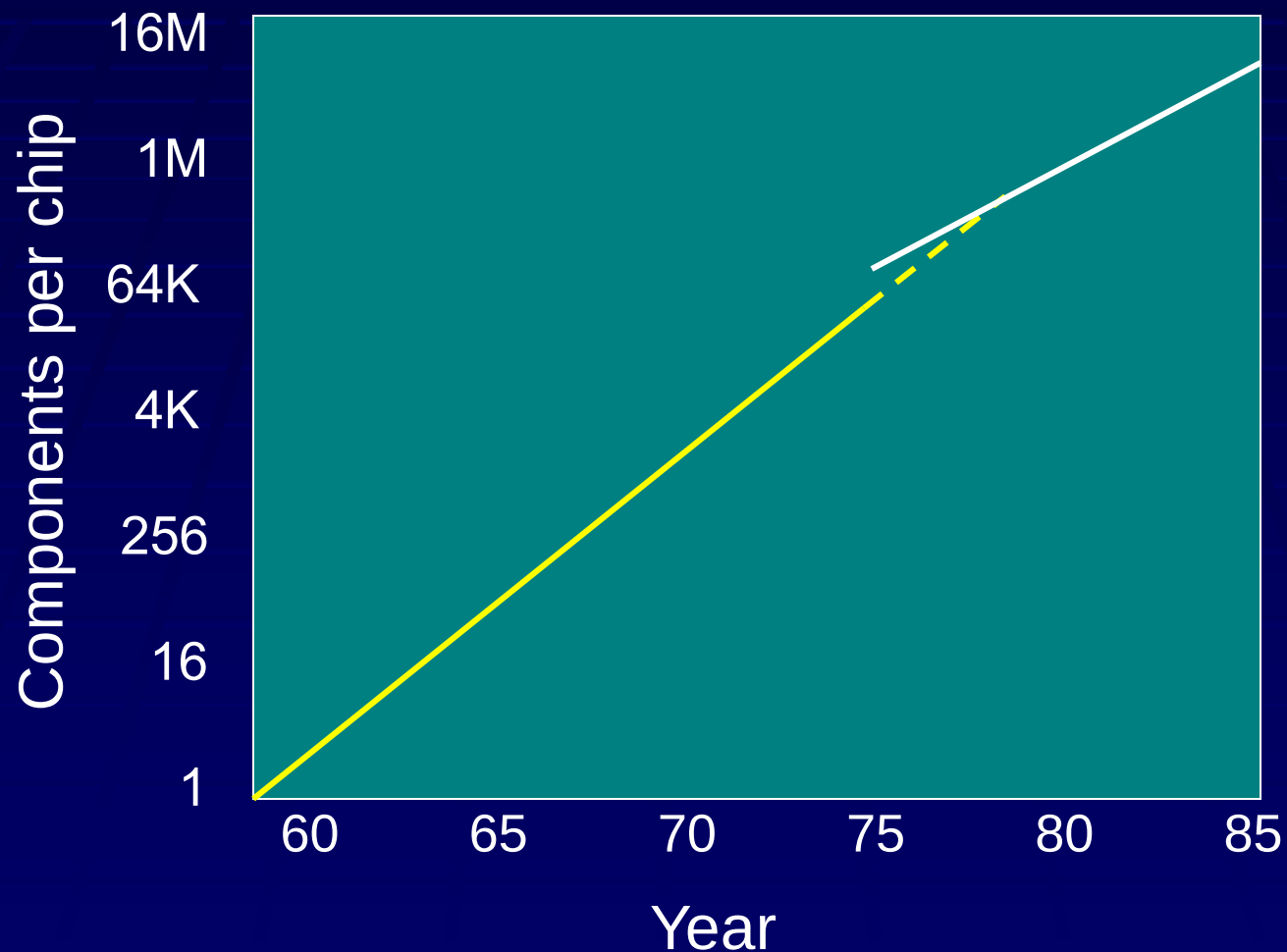
Technology Advances



1975

- “Progress in Digital Integrated Electronics,” *IEDM Tech. Digest*, 1975, pp. 11-13.
- . . . the rate of increase of complexity can be expected to change slope in the next few years as shown in Figure 5. The new slope might approximate a doubling every two years, rather than every year, by the end of the decade.

Figure 5 of Moore's 1975 Paper



1995

- “Lithography and the Future of Moore’s Law,” *Proc. SPIE*, vol. 2437, May 1995.
- By making things smaller, everything gets better simultaneously. There is little need for trade-offs. The speed of our products goes up, the power consumption goes down, system reliability, as we put more of the system on a chip, improves by leaps and bounds, but especially the cost of doing thing electronically drops as a result of the technology.

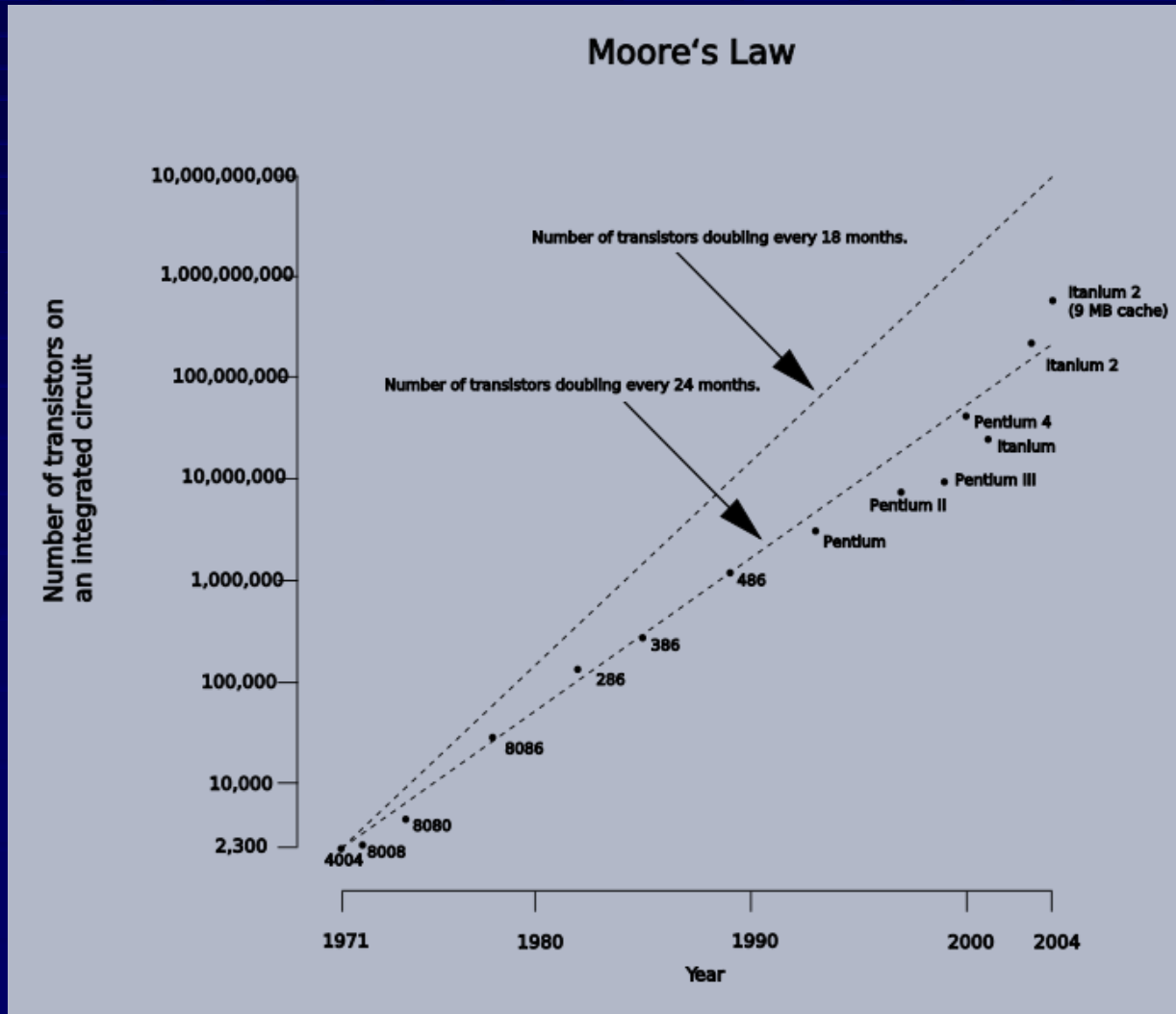
(SPIE – Society of Photonic Instrumentation Engineers)

Also in the 1995 Paper

. . . I have no idea what will happen beyond 0.18 microns.

In fact, I still have trouble believing we are going to be comfortable at 0.18 microns using conventional optical systems. Beyond this level, I do not see any way that conventional optics carries us any further. Of course, some of us said this about the one micron level. This time, however, I think there are fundamental materials issues that will force a different direction. The people at this conference are going to have to come up with something new to keep us on the long term trend.

Moore's Law



Source: Wikipedia

2016

- Problems arising with technology advances:
 - High power consumption
 - Power density
 - Leakage
 - Process variation – larger as a fraction of feature size
 - Increased noise sensitivity
- Problems with design:
 - Verification of correctness – logic and timing
 - Ensuring reliable operation
 - Testing complexity – time and cost